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Methodological Approaches to Phylogenetics and Macroevolution: Integrating Fossil and Extant Data in a Case Study of Cricetinae (Rodentia)

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Abstract

Hamsters (Cricetinae, Rodentia), widely known in science as laboratory animals and in public as popular pets, are mouse- to rat-sized rodents characterized by their internal cheek pouches and short tails. The group currently comprises 19 extant species distributed across the Palearctic realm, but its known fossil diversity, which dates back to the Middle Miocene, is substantially higher. Nonetheless, the evolutionary history of hamsters, including the phylogenetic relationships between their extinct and extant members, as well as the processes and drivers that shaped their diversification remain largely understudied. The main goal of this thesis is to contribute to filling these gaps, with an emphasis on incorporating fossil data. First, I reconstructed a robust phylogenetic tree of hamsters, based on their molar morphologies. With the insights gained in the classification of hamsters, I was able to contribute to several longstanding taxonomic debates concerning extinct hamsters. By integrating molecular data from extant species, I performed the first total-evidence dating analysis on hamsters, revealing a Pliocene age of its crown group, which is markedly younger than previously estimated. Based on this phylogenetic hypothesis, I traced the biogeographic history of hamsters and found support for a geographic origin of the group in Central to Eastern Europe. Additionally, I compared several approaches for reconstructing paleodiversity and macroevolutionary rate dynamics based on hamster phylogenetic trees and fossil occurrence data, which I compiled for this purpose. The results evidence a hump-shaped diversity trajectory, with a peak of approximately 35 contemporaneous hamster species during the Messinian and Pliocene. The inferred speciation rate is relatively constant but slightly decreasing through time and depends mainly on geographical distribution, as well as global paleotemperature. I found no evidence for an impact of these predictors on the extinction rates, suggesting a certain degree of resilience in hamsters against unfavorable environmental conditions. The differences in the estimated extinction rate and paleodiversity between the different approaches highlight the difficulty of accounting for the incompleteness and biases inherent in the fossil record.

Résumé

Les hamsters (Cricetinae, Rodentia), largement connus dans le monde scientifique comme animaux de laboratoire et auprès du grand public comme animaux de compagnie populaires, sont des rongeurs de la taille d'une souris ou d'un rat, caractérisés par leurs poches internes dans les joues et leur queue courte. Le groupe comprend actuellement 19 espèces existantes réparties dans le domaine paléarctique, mais la diversité fossile connue, qui remonte au Miocène moyen, est nettement plus importante. Néanmoins, l'histoire évolutive des hamsters, y compris les relations phylogénétiques entre leurs membres éteints et existants, ainsi que les processus et les facteurs qui ont façonné leur diversification, restent largement méconnus. L'objectif principal de cette thèse est de contribuer à combler ces lacunes, en mettant l'accent sur l'intégration des données fossiles. Tout d'abord, j'ai reconstruit un arbre phylogénétique robuste des hamsters, basé sur la morphologie de leurs molaires. Grâce aux connaissances acquises dans le cadre de la classification des hamsters, j'ai pu contribuer à plusieurs débats taxonomiques de longue date concernant les hamsters éteints. En intégrant les données moléculaires des espèces existantes, j'ai réalisé la première analyse de datation « total-evidence » des hamsters, révélant un âge pliocène de son groupe couronne, nettement plus jeune que ce qui avait été estimé précédemment. Sur la base de cette hypothèse phylogénétique, j'ai retracé l'histoire biogéographique des hamsters et trouvé des éléments corroborant l'origine géographique du groupe en Europe centrale et orientale. J'ai également comparé plusieurs approches de reconstruction de la paléodiversité et de la dynamique des taux macroévolutifs à partir d'arbres phylogénétiques des hamsters et de données sur la présence de fossiles, que j'ai compilées à cette fin. Les résultats montrent une trajectoire de diversité en forme de bosse, avec un pic d'environ 35 espèces de hamsters contemporaines pendant le Messinien et le Pliocène. Le taux de spéciation déduit est relativement constant, mais diminue légèrement au fil du temps et dépend principalement de la répartition géographique, ainsi que de la paléotempérature mondiale. Je n'ai trouvé aucune preuve d'un impact de ces prédicteurs sur les taux d'extinction, ce qui suggère un certain degré de résilience des hamsters face à des conditions environnementales défavorables. Les différences entre les taux d'extinction et la paléodiversité estimés selon les différentes approches soulignent la difficulté de tenir compte du caractère incomplet et des biais inhérents aux archives fossiles.

Table of Contents

Chapter 1. Introduction	1
1.1 Cricetinae – the subfamily of true hamsters	1
1.2 Reconstructing phylogenies including fossil taxa	5
1.3 Macroevolutionary analyses based on fossil occurrences and phylogenies	12
1.4 Main objectives and methodologies of the following chapters	14
Chapter 2. Phylogenetic relationships of Neogene hamsters (Mammalia, Rodentia, Cricetinae) revealed under Bayesian inference and maximum parsimony	15
Chapter 3. Total-evidence phylogeny reveals recent crown group radiation and biogeographical history of hamsters	45
3.1 Introduction	48
3.2 Material and Methods	50
3.3 Results	54
3.4 Discussion	56
Chapter 4. Exploring hamster diversification and diversity with Bayesian and deep learning approaches	77
4.1 Introduction	80
4.2 Material and Methods	82
4.3 Results	89
4.4 Discussion	95
Chapter 5. Discussion	105
5.1 Building and using a robust phylogeny of extinct and extant hamsters	105
5.2 Reconstructing macroevolutionary rate dynamics of hamsters	109
5.3 Challenges in integrating fossils into phylogenies	114
5.4 Conclusion	116
Chapter A. Appendix	119
A.1 Bayesian tip-dated timeline for diversification and major biogeographic events in Muroidea (Rodentia), the largest mammalian radiation	119
A.2 French Summary	140
References	147
Supplementary Material	185

Chapter 1

Introduction

1.1 Cricetinae – the subfamily of true hamsters

Cricetinae, commonly known as hamsters, form a subfamily within the family Cricetidae, one of the most diverse groups of mammals. With over 700 extant species, Cricetidae is the second-largest mammalian family and the sister group to Muridae, the largest. Together, these two families include the vast majority of species within the superfamily Muroidea, which includes more than 1,695 species and accounts for over a quarter of all mammalian diversity (Burgin et al. 2018). Within Cricetidae, molecular studies have identified five subfamilies (Fig. 1.1): the New World rats and mice (Sigmodontinae, Neotominae, and Tylomyinae) and the Old World true hamsters, and voles and lemmings (Cricetinae and Arvicolinae), the latter forming a sister-group relationship (Musser and Carleton, 2005; Neumann et al., 2006; Steppan and Schenk, 2017). While Arvicolinae have achieved broad diversity across the Northern Hemisphere, Cricetinae remain confined to the Palearctic region, inhabiting a wide range of environments from steppes and grasslands to deserts and urban areas (Pardiñas et al., 2017; Kryštufek and Shenbrot, 2025). Cricetine size ranges from ~30 cm in the European or Common Hamster (*Cricetus cricetus*), down to ~7 cm in the Roborovski Dwarf Hamster or Desert Hamster (*Phodopus roborovskii*) (Ross 1994; Kryštufek et al. 2020). Hamsters have an opportunistic, omnivorous diet, including weeds and grains, as well as smaller animals, which they can carry in their characteristic cheek pouches to their burrows. Studies on several hamster species in the wildlife showed diurnal or crepuscular activity outside their burrows in contrast to nocturnal activity in laboratories (Fritzsche et al. 2017 and references within). They are generally well known to the public as the group includes popular pets, namely the dwarf hamsters belonging to the genus *Phodopus*, the Syrian or Golden Hamster (*Mesocricetus auratus*), and the Chinese hamster (*Cricetulus (barabensis) griseus*). In science, the latter two have been used for decades as laboratory animals in medicinal or behavioral studies (Cohen et al. 1961; Valentine et al. 2012; Yapijakis et al. 2019; Wang and Cormier 2022). Additionally, the European Hamster has been the focus of studies on population dynamics and the conservation of endangered species,

as its numbers were recently drastically reduced, due to hunting and habitat reduction (Feoktistova et al. 2013, 2023; La Haye et al. 2014).

The status of hamsters as a subfamily within the Cricetidae is today generally accepted (Musser and Carleton 2005; Pardiñas et al. 2017a; Kryštufek and Shenbrot 2025), owing to studies of the molecular phylogenetics of muroid rodents (Michaux et al. 2001; Steppan et al. 2004). Regarding their internal systematics, there have been a few phylogenetic analyses based on morphological and karyotypical characters (summarized in Kryštufek and Shenbrot 2025). In recent years, however, the most important advances in hamster systematics have been achieved through analyses of selected nuclear and mitochondrial gene sequences (Neumann et al. 2006; Lebedev et al. 2018), and more recently, through studies including genome- or mitogenome-wide data (Ding et al. 2020; Pan et al. 2025). Summarizing these and additional phylogenetic studies (Romanenko et al. 2021; Jiang et al. 2024), 19 species in 9 or 10 genera (Kryštufek and Shenbrot 2025; Pan et al. 2025) are currently recognized, with phylogenetic relationships as shown in Figure 1.1. Taxonomic discrepancies in the recent literature concern *Phodopus* (partly *Cricetiscus* in Kryštufek and Shenbrot 2025), *Cricetulus barabensis griseus* (see Poplavskaya et al. 2019; Ivanova et al. 2022), and *Tscherskia collina* (see Jiang et al. 2024; Pan et al. 2025).

Besides their internal cheek pouches and short tails, hamsters are characterized by several features related to the digestive and circulatory systems, as well as cranial and skeletal traits (Kryštufek and Shenbrot 2025). However, the most important characters for taxonomy, when working with fossil rodents in general and hamsters in particular, are found in the teeth. In fact, molars are often the only remains available for extinct hamsters (Fig. 1.2) and they have a long history of being used in classifications of both extant and extinct hamsters (e.g., Schaub 1930; Argyropulo 1933). The molars of cricetines show the so called ‘cricetid plan’ which refers to brachydont, rooted molars with cusps arranged in two longitudinal rows and usually bifurcated anterocones on the first upper molar (Flynn et al. 1985; Kryštufek and Shenbrot 2025). The different genera and species of extinct hamsters are distinguished by details in their dental morphology, as depicted in Figure 1 of Chapter 2. The ‘cricetid dental plan’ is found in a variety of fossil muroid groups, which led to extensive debates about the taxonomic scope of the Cricetidae and to several changes regarding the rank, position and content of fossil but also extant hamsters (e.g., Ellerman 1941; Simpson 1945; Mein and Freudenthal 1971; Chaline et al. 1977; McKenna and Bell 1997; Kálin 1999).

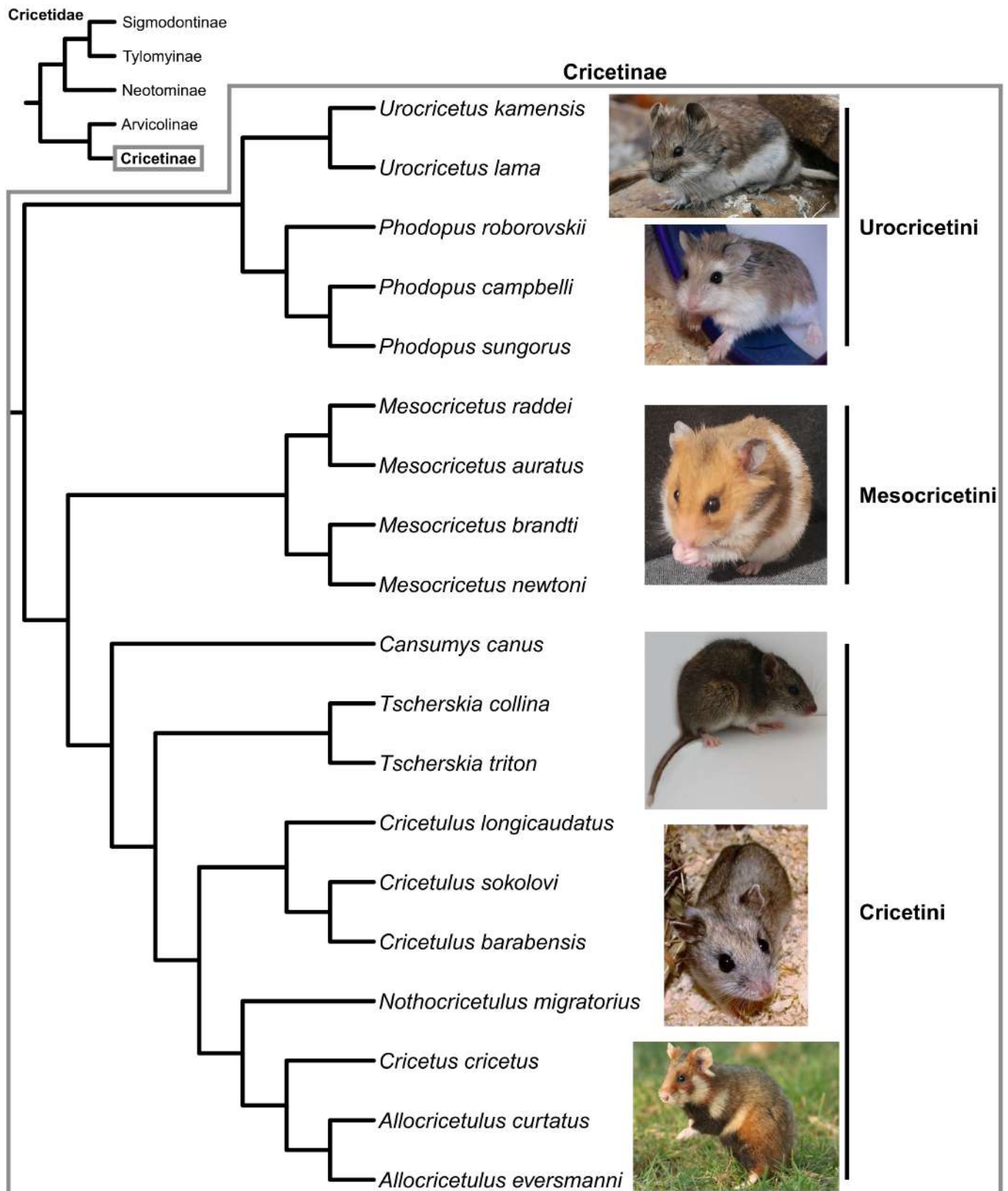


Figure 1.1. Cladogram showing the currently accepted phylogenetic relationships of the group of extant Cricetidae (after Steppan and Schenk 2017) and Cricetinae (after Lebedev et al. 2018; Pan et al. 2025), including their tribes (after Kryštufek and Shenbrot 2025). Depicted taxa from top to bottom: *Urocrinetus kamensis*, *Phodopus roborovskii*, *Mesocricetus auratus*, *Tscherskia triton* (photo by Irina V. Kartavtseva, in Kryštufek and Shenbrot 2025), *Cricetulus barabensis*, *Cricetus cricetus*.

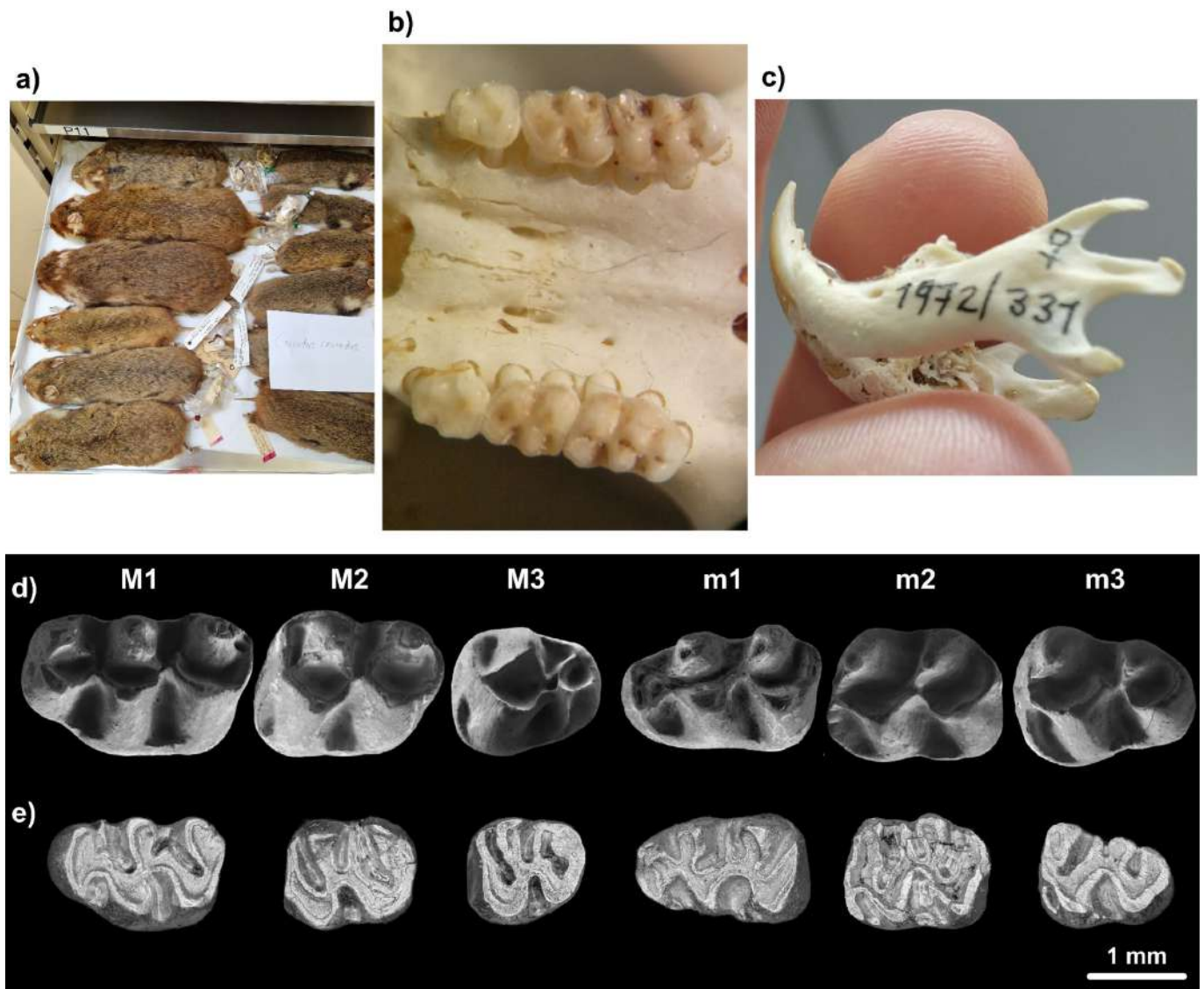


Figure 1.2. Extant hamsters in the collection of the **a)** MNHN (*C. cricetus*) and **b–c)** SNSB-ZSM (**b)** *P. sungorus*, maxilla; **c)** *M. auratus* mandible). Fossil upper (M1–M3) and lower (m1–m3) molars of **d)** †*Apocricetus alberti* (adapted from Mansino et al. 2014) and **e)** †*Collimys hiri* SNSB-BSPG 1973 XIX 182–185, 265, 266 (photos by Imelda Hausmann, SNSB-BSPG).

In this thesis, the group of Cricetinae is defined following the definition outlined in Chapter 2 (Dirnberger et al. 2024b), with *Collimys* as the earliest known member (see also Fejfar et al. 2011). Democricetinae, which are sometimes included in Cricetinae (e.g., Mein and Freudenthal 1971), are here seen as stem Cricetidae and potential ancestors of the Cricetinae, following Chapter A.1 (López-Antoñanzas et al. 2024). Based on this definition, the fossil record of the group extends back to the Middle Miocene (Kälin 1999; Prieto and Rummel 2016). The group includes around 90 known fossil species (see Supplementary Material of Chapter 3) and thousands of fossil occurrences (see Supplementary Material of Chapter 4), with a biogeographical distribution that is largely unchanged compared to that of extant hamsters (with

a few exceptions, such as the today reduced extension to Western Europe, i.e., the Iberian Peninsula or Italy).

Due to their abundance and high evolutionary rate during the Late Miocene, Pliocene and Pleistocene, fossil Cricetinae play an important role in European and Asian biostratigraphy (Tchernov 1998; Hír 2002; Casanovas-Vilar et al. 2016; Wu and Flynn 2017). In contrast, attempts to reconstruct phylogenies of fossil hamsters have been limited in taxon sampling, including only one or two genera, and exclusively relying on maximum parsimony approaches (Cuenca-Bescós 2003; López-Antoñanzas et al. 2014; Sinitsa and Delinschi 2016; Feroz et al. 2024). Additionally, divergence dates of hamsters have been estimated so far only on molecular phylogenies of extant taxa, with fossil data only used as calibration points (Neumann et al. 2006; Steppan and Schenk 2017; Lebedev et al. 2018; Jiang et al. 2024; Pan et al. 2025). Accordingly, the phylogenetic relationships among fossil hamster genera, the connection between extant hamsters and their fossil ancestors, and the timing and geographical origin of the main hamster groups have remained largely unresolved. Therefore, the main objectives of Chapters 2 and 3 of this thesis are to address these questions. In addition, the macroevolutionary patterns that shaped the group of hamsters are equally unclear, apart from a possible unspecified influence of aridification, vegetation opening and the emergence of closely related murids and arvicolines (Kälin 1999; Neumann et al. 2006; Pardiñas et al. 2017a). Consequently, the objective of Chapter 4 is to explore the diversification patterns and trends within the Cricetinae and to assess the impact of potential drivers such as paleotemperature or biogeography. By comparing several state-of-the-art methods that have rarely been applied to empirical datasets to date, we can also draw practical conclusions about these methodologies when used with groups that exhibit relatively abundant fossil records.

1.2 Reconstructing phylogenies including fossil taxa

Phylogenetic trees have been a popular approach to illustrate relationships among taxa for more than a century (Darwin 1859; Haeckel 1866). Early phylogenetic trees were created manually, not based on clearly defined and published methodology. Following the phylogenetic concepts of Hennig (1950, 1966) and the advent of computers, however, the first numerical approaches to infer phylogenies from characters were developed (summarized in Felsenstein 1982, 2004). These characters can comprise morphological traits or molecular ones, like sequences of nucleotides in DNA or RNA. In both cases it is essential to adhere to the concept of homology (Hennig 1966; Nixon and Carpenter 2012), which are those features inherited from common

ancestors. Homology assessment is achieved through careful character construction for morphological data (Simões et al. 2017) or through sequence alignment for molecular data (e.g., Edgar 2004; Katoh and Standley 2013). In this thesis, the morphological data refer to hamster molars. Although preserved skull fragments and mandibles can also provide valuable taxonomic information at various levels (Mein and Freudenthal 1971; Topachevsky and Skorik 1992; Xie et al. 2023), molars are usually the only available material for many fossil taxa, as mentioned above. Rodent teeth are thought to exhibit high degrees of homoplasy, as they are constrained by narrow functional requirements related to occlusion and diet (Jansa and Weksler 2004; Lazzari et al. 2008; Ungar 2010). Therefore, homologous structures have to be carefully identified and coded (e.g., Barbière et al. 2019; López-Antoñanzas and Peláez-Campomanes 2022). There are also certain pitfalls when working with molecular data (e.g., incomplete lineage sorting, horizontal gene transfer), which are usually mitigated by using multiple genes or genome-wide data (e.g., Liu et al. 2022). In Chapter 3 of this thesis, a relatively small set of genes is analyzed. This selection is justified by their broad taxonomic coverage for hamsters, as they have been used in previous studies (Neumann et al. 2006; Lebedev et al. 2018) and by the highly congruent results obtained from a recent genome-wide analysis (Pan et al. 2025).

Among the earliest approaches to numerically reconstruct phylogenies is the method of maximum parsimony. The principle of parsimony states that the best phylogenetic tree is the one that explains the observed differences among taxa with the fewest evolutionary changes (Edwards and Cavalli-Sforza 1963, 1964). Several methods and algorithms were developed in the following years to accommodate both morphological and molecular data (e.g., Camin and Sokal 1965; Farris 1970; Fitch 1971). The approach is still widely used today, especially for morphological data (Goloboff et al. 2018, 2019). The here used program TNT (Tree analysis using New Technology) (Goloboff and Morales 2023) allows also to incorporate character weighting to account for homoplastic characters (Farris 1969; Goloboff 1993; Goloboff et al. 2008) and to assess support for the recovered topology by e.g., bootstrapping (Felsenstein 1985) or Bremer support (Bremer 1988).

Other widely used methods of phylogenetic inference are based on probabilistic modelling. Maximum likelihood approaches were already applied early in the history of phylogenetic inference (Edwards and Cavalli-Sforza 1964; Felsenstein 1973, 1981), whereas Bayesian inference was adopted later, due to computational difficulties (Rannala and Yang 1996). Since phylogenetic Bayesian approaches form the focus of this thesis, they are explained in more detail in the following. One of the main advantages of Bayesian inference is the direct quantification of uncertainty in the results. In maximum likelihood approaches, the specific tree

with the highest likelihood, the probability of the data given the tree is sought ($\text{Pr}[\text{Data} \mid \text{Tree}]$), whereas in Bayesian inference, the probability of the tree given the data is calculated ($\text{Pr}[\text{Tree} \mid \text{Data}]$, the posterior probability distribution). This is possible thanks to the so-called Bayes' theorem, which can be formulated for phylogenetic inferences as: $\text{Pr}[\text{Tree} \mid \text{Data}] = \frac{\text{Pr}[\text{Data} \mid \text{Tree}] \times \text{Pr}[\text{Tree}]}{\text{Pr}[\text{Data}]}$ (Huelsenbeck et al. 2001). The equation includes $\text{Pr}[\text{Tree}]$, the prior probability of the phylogeny and $\text{Pr}[\text{Data}]$, the marginal probability of the data. The prior probability represents the knowledge about the tree before considering the data, while the marginal probability represents the probability of the data, averaged over all possible parameter values in the model (here the tree) and following the priors (Oaks et al. 2019). Even in the simplest case, calculating the marginal probability requires summing over all possible topologies and performing a multidimensional integration over all possible branch lengths (Huelsenbeck et al. 2001), which explains the computational difficulties encountered in early Bayesian inferences. These challenges have been addressed by Markov Chain Monte Carlo (MCMC) methods, primarily the Metropolis-Hastings algorithm (Metropolis et al. 1953; Hastings 1970), which is widely used in phylogenetics (Yang and Rannala 1997; Barido-Sottani et al. 2024). It enables estimating the posterior probability without having to calculate the marginal probability. Starting from a randomly drawn value for each parameter, slightly changed values are proposed for one or multiple parameters, and are either accepted or rejected according to the Metropolis-Hastings algorithm. This algorithm is based only on the ratio of the corresponding posterior probabilities (R) of the starting parameter values and the proposed values. Proposed values are always accepted if the resulting posterior probability is higher than before ($R > 1$). If the posterior probability is smaller than before ($R < 1$), the new values are only accepted with a certain probability (R) (Barido-Sottani et al. 2024). Either the accepted slightly changed parameter values or, if rejected, the original values are then used as starting points for the next step. Repeating this procedure a sufficient number of times, while sampling the parameter values, produces an approximation of the posterior. For the calculation of the so-called posterior odds ratio R , which is required for the Metropolis-Hastings algorithm, the marginal probability cancels out. Consequently, MCMC allows obtaining the posterior distribution of the parameter values without having to calculate the marginal probability.

The formula given above to illustrate the Bayes' theorem in the context of phylogenetic reconstructions is highly simplified. In practice, the underlying model has three parts addressing different aspect of phylogenetic reconstruction and additional data besides morphological and/or molecular characters can be incorporated (Warnock and Wright 2021; Mulvey et al.

2025b). While there are many variations of these models, in this thesis, the goal is total-evidence dating (TED), which integrates extinct and extant taxa with morphological and molecular data within a single framework to produce dated trees (Ronquist et al. 2016). When reconstructing undated trees, either with maximum parsimony or possible with Bayesian inference, branch lengths reflect the expected changes or substitutions per character or site. To estimate divergence times or to perform several downstream analyses based on phylogenies, the tree must be dated with branch lengths reflecting time. The resulting Bayes' theorem incorporating these different aspects for dated trees including fossil taxa is represented in Figure 1.3. Dating can also be performed a posteriori (see e.g., Bell and Lloyd 2015), which allows also maximum parsimony trees to be reflecting time instead of expected character changes.

The first part of the underlying model, equally important to reconstruct both undated and dated phylogenies, is the substitution model, that describes the evolution of characters along the phylogeny. A wide variety of approaches exist to model the substitution between nucleotide bases, as nucleotide data exhibits well-defined properties. The two important parameters are the exchangeability, the probability of change from one specific nucleotide to another and the equilibrium frequency of the nucleotides. Different assumptions about these two parameters can be combined, e.g., the exchangeability and the equilibrium frequency is the same for all bases (Jukes and Cantor 1969), or different for all bases (Tavaré 1986), or in-between, following for example the different transition/transversion rates of nucleotides (Kimura 1980; Felsenstein 1981; Hasegawa et al. 1985; Zharkikh 1994). As the resulting substitution rates can vary across sites, impacting phylogenetic inferences (Yang 1996), substitution models were extended to account for this variation using a discrete Gamma distribution (Yang 1993). For discrete morphological data developing substitution models is more challenging due to differences between characters and their underlying processes in one dataset (Wright 2019). Therefore, the rather simple Mk model (Lewis 2001) is commonly used, which adopts assumptions of the molecular Jukes-Cantor model (Jukes and Cantor 1969) for morphological data, i.e., equal transition rates between states and equal state frequencies. The model has been further extended to account for ascertainment (or acquisition) bias of only including variable characters (Lewis 2001) and to incorporate among-character rate variation, similarly as with molecular data (Yang 1993). Despite its simplicity, the model performs well in many cases (Mulvey et al. 2025a). The incorporation of morphological data is especially important when working with fossils, as molecular data are usually unavailable for them. Both data types can and possibly should be combined when working with only extant or both extant and extinct taxa (Nylander et al. 2004; Cabra-García and Hormiga 2020; Keating et al. 2023). Even when morphological characters

are clearly outnumbered by molecular site, as it is usually the case, it has been proved that they can substantially impact phylogenies (Neumann et al. 2021).

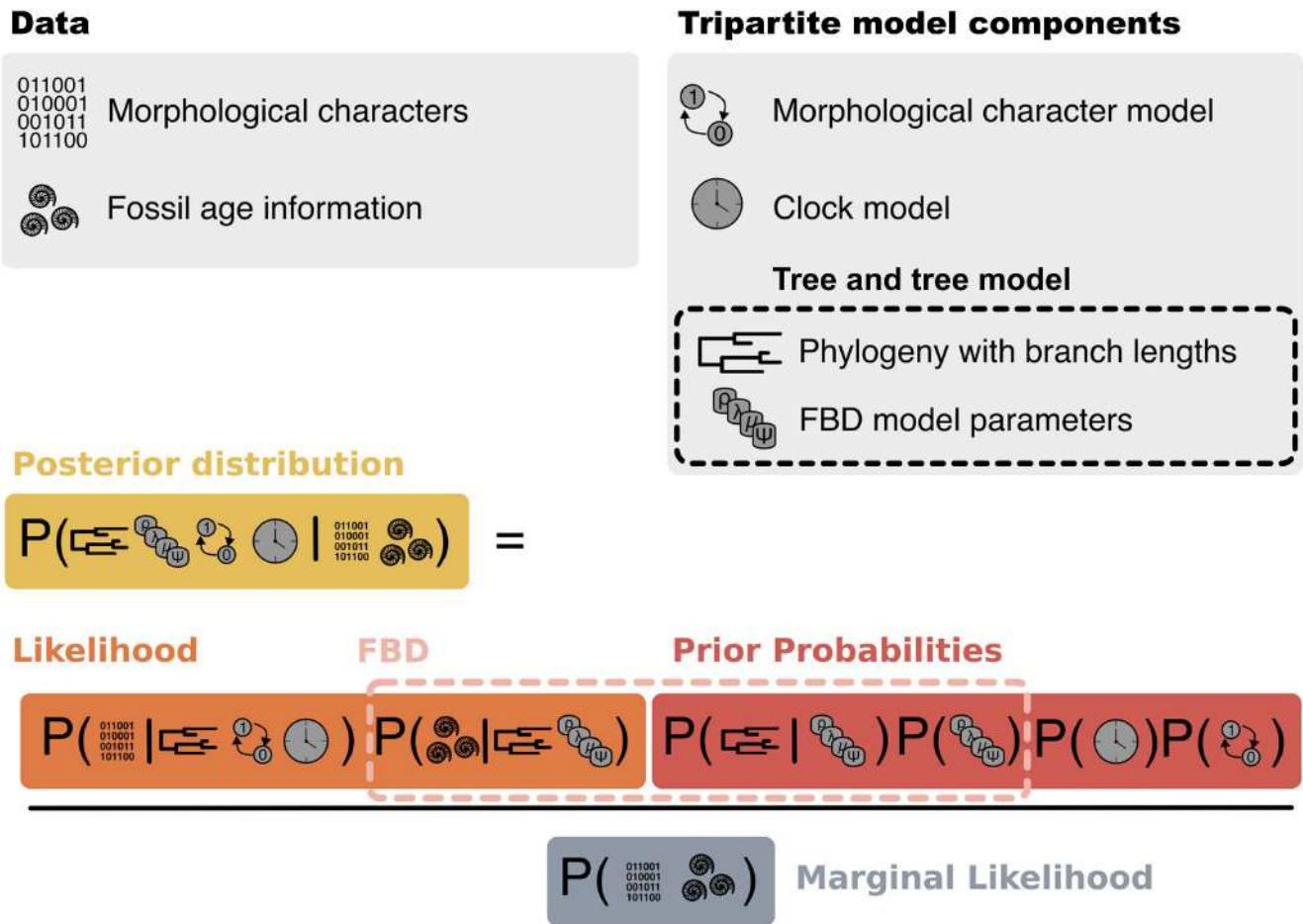


Figure 1.3. Representation of the Bayes' theorem in the case of dated phylogenetic reconstruction, based on morphological characters and the Fossilized Birth-Death (FBD) tree model. Additional to morphological data, molecular data can be analyzed in a total-evidence framework and the uncertainty in fossil ages can be considered in the use of priors. The figure is adapted from Mulvey et al. (2025b) and Warnock and Wright (2021).

The following two components of the underlying model, the tree model and the clock model (Fig. 1.3), are especially important for producing dated phylogenetic trees (Warnock and Wright 2021). The clock model is originally based on the idea of molecular clocks, which assume that evolutionary rates remain constant through time and across lineages (Zuckerkandl & Pauling, 1962). Consequently, by incorporating a clock rate, phylogenies can be reconstructed with branch lengths proportional to time, rather than representing morphological character changes or molecular substitutions. However, later studies showed that allowing evolutionary rates to vary among branches of a phylogenetic tree, known as the relaxed clock assumption, provides

a more realistic representation of many evolutionary processes (see Ho 2021 for an overview of the evolutionary clock). Different clock models allow the substitution rate to vary between lineages in different ways to relax the assumption of a constant clock. In the program for phylogenetic Bayesian inference, which is used in this thesis, MrBayes (Ronquist et al. 2012b), there are four different relaxed clock models incorporated, that are compatible with our choice of tree model (Zhang 2022). These models can be grouped into two categories. The first one consists of autocorrelated models, which assume a correlation of substitution rates between branches, enforcing more gradual changes in evolutionary rates (Thorne and Kishino 2002). The second group, comprises the uncorrelated relaxed clocks, which allow rates to vary independently between branches, enabling greater changes in evolutionary rates of neighboring branches (Drummond et al. 2006; Lepage et al. 2007). As the choice of clock model is known to impact divergence time estimates, tree topology, and the capability to retrieve direct ancestors, careful selection of the model that best fits the data has been strongly recommended (Ronquist et al. 2012a; Simões et al. 2020; Mongiardino Koch et al. 2025). Multiple, unlinked clocks can be applied within the same analysis to different partitions of the dataset, e.g., separating morphological and molecular data, constituting another aspect to consider when choosing the best fit clock model (Lee et al. 2013; Lee 2016; Simões et al. 2020; Barba-Montoya et al. 2021).

The final part of the model (Fig. 1.3) is the tree model, which specifies a distribution of tree topologies as well as node and tip ages by incorporating tree-generating processes, calibration points and potential topological constraints. Early implementations of TED used tree models with several limitations, as they did not specifically model fossil sampling and did not account for the inclusion of direct ancestors (e.g., Pyron 2011; Ronquist et al. 2012a). These limitations were overcome by the so-called Fossilized Birth-Death (FBD) model (Stadler 2010; Heath et al. 2014). In the context of macroevolution, the FBD model was initially applied without inferring the phylogenetic position of the fossil taxa in the process (Gavryushkina et al. 2014; Heath et al. 2014), but full TED analyses were subsequently performed in the following years, inferring trees and divergence times in a combined analysis (Zhang et al. 2016; Gavryushkina et al. 2017). The FBD model incorporates the widely used birth-death process (Yule 1925; Feller 1939; Kendall 1948; Thompson 1975; Nee et al. 1994; Stadler 2009), which describes the generation of phylogenies with lineages originating under a speciation rate λ , or going extinct under a extinction rate μ . As usually not all species are sampled in phylogenetic inference, sampling of extant taxa can be modeled following a (often fixed) proportion p under uniform sampling (Stadler 2009) or considering a sampling scheme that aims to maximize sampled

diversity (Höhna et al. 2011; Zhang et al. 2016). The FBD model incorporates the sampling of extinct taxa under a continuous process with a fossil sampling rate ψ (Stadler 2010), representing also the process of fossilization, excavation and identification. The last FBD parameter is the root age or the origin time (see Zhang et al. 2016; Gavryushkina et al. 2017), which is usually constrained by fossil data or previously estimated divergence dates. Another essential aspect of the tree model is the incorporation of additional temporal data as calibration points. This can be achieved through two main approaches, node- or tip-dating (reviewed in Heckeberg et al. 2025), which can also be combined in one analysis (O'Reilly and Donoghue 2016). Node calibrations have a long history in divergence time estimation, as they can also be applied to trees containing only extant taxa but they come with certain pitfalls (Parham and Irmis 2008; Parham et al. 2012; Lee and Palei 2015; Wright et al. 2022). The main one is probably the necessity to make precise assumptions about fossil positions in the tree, which can be difficult (see e.g., Kimura et al. 2015). In contrast, tip-dating uses the ages of fossils included in the phylogenetic reconstruction, which is enabled by the FBD model, and therefore reduces the need for strict a priori fossil placement assumptions. If no morphological data are available for some or all fossil taxa, topological constraints based on taxonomy can still be utilized and can also incorporate uncertainty in the placement (Barido-Sottani et al. 2023; Nikolic et al. 2025). As ages of fossil are usually uncertain, incorporating an age range is critical to avoid bias in the inferred divergence times or tree topologies (Barido-Sottani et al. 2019, 2020a).

One of the main limitations of the earliest versions of the FBD model is the assumption of constant diversification and preservation rates. Attempts to model diversification rate variation among clades or lineages, or over time, and depending on specific characters (Maddison et al. 2007; Rabosky and Lovette 2008; Alfaro et al. 2009; FitzJohn 2010; Morlon et al. 2011; Rabosky 2014), resulted in a variety of extensions to the birth-death process (summarized in Morlon et al. 2024). Most of these models consider, however, only trees of extant taxa and are often applicable only to fixed phylogenies. For phylogenetic inference based on the FBD model, piecewise changes of diversification and fossil sampling rates over time have been soon incorporated (Stadler et al. 2013; Gavryushkina et al. 2014; Zhang et al. 2016, 2023). Recent developments extended the FBD framework to accommodate across-lineage variations in diversification and/or fossil sampling rates (Barido-Sottani and Morlon 2025; Chabrol et al. 2025). Other methods to infer diversification rates including extinct taxa, are only applicable to fixed trees (e.g., Mitchell et al. 2019; Beaulieu and O'Meara 2023; do Rosario Petrucci et al. 2025; Quintero et al. 2025). Some of these approaches are explored further in Chapter 4 of this thesis, and in the next paragraph.

1.3 Macroevolutionary analyses based on fossil occurrences and phylogenies

One of the most important endeavors in biology is to connect biodiversity to evolutionary processes (Darwin 1859; Simpson 1944, 1953). In general, the interplay of speciation and extinction shapes the tree of life in all its variations in biodiversity across time and lineages (Quental and Marshall 2013; Marshall 2017; Quintero et al. 2025). Thus, studying these diversification rates, a long-standing pursuit among biologists and paleontologists (see e.g., Newell 1952), is key to understanding all kinds of heterogeneous diversity patterns and the underlying processes. Famous examples of such heterogeneous patterns include mass extinction events (Raup and Sepkoski 1982, 1984; Jablonski 2002; Cermeño et al. 2022), or the latitudinal diversity gradient (Willig et al. 2003; Powell 2007; Rolland et al. 2015; Fenton et al. 2023). Initial reconstructions of macroevolutionary rates over geological time have been based directly on the fossil record (Newell 1952; Raup 1972; Sepkoski 1981). However, the fossil record is well known to be incomplete and biased, due to several geological factors, socio-economic reasons or differences in fossilization potential between taxa (Raup 1979; Donoghue et al. 2006; Close et al. 2020; Raja et al. 2022; Krone et al. 2024). This has led to an increased emphasis on overcoming the effects of these biases (Raup 1975; Sepkoski 1984; Alroy 2010, 2014, 2015). Model-based approaches analyzing fossil occurrences with speciation and extinction as stochastic processes are usually based on the birth-death process (Raup et al. 1973; Raup 1985; Foote 2000), as discussed above. Notably, PyRate, which is among the most widely used programs to infer macroevolutionary rates on fossil occurrence data, models a preservation process to obtain estimations for the origin and extinction of species. These estimates are then used to infer speciation and extinction rates including rate shifts over time, following a birth-death process (Silvestro et al. 2014, 2019). The preservation rate can hereby vary across lineages and through time. A limitation of this model is that it assumes complete sampling for the applied birth-death process, which leads to an underestimation of true species richness and can be problematic in the case of nonrandom incomplete sampling. In such cases, accounting for incomplete sampling, for instance with a “treeless” FBD range model, can be advantageous, especially in cases of limited fossil data (Warnock et al. 2020). Other methods also aim to estimate true diversity through geological time while accounting for unsampled lineages (e.g., Starrfelt and Liow 2016; Flannery-Sutherland et al. 2022), including DeepDive which uses a deep learning model to address the incompleteness and bias of the fossil record (Cooper et al. 2024b).

Phylogenies have emerged as the second primary data source for inferring macroevolutionary rates, following the integration of the birth-death process into phylogenetics of extant taxa (Nee

et al. 1992, 1994). As explained previously, the initial assumption of constant speciation and extinction rates were relaxed subsequently which allowed to assess how diversification depends on various factors such as time, diversity, or species specific traits (Maddison et al. 2007; Rabosky 2013; Beaulieu and O’Meara 2016; Henao Diaz et al. 2019). The validity of reconstructed extinction rates based on extant-only phylogenies has been questioned, however, including identifiability concerns (Rabosky 2010; Louca and Pennell 2020, 2021). This has sparked considerable debate (Beaulieu and O’Meara 2015; Rabosky 2016; Morlon et al. 2020; Helmstetter et al. 2022; Kopperud et al. 2023) and motivated the increasing integration of approaches including fossil data. Apart from analyses, that integrated extinction rates inferred from the fossil record into their phylogenetic inferences (Upham et al. 2021; Quintero et al. 2024), there are also methods that utilize trees that include extinct taxa, predominantly under the FBD model (Beaulieu and O’Meara 2023; do Rosario Petrucci et al. 2025; Quintero et al. 2025). Conversely, approaches based on fossil occurrences have also begun to incorporate phylogenetic information (e.g., Brée et al. 2022; but see Calderón del Cid et al. 2024). A particularly useful approach is hereby the birth-death neural network (BDNN), which uses an unsupervised Bayesian neural network to incorporate not only phylogenetic relatedness but a variety of species-specific traits or time-dependent variables in the reconstruction of species-specific diversification rates (Hauffe et al. 2024).

These methods to infer diversification rates from phylogenetic trees belong to the vast field of phylogenetic comparative methods (Felsenstein 1985; Harvey and Pagel 1991; Harmon 2019). They can be used to explore various aspects of macroevolution (López-Antoñanzas et al. 2022) and several approaches based on trees that include extinct taxa are applied through this thesis, including the reconstruction of discrete ancestral states (reviewed in Revell 2025). While ancestral state reconstructions were traditionally based on maximum parsimony, assessing uncertainty under this framework is not straight forward, promoting the use of maximum likelihood approaches (see Felsenstein 2004). Here, a Mk model (Pagel 1994; Lewis 2001) is fitted to the tree and data, with different configurations of transition rates, e.g., equal rates (ER) or all rates different (ARD). The best fitting model (or a mixture) is then used for marginal or joint ancestral state estimation or for stochastic character mapping (Pagel 1999; Bollback 2006; Revell 2025). Similarly, the biogeographic history of a group can be explored by reconstructing ancestral ranges (Ree et al. 2005; Ree and Smith 2008). There are a variety of models that incorporate different combinations of anagenetic and/or cladogenetic range changes, reflecting biogeographic events, such as anagenetic dispersal and extirpation, as well as different forms of sympatry, allopatry or jump dispersals (summarized in Matzke 2013). Robust

biogeographical histories can be inferred following thorough model testing and potentially incorporating constraints on geographic connectivity, which can also change over geological time scales, (Matzke 2018, 2022).

1.4 Main objectives and methodologies of the following chapters

The overall goal of this thesis is to fill gaps in our knowledge about the evolutionary history of Cricetinae. As the known extinct diversity of hamsters far exceeds the extant one, the incorporation of fossil hamsters is an integral part of this work. The phylogenetic relationships between extinct hamsters are mostly uncertain and several taxonomical debates linger. Therefore, in the first two chapters of this work, a robust phylogeny that includes extinct hamsters is built. This phylogeny serves both to contribute to our understanding of hamster evolution and as a basis for several downstream analyses. To this end, in Chapter 2, the phylogenetic relationships of the first important radiations of European hamsters are studied based on their dental morphology. Here, the results from a maximum parsimony approach and Bayesian inference are compared and both trees are used to reconstruct ancestral dental configurations to define synapomorphies. Another key question in the group of hamsters, which has not been answered with certainty, is the phylogenetic position of the extant hamsters compared to their fossil relatives. Therefore, in Chapter 3, this phylogeny is expanded to include several Quaternary and extant hamsters, incorporating molecular data, to perform the first total-evidence dating analysis for hamsters. Based on the resulting comprehensive phylogeny, the geographical origin of hamsters and the subsequent dispersal events are investigated. Due to the limited extant diversity of hamsters and the lack of a phylogeny including extinct species, the variation in diversification of hamsters has not yet been analyzed in detail. Consequently, the factors that impacted the speciation or extinction of the group have only been vaguely understood. To answer these questions, in Chapter 4, an extensive fossil occurrence database for hamsters is compiled through a thorough literature review. Based on this dataset and the phylogenetic tree reconstructed earlier, a suite of analyses is conducted with the aim of detecting heterogeneity in the diversification rates, reconstructing the paleodiversity, and gaining insights into the abiotic or biotic drivers impacting their macroevolutionary dynamics. By comparing the results, uncertainties inherent in relying on a single approach are mitigated and conclusions can be drawn about the applicability of the different methods, in the case of hamsters. Overall, this thesis contributes to our understanding of the evolutionary pathways that have shaped the hamster group, leading to the diversity observed today.

Chapter 2

Phylogenetic relationships of Neogene hamsters (Mammalia, Rodentia, Cricetinae) revealed under Bayesian inference and maximum parsimony

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Previous phylogenetic reconstructions regarding fossil hamsters have been limited in their taxonomic sampling, focusing on one or two genera. In this chapter the relationships between seven important Neogene hamster genera are explored by comparing two phylogenetic approaches, namely maximum parsimony and Bayesian inference after thorough model testing. Additionally, rogue taxa are identified and excluded from the consensus trees to improve resolution and clade support, and the stratigraphic fit of the resulting phylogenies is evaluated. Based on the resulting trees, synapomorphies are defined for the genera that are retrieved as monophyletic following ancestral character reconstruction.

Phylogenetic relationships of Neogene hamsters (Mammalia, Rodentia, Cricetinae) revealed under Bayesian inference and maximum parsimony

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ABSTRACT

There is an ongoing debate about the internal systematics of today's group of hamsters (Cricetinae), following new insights that are gained based on molecular data. Regarding the closely related fossil cricetids, however, most studies deal with only a limited number of genera and statements about their possible relationships are rare. In this study, 41 fossil species from the Late Miocene to the Pliocene, belonging to seven extinct cricetine genera, *Collimys*, *Rotundomys*, *Neocricetodon*, *Pseudocricetus*, *Cricetulodon*, *Apocricetus* and *Hattomys* are analysed in a phylogenetic framework using traditional maximum parsimony and Bayesian inference approaches. Following thorough model testing, a relaxed-clock Bayesian inference analysis is performed under tip-dating to estimate divergence times simultaneously. Furthermore, so-called 'rogue' taxa are identified and excluded from the final trees to improve the informative value of the shown relationships. Based on these resulting trees, the fit of the topologies to the stratigraphy is assessed and the ancestral states of the characters are reconstructed under a parsimonious approach and stochastic character mapping. The overall topologies resulting from Bayesian and parsimonious approaches are largely congruent to each other and confirm the monophyly of most of the genera. Additionally, synapomorphies can be identified for each of these genera based on the ancestral state reconstructions. Only *Cricetulodon* turns out to be paraphyletic, while '*Cricetulodon*' *complicidens* is a member of *Neocricetodon*. Lastly, this work makes a contribution to a debate that went on for decades, as the genus *Kowalskia* can be confirmed as junior synonym of *Neocricetodon*.

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INTRODUCTION

Cricetidae, with more than 700 living species, is the second most speciose family inside Muroidea. According to molecular studies (*Musser & Carleton, 2005; Neumann et al., 2006; Steppan & Schenk, 2017*), it comprises the following subfamilies: the new World rats and mice (Sigmodontinae, Neotominae and Tylomyinae), the group of voles, musk rats

and lemmings (Arvicolinae), and the Old World hamsters (Cricetinae). Cricetinae (commonly known as hamsters), is a group of mouse to rat-sized rodents with cheek pouches and short tails, which comprises nowadays 18 species distributed in seven genera. Today, they live in the Palearctic realm, mostly in steppe and grassland habitats but also in desert areas and urban environments ([Pardiñas et al., 2017](#)). Over the history, hamsters have been considered either as a tribe (e.g., [Simpson, 1945](#)) or a subfamily (e.g., [Mein & Freudenthal, 1971](#)). The different taxonomic ranks attributed to this group have resulted from the lack of consensus concerning the taxonomic rank of Cricetidae, which has been classified as a subfamily inside Muridae with all its main clades treated as tribes instead of subfamilies ([McKenna & Bell, 1997](#)) or it has been considered as a family on its own ([Chaline, Mein & Petter, 1977](#)), which agrees with morphological and molecular reconstructions, with the exclusion of some genera from the group, however ([Steppan & Schenk, 2017](#); [López-Antoñanzas et al., in press](#)).

Depending on the fossils attributed to Cricetinae, its temporal range varies from the Early Miocene (e.g., [Mein & Freudenthal, 1971](#) who included *Democricetodon* within cricetines) or from the Middle Miocene (e.g., [Fejfar et al., 2011](#) with *Collimys* as the earliest cricetine) until nowadays. In this work, we consider Democricetodontinae (including *Democricetodon* and *Copemys*, among other genera) to be stem Cricetidae among which we may find its potential ancestors ([Lindsay, 2008](#); [López-Antoñanzas et al., in press](#)). Therefore, as a working hypothesis, we treat the cricetines as having a temporal range that spans from the Middle Miocene until today as considered by [Daxner-Höck \(1972\)](#) and [Fejfar et al. \(2011\)](#).

This study does not include these stem cricetids but focuses on the earliest representatives of the subfamily Cricetinae: *Apocricetus* [Freudenthal, Mein & Martín Suárez, 1998](#), *Collimys* [Daxner-Höck, 1972](#), *Cricetulodon* [Hartenberger, 1965](#), *Hattomys* [Freudenthal, 1985](#), *Neocricetodon* [Schaub, 1934](#), *Pseudocricetus* [Topachevsky & Skorik, 1992](#) and *Rotundomys* [Mein, 1965](#). Its objective is to elucidate the phylogenetic relationships inside this group, for which these early forms represent the most important initial radiation.

Previous phylogenetic reconstructions merely focused on species belonging to one or two genera and were based on maximum parsimony solely ([Cuenca Bescós, 2003](#); [López-Antoñanzas, Peláez-Campomanes & Álvarez-Sierra, 2014](#); [Sinitsa & Delinschi, 2016](#)). Moreover, molecular phylogenetic studies dealing with extant Cricetinae incorporated fossil data solely to calibrate the nodes ([Steppan, Adkins & Anderson, 2004](#); [Neumann et al., 2006](#); [Steppan & Schenk, 2017](#); [Lebedev et al., 2018](#)). However, additional approaches based on Bayesian methods have to be explored (see [López-Antoñanzas et al., 2022](#)) to shed light on the diversification processes of the studied groups and to be able to accurately estimate divergence times. Recent advances in this field include the so-called morphological clock, which refers to the rate of morphological changes through time. This rate together with the incorporation of fossils as tips, in order to calibrate the tree in a tip-dating approach, allows estimating divergence times, even in completely extinct clades ([Turner, Pritchard & Matzke, 2017](#)). The position of the fossil taxa on the tree is hereby simultaneously reconstructed. In this way, it is not necessary to rely on possibly wrong assumptions about

the position of fossil taxa, as is the case when applying the node-dating method ([Near, Meylan & Shaffer, 2005](#); [Parham & Irmis, 2008](#)). Based on these ideas, more complex ways of modelling different aspects regarding a more accurate reconstruction of phylogenetic trees have been explored. This includes for example, relaxing the morphological clock rate ([Zhang, 2022](#)), incorporating a fossilized birth-death tree model ([Stadler, 2010](#)), or accounting for different taxon sampling strategies ([Höhna et al., 2011](#)).

In this study, we present the first reconstructed phylogeny based on dental morphological data, from a selected series of Late Miocene to Pliocene cricetine genera, mainly distributed in Europe, which exhibit high levels of species diversity. We compare the results obtained by applying different phylogenetic techniques, such as maximum parsimony and Bayesian inference approaches to propose the most robust phylogenetic hypothesis. Overall, the study contributes to a better understanding of the early evolution of the group and help to clear up previous systematic and taxonomic questions.

MATERIALS AND METHODS

Upper molars are indicated with upper-case letters (M1, M2, M3), lower molars with lower-case letters (m1, m2, m3). The dental terminology used in this work is shown in [Fig. 1](#).

Taxon set

Included taxa depend on data availability and general completeness of the material. Within the seven extinct genera, *Apocricetus*, *Collimys*, *Cricetulodon*, *Hattomys*, *Neocricetodon* (including species assigned to *Kowalskia*), *Pseudocricetus* and *Rotundomys*, a total number of 41 species could be coded, which makes up around 77% of the total number of species within these genera (53), that were found in the literature. Additionally, two extant taxa were included as well, *Cricetus cricetus* ([Linnaeus, 1758](#)) and *Nothocricetulus migratorius* ([Pallas, 1773](#)). As outgroup, *Eucricetodon wangae* [Li, Meng & Wang, 2016](#) was added, as coded in [López-Antoñanzas & Peláez-Campomanes \(2022\)](#). For additional information about the included taxa, e.g., age interval, references, observed material, see [Supplemental Material S1](#).

Morphological characters and matrix construction

The matrix was constructed in Mesquite v. 3.81 ([Maddison & Maddison, 2023](#)). It is based on the morphological matrix from [López-Antoñanzas & Peláez-Campomanes \(2022\)](#), and expanded here from 82 characters to 116 characters, introducing additional characters corresponding to the structures allowing to differentiate cricetine genera and species. Four characters concern the whole molar row, six refer to morphometrics, and the remaining 106 are related to the morphology of each dental element (M1: 37; M2: 11; M3: 16; m1: 23; m2: 9; m3: 10). In cases of intraspecific variability between different locations, only the condition found in the type locality has been considered. In case of variability in the type location, only the character state present in most of the specimens was taken into account, except for specimens, for which no clear majorities were seen. The morphological matrix is provided in [Supplemental Material S2](#).

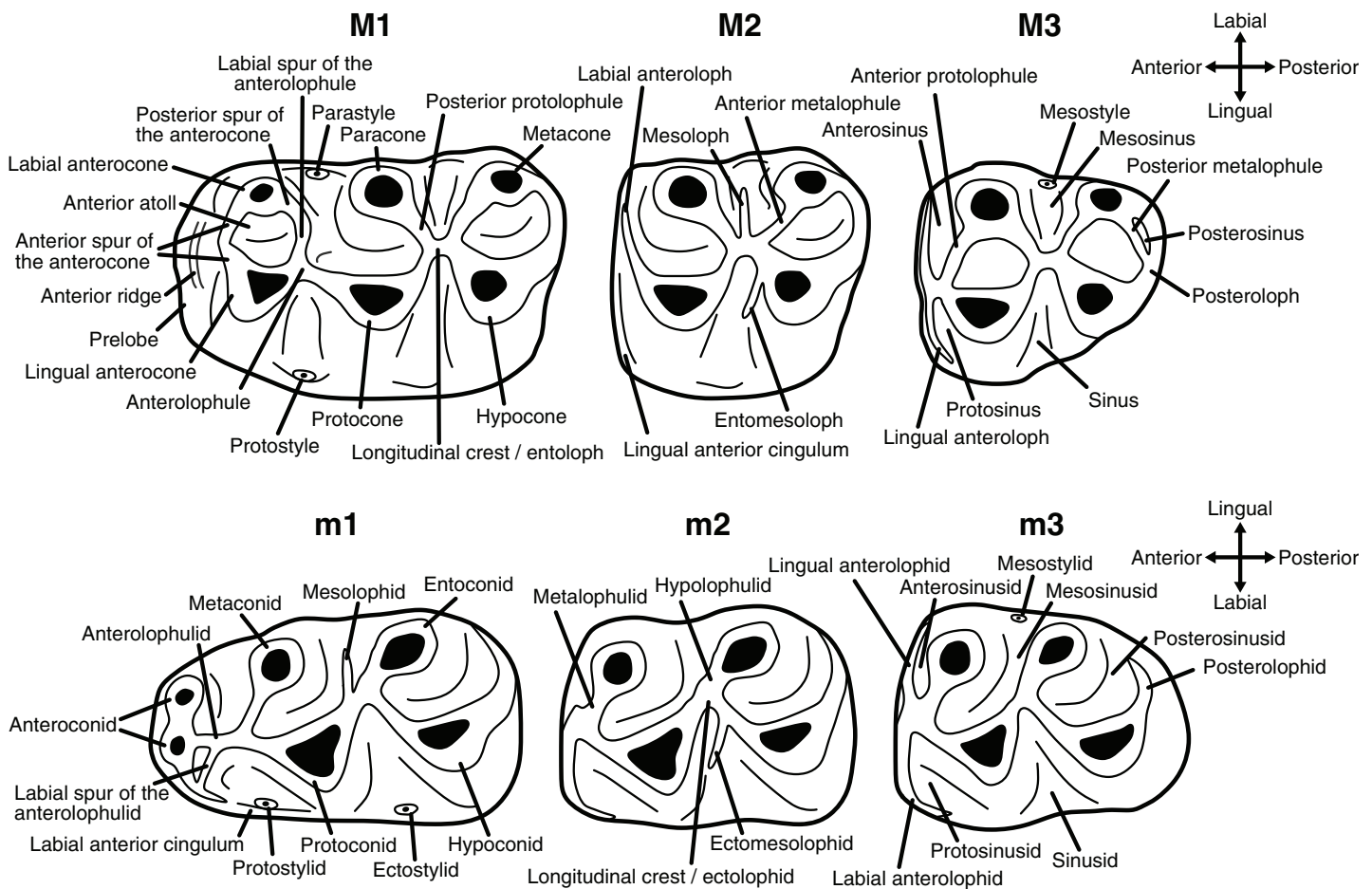


Figure 1 Dental terminology used in this study.

Full-size [DOI: 10.7717/peerj.18440/fig-1](https://doi.org/10.7717/peerj.18440/fig-1)

Phylogenetic reconstructions

All final trees were annotated and visualized in R with the packages *treeio*, *ggtree* and *deeptime* (Wang et al., 2020; Yu, 2022; Gearty, 2023). For input and output files of the Bayesian inference and maximum parsimony analyses, see [Supplemental Material S3](#).

Maximum parsimony analyses. All maximum parsimony analysis were run with TNT v. 1.6 (Goloboff & Morales, 2023) with all characters treated as unordered.

Equal weights analysis. The analysis under equal weights (MP-EW) was conducted with new technology algorithms using initial trees from 1,000 rounds of random addition sequence, with 100 iterations or rounds for sectorial search, ratchet, and tree fusing. The resulting 106 most parsimonious trees (256 steps, consistency index (CI): 0.414, retention index (RI): 0.715) were used to calculate a 50% majority consensus tree (258 steps, CI: 0.411, RI: 0.711). Clade support (given in %) was calculated based on 1,000 bootstrap (BS) replicates under the same parameters (Felsenstein, 1985).

Implied weights analysis. An additional analysis was run under the same options, as before but including implied weighting (MP-IW) (Goloboff et al., 2008). Following recent suggestions, a larger concavity index (k) of 12 was used (see Goloboff, Torres & Arias,

2018). The resulting two most parsimonious trees (257 steps, CI: 0.412, RI: 0.713) were used to calculate a consensus tree, (258 steps, CI: 0.411, RI: 0.711) with clade support based on 1,000 BS replicates (Felsenstein, 1985).

Bayesian inference analyses. Bayesian analyses were run with the parallel version of MrBayes v. 3.2.7a (Altekar et al., 2004; Ronquist et al., 2012b) using the Cyber-Infrastructure for Phylogenetic Research (CIPRES) Science Gateway version 3.3 (Miller, Pfeiffer & Schwartz, 2010).

Non-clock analysis. The Mkv model (Lewis, 2001) was used with among character rate heterogeneity modelled under a gamma distribution (Yang, 1993). All characters were treated as unordered. The analyses were run with four independent Metropolis-Coupled Markov chain Monte Carlo (MCMCMC) runs with six chains and 30,000,000 generations, sampling every 1,000 steps and a burn-in of 30%. Convergence and sufficient length of the runs were checked, using the R package Convenience v. 1.0.0 (Fabreti & Höhna, 2022). Based on the posterior tree sample a maximum clade compatibility (MCC) tree was calculated, as a consensus tree.

Time-calibrated relaxed-clock analysis. All settings of the non-clock analysis were adopted, except the number of MCMCMC generations, which was increased to 50,000,000. Time-calibrated relaxed-clock analyses were performed under a fossilized birth-death (FBD) tree prior (Stadler, 2010; Zhang et al., 2016). To model the way in which extant and extinct taxa are sampled in the construction of the tree, different strategies can be used (Simões, Caldwell & Pierce, 2020). To avoid problems when inferring speciation or extinction rates (Höhna et al., 2011), we have tested two of the three strategies, that are compatible with the FBD tree prior. The option ‘diversity’, that assumes a sampling strategy to maximize the diversity of extant taxa, was excluded, as our database only includes two extant species. Consequently, we have tested the two models that assume randomly sampled extant species. The first one, with sampled ancestors, SA-FBD (‘random’), allows the fossil taxa to be tips or ancestors of other taxa, while in the second one, the so-called noSA-FBD (‘fossil tip’), the fossil taxa have to be tips. The use of one or another can have an impact in the estimations of divergence times (Gavryushkina et al., 2014; Simões, Caldwell & Pierce, 2020). For the extant sampling probability, the number of included extant taxa (2) is divided by the total number of extant Cricetinae species (18 after Musser & Carleton, 2005).

In order to time-calibrate the tree, the tip-dating approach was used (Ronquist et al., 2012a; Ronquist, Lartillot & Phillips, 2016). Age ranges of the fossil taxa, resulting from age uncertainties of one or multiple locations of one taxon, were addressed by assigning uniform prior distributions to the tip calibrations, which can help to avoid erroneous divergence time estimations (O’Reilly, Dos Reis & Donoghue, 2015; Barido-Sottani et al., 2019). For the root age, an offset exponential distribution was set as a prior, with a minimum of 33 Ma (= minimal age of the oldest included fossil *Eucricetodon wangae*) and a mean of 41.2 Ma (following López-Antoñanzas & Peláez-Campomanes, 2022).

To give an informative prior to the base rate of the clock, the median tree length, calculated by a preceding non-clock analysis, was divided by the median of the root age prior ($3.189768/37.1 = 0.085978$) (following Simões et al., 2018, 2020). This estimated rate

in natural log scale ($= -2.45367$) was used as the mean of a log-normal distribution with the exponent of the mean ($e^{0.085978} = 1.08978$) as the standard deviation (following [Pyron, 2017](#)). To enforce proper rooting of the tree and facilitate reaching convergence, the ingroup was constrained to be monophyletic.

For relaxing the clock, two different models, compatible with the FBD prior are implemented in MrBayes v. 3.2.7a. The IGR (Independent gamma rate) model draws substitution rates from a gamma distribution, uncorrelated between branches, which allows more dramatic rate changes (more punctuated mode of evolution) ([Drummond et al., 2006](#)). The second model, TK02, samples from a lognormal distribution and is autocorrelated between branches, which represents a more gradual mode of evolution ([Thorne & Kishino, 2002](#)). Both models were used here, resulting in a total of four different models with all combinations of ‘fossil tip’ vs. ‘random’ and IGR vs. TK02. To choose the best fit model, stepping-stone sampling was done to estimate the marginal likelihoods ([Xie et al., 2011](#)). These can be used to calculate Bayes factors to compare the fit of two models to the data. For the stepping-stone sampling, the number of MCMCMC generations was increased by a factor of 10 to 500,000,000 as suggested by [Ronquist et al. \(2020\)](#).

All analyses were checked for convergence by the R package Convenience v. 1.0.0 ([Fabreti & Höhna, 2022](#)), as mentioned for the non-clock analysis.

Rogue taxon identification and tree set pruning. To improve posterior probabilities of the resulting trees, so-called rogue (‘wildcard’) taxa, were identified. These taxa are characterised by an unstable position in the tree, as they are resolved in varying clades in the trees of a tree set, e.g., the posterior tree sample of a Bayesian inference analysis. This leads to decreased posterior probabilities or even less resolved consensus trees. Equally, in the case of maximum parsimony analyses, they can affect the consensus tree calculated by several most parsimonious trees or the support values, given by a set of bootstrap trees. Deletion of the rogue taxa from the tree sets before calculating the consensus trees, can therefore lead to better resolved and supported trees, while they are still incorporated in the actual reconstruction. The deletion of the rogue taxa from the taxon set followed by a re-run of the analysis (e.g., in [Aberer & Stamatakis, 2011](#); [Simões, Caldwell & Pierce, 2020](#)) is seen critically by some authors as it means disregarding available and potentially important information (as discussed in [Goloboff & Szumik, 2015](#)).

In this study, the posterior tree samples of both clock trees were used to identify rogue taxa utilising the R package Rogue v. 2.1.6 ([Smith, 2022, 2023](#)). An additional examination of the 106 most parsimonious trees of the equal weighting maximum parsimony analysis, using the web interface of RogueNaRok ([Aberer, Krompass & Stamatakis, 2013](#)), did not result in any identified rogue taxa. For the R-code used to identify the rogue taxa, see [Supplemental Material S4](#).

This study does not include the Pleistocene cricetine taxa, for which a revision beyond the scope of this paper is needed. The lack of the youngest cricetine fossil taxa from our analysis, makes the inferred position of the two included extant species uncertain. For this reason, they were removed from the resultant trees ([Figs. 1, 2, S5.1–3](#)) but are shown in [Figs. S5.4–8](#) together with a discussion on the identified rogue taxa in [Supplemental Material S5](#).

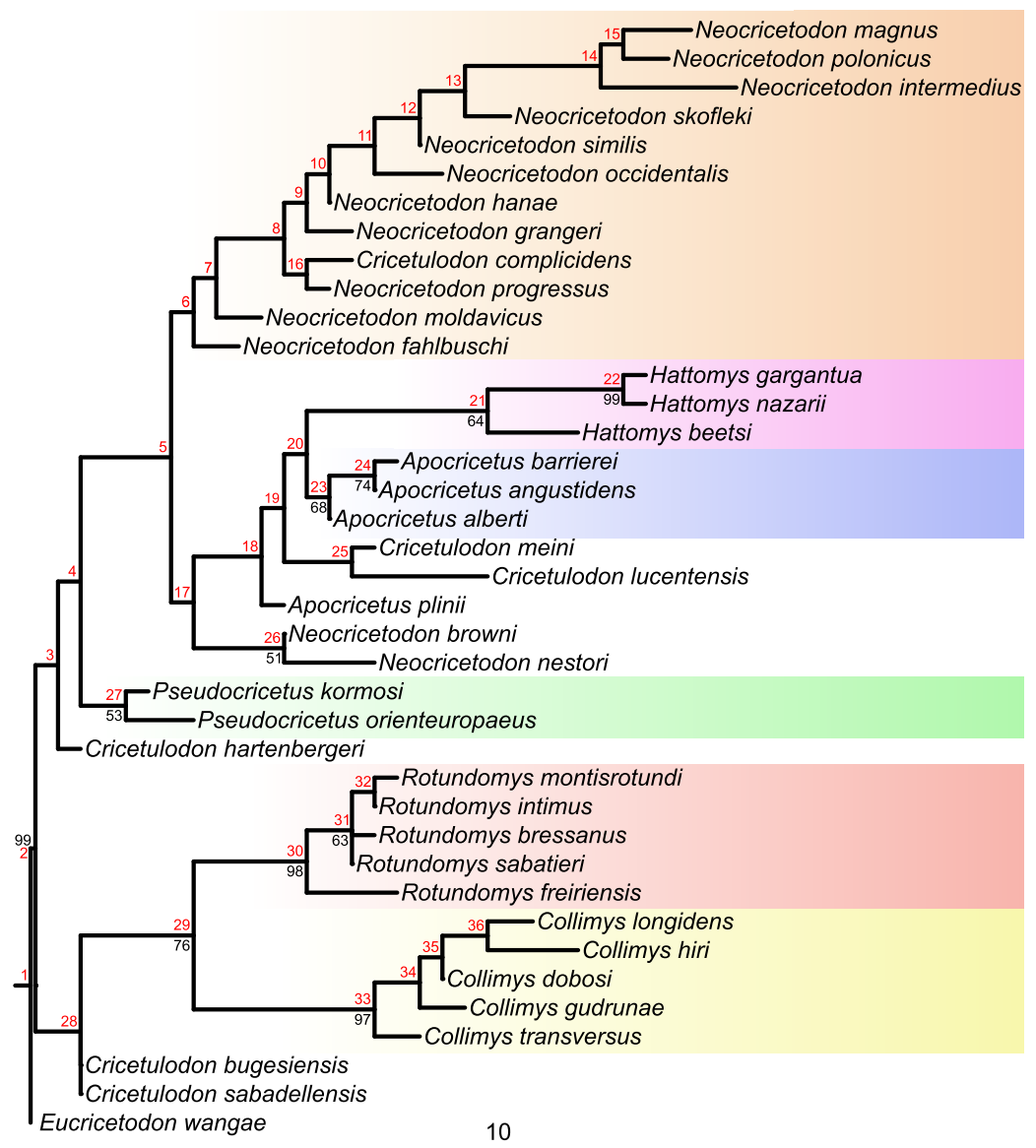


Figure 2 Majority consensus tree (phylogram), calculated from the two most parsimonious trees of the implied weighting maximum parsimony analysis. Bootstrap values over 50% are indicated at respective nodes in black, node numbers in red. The scale bar represents character state changes.

Full-size [DOI: 10.7717/peerj.18440/fig-2](https://doi.org/10.7717/peerj.18440/fig-2)

In total, the tree sets of all analyses were pruned by the same rogue taxa and the extant taxa, to obtain comparable trees. These trees, that are based on the pruned taxon set were then used for the subsequent analyses.

Assessing stratigraphic congruence

To compare the fit of the topologies resulting from the different reconstruction methods, the R package strap v. 1.6 (Bell & Lloyd, 2015) was used to first time calibrate the non-clock trees, and then to calculate the following stratigraphic fit indices: (i) the relative completeness index (RCI) assesses the amount of gaps in the fossil record in relation to the

observed fossil ranges in the tree ([Benton & Storrs, 1994](#)); (ii) the gap excess ratio (GER) indicates the sum of ghost ranges in the tree scaled in relation to the possible minimum and maximum sum of ghost ranges in theoretical topologies ([Wills, 1999](#)); (iii) the modified Manhattan stratigraphic measure (MSM^*) indicates the sum of ghost ranges in the tree in relation to the sum of ghost ranges of the theoretical tree of best fit to the stratigraphy ([Siddall, 1998](#); [Pol & Norell, 2001](#)). For all indices, significance tests were carried out, that resulted in very small p -values. Resulting indices of all analyses are listed in [Supplemental Material S6](#).

Character mapping

To ensure reliable results, two methods and two trees were used to map ancestral character states. Following the results of the stratigraphic congruence analyses, the implied weighting maximum parsimony tree was used to map characters in TNT under a parsimonious approach (see [Supplemental Material S7](#)) and the time-calibrated tree under the IGR clock model was used for stochastic character mapping with the R package phytools v. 2.1.1 ([Revell, 2024](#)). For the stochastic character mapping, three different models were fit to each character, with rates between states being either equal, symmetrical, or all different. The respective best fitting model was chosen using the Akaike information criterion and then used to map the character on the tree. Finally, for each character, the posterior probabilities of the different states were plotted on the nodes of the tree. For the reconstructed maps of characters, that are mentioned in the results, see [Supplemental Material S8](#).

RESULTS

Three taxa, *Apocricetus darderi*, *Neocricetodon ambarrensis* and *Pseudocricetus polgardiensis* were identified as rogue taxa, based on the clock trees and were pruned from the taxon set before calculating the consensus trees (MP-EW: 234 steps, CI: 0.453, RI: 0.757; MP-IW: 235 steps, CI: 0.451, RI: 0.755).

Maximum parsimony analysis

Maximum parsimony analyses of morphological data sets including highly homoplastic characters can be improved by weighting characters according to their homoplasy ([Goloboff et al., 2008](#)). Consequently, implied weighting parsimony analyses can produce more resolved and accurate trees than standard equal weights ([Smith, 2019](#)). To assess which tree, from equal or implied weighting, fits better with the chronostratigraphy, stratigraphic congruence indices were calculated. While the topologies of both parsimony trees are quite similar (see [Figs. 1](#) and [S5.3](#)), our results evidence a better stratigraphic congruence of the implied weighting tree than of the equal weighting tree (see [Supplemental Material S5](#)). Therefore, we discuss below the topology retrieved by applying implied weighting ([Fig. 2](#)).

The topology of the consensus tree of the two most parsimonious trees shows a major basal split into two main clades. The first one (stemming from node 28) consists of *Collimys*, *Rotundomys*, *Cricetulodon bugesiensis* and *Cricetulodon sabadellensis* whereas the other one (stemming from node 3) includes the remaining cricetines. The most basal

taxon of this latter clade is *Cricetulodon hartenbergeri*, followed by *Pseudocricetus* (node 27), which is in turn sister to a large clade (node 5) that splits into two lineages. The first of them (stemming from node 6) includes *Cricetulodon complicidens* and most of the species belonging to the genus *Neocricetodon*. The second one (stemming from node 17) has as most basal taxa the sister species *N. browni* and *N. nestori*, as sister clade to the group stemming from node 18: *Apocricetus plinii*, followed by *Cricetulodon meini* plus *Cricetulodon lucentensis*, and the sister clades *Hattomys* and *Apocricetus sensu stricto* (s.s.).

Tip-dated Bayesian analysis

The results of the stepping stone sampling strongly evidence a better fit for the ‘fossiltip’ sampling strategy under both clock models ($2 \log_e(B_{10}) > 10$, see [Kass & Raftery, 1995](#)). Comparing Tk02 with IGR under ‘fossiltip’, the better fit model is the Tk02 ($\log_e(B_{10}) = 4.6$) but without strong evidence. However, the analysis under the Tk02 model showed some problems in reaching convergence (ESS below 200 for two parameters, ‘convergence failed’ according to Convenience). The IGR model showed a considerably better performance (lowest ESS $> 3,300$, ‘convergence reached’ according to Convenience). Moreover, the results of the strap analyses show a better stratigraphic congruence for the IGR model than for the TK02 model (see [Supplemental Material S6](#)). As both calibrated Bayesian inference analyses resulted practically in the same topology of the trees (see [Figs. 2](#) and [S5.2](#)) with only slight differences in the posterior probabilities (generally a bit higher in Tk02) and in the relationships among the species belonging to *Neocricetodon* (particularly in that concerning the clades with low posterior probabilities), we describe below only the results obtained by applying the ‘fossiltip’ strategy under the IGR model ([Fig. 3](#)).

The topology of our tree evidences four major clades. The most basal one (stemming from node 35) consists of all species belonging to *Collimys*. Its sister clade (stemming from node 3) includes all remaining cricetines, with its most basal clade (stemming from node 31) including all *Rotundomys* species on the one side and the remaining two major clades (stemming from nodes 5 and 18) on the other. The first of the latter (stemming from node 5) consists of all species of *Neocricetodon*, as well as *Cricetulodon complicidens*. The second one (stemming from node 18) has *Cricetulodon hartenbergeri* as sister of two clades: a small one (node 30) including *Cricetulodon bugesiensis* and *Cricetulodon sabadellensis* and a larger one (stemming from node 19) constituted by a succession of clades with *Pseudocricetus* (node 29) at the base, followed by the sister taxa *Cricetulodon meini* and *Cricetulodon lucentensis* (node 28). One node up (node 22) inserts *A. plinii*, followed by the sister lineages *Apocricetus* s.s. and *Hattomys*. The topology of the tree supports the monophyly of *Collimys* (node 35), *Rotundomys* (node 31), *Neocricetodon* (node 5), *Pseudocricetus* (node 29), *Apocricetus* s.s. (node 26), and *Hattomys* (node 24). In contrast, the genus *Cricetulodon* is paraphyletic and only the type species *Cricetulodon sabadellensis* and *Cricetulodon bugesiensis* should be included in this genus.

Divergence times. The divergence times estimated using the two different clock models IGR and TK02, vary only slightly, with differences of mostly less than 500,000 years. In the same way, uncertainties on divergence times (measured by 95% highest probability

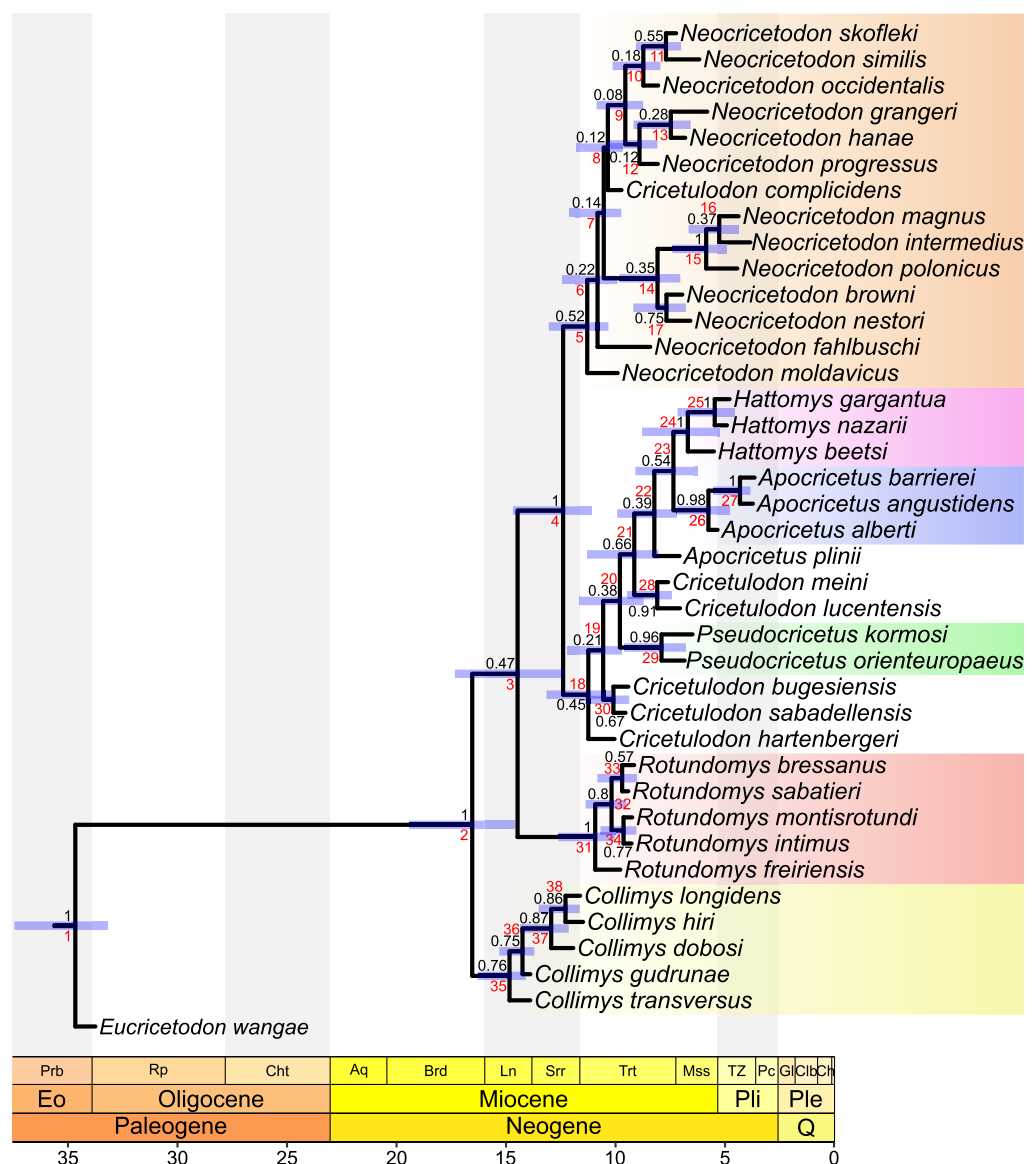


Figure 3 MCC tree of the time-calibrated relaxed-clock IGR Bayesian inference analysis. Posterior probabilities of clades are indicated at respective nodes in black, node numbers in red, node bars indicate the 95% highest posterior density for divergence times. The scale axis is in Ma, the chronostratigraphic chart follows [Cohen, Harper & Gibbard \(2022\)](#). [Full-size](#) DOI: 10.7717/peerj.18440/fig-3

densities (HPD) ranges) are quite similar independently of the clock model applied. In the following, the results of the analysis under the IGR model are reported. They reveal a late Early Miocene age for the first split within the ingroup (16.54 Ma, 95% HPD: 14.53–19.12 Ma). *Collimys* is recovered as the oldest genus, diverging during the Middle Miocene (14.81 Ma, 95% HPD: 14.07–15.85 Ma). The remaining five monophyletic genera diverged later, during the Late Miocene: *Neocricetodon* (11.28 Ma, 95% HPD: 10.26–12.64 Ma), *Rotundomys* (10.91 Ma, 95% HPD: 10.01–12.23 Ma), *Pseudocricetus* (7.9 Ma, 95% HPD:

6.69–9.29 Ma), *Hattomys* (6.7 Ma, 95% HPD: 5.17–8.47 Ma), and *Apocricetus s.s.* (5.7 Ma, 95% HPD: 4.59–6.81 Ma).

Ancestral character state reconstructions

The slight differences in the topology of both trees (particularly regarding the phylogenetic position of *Rotundomys* and *Collimys*) result in different reconstructed synapomorphies for some clades. However, concerning the genera, most of the synapomorphies found when analysing the results of the stochastic character mapping are also retrieved by the parsimonious mapping of synapomorphies (see [Supplemental Materials S7](#) and [S8](#)). Only a few synapomorphies are not found with the latter approach, due to the coding of polymorphic states as ‘ambiguous’ in the mapping process in TNT. Therefore, below we list mainly the synapomorphies obtained by applying stochastic character mapping.

All species belonging to *Collimys* (stemming from node 35) share the exclusive synapomorphy of having an ectomesolophid on the m1 (68: 0→1). Additional ambiguous synapomorphies are: e.g., a labial spur of the anterolophule reaching the labial border of the M1 (103: 0→2), a long mesoloph on the M3 (49: 2→0) and the presence of a labial spur of the anterolophulid on the m1 (64: 0→1).

The clade *Rotundomys* (stemming from node 31) is defined by the following exclusive synapomorphies: a very weak to absent mesoloph on M1 and M2 (20: 1→2, 37: 1→2), a posteroloph that is merged with the posterior metalophule on the M1 (26: 1→3) and the presence of a short or hanging labial anterolophid on the m3 (88: 0→1), as well as many non-exclusive synapomorphies.

Representatives of the clade stemming from node 5 including *Neocricetodon* and *Cricetulodon complicidens* share non-exclusive synapomorphies such as the presence of the anterior metalophule on the M2 (38: 2→1/0). Additional non-exclusive synapomorphies, only shared with *Collimys*, are the presence of a long mesolophid on the m1 (66: 0/1→2) and having a medium to long labial spur of the anterolophule on the M1 that reaches the molar border in most of the taxa belonging to *Neocricetodon* (103: 0→1/2).

Representatives of *Pseudocricetus* (stemming from node 29) are characterised by having a short but distinct posteroloph on the M3 (56: 0→1) and the tendency to form a very small mesolophid on the m2 (73: 2→1).

Apocricetus s.s. (stemming from node 26) is characterised by the two non-exclusive synapomorphies of having a multi-lobed, crestiform anteroconid (57: 2→4) with a poorly developed labial anterolophid (60: 0→1) on the m1.

Representatives of *Hattomys* (stemming from node 24) are clearly distinct from those belonging to its sister group *Apocricetus s.s.* They are characterised by sharing the two exclusive synapomorphies of having large M1, longer than 3.2 mm (1: 1→3) and an hypolophulid connected to a medium sized mesolophid on the m1 (109: 2→5).

DISCUSSION

Maximum parsimony vs. Bayesian trees

All maximum Parsimony and Bayesian tip-dated and undated trees ([Figs. 1, 2, S5.1–3](#)) support the monophyly of *Collimys*, *Rotundomys*, *Pseudocricetus* *Apocricetus s.s.* and

Hattomys. *Neocricetodon* includes *Cricetulodon complicitens* in all trees. It is monophyletic in the Bayesian trees but not in the parsimonious ones, as two species branch outside of the clade (*N. browni* and *N. nestori* in the implied weighting analysis and *N. moldavicus* and *N. fahlbuschi* in the equal weighting one).

Cricetulodon splits up in the same groups in both, the IGR Bayesian clock analysis and the implied weighting parsimony analysis (Figs. 1 and 2). However, some differences are observed concerning the phylogenetic position of the clades. In this way, the topology of the parsimony tree shows *Cricetulodon bugesiensis* and *Cricetulodon sabadellensis* as the most basal taxa of the clade including *Collimys* and *Rotundomys* (stemming from node 28, Fig. 2), whereas in the Bayesian topology this lineage belongs to the clade that comprises the remaining species of *Cricetulodon*, *Pseudocricetus*, *Apocricetus* and *Hattomys* (stemming from node 18, Fig. 3). Moreover, *A. plinii* is basal to *Cricetulodon meini* and *Cricetulodon lucentensis* in the parsimony tree, but sister to *Apocricetus s.s.* and *Hattomys* in the Bayesian one (inserting at node 18, Fig. 2 vs. node 22, Fig. 3).

The most striking difference when comparing the Bayesian clock tree to the parsimony one concerns the relationship between *Collimys* and *Rotundomys*, for which the parsimony analysis found a sister relationship (see node 29, Fig. 2, BS = 76) whereas in the Bayesian tree they insert sequentially (at node 2 and 3, Fig. 3). Therefore, depending on the topology, contrasting ancestral character state reconstructions are found. The characters that are different in *Collimys* and *Rotundomys* compared to the remaining cricetines are the following: M1 anterocone not clearly divided, but more crestiform vs. divided in two (5: 6 vs. 2); M1 lingual anteroloph clearly present vs. weak or absent (11: 0 vs. 1); upper molars protolophule posterior vs. double (7: 0 vs. 1, 34: 2 vs. 1, 47: 3 vs. 0); m3 lingual anterolophid well-developed vs. weak or absent (76: 1 vs. 0, reversed in *Apocricetus s.s.* and *Hattomys* (76: 0 vs. 1)). According to the stochastic character mapping, all these characters serve as synapomorphies for the clade *Neocricetodon* + *Cricetulodon* + *Pseudocricetus* + *Apocricetus* + *Hattomys* (stemming from node 4, Fig. 3). In the parsimony analysis, on the other side, the results are reversed. Only having a double protolophule on the M1 is reconstructed as a synapomorphy for this clade (stemming from node 3, Fig. 2), while the other above-mentioned characters serve here as synapomorphies for the clade *Collimys* + *Rotundomys* (node 29, Fig. 2).

The genera *Rotundomys* and *Collimys* were informally grouped by Kälin (1999) on the basis of emerging hypsodonty, whereas Heissig (1995) excluded *Collimys* as potential ancestor of *Rotundomys*, stating that the tendency of acquiring hypsodonty evolved independently.

In the parsimony analysis, *Cricetulodon sabadellensis* and *Cricetulodon bugesiensis* are additionally positioned as sister taxa to *Collimys* and *Rotundomys*. Considering the much younger age of the *Cricetulodon* taxa compared to *Collimys*, the arrangement in the Bayesian tree seems more plausible.

Collimys

According to our results, *Collimys* forms a monophyletic group, from which *Collimys transversus* and *Collimys gudrunae* are splitting first. Prieto & Rummel (2009a, 2009b)

proposed three different lineages within this genus. An early lineage *Collimys transversus*–*Collimys gudrunae*, a second temporally intermediate lineage involving *Collimys* sp. 1, 2 (from Petersbuch 10, 18, 6 and 48) and a later lineage *Collimys hiri*–*Collimys longidens*–*Collimys dobosi*. Our results, nevertheless, do not support the lineages proposed by [Prieto & Rummel \(2009a, 2009b\)](#). Moreover, according to the topology of our tree, the clade *Collimys dobosi* plus more derived taxa (stemming from node 37, [Fig. 3](#)) differs significantly from the *Collimys hiri*–*Collimys longidens*–*Collimys dobosi* lineage proposed by [Prieto & Rummel \(2009a, 2009b\)](#). The latter authors proposed the lineage mainly on the basis of an increase in size, in hypsodonty and in mesoloph length on the M1 and M2, together with other minor morphological variations but the validity of the lineages has been doubted later, due to regional variation of the taxa ([Prieto et al., 2014](#); [Hír et al., 2017](#)). Our results show that the most basal position of *Collimys transversus* and *Collimys gudrunae* is supported by the smaller size of *Collimys transversus* and the presence of a slightly more developed lingual anteroloph on the M3, in both reconstructed phylogenies. *Collimys dobosi* is the next splitting species, sister to *Collimys hiri* and *Collimys longidens*, due to a more square-shaped M2, which is elongated in the latter taxa, and the presence of an anterior metalophule in the M3. The absolute size differences between the species are too small to result in different states in the phylogenetic matrix. The length of the mesoloph on the M1 is variable in *Collimys dobosi*, *Collimys hiri* and *Collimys longidens* but it does not reach the border of the teeth in the majority of the specimens, whereas it usually does on the M2 of all three taxa ([Kälin & Engesser, 2001](#); [Hír, 2005](#); [Prieto & Rummel, 2009b](#)). Therefore, the reconstructions resulted in a more basal *Collimys dobosi*, in both dated and undated analyses, due to above mentioned reasons. This basal position of *Collimys dobosi* compared to *Collimys hiri* and *Collimys longidens* is congruent to the slightly older age attributed to the former species. It has been recorded from Felsőtárkány, Hungary (~12.2–11.6 Ma), while *Collimys longidens* and *Collimys hiri* have been found in Nebelbergweg, Switzerland and Hammerschmiede, Germany, respectively (~11.9–11.3 Ma) ([Hír et al., 2016, 2017](#); [Prieto & Rummel, 2016](#)).

Rotundomys

Rotundomys freiriensis as the basal-most taxon within *Rotundomys* is the best supported split in the reconstructed phylogenies. It is based on the absence of the mesoloph (or anterior metalophule) on the M3, which is present in the remaining species of the genus and in having the lingual anterolophid on m1 and m2 better developed than in more derived taxa. This arrangement follows the proposal of several previous studies ([Antunes & Mein, 1979](#); [Freudenthal, Mein & Martín Suárez, 1998](#); [López-Antoñanzas, Peláez-Campomanes & Álvarez-Sierra, 2014](#)). *Rotundomys montisrotundi* and *R. intimus* form a clade characterised by a poorly developed lingual anteroloph on the M2. [López-Antoñanzas, Peláez-Campomanes & Álvarez-Sierra \(2014\)](#) addressed the similarity between these two species and with *R. sabatieri*, and separated them on the basis of size, as well as some minor morphological differences and different proportions of morphotypes. The results of their phylogenetic analysis showed *R. montisrotundi* and *R. bressanus* as sister species, based on the absence of the lingual anteroloph on the M2 in most specimens.

However, their phylogenetic analysis only included the genera *Rotundomys* and *Cricetulodon* and therefore these results should be taken with caution before drawing general conclusions. [Mein \(1975\)](#) proposed *R. bressanus* to be derived from *R. montisrotundi*, mainly based on size differences. Intraspecific variation of *R. montisrotundi* complicates, however, confirming or refuting this hypothesis ([Freudenthal, Mein & Martín Suárez, 1998](#)).

In general, discrimination between species often relies on differences in size, that can, however, overlap in their ranges (see *R. sabatieri* vs. *R. bressanus* in [Aguilar, Michaux & Lazzari, 2007](#)), resulting in short branch lengths in the maximum parsimony phylogram and in collapsed clades, due to zero-size branch lengths (see *R. sabatieri*). Consequently, the only reliable relationships within the genus seem to be the basal position of *R. freiriensis* and the closely related *R. montisrotundi* and *R. intimus*.

Neocricetodon

The name *Neocricetodon* [Schaub, 1934](#) was validated by [Daxner-Höck et al. \(1996\)](#) and followed by a majority of authors afterwards.

Our results show three synapomorphies for *Neocricetodon*: (i) the presence of a long mesolophid on the m1, (ii) the presence of a labial spur of the anterolophule on the M1 and (iii) the presence of an anterior metalophule on the M2. These results agree with those of [Freudenthal, Mein & Martín Suárez \(1998\)](#), regarding the synapomorphies (i) and (ii). Additionally, they also mentioned that the species belonging to *Neocricetodon* show a labial anterolophulid on the m1 and elongated mesolophids on the upper molars. Our results are also in line with those of [Sinita & Delinschi \(2016\)](#), agreeing on the synapomorphy (iii) the presence of the anterior metalophule on the M2. The latter authors additionally proposed as synapomorphies of this clade an expanded anterocone and the presence of a labial anterolophule on the M1, a labial anterolophulid on the m1, and a four rooted M2. They have, however, only included *Cricetulodon sabadellensis*, '*Kowalskia cf. schaubi*' ([Kretzoi, 1951](#)) and *Democricetodon* as outgroup taxa but no other cricetines, therefore these proposed synapomorphies must be treated carefully. A four rooted M2, for example, is reconstructed as plesiomorphic for *Neocricetodon* by the stochastic character mapping.

Interestingly, the first two synapomorphies we have proposed for *Neocricetodon*, (i) the presence of a mesolophid on the m1 (reaching the molar border in most of the taxa) and (ii) the presence of a labial spur of the anterolophule on the M1 (reaching the border of the molar in most taxa) are not considered as synapomorphies by [Sinita & Delinschi \(2016\)](#). In fact, these authors, coded *N. occidentalis*, *N. progressus* and *N. moldavicus* as lacking or having a short mesolophid on the m1 ([Sinita & Delinschi, 2016](#); Table 2). However, previous studies have described *N. occidentalis* and *N. progressus* as having usually long mesolophids ([de Bruijn et al., 1975](#); [Freudenthal, Lacomba & Martín Suárez, 1991](#); [Topachevsky & Skorik, 1992](#); [Freudenthal, Mein & Martín Suárez, 1998](#); [Sinita, 2012](#)) and *N. moldavicus* as having short to medium mesolophids. Moreover, the holotype of this latter species shows a clearly well-developed mesolophid ([Lungu, 1981](#); [Sinita & Delinschi, 2016](#)). Regarding (ii) the spur of the anterolophule on the M1, [Sinita & Delinschi \(2016\)](#) coded having short or absent spurs as one single state of character. This could be the reason

why the presence of this structure has not been expressed in their matrix of characters in several taxa (e.g., *N. nestori*, see [Engesser, 1989](#)), impeding its identification as a possible synapomorphy.

Both the above-mentioned structures, the mesolophid on the m1 and the labial spur of the anterolophule on the M1, are relatively poorly developed in *N. moldavicus*, justifying its basal position within the clade. Only the third synapomorphy, (iii) the presence of the anterior metalophule on the M2 (coded here as metalophule either anterior or double), is clearly observable in *N. moldavicus*. However, as noticed by [Freudenthal, Mein & Martín Suárez \(1998\)](#), who did not include this character as a diagnostic trait of the genus, it is quite variable in several taxa (e.g., *N. nestori*, *N. progressus*, *N. hanae*, *N. browni*), although there is a strong tendency towards its presence. [Sinita & Delinschi \(2016\)](#) termed the character as ‘phylogenetically irrelevant’, due to homoplasy and reversals in some clades. They specifically mentioned the loss of this structure in *N. grangeri* but the single M2 from the original material is too heavily damaged to make any statement concerning the metalophule ([Daxner-Höck et al., 1996](#)). Yet, additional found material of this species, that includes several complete M2s, evidences that most of them have an anterior metalophule ([Wu & Flynn, 2017](#)).

Several phylogenetic hypotheses within *Neocricetodon* have been proposed ([Wu, 1991](#); [Daxner-Höck, 1992](#); [Freudenthal, Mein & Martín Suárez, 1998](#); [Qiu & Li, 2016](#); [Sinita & Delinschi, 2016](#)). However, the evolutionary history of this taxon, that includes numerous species with wide geographical and temporal distribution, is complex to untangle. This complexity is also reflected in our results, which put in evidence low posterior probabilities for most of the clades within *Neocricetodon* and some differences in the topologies of the trees, which mostly prevents reliable statements about the proposed lineages. However, the clade combining *N. magnus*, *N. intermedius* and *N. polonicus* (stemming from node 14, [Fig. 2](#), or node 15, [Fig. 3](#)), which were dominantly distributed in Eastern Europe, in Hungary, Poland, Slovakia and Ukraine, ([Fahlbusch, 1969](#); [Jánossy & Kordos, 1977](#); [Pevzner et al., 1996](#)) is consistent in all our analyses and very well supported in the IGR Bayesian tree (PP = 1). This result is in disagreement with the hypothesis of [Wu \(1991\)](#), who separated the three species into three different lineages mainly on the basis of their molar size. Conversely, it corroborates the close relationship between *N. polonicus* and *N. intermedius*, as recovered by [Sinita & Delinschi \(2016\)](#), who did not include *N. magnus* in their analysis.

According to the topology of our tree, *Neocricetodon* also includes *Cricetulodon complicidens*. Difficulties regarding the genus assignment of this species were already mentioned in several papers ([Freudenthal, Mein & Martín Suárez, 1998](#); [Kälin, 1999](#)). [Topachevsky & Skorik \(1992\)](#), who coined this species, described some similarities with *Neocricetodon*, such as the presence of a long mesolophid on the m1, a long labial spur of the anterolophule on the M1, and an anterior metalophule on the M2. These are, in fact, the three above mentioned synapomorphies that define this group. Consequently, the reallocation of *Cricetulodon complicidens* into the genus *Neocricetodon* seems to be justified.

This reallocation could seem to cause confusion considering another species coined by [Topachevsky & Skorik \(1992\)](#) as '*Kowalskia complicidens*' due to the fact that *Kowalskia* is considered a junior synonym of *Neocricetodon* by several authors ([Freudenthal, Mein & Martín Suárez, 1998](#); [Sinita & Delinschi, 2016](#)). However, '*Kowalskia*' *complicidens* is thought not to belong to *Neocricetodon* (or *Kowalskia*) but rather to *Sinocricetus* [Schaub, 1930](#) ([Daxner-Höck et al., 1996](#); [Qiu & Li, 2016](#); [Sinita & Delinschi, 2016](#)).

This leads to the question about the validity of the genus *Kowalskia* or its synonymy with *Neocricetodon*. The scarce material of the type species *N. grangeri* did not help to clarify this issue and therefore some authors keep the genera separated ([Daxner-Höck et al., 1996](#)), whereas others consider *Kowalskia* as junior synonym of *Neocricetodon*, until the discovery of additional material of this species would allow to either detect clear similarities or differences ([Freudenthal, Mein & Martín Suárez, 1998](#)). [Sinita & Delinschi \(2016\)](#) reconstructed the phylogeny of the group. Their work retrieved *K. polonica*, the type species of *Kowalskia*, branching in the same clade as *N. grangeri* and other species of *Neocricetodon*. However, due to the limited material of *N. grangeri*, they could not code any of the characters related to the M1 of this species, which make up nearly half of the total characters of their matrix. Soon afterwards, [Wu & Flynn \(2017\)](#) published additional material of *N. grangeri*, which helped them to conclude that the synonymy of *Kowalskia* with *Neocricetodon* is strongly supported.

The topologies of the parsimony, undated and tip dating TK02 clock trees ([Figs. 1, S5.1, 2](#)) agree with [Sinita & Delinschi \(2016\)](#) in the phylogenetic position of the type species of *Kowalskia* and *Neocricetodon*, as nesting within a clade that includes the remaining species of *Neocricetodon* (stemming from node 6, [Fig. 2](#)). All these trees evidence a derived position of *K. polonica*, which shares the three synapomorphies mentioned above that characterise the genus *Neocricetodon*. Only the Bayesian IGR clock tree ([Fig. 3](#)) shows two main clades inside the clade *Neocricetodon*, one of which (stemming from node 8, [Fig. 3](#)) includes the type species *N. grangeri* and the other one (stemming from node 14, [Fig. 3](#)) includes *K. polonica*. These two clades could be interpreted as two separated genera, with all species in the same clade as *K. polonica* reallocated to *Kowalskia*. However, taking into account the very low posterior probabilities of these clades (0.12 and 0.35), the absence of clear synapomorphies for both of the clades, and that these clades are not recovered in any of the remaining analyses presented here, we consider the synonymy of *Kowalskia* with *Neocricetodon* to be justified.

Cricetulodon

The difficulty of defining the genus *Cricetulodon* is exemplified by previous proposals of synonymising it with *Rotundomys* or with *Neocricetodon* ([Freudenthal, 1967, 1985](#)). [Freudenthal, Mein & Martín Suárez \(1998\)](#) eventually separated *Cricetulodon* from *Neocricetodon* mainly based on the presence of a mostly lingual anterolophulid on the m1 of the former taxon. Due to the high variability observed on the anterior part of the m1 of these two taxa, the determination of a dominantly lingual or labial anterolophulid can be problematic, particularly, when the anterolophid is double or more centrally positioned ([Engesser, 1989](#); [Wu, 1991](#); [Daxner-Höck & Höck, 2015](#)). The problems of relying on this

variable character to allocate a species into a genus are exemplified by the above-mentioned *Cricetulodon complicidens*.

The topology of our trees (Figs. 1 and 2) does not support the monophyly of *Cricetulodon*, which is in agreement with previous phylogenetic studies (López-Antoñanzas, Peláez-Campomanes & Álvarez-Sierra, 2014). However, the work of these authors was focused on a new species of *Rotundomys* and only the species belonging to *Rotundomys* and *Cricetulodon* were analysed. The three clades they recovered, *Cricetulodon hartenbergeri* plus *Cricetulodon sabadellensis*, *Cricetulodon bugesiensis* plus *Cricetulodon lucentensis*, and *Cricetulodon meini*, all basal to *Rotundomys*, are not found in our trees. Our results show *Cricetulodon meini* and *Cricetulodon lucentensis* as sister species, which is consistent with the hypothesis of a potential ancestor-descendant relationship between these two taxa suggested by Freudenthal, Mein & Martín Suárez (1998). After Freudenthal (1967), the position of *Cricetulodon hartenbergeri* and *Cricetulodon sabadellensis* as potential ancestors of *Rotundomys*, was adopted and discussed by several authors (Fejfar, 1970; Daxner-Höck, 1972; Kälin, 1999; Fejfar et al., 2011; López-Antoñanzas, Peláez-Campomanes & Álvarez-Sierra, 2014). While *Cricetulodon sabadellensis* and *Cricetulodon bugesiensis* are recovered as possible ancestors of *Rotundomys* in the undated analyses (see node 28, Fig. 2), they are quite distant in the clock trees, which is likely resulting from their similar or even younger age compared to *R. freiriensis*.

Be that as it may, the clade consisting of *Cricetulodon lucentensis* and *Cricetulodon meini* is not closely related to the type species of the genus *Cricetulodon sabadellensis* in any of our trees. Consequently, these species should be excluded from the genus and transferred into a new one.

Pseudocricetus

This genus was partly defined on the basis of some characters of the mandible, the skull and the incisors by Topachevsky & Skorik (1992) and Sinita (2010). Some dental morphological characters, such as the presence of reduced mesolophs and mesolophids, anterior protolophules or the split of the anterocone that were proposed to define *Pseudocricetus*, are in fact also present in several other genera (Daxner-Höck et al., 1996; Freudenthal, Mein & Martín Suárez, 1998). According to our results, *Pseudocricetus* is monophyletic and, in the Bayesian tree, sister clade to the lineage of *Cricetulodon lucentensis*, *Cricetulodon meini*, *Apocricetus* and *Hattomys* (stemming from node 21, Fig. 3). This agrees with previous hypotheses, according to which there were morphological similarities between *Pseudocricetus* and *Apocricetus* (Kälin, 1999), or that considered *Pseudocricetus* as a possible ancestor of *Hattomys* (Freudenthal & Martín Suárez, 2010).

Apocricetus

Freudenthal, Mein & Martín Suárez (1998) proposed the phyletic lineage, *Apocricetus plinii*–*A. alberti*–*A. barrierei*–*A. angustidens*. According to these authors, the changes

along this lineage, *e.g.*, the development of the anterior protolophules or the presence of an anterior ridge in the M1, are gradual and refer to size as well as to morphological features (see also [Ruiz-Sánchez et al., 2014](#); [Mansino et al., 2014](#)). The topologies of our trees mostly agree with the thoughts of [Freudenthal, Mein & Martín Suárez \(1998\)](#) except for *A. plinii*, which, despite being basal to *Apocricetus s.s.*, does not belong to this clade. Instead, the results of the Bayesian analysis show *A. plinii* (inserting at node 22, [Fig. 3](#)) as sister species to the sister clades *Apocricetus s.s.* and *Hattomys*. The topology of the maximum parsimony tree shows *A. plinii* (inserting at node 18, [Fig. 2](#)) as basal to the clade (stemming from node 19, [Fig. 2](#)) consisting in *Cricetulodon meini* plus *Cricetulodon lucentensis* and the sister clades *Apocricetus s.s.* and *Hattomys*.

Apocricetus plinii and *A. alberti* show less derived features such as a better developed anterior protolophule on the M1 when comparing with *A. angustidens* and *A. barrierei*. However, *A. plinii* differs from *A. alberti* by its less derived morphology of the anteroconid on the m1, which is not crest-like but split into two anteroconids. In addition, the labial spur of the anterolophule on the M1 of *A. plinii* is usually free and not connected to the labial anterocone as is the case of *A. alberti* ([Freudenthal, Mein & Martín Suárez, 1998](#)). Therefore the ‘anterior atoll’ that is formed between the two anterocones in all species belonging to *Apocricetus s.s.* and *Hattomys*, is often absent in *A. plinii*, which could explain its phylogenetical position in the tree.

Hattomys

Regarding the characters that have been used to define *Hattomys*, special attention was paid to the mesoloph(id) and the so-called ‘preloph(id)’ ([Freudenthal, 1985](#); [Savorelli, 2013](#)). Due to the direction and position of the structure that connects the ectoloph and the entoconid, it is difficult to know whether it represents the mesolophid, the anterior hypolophulid or a combination of both ([Freudenthal, 1985](#); [Savorelli, 2013](#)). This structure is here interpreted as a mesolophid of medium length (or long in the case of the m3), which is fused with the anterior hypolophulid to some extent (well visible in [Savorelli, 2013](#), [Fig. 5.6](#)). The so-called ‘prelophid’ is here interpreted as a lingual spur of the anterolophulid that is connected to the anterior metalophulid and to the posterior spur of the lingual anteroconid. In the upper molars, the ‘preloph’ is, accordingly, the labial spur of the anteroloph, which is either connected to a longitudinal running anterior protolophule and the posterior spur of the labial anterocone, or runs freely, as frequently seen in *H. beetsi*. This spur can also continue towards the labial border after its connection to the labial anterocone (see *e.g.*, [Freudenthal, 1985](#), plate 3.1). On all upper molars, the anterior metalophule seems to be lacking and there is only a long mesoloph, that usually reaches the labial border of the tooth. It can sometimes connect to the metacone.

The presence of a ‘preloph(id)’ is, independently of its interpretation, not a synapomorphy for *Hattomys* considering its frequent presence in *Apocricetus s.s.* ([Ruiz-Sánchez et al., 2014](#); [Mansino et al., 2014](#)). Instead, two non-exclusive synapomorphies, the presence of mesolophids of medium length on the m1 and the m2, and a single exclusive synapomorphy, the connection of the hypolophulid to the mesolophid on the m1, are

identified by the stochastic character mapping. Moreover, synapomorphies proposed in previous studies ([Freudenthal, 1985](#); [Savorelli, 2013](#)), such as the presence of a long mesoloph on the M2 and the M3 that reaches the labial border of the tooth and the well-developed ‘flanges’ on the cusps were only identified in *H. nazarii* and *H. gargantua*, here. These characters turned out to be plesiomorphic in *H. beetsi*, which could explain the basal-most position of this taxon inside the clade.

Hattomys is found in Gargano peninsula, Italy, as part of a clearly insular fauna. There is uncertainty regarding the timing and modes of colonisation of the island ([Mazza & Rustioni, 2008](#); [van den Hoek Ostende, Meijer & van der Geer, 2009](#); [Freudenthal & Martín Suárez, 2010](#); [Freudenthal, van den Hoek Ostende & Martín-Suárez, 2013](#); [Savorelli & Masini, 2016](#)). Due to the uncertainty of the age of the fauna, a relatively large interval of time was chosen for these three species, regarding the tip-dating. This could explain the relatively large 95% HPD range of the estimated divergence time of this clade, when compared with other taxa such as *Apocricetus s.s.*

Possible ancestors of *Hattomys* were assumed to be found in *Neocricetodon*, *Pseudocricetus* or *Apocricetus* ([Freudenthal & Martín Suárez, 2010](#)). [Freudenthal \(1985\)](#) especially emphasized the similarity between *Hattomys* and *A. alberti*, which is congruent with the close relationship between *Hattomys* and *Apocricetus s.s.*, that we retrieved in this study. According to the topology of our Bayesian tree, the common ancestor of *Hattomys* and *Apocricetus s.s.* could be a species close to *A. plinii*. The timing of the split between these two genera (7.41 Ma, 95% HPD: 6.23–8.8 Ma), and to *A. plinii* (8.22 Ma, 95% HPD: 7.1–9.56 Ma) could have an impact in the estimations of the age of the Gargano fauna. [Freudenthal, van den Hoek Ostende & Martín-Suárez \(2013\)](#) assume a single colonisation event at around 8.8–7.5 Ma, which fits quite well with the here reconstructed divergence estimations.

CONCLUSION

This study is the first to analyse the origin and early diversification of cricetine rodents based on a morphological only dataset of late Miocene and Pliocene fossils applying Bayesian and parsimony methods. Our results unravel the relationships within and between several of its genera, providing answers to their systematic uncertainties. This work evidences that the genera *Collimys*, *Rotundomys*, *Pseudocricetus*, *Apocricetus s.s.* and *Hattomys* are monophyletic whereas *Cricetulodon* is paraphyletic. The species *Apocricetus plinii* does probably not belong to *Apocricetus*, being basal to the sister clades *Apocricetus s.s.* and *Hattomys*. *Pseudocricetus* is closer to the *Apocricetus-Hattomys* clade than to *Neocricetodon*. Finally, *Kowalskia* is confirmed as a junior synonym of *Neocricetodon* with ‘*Cricetulodon*’ *complicidens* being most likely a member of this genus. The new insights into the relationships between these extinct genera, help to gain a better understanding of the evolutionary history of the Cricetidae. Based on the expanded morphological matrix, additional extinct and also extant members of the group can be rapidly added to the phylogeny in future studies. Hence, this work provides the first basis for the still relatively poorly understood origin of today’s hamsters.

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ADDITIONAL INFORMATION AND DECLARATIONS

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Competing Interests

Raquel López-Antoñanzas is an Academic Editor for PeerJ.

Author Contributions

- Moritz Dirnberger conceived and designed the experiments, performed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Pablo Peláez-Campomanes conceived and designed the experiments, analyzed the data, authored or reviewed drafts of the article, and approved the final draft.
- Raquel López-Antoñanzas conceived and designed the experiments, analyzed the data, authored or reviewed drafts of the article, and approved the final draft.

Data Availability

The following information was supplied regarding data availability:

The morphological matrix on which the phylogenetic reconstructions are based, the input and output files of the stepping stone sampling and the phylogenetic reconstructions in MrBayes and TNT, the R code used for the identification of rogue taxa, and the assessment of the stratigraphic fit of the tree topologies and the stochastic character mapping are available in the [Supplemental File](#).

Supplemental Information

Supplemental information for this article can be found online at <http://dx.doi.org/10.7717/peerj.18440#supplemental-information>.

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Summary

The phylogenies retrieved by Bayesian inference and maximum parsimony are largely congruent in their topologies, thus resulting in a robust phylogenetic hypothesis regarding several Neogene hamsters. Based on these results several lingering taxonomic and systematic questions for the group of hamsters are clarified with the classification of well-defined monophyletic groups. This study is the first to use a tip-dating approach in the case of hamsters, providing novel divergence date estimates for the included extinct clades. This chapter focuses on the earliest radiation within the group of hamsters in the Neogene, that was relatively understudied in terms of phylogenetics. Another issue in hamster systematics refers to the unknown relationship between fossil hamsters and their extant descendants, which is the topic of the next chapter.

Chapter 3

Total-evidence phylogeny reveals recent crown group radiation and biogeographical history of hamsters

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To answer the question of the phylogenetic position of extant hamsters, in this chapter, based on the framework in the previous chapter, the taxon set is considerably expanded, by including Asian, and Quaternary fossil hamsters, as well as extant ones. Additionally, molecular data of extant hamsters is incorporated to perform the first total-evidence dating on hamsters. The reconstructed phylogeny is then used to contribute to the taxonomical questions regarding extinct hamsters, compare divergence time estimates to previous studies and to explore the biogeographical history of fossil and extant hamsters.

Total-evidence phylogeny reveals recent crown group radiation and biogeographical history of hamsters

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Abstract

Background. The systematics of extant hamsters (Cricetinae) are continuously clarified, due to advances in molecular phylogenetics. In contrast, the relationships with their fossil relatives have remained relatively unclear. Furthermore, studies on the biogeographical history and divergence times of the main groups of extant hamsters have so far been limited to molecular phylogenies and node dating approaches.

Results. Here, we present the first ‘total-evidence’ analysis of hamsters that combines extinct and extant taxa, based on a comprehensive dataset covering 82 species (~75% of the total known diversity of 109 species). We performed a relaxed-clock Bayesian phylogenetic reconstruction and used the resulting tip-dated tree to estimate ancestral geographic ranges. Our results confirm ‘†*Kowalskia*’ as a synonym of †*Neocricetodon*, support the previously suggested non-monophyly of †*Allocricetus*, †*Cricetulodon*, and ‘†*Cricetinus*’ and reveal several fossil taxa as potential close relatives of the crown group. We recover a Pliocene origin of the crown hamsters, considerably younger than previous estimates of a Late/Middle Miocene origin. Our biogeographic reconstructions suggest a Central and Eastern European origin of the entire group, with crown hamsters emerging in the region around the Black Sea and the Eastern Mediterranean. Subsequent dispersal events into Western Europe and East Central Asia may be linked to the expansion of open vegetation.

Conclusions. Based on a total-evidence phylogenetic reconstruction, we highlight necessary taxonomical revision for several fossil cricetine taxa and explore the biogeographical evolution of the group. Importantly, our estimated divergence dates reveal a substantially younger group of crown hamsters than previously assumed.

Keywords: Systematics, Phylogenetics, Bayesian inference, Total-evidence dating, Divergence dates, Paleobiogeography, Cricetinae

3.1 Introduction

Hamsters (Cricetinae) are a subfamily of small- to medium-sized rodents within the family of 'hamster-like' Cricetidae (López-Antoñanzas et al. 2024). They are distributed exclusively in the Palearctic, mostly in open steppe and grassland habitats or semideserts but also on agricultural lands (Pardiñas et al. 2017a). In addition to their typical internal cheek pouches and generally short tails, they can be characterized by the morphology of their brachydont molars, with cusps organized in two longitudinal rows and usually bifurcated anterocones or anteroconids in the first molars (Kryštufek and Shenbrot 2025) (Fig. 3.1). Although popular as pets and used in medicinal or behavioral studies for decades (Wynne-Edwards 2003; Valentine et al. 2012; Yapijakis et al. 2019), the taxonomy and systematics of extant hamsters have been subject of debate, with novel phylogenetic hypotheses and conflicting classifications of taxa proposed in recent years (Lebedev et al. 2018; Ivanova et al. 2022; Jiang et al. 2024; Kryštufek and Shenbrot 2025; Pan et al. 2025). Currently, 19 species are accepted in either nine (Pan et al. 2025) or ten (Kryštufek and Shenbrot 2025) genera. Advances in extant hamster phylogenetics in the last decades have been enabled by increased availability of molecular data (Neumann et al. 2006), whereas phylogenetic analyses of extinct cricetines, based on morphological data, have focused mainly on limited taxonomic sampling or specific time periods (Cuenca-Bescós 2003; Sinitza and Delinschi 2016; Dirnberger et al. 2024b). To date, there has been no attempt to combine fossil and extant species in one global phylogenetic analysis of hamsters, even though the number of fossil taxa exceeds the extant diversity by far. Furthermore, divergence time estimates of hamsters using molecular clocks have always been limited to node-calibrated approaches, which are based on fossil ages and/or ages taken from previous molecular estimations (e.g., Stepan and Schenk 2017; Lebedev et al. 2018; Ding and Liao 2019; Jiang et al. 2024; Pan et al. 2025) (but see Dirnberger et al. 2024b).

To overcome potential downsides of node calibrations, such as errors in the assumed phylogenetic position of fossils (see, e.g., Lee et al. 2009; López-Antoñanzas et al. 2022), fossil taxa can be analyzed with extant species by integrating their phylogenetic positions and stratigraphic ages through tip-dating (Pyron 2011; Ronquist et al. 2012a). In this process, ages of fossils used as operational taxonomic units work as calibration points, removing the requirement of constraining calibrated nodes as monophyletic and following the placement of fossils determined by the joint inferential analysis rather than established a priori. While for extant taxa, molecular data are commonly used to reconstruct phylogenetic relationships, the incorporation of extinct taxa in the analysis requires the inclusion of morphological characters

that are available for both living and fossil taxa. Thus, tip-dating is frequently applied in combined datasets where both morphological and molecular data are analyzed jointly, in a so-called total-evidence dating (TED) framework. Although there are usually far fewer morphological characters than molecular loci in such analyses, they can still have a significant impact on the resulting phylogenetic topology. For this reason, if working with morphological and molecular data, integrating both data partitions is often preferable to analyzing them separately (Neumann et al. 2021; Keating et al. 2023).

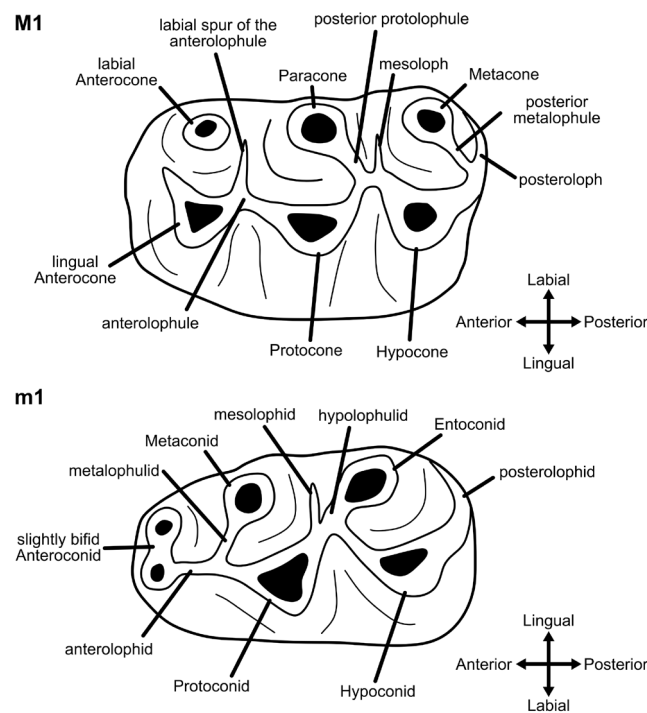


Figure 3.1. First upper (**M1**) and lower (**m1**) molar of cricetines showing the typical ‘cricetid plan’ and including the structures discussed in this study. Nomenclature and drawing following Dirnberger et al. (2024b).

In this study, we performed the first total-evidence dating analysis for hamsters that includes extant and extinct cricetines, providing a comprehensive phylogenetic hypothesis of this group of rodents. Based on the resulting phylogeny, we present potential taxonomic revisions for several fossil taxa and the closest fossil relatives of extant hamsters, along with new estimates of the biogeographical origins and divergence dates.

3.2 Material and Methods

Assembling the combined morphological and molecular data matrix

Taxonomic sampling. The morphological dataset used here expands upon Dirnberger et al. (2024b, suppl. mat. S2). The latter focused on the earliest, diverse European Neogene Cricetinae, including 41 species in seven genera, plus the outgroup †*Eucricetodon wanga* Li, Meng & Wang, 2016, as coded in López-Antoñanzas and Peláez-Campomanes (2022). Here, we considerably expanded the taxon sampling by including: (1) non-European species, (2) species of monospecific genera and (3) species ranging from the Pleistocene to the present. The full diversity of both extinct and extant Cricetinae comprises around 109 species across 28 genera (Additional file 1). For 82 species across 24 genera (75% of the total diversity), adequate material and sufficiently detailed descriptions were available to us, allowing their inclusion in this analysis, thus doubling the previous taxonomic coverage. Among these 82 species, 69 are extinct (77% of all extinct species), and 13 are extant (68%). For details of all included taxa, including stratigraphic ranges, examined material, or references used for morphological data, as well as for general taxonomic information of the group, see Additional file 1.

Morphological characters. The morphological matrix, edited in Mesquite v.3.81 (Maddison and Maddison 2023), includes discrete morphological and morphometric characters of all six molars, following López-Antoñanzas and Peláez-Campomanes (2022), and Dirnberger et al. (2024b), as well as their dental terminology. For the most important characters discussed here, see Figure 3.1. The morphological data for the newly incorporated taxa were obtained from published descriptions and images, a micro-CT scan provided by the Natural History Museum in London, and direct examination of specimens housed in the following collections and institutes: Bavarian State Collection for Palaeontology and Geology, Munich; Bavarian State Collection for Zoology, Munich; Faculty of Sciences, University Claude Bernard, Lyon; Institute of Evolutionary Science of Montpellier, University of Montpellier; National Museum of Natural History, Paris (for details, see Additional file 1). The character composition of the matrix was slightly revised, compared to Dirnberger et al. (2024b). We removed invariable characters, that had remained from the original analysis (López-Antoñanzas and Peláez-Campomanes 2022), and added two characters: the presence or absence of a direct connection between the posterior spur of the lingual anteroconid and the metalophulid in the m1 (character 86) and the presence or absence of a small extra ridge between the middle of the mesolophid and the metaconid in the m3 (character 88). Additionally, we changed and recoded the character states of the labial posteroloph in the M1 (character 87, in Dirnberger et al. (2024b): character

26). Furthermore, a contingent coding scheme was used, as it avoids biases introduced by multistate coding and, under Bayesian inference, the negative impact introduced by inapplicable characters (Simões et al. 2023). The matrix with descriptions of the characters is available in Additional file 2.

Molecular data. For all 13 extant species, molecular data were obtained from GenBank (Clark et al. 2016). Following previous molecular phylogenetic analyses (e.g., Neumann et al. 2006; Lebedev et al. 2018), the data include sequences of six genes: Cytb, VWF, IRBP, GHR, RAG1, and BRCA1. These genes show high taxonomic coverage for hamsters and have been shown to resemble phylogenetic relationships from nuclear genome-wide and mitogenomic analyses (Ding et al. 2020; Pan et al. 2025). For details of the sources and GenBank accession numbers, see Additional file 3. The sequences were aligned in AliView v.1.28 (Larsson 2014) with MUSCLE (Edgar 2004), manually trimmed, and concatenated in SeaView v.5.1 (Gouy et al. 2010), resulting in a total of 5,828 base pairs. The alignment is available in Additional file 4. The molecular data were then combined with the morphological ones in Mesquite, resulting in a total-evidence matrix of 5916 characters.

Removal of rogue taxa. Before obtaining the final phylogenies, a set of eleven so-called rogue (‘wildcard’) taxa (Wilkinson 1996) was identified to be removed from the taxon set. These taxa are unstable in their position in the tree set, and their removal therefore increases the overall posterior probabilities of the clades of the consensus tree (Aberer et al. 2013; Smith 2022). For information on the removed taxa and their identification, see Additional file 5. The following analyses were then run with the reduced taxon set of 71 cricetine taxa plus the outgroup, †*Eucricetodon wangae*.

Tip-dated relaxed-clock analysis

Relaxed-clock Bayesian inference analyses were performed using the developer version of MrBayes (future MrBayes 3.2.8 release) compiled from source code available at (<https://github.com/NBISweden/MrBayes>) (Ronquist et al. 2012b; Zhang 2022). Analyses were conducted in the DELLA cluster at Princeton University.

Character evolution model. For the evolution of morphological characters, the Mkv model (Lewis 2001) was used, which also corrects the ascertainment bias of including only variable characters. In our dataset, autapomorphies are present to avoid a sampling bias that could affect dating (Matzke and Irmis 2018). Rate variation across sites was considered by allowing for four

discrete rate categories that approximate a gamma distribution (Yang 1993), with shape (α) sampled from an exponential distribution with mean = 1.0.

The best fitting partitioning scheme and substitution models for the molecular data were found with the Partitionfinder2 algorithm (Lanfear et al. 2017) implemented in IQTREE (Nguyen et al. 2015) under the ‘greedy’ algorithm, selected with the corrected Bayesian information criterion (BIC). The resulting partitions and the respective substitution models (HKY+G and GTR+G (Hasegawa et al. 1985; Tavaré 1986; Yang 1993)) can be found in the MrBayes input files in Additional file 6.

Clock models. To relax the assumption of a constant clock rate, several relaxed clock models have been proposed, which allow relative rates to change between lineages. In the development version of MrBayes, four different ways to model relaxed clock rates are implemented (Zhang 2022). One of them (TK02) assumes an autocorrelation of rates between branches (Thorne and Kishino 2002), whereas the other three allow the relative rate of each branch to be drawn independently, either from independent log-normal distributions (ILN), from independent gamma (IGR) distributions (Drummond et al. 2006) or from gamma distributions with variance proportional to the branch length (Lepage et al. 2007). The latter model is called the white noise model (WN) in the latest MrBayes version. However, it has been named ‘IGR model’ in previous versions of MrBayes (< v.3.2.7a) (Zhang 2022). The choice of clock model has been shown to impact estimated divergence times and evolutionary rates for morphological and total evidence datasets, thus emphasizing the importance of finding the best fitting model to the respective dataset (Simões et al. 2020; Mongiardino Koch et al. 2025).

Here, we calculated the best fit clock model to the data using stepping-stone sampling with 30 steps and a burn-in of 5 steps for 200 million generations (Xie et al. 2011). As stepping-stone analyses struggled to reach convergence for the present dataset, we calculated Bayes factors using marginal likelihoods from both stepping-stone sampling and harmonic means (Additional file 6). Although the latter is less accurate than the former (Xie et al. 2011), the runs reached stationarity and convergence. The results from both approaches show that the TK02 model had the worst fit. Among the other three models, the IGR and WN models are better supported than the ILN model. As the WN model had substantially better convergence diagnostics than IGR, we report here the results of the WN clock in the main text (but see Additional file 5 for the resulting phylogeny from the IGR clock).

For the prior for the base clock rate, we used a log-normal distribution, with the mean calculated by dividing the median tree length of a preceding non-clock analysis by the median root age

prior ($6.12/37.1 = 0.165$, in natural log scale: -1.802) (following Simões et al. 2018). For a broad standard deviation, the exponent of the mean ($e^{0.165} = 1.179$) was chosen (following Pyron 2017). The morphological and molecular clocks are unlinked (but linked among molecular partitions), which has been shown to be generally better fitting than a single linked clock (Lee et al. 2013; Lee 2016; Simões et al. 2020; Barba-Montoya et al. 2021).

Tree model and calibrations. The fossilized birth-death (FBD) tree model was used, which allows the inclusion of fossil taxa as tips while modeling the process of fossil recovery and incomplete species sampling over time (Stadler 2010; Heath et al. 2014; Zhang et al. 2016). We kept the default, flat prior distributions for the relative extinction rate (turnover) and relative fossilization rate (Beta(1.0,1.0)) but used an exponential distribution with mean = 1.0 for the net diversification rate. Following our sampling of extant species, which was aimed at maximizing the included diversity, we used the ‘diversity’ sampling strategy since accounting for different sampling strategies can have an impact on the estimated diversification rates (Höhna et al. 2011). The extant sampling probability (ρ) is fixed to $13/19 = 0.68$.

As prior for the root age, an offset exponential distribution was set with a minimum age of 33 Ma (the minimal age of the oldest included fossil species, *Eucricetodon wangae*) and a mean age of 41.2 Ma (López-Antoñanzas and Peláez-Campomanes 2022). All other calibration points were based on fossil tip-dating (Ronquist et al. 2012a, 2016), constraining the ages of the fossil tips to uniform distributions that follow the stratigraphic ranges of the fossil species and uncertainties in the dates of localities (for more information about the specific ranges, see Additional file 1). Assigning range calibrations instead of points avoids possible errors in the estimated divergence times (Barido-Sottani et al. 2019). Lastly, the ingroup was constrained to be monophyletic to help with correct rooting and reaching convergence.

MCMC and output processing. The analyses were run with four independent Metropolis-Coupled Markov chain Monte Carlo runs (MCMCMC), with four chains each, sampling every 1,000 generations for a total of 120 million generations, and a burn-in of 25%. Convergence of runs was checked by the potential scale-reduction factor ($\text{PSRF} \approx 1$) (Gelman and Rubin 1992), the average standard deviation of split frequencies ($\text{ASDSF} < 0.01$) (Lakner et al. 2008), the estimated sample size ($\text{ESS} > 200$), and visually in Tracer v.1.7.2 (Rambaut et al. 2018). The resulting maximum clade compatibility consensus trees (‘allcompat’) were visualized in R with the treeio v.1.30.0, ggtree v.3.14.0 and deeptime v.2.1.0 packages (Yu et al. 2017; Wang et al. 2020; Yu 2022; Gearty 2024). The input and output files of the stepping-stone sampling and the

WN and IGR analyses are available in the Additional file 6 and in the online Supplementary data (Dirnberger et al. 2025a).

Estimation of ancestral biogeography

To reconstruct ancestral geographical ranges, each taxon was assigned to one or more of five large geographical areas, based on their fossil occurrences and/or extant biogeography (following Pardiñas et al. 2017b; Kryštufek and Shenbrot 2025). The ancestral ranges were inferred, using the maximum-likelihood framework of the R package BioGeoBEARS v.1.1.3 (BioGeography with Bayesian (and likelihood) Evolutionary Analysis with R Scripts) (Matzke 2013, 2014, 2018), which allows also for testing which model of geographical range evolution fits the data best. All compared models allow for anagenetic range expansion and contraction but differ in the way cladogenesis events are modeled. BayArea (Landis et al. 2013) (here, likelihood version: BAYAREALIKE) assumes inheritance of the geographical range identically to both daughter lineages, whereas the dispersal-extinction-cladogenesis (DEC) model (Ree and Smith 2008) and the dispersal-vicariance analysis (Ronquist 1997) (here, likelihood version: DIVALIKE) take slightly varying events of sympatry and vicariance into account (summarized in Ronquist and Sanmartín 2011). Additionally, founder-event speciation (jump dispersals) can be included in all three models (model + J), resulting in six different models to be tested. For all models, ranges of non-adjacent geographical areas and dispersal between such areas were excluded.

The best fitting model is the DEC+J model, based on the corrected Akaike information criterion (Akaike weight: 0.9, see Additional file 5). The resulting most likely ancestral ranges are shown in the main text. As the implementation of jump dispersals is under discussion (Ree and Sanmartín 2018; Matzke 2022), the results of the remaining models are shown in Additional file 5, including the most likely estimates, as well as pie charts to show uncertainties. The R code used for the analysis, and the input and output files are available in Additional file 7.

3.3 Results

Phylogenetic reconstruction

The phylogenetic tree described here is the first to include a combination of extinct and extant cricetines and is based on the most comprehensive hamster taxon sample to date.

Cricetinae systematics. In the maximum clade compatibility tree of the tip-dated, relaxed clock analysis, nine important clades A–I can be identified (Fig. 3.2). The earliest diverging clade A represents the genus †*Collimys*, followed by clade B, which includes three genera: †*Pseudocollimys*, and the sister taxa †*Colloides* and †*Rotundomys*. A clade of †*Nannocricetus primitivus*, †*Cricetulodon bugesiensis* and †*C. sabadellensis* precedes clade C, which comprises all species of †*Neocricetodon*, (including species of ‘†*Kowalskia*’), †*Cricetulodon complicidens*, †*Cricetulus mesolophidos* and †*Sinocricetus major*. Despite the high support for the clade itself (posterior probability (PP) = 0.88), most relationships within the clade remain uncertain. The remaining cricetines consist of several subsequently deriving lineages, the basal-most of which (clade D) includes all species of †*Allocricetus*, except for †*A. ehiki* and †*A. bursae* (the type species), as well as the taxa †*Nannocricetus qiui* and †*Cricetulus koufosi*. Clade E (PP = 0.45) is represented by †*Stylocricetus meoticus*, and a well-supported clade (PP = 0.84) of †*Pseudocricetus* including †*Cricetulus europaeus*. The evolutionary relationships between the following three clades (D, E, and (†*Cricetulodon meini* plus †*Cricetulodon lucentensis*)) are poorly supported (PP = 0.3 and 0.46). Therefore, any of these clades are potential sister groups of the remaining cricetines. The next diverging clade F that combines all Western European insular species, is split up in two groups, †*Hattomys* on the one side and †*Tragomys macpheeii* and †*Apocricetus darderi* on the other. Clade G shows a poorly supported sister relationship (PP = 0.44) between †*Hypsocricetus strimonis* and the well supported monophyletic group †*Apocricetus* (excluding †*A. darderi*). Similarly, clade H includes the well-supported genus †*Odessamys*, as sister to †*Mesocricetus primitivus* with low support (PP = 0.49). †*Cricetus lophidens* splits between clades G and (H,I). Lastly, †*Allocricetus ehiki* is sister taxon to the crown group I, which comprises all extant cricetines and †*Allocricetus bursae*. The evolutionary relationships between clades G, H, †*Cricetus lophidens* and clade I are poorly supported (PP ≤ 0.42), so any of these clades could be potential sister taxa to the crown group. Within clade I, the monophyly of all genera, the sister relationship between *Phodopus* and *Urocricetus*, and the clade of *Nothocricetus migratorius*, *Allocricetulus eversmanni* and *Cricetus cricetus* are strongly supported (PP ≥ 0.84). The positions of †*Allocricetus bursae*, *Tscherskia triton* and *Cricetulus*, diverging sequentially as part of the latter clade, show relatively low support (PP = 0.43, 0.49, 0.62). The phylogenetic position of *Mesocricetus* is not reliably resolved (PP = 0.22).

Divergence times. All median divergence times following the white noise (WN) clock model are depicted in Figure 3.2. The ingroup diverged at 16.69 Ma (95% highest probability density range (HPD): 15.09–18.8 Ma). Most of the fossil genera originated in the Late Miocene,

whereas the earliest split in the extant group occurred around the transition from Early to Late Pliocene, with extant genera diverging in the Pleistocene. The estimates for the earliest splits in major fossil clades are as follows: clade A: 14.74 Ma (95% HPD: 14.16–15.51 Ma), clade B: 13.15 Ma (95% HPD: 11.64–15.05 Ma), clade C: 11.4 Ma (95% HPD: 10.59–12.46 Ma), clade D: 8.43 Ma (95% HPD: 7.71–9.35 Ma), clade E: 9.32 Ma (95% HPD: 7.89–10.8 Ma), clade F: 7.76 Ma (95% HPD: 6.61–9.13 Ma), clade G: 6.82 Ma (95% HPD: 5.86–7.94 Ma), and clade H: 5.45 Ma (95% HPD: 4.33–6.68 Ma). For the divergence ages of extant hamsters, refer to Table 3.1.

Biogeographical history

According to the dispersal-extinction-cladogenesis analysis including jump dispersals (DEC + J), the most likely inferred geographical origin of the total group is found in Central to Eastern Europe (Fig. 3.3), from where dispersal towards Western Europe and towards the East occurred multiple times. Entrance into Eastern Central Asia is mostly connected to a preceding range expansion to South Eastern Europe/South Western Asia. Regarding extant hamsters, the most probable area of the last common ancestor is South Eastern Europe to South Western Asia, from where two lineages dispersed towards Eastern Central Asia, where most of the extant genera subsequently diversified. The rate of anagenetic range expansion is higher than the rate of jump dispersal, and clearly higher than the rate of range contraction (Additional file 5).

3.4 Discussion

Systematics and phylogeny of Cricetinae

The final phylogeny (after removing rogue taxa) includes 71 cricetine species in 24 genera, representing an addition of 33 species in 17 genera in this study.

Compared with the results retrieved by Dirnberger et al. (2024b), the topology of the phylogeny remains largely consistent for the taxa they included: †*Collimys*, †*Rotundomys*, †*Neocricetodon*, †*Cricetulodon*, †*Pseudocricetus*, †*Apocricetus*, and †*Hattomys*. We confirm their six monophyletic genera (excluding †*Cricetulodon*) with strong support (PP $\geq$ 0.84), as the posterior probabilities of †*Collimys* (0.99 vs. 0.76) and †*Neocricetodon* (0.88 vs. 0.52) increased clearly.

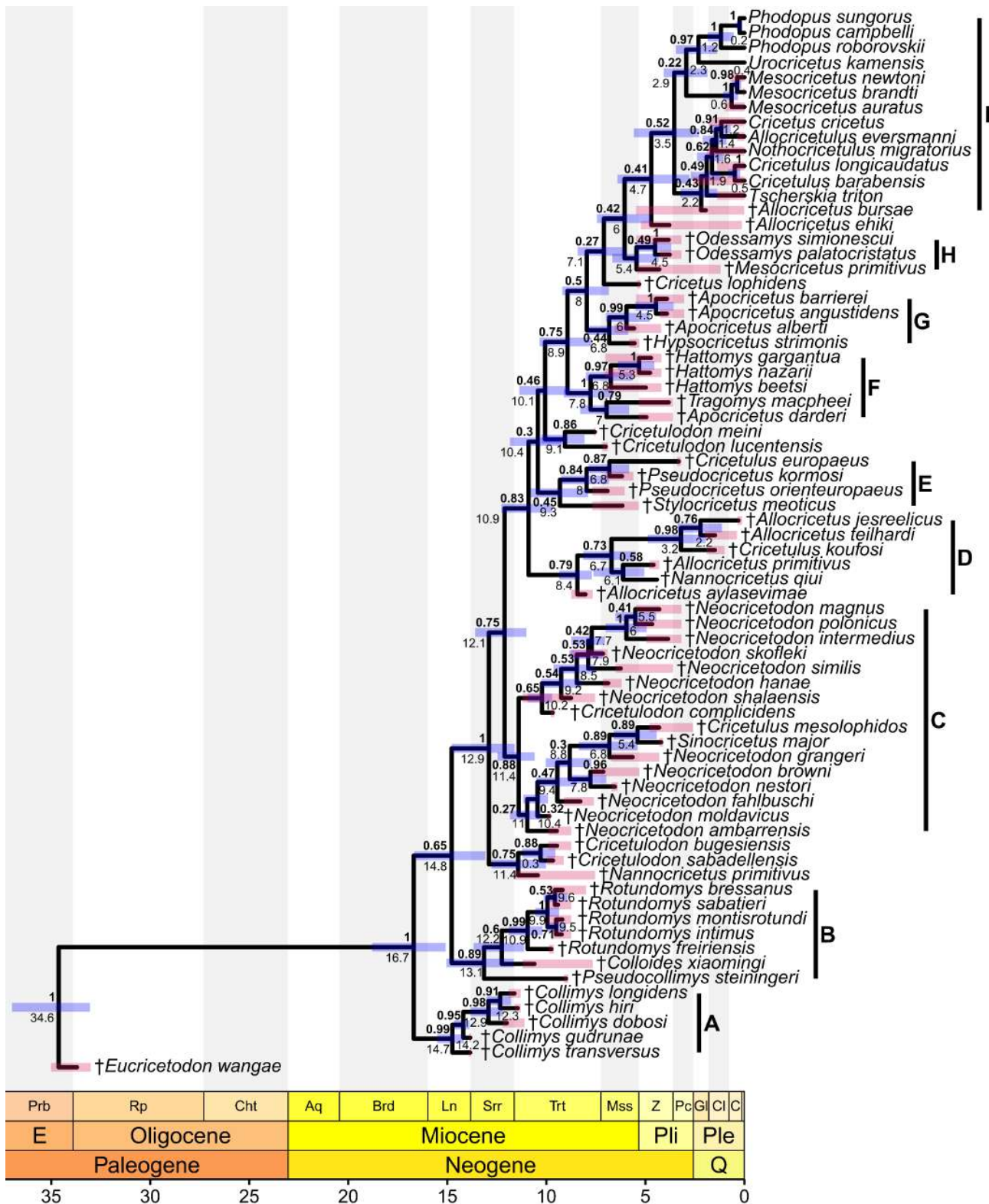


Figure 3.2. Maximum clade compatibility tree from the tip-calibrated relaxed-clock WN Bayesian inference analysis combining molecular and morphological data. Posterior probabilities (in bold) and median ages of clades are indicated at respective nodes, blue node bars indicate the 95% highest posterior density for divergence times, red tip bars indicate the stratigraphic range of the respective taxon. The scale axis is in Ma, the chronostratigraphic chart follows Cohen et al. (2013; updated 2024/10).

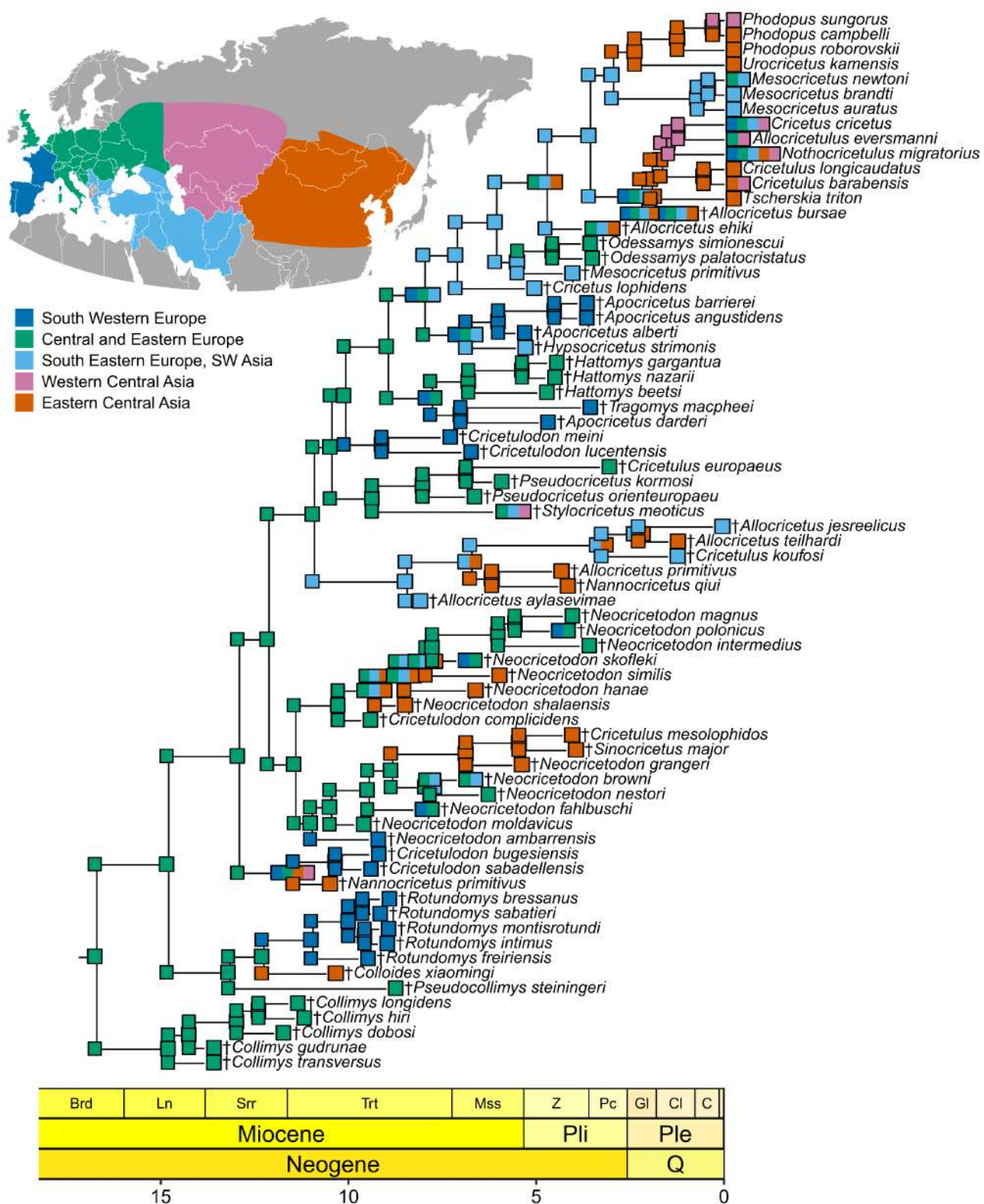


Figure 3.3. Reconstruction of the biogeographical history of Cricetinae based on the DEC+J model. Tip ranges and estimated most likely ancestral ranges are represented by colors, explained by the top left map showing the respective areas. The scale axis is in Ma, the chronostratigraphic chart follows Cohen et al. (2013; updated 2024/10).

Differences, following from the expanded taxon set, are observed in the phylogenetic position of †*Cricetulodon bugesiensis* and †*C. sabadellensis* that diverge more basally (together with †*Nannocricetus primitivus*), and the internal relationships of †*Neocricetodon*.

While most nodes within †*Neocricetodon* (clade C) remain poorly supported, the more strongly supported clades identified by Dirnberger et al. (2024b), such as (†*N. magnus*, †*N. polonicus*, †*N. intermedius*) and (†*N. browni*, †*N. nestori*) are also recovered here. In addition to †*Cricetulodon complicidens*, which was also previously identified as a potential member of †*Neocricetodon* (Dirnberger et al. 2024b), our analysis places †*Cricetulus mesolophidos* and †*Sinocricetus major* within this clade, as both taxa are included here for the first time. Our results thus support the earlier tentative reassignment of †*C. mesolophidos* to †*Neocricetodon*, which was based mainly on the presence of elongated mesolophids (Xie et al. 2021, 2023). †*Sinocricetus major* likewise exhibits a medium to long mesolophid on the m1. This species was originally assigned to †*Sinocricetus* due to the presence of a mesolophid (or ‘pseudomesolophid’) on the m2 that connects to the metaconid (Li 2010). However, in the type species †*S. zdanskyi* (Schlosser 1924; Schaub 1930; Wu 1991; Qiu and Li 2016) and in †*S. progressus* (Qiu and Storch 2000; Qiu and Li 2016), the mesolophid of the m2 seems either absent or weak. Since both †*S. zdanskyi* and †*S. progressus* were identified as rogue taxa and excluded from the analysis, the position of †*S. major* within †*Neocricetodon* should be considered with caution.

Dirnberger et al. (2024b) also addressed the question of the seniority of †*Neocricetodon* over ‘†*Kowalskia*’, as demonstrated by Freudenthal et al. (1998). In line with their findings, our results recover two clades within the group, potentially supporting a distinction between the two genera, as †*Neocricetodon grangeri* and †*Neocricetodon polonicus* (the type species of †*Neocricetodon* and ‘†*Kowalskia*’, respectively) fall into different clades. However, one of these clades (including †*N. grangeri*) is not supported (PP = 0.27), and the species composition differs from the results Dirnberger et al. (2024b), underlining that these clades are not reliable. The support for clade C, including all species of †*Neocricetodon*, increased (PP = 0.88 vs. 0.52), which reinforces this genus as a stable and distinct monophyletic lineage.

Recently, Feroz et al. (2024) revisited the question of separating †*Neocricetodon* and ‘†*Kowalskia*’, arguing for two distinct genera. They stated that Dirnberger et al. (2024b) focused on “European species, such as *N. grangeri*, *N. complicidens*, and *N. progressus*” (Feroz et al. 2024, p. 3) and that these species have already been rejected as ‘†*Kowalskia*’ by Qiu and Li (2016). The mentioned taxon ‘†*N. complicidens*’ (addressed by Qiu and Li 2016) was,

however, not part of the analysis of Dirnberger et al. (2024b) (see p. 16 there for clarification). Although still not all †*Neocricetodon* species are included in our analysis, such as the recently described ‘†*Kowalskia weni*’ Feroz, Li, Fazal, Qiu & Ni, 2025, we incorporate now a broad range of taxa, including four species from China, namely †*N. hanae*, †*N. shalaensis*, and †*N. similis*, as well as †*N. grangeri*, which is in fact a Chinese species (Daxner-Höck et al. 1996; Wu and Flynn 2017) (potentially five, if †*Cricetulus mesolophidos* is also considered). We also want to point out that Wu and Flynn (2017) emphasized the wide morphological variability within †*Neocricetodon*, encompassing ‘†*Kowalskia*’, and that Sinita and Delinschi (2016) presented a phylogeny with both †*N. grangeri* and †*N. polonicus* clearly belonging to the same clade, stating: “The congeneric status of *Neocricetodon grangeri* and *Kowalskia polonica* was clearly demonstrated by Freudenthal et al. (1998) [...]. Although this matter is not accepted by some specialists [...], we follow most recent authors [...] in recognizing the priority of the name *Neocricetodon*” (p. 773) (see also Prieto and Rummel 2009a; de Bruijn et al. 2012; Maridet et al. 2014; Xie et al. 2023; Badea et al. 2025). Thus, following Wu and Flynn (2017) and Sinita and Delinschi (2016), as well as the results presented here and in Dirnberger et al. (2024b), the seniority of †*Neocricetodon* over ‘†*Kowalskia*’ is supported.

In contrast to the work of Dirnberger et al. (2024b), †*Apocricetus darderi* is not identified as a rogue taxon but is recovered here as the sister taxon to †*Tragomys macpheeii*. Both species are found in Mallorca (Agustí et al. 2012; Torres-Roig et al. 2019). Together with †*Hattomys*, they form a clade (Fig. 3.2, clade F), which thus includes all Western Mediterranean insular endemic cricetines. We agree with the hypothesis of Torres-Roig et al. (2019) in considering that †*A. darderi* is either a sister taxon or a direct ancestor of †*T. macpheeii*. However, we cannot confirm their assignment of †*A. darderi* to †*Apocricetus*. Following their reasoning to not combine both insular species within a single genus, due to clear morphological differences, †*A. darderi* requires a new genus name.

The genus †*Pseudocricetus* (in clade E), represented here by †*P. kormosi* and †*P. orienteuropaes* also includes †*Cricetulus europaeus*. Xie et al. (2023) provisionally reassigned †*Cricetulus europaeus* to *Tscherskia* but noted that this species shows a lower tendency to develop mesolophids on the m1 and m2 compared to *Tscherskia triton*. Specifically, a short- to medium-length mesolophid is present on most m2s, whereas the m3 consistently exhibits a mesolophid of variable length (Hir 1994; Xie et al. 2023). This dental morphology closely resembles that of †*Pseudocricetus* (Topachevsky and Skorik 1992; Daxner-Höck et al. 1996; Freudenthal et al. 1998) (Fig. 3.4a). In particular, the presence of a short mesolophid on the m2

(Dirnberger et al. 2024b) and the frequent presence of a mesolophid on the m3 (Freudenthal and Kordos 1989; Topachevsky and Skorik 1992; Sinitsa 2010).

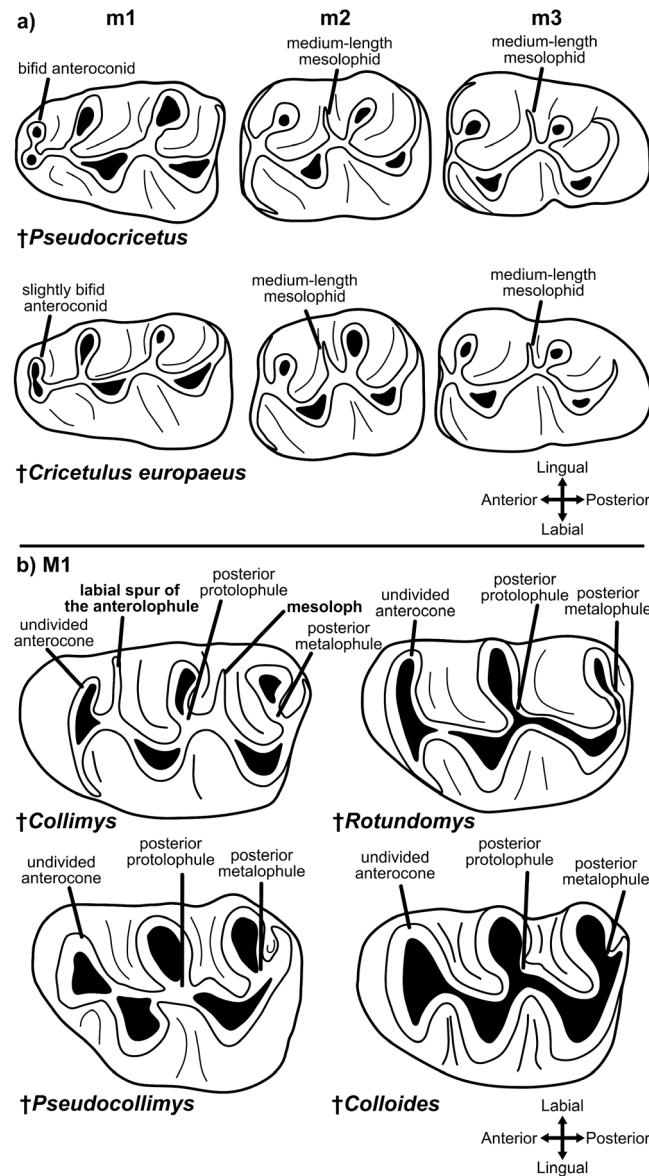


Figure 3.4. Selected dental characters important for cricetine systematics regarding **a)** clade E, and **b)** clades A and B. Drawings follow data in the following references: †*Pseudocricetus* (Topachevsky and Skorik 1992; Sinitsa 2010); †*Cricetulus europaeus* (Hír 1994); †*Collimys* (Hír 2005; Prieto and Rummel 2009b, 2009a); †*Rotundomys* (Mein 1965, 1975; López-Antoñanzas et al. 2014), †*Pseudocollimys* (Daxner-Höck 2004), †*Colloides* (Qiu and Li 2016).

Other characteristics and proposed synapomorphies of †*Pseudocricetus* (Topachevsky and Skorik 1992; Freudenthal et al. 1998; Dirnberger et al. 2024b), such as the presence of an anterior protolophule on the upper molars and a short but distinct posteroloph on the M3, are also observed in †*C. europaeus*. An argument against the reallocation of †*C. europaeus* to

†*Pseudocricetus*, is the presence of a single or only slightly divided anteroconid on the m1 of †*C. europaeus* (Kretzoi 1959, 1962; Hír 1994), compared to the bifid anteroconid of †*Pseudocricetus* (Topachevsky and Skorik 1992) (Fig. 3.4a). This character can be variable within genera and even within species (Xie et al. 2023). Therefore, it would be necessary to assess the intraspecific variability of this character in †*C. europaeus* to confirm or refute this argument. This is, however, not possible, due to the limited available material of †*C. europaeus*. Given the high support for the placement of †*C. europaeus* within †*Pseudocricetus* in the phylogeny (Fig. 3.2, clade E) and the above-mentioned morphological similarities, we tentatively propose transferring this taxon to †*Pseudocricetus*.

Two here included monospecific genera, †*Pseudocollimys* and †*Colloides*, were described as morphologically similar to †*Collimys* by the original authors (Daxner-Höck 2004; Qiu and Li 2016), owing to an undivided anterocone on the M1, only posterior protolophule and metalophule on M1 and M2 (Fig. 3.4b), as well as increased hypsodonty compared to other cricetines. These similarities led Tesakov et al. (2017) to suggest that both taxa are synonymous with †*Collimys*. In our analysis, †*Pseudocollimys* and †*Colloides* clustered within the same clade as the monophyletic †*Rotundomys* (Fig. 3.2, clade B). Interestingly, a relatively close relationship between †*Pseudocollimys* and †*Colloides* was already suspected by Qiu and Li (2016), and similarities between †*Pseudocollimys* and †*Rotundomys* were mentioned by Daxner-Höck and Höck (2015). The main arguments for separating †*Pseudocollimys* and †*Colloides* from †*Collimys* refer to the reduced mesoloph(id)s and the absence of the labial spur of the anterolophule in the two former taxa (Fig. 3.4b). In †*Rotundomys*, however, these characteristics are also found, as are the above-mentioned single anterocone on the M1, and the only posterior protolophule and the metalophule on the M1 and M2 (Mein 1965; López-Antoñanzas et al. 2014) (Fig. 3.4b). These similarities explain the close relationship between the genera in clade B.

The genus †*Nannocricetus* was not resolved as monophyletic, in our analysis. The phylogenetic position of the type species †*N. mongolicus* was uncertain, and as a result, it was identified as a rogue taxon and excluded from the final dataset. Several synapomorphies have been proposed for this genus. These include (1) a single or slightly divided anteroconid on the slender m1; (2) short and transverse metalophulids and hypolophulids on the lower molars; (3) a general absence of mesoloph(id)s; (4) a bilobed anterocone and the lack of a protolophule on the M1; and (5) specific shapes of sinuses and sinusids (Wu 1991; Zhang et al. 2008; Qiu and Li 2016; Li et al. 2017). Some of these characters are also commonly found in other cricetine genera or show considerable variability within †*Nannocricetus* itself. Moreover, when examining

published figures of †*N. mongolicus* (Wu 1991, pl. 5 & 6, figs. 62–69 & 79–81), we could not clearly identify consistent differences in the direction and length of meta- and hypolophulids compared to other cricetines. Nonetheless, some of these potentially diagnostic characters, such as the width of the m1, are not represented in our current character matrix. Including these traits in a future version of the matrix may help in recovering †*Nannocricetus* as a monophyletic group.

The phylogenetic position of the monospecific genus †*Stylocricetus* as sister to †*Pseudocricetus* (Fig. 3.2, clade E) is only weakly supported (PP = 0.45). Based mainly on features of the skull and mandible, especially referring to the position and shape of the incisive foramen, Topachevsky and Skorik (1992) distinguished †*Stylocricetus* from †*Pseudocricetus* and other cricetines and proposed a closer affinity to *Phodopus* and †*Odessamys*, which is, however, not supported by our results. An affinity between †*Stylocricetus* and ‘†*Kowalskia complicidens*’ was suggested by Topachevsky & Skorik (1992), based on the strong ectoloph (or ‘pseudoectoloph’), the crest connecting the paracone and the metacone on the upper molars, which is present in both taxa. This character was the main reason for excluding ‘†*K. complicidens*’ from †*Neocricetodon* (or ‘†*Kowalskia*’) in subsequent studies (Daxner-Höck et al. 1996; Qiu and Li 2016; Sinita and Delinschi 2016) and can therefore not be used to argue for an affinity of †*Stylocricetus* to †*Neocricetodon*. As this strong ectoloph is absent in †*Pseudocricetus* (Freudenthal and Kordos 1989; Topachevsky and Skorik 1992; Sinita 2010), its presence in †*Stylocricetus* justifies, however, the separation of †*Stylocricetus* and †*Pseudocricetus*.

†*Hypsocricetus* is distinguished from other late Neogene to extant cricetines by its relatively high tooth crowns (Daxner-Höck 1992). The close affinity between †*Hypsocricetus* and †*Apocricetus* in our phylogeny is only very weakly supported (Fig. 3.2, clade G). In addition to the lower tooth crown of †*Apocricetus*, the presence of a bifid anteroconid on the m1 in †*Hypsocricetus* further separates the two genera. In †*Apocricetus s.s.*, the anteroconid is multi-lobed and more crest-like (Freudenthal et al. 1998). This character had also supported the exclusion of †*A. plinii* from †*Apocricetus s.s.* in the analysis of Dirnberger et al. (2024b), as the latter also has a bifid anteroconid.

The genus *Cricetulus* deserves special attention, due to frequent changes in its scope. Regarding the extant taxa, this genus formerly comprised several smaller forms with similar dental morphology (McKenna and Bell 1997; Musser and Carleton 2005), which could later be split up based on advances in molecular phylogenetics, resulting in the erection of *Urocrinetus* and

Nothocricetulus alongside *Cricetulus* (Neumann et al. 2006; Lebedev et al. 2018). While the status and content of these three genera are now widely accepted (Kryštufek and Shenbrot 2025), there are still frequently used fossil genera whose status remains controversial, as some authors consider them synonymous with *Cricetulus* (e.g., McKenna and Bell 1997; van den Hoek Ostende et al. 2015). One of these genera is ‘†*Cricetinus*’ Zdansky, 1928, which was synonymized with *Cricetulus* by various authors and databases (McKenna and Bell 1997; Musser and Carleton 2005; The NOW Community 2023). The three fossil *Cricetulus* species included in our phylogeny, †*Cricetulus europaeus*, †*C. koufosi* and †*C. mesolophidos* were originally described as part of ‘†*Cricetinus*’ (Kretzoi 1959; Koliadimou 1996; Wu and Flynn 2017). Recently, the type species ‘†*Cricetinus varians*’ was identified as a chrono-subspecies of *Tscherskia*, rendering ‘†*Cricetinus*’ a synonym of *Tscherskia* (Xie et al. 2021, 2023). Of the six remaining species originally placed in ‘†*Cricetinus*’, four, †*C. europaeus*, †*C. gritzai*, †*C. janossyi*, and †*C. koufosi*, were provisionally reassigned to *Tscherskia*, whereas the other two, †*C. beremendensis* and †*C. mesolophidos*, were reallocated to †*Allocricetus* and †*Neocricetodon*, respectively (Xie et al. 2023). Our analysis supports the transfer of †*C. mesolophidos* to †*Neocricetodon* (Fig. 3.2, clade C). †*C. beremendensis*, is identified as a rogue taxon but shows a provisional position close to †*Allocricetus ehiki* (Additional file 5), which also supports its tentative assignment to †*Allocricetus* (Xie et al. 2023). Regarding the above-mentioned species reassigned to *Tscherskia* by Xie et al. (2023), †*C. janossyi* was excluded from our analysis due to the limited material available (no M1, see Hír 1996), as was †*C. gritzai*, which was identified as a rogue species. The position of †*C. europaeus* within †*Pseudocricetus* (Fig. 3.2, clade E), together with their morphological similarities, hint toward a transfer of †*C. europaeus* to †*Pseudocricetus*, as discussed above. Finally, †*C. koufosi* falls within the clade that includes most species of †*Allocricetus* (Fig. 3.2, clade D). Its provisional transfer to *Tscherskia* was mainly justified by the absence of a mesolophid on the m1, a character that is, however, also absent in †*Allocricetus* (Xie et al. 2023).

Hír (1993) noted a close morphological similarity between †*Allocricetus bursae*, the type species of †*Allocricetus*, †*A. ehiki*, and ‘*Cricetulus migratorius*’ but kept them as separate taxa. In contrast, Kowalski (2001) considered †*Allocricetus bursae* to be a synonym of ‘*Cricetulus migratorius*’. Several authors subsequently treated the genus †*Allocricetus* as a junior synonym of *Cricetulus* (van den Hoek Ostende et al. 2015; Horáček and Lebedová 2022). Our analysis reveals a close affinity of †*A. ehiki* and †*A. bursae* to the extant species, but not specifically to *Nothocricetulus migratorius* (Fig. 3.2). Our results are partly consistent with those of Cuenca-Bescós (2003), as they support the non-monophyly of the genus †*Allocricetus*, i.e., the

placement of †*A. bursae*, †*A. ehiki*, and †*A. jesreelicus* + †*A. teilhardi* in separate clades. According to our results, †*A. aylasevimae* and †*A. primitivus*, which were described later, also belong to the latter clade, along with †*Nannocricetus qiui* and †*Cricetulus koufosi* (Fig. 3.2, clade D). While Cuenca-Bescós (2003) considered this clade to be closely related to *Nothocricetulus migratorius*, our results do not confirm such a relationship. Consequently, the proposed transfer of these taxa to *Cricetulus* (or *Nothocricetulus*) is not corroborated by our study. Since †*A. bursae* is the type species of †*Allocricetus*, the establishment of a new genus name would be required for this group.

Following the original description of ‘†*Cricetulus simionescui*’ by Schaub (1931) on the basis of three hemi-mandibles, this taxon was subsequently transferred to a new subgenus, called *Moldavimus* Samson & Rădulescu, 1973, due to a slightly shorter m1 than in other *Cricetulus* species and an undivided anteroconid (Samson and Rădulescu 1973). Topachevsky and Skorik (1992) described new material of the species from Ukraine and transferred it to a new genus, called †*Odessamys* with type species †*O. palatocristatus*. Recently, Crespo et al. (2023) re-examined parts of the original material and transferred it to †*Allocricetus*. Unfortunately, they did not take into account the previous transfer to †*Odessamys*, as they only compared the material to †*Allocricetus* and †*Neocricetodon*. Our analysis supports a close phylogenetic relationship between ‘†*Cricetulus simionescui*’ and †*Odessamys palatocristatus*, and we therefore concur with the transfer to the genus †*Odessamys*, as proposed by Topachevsky and Skorik (1992).

There are two fossil members of extant genera included in our analysis, †*Mesocricetus primitivus* and †*Cricetus lophidens*. In our results, none of them are recovered within their respective genera. However, their positions in the tree are not well supported. Both taxa were originally described from the Mio-Pliocene transition of Greece (de Bruijn et al. 1970). †*Mesocricetus primitivus* was described as “less specialized” than extant *Mesocricetus* taxa, which explains its basal position in our analysis, suggesting that it probably does not belong to *Mesocricetus*. †*Cricetus lophidens* was later considered an “enigmatic” taxon, and its classification as *Cricetus* was questioned by Kálin (1999) and de Bruijn et al. (2013), with the latter referring to it as “*Cricetus*” in quotation marks. This interpretation is further supported by the selenodont morphology of †*C. lophidens*, which contrasts sharply with *Cricetus cricetus* (Pardiñas et al. 2017a). Additionally, four taxa from the Pleistocene are recognized: †*C. major* (Woldřich, 1880), †*C. nanus* (Schaub, 1930), †*C. praeglacialis* (Schaub, 1930) and †*C. runtonensis* (Newton, 1909). The status of these taxa has been extensively discussed over the last century (see summaries in (Hír 1997a; Kryštufek et al. 2020)). The main issues concern

whether they represent distinct species or subspecies (Hír 1997a; Sabol and Zervanová 2014; Horáček and Lebedová 2022) and the possible synonymy of the two largest forms, †*C. runtonensis* and †*C. major* (Fahlbusch 1976; Pradel 1985; Maul and Parfitt 2010). A revision of these taxa is beyond the scope of this study, and they are therefore not included in our analysis.

Relationships with and within extant Cricetinae. All extant hamsters form a monophyletic clade, including a single extinct taxon, †*Allocricetus bursae* (Fig. 3.2, clade I). Within this clade, we recognize the same three groups (tribes according to Kryštufek and Shenbrot 2025), as previous nuclear, mitochondrial and cytogenetic analyses: Urocricetini: *Phodopus* and *Urocricetus*; Mesocricetini: *Mesocricetus*; and Cricetini: *Cricetus*, *Allocricetulus*, *Nothocricetulus*, *Cricetulus*, and *Tscherskia* (Neumann et al. 2006; Romanenko et al. 2007; Lebedev et al. 2018; Ding et al. 2020). The remaining genus, *Cansumys*, which is not included in this analysis due to the scarcity of available morphological and molecular data, was recently shown to also belong to the Cricetini (Pan et al. 2025). The internal relationships within these clades follow most of the mentioned studies, as well. A difference is found, however, in the relationships between the clades, as *Mesocricetus* is recovered as sister to the Urocricetini and not to the Cricetini. The support for this relationship is, however, negligible (PP = 0.22). This uncertainty of the position of *Mesocricetus* is unsurprising, considering that most of the gene sequences are taken from two studies, in which the position of *Mesocricetus* also varied, depending on the analyzed gene (Neumann et al. 2006; Lebedev et al. 2018). Additionally, there was lower support for the position of *Mesocricetus* in the concatenated tree than for other nodes (see Neumann et al. 2006, fig. 2, node 2; Lebedev et al. 2018, fig. S1). The same problems arose, however, with sequences and/or genes not included (see Steppan and Schenk 2017, figs. S4, S6; Lebedev et al. 2018, fig. 1), as well as in an analysis of genome-wide single nucleotide polymorphisms (Pan et al. 2025). It appears that the position of *Mesocricetus* needs to be verified in future studies.

The most frequently proposed ancestor of extant hamsters is †*Neocricetodon* (Fahlbusch 1969; Zheng 1984; Xie et al. 2023), although Fejfar et al. (1970) considered †*Neocricetodon* to represent a separate evolutionary lineage, which is in agreement with our results. There has been general agreement on a close relationship between †*Allocricetus* (considering †*A. bursae* and †*A. ehiki*) and the extant Cricetini, a relationship that is supported by our results. In contrast, our analysis does not confirm a close phylogenetic relationship between other extant and fossil genera, such as the relationship between *Cricetulus* and †*Nannocricetus* proposed by Qiu and Li (2016) or between *Cricetulus* (which at the time also included *Nothocricetulus* and

Urocricetus) and †*Odessamys* suggested by Topachevsky and Skorik (1992). The latter hypothesis is reflected in our results through the affinity of †*Odessamys* with the entire group of extant taxa. However, the phylogenetic relationships of the closest fossil relatives to extant cricetines are recovered with low support. Thus, besides †*A. ehiki* and †*Odessamys*, also †*Cricetus lophidens*, †*Mesocricetus primitivus*, and clade G could be potential sister groups of the extant hamsters.

Divergence dates

The estimated divergence times of the fossil genera obtained in our previous partial study (Dirnberger et al. 2024b) remain mostly unchanged but show here slightly higher precision. Regarding the crown group, our estimates are more precise and, in some cases, clearly younger than previous results based on extant molecular phylogenies with node calibrations, as shown in Table 3.1 (Steppan et al. 2004; Neumann et al. 2006; Lebedev et al. 2018; Romanenko et al. 2021; Jiang et al. 2024; Pan et al. 2025).

The largest difference can be seen in the age of crown hamsters (see also Steppan et al. (2004): 13.5–14.1 Ma; Romanenko et al. (2021): 11.2–12 Ma; Jiang et al. (2024): 12.73 Ma). To explore the reasons for these differences, the calibration points of these studies are examined. According to Romanenko et al. (2021), their only calibration point, regarding the root of Cricetidae, was taken from Lebedev et al. (2020), who focused on Arvicolinae and the genetic variation of *Ellobius*. The two calibration points used by Neumann (2006), i.e., the splits between *Mus* and *Rattus* (12 Ma) and between *Gerbillus* and *Tatera* (8 Ma), are outside of the Cricetinae. The former calibration point has been controversial (Kimura et al. 2015; Aghová et al. 2018), and its usage has been shown to possibly lead to overestimated results (Steppan et al. 2004). Subsequent studies that estimated divergence ages of specific groups within Cricetinae were based on the possibly too old ages of Neumann (2006), which again influenced their results (Meshchersky and Feoktistova 2009; Neumann et al. 2017).

All other above-mentioned studies built their analyses directly or indirectly upon the results of Steppan et al. (2004). Lebedev et al. (2018) took the age inferred by Steppan et al. (2004) for Cricetidae (18.7–19.6 Ma, 95% CI 16.6–21.1 Ma) as basis of their root age prior. This was followed by Jiang et al. (2024), although their root corresponds to the split of Arvicolinae and Cricetinae (outgroup *Lagurus lagurus*), which is considerably younger (see Steppan and Schenk 2017). Pan et al. (2025) utilized three calibration points, based on the inferred ages and

calibration points of Lebedev et al. (2018). Consequently, all the applied calibration strategies depend on each other, which can be problematic (see Graur and Martin 2004). In this case, the age of Cricetidae reconstructed again in later studies, turned out to be noticeably younger (14.6 Ma in Steppan and Schenk 2017).

Table 3.1. Divergence ages (in Ma) within crown cricetines, estimated in this and other studies. The age ranges of Neumann et al. (2006) follow their implementations of different clocks and calibration points. For the other studies, the median (this study) or mean (Lebedev et al. 2018; Pan et al. 2025) ages and 95% HPD ranges are given.

	This study ^a	Neumann et al. (2006) ^b	Lebedev et al. (2018) ^c	Pan et al. (2025) ^d
Crown group	3.55 (2.28–5.58)	8.5–12.2	12.25 (10.24–14.54)	12.25 (10.24–14.54)
Urocrinetini	2.32 (1.37–3.47)	-	10.37 (8.37–12.52)	11.23 (8.83–13.77)
<i>Phodopus</i>	1.19 (0.54–1.87)	4.9–6.9	5.69 (4.38–7.04)	7.45 (4.03–10.79)
<i>P. sungorus</i> / <i>P. campbelli</i>	0.24 (0.1–0.41)	0.8–1.0	-	-
<i>Mesocricetus</i>	0.65 (0.32–1.06)	2.5–2.7	1.81 (1.04–2.61)	-
<i>M. newtoni</i> / <i>M. brandti</i>	0.37 (0.16–0.64)	1.7–1.8	-	-
Cricetini	1.9 (1.21–2.7)	5.8–7.4	6.36 (4.83–8.01)	8.45 (5.64–11.39) ^e
<i>Cricetulus</i> / <i>Nothocricetulus</i> + <i>Allocricetulus</i> + <i>Cricetus</i>	1.63 (1.04–2.39)	-	5.61 (4.43–6.99)	7.78 (4.93–10.7)
<i>Nothocricetulus</i> / <i>Allocricetulus</i> + <i>Cricetus</i>	1.43 (0.9–2.1)	-	4.46 (3.41–5.88)	6.56 (3.61–9.52)
<i>Cricetus</i> / <i>Allocricetulus</i>	1.17 (0.69–1.75)	-	2.65 (1.60–4.00)	5.46 (2.67–8.35)
<i>Cricetulus</i>	0.48 (0.21–0.82)	2.9–3.1	1.06 (0.50–1.64)	4.02 (2.43–6.51)

^aexcluding *A. curtatus*, *C. canus*, *C. sokolovi*, *M. raddei*, *T. collina*, *U. lama*

^bexcluding *A. curtatus*, *C. canus*, *C. sokolovi*, *T. collina*, *U. kamensis*, *U. lama*

^cexcluding *C. canus*, *M. newtoni*, *M. raddei*, *P. campbelli*, *T. collina*

^dexcluding *A. curtatus*, *M. brandti*, *M. newtoni*, *M. raddei*, *P. sungorus*

^efor comparison, age of Cricetini taken excluding *C. canus*

The general problem in dating the origin of crown cricetines lies in the difficulty of finding reliable calibration points, as the phylogenetic relationships between fossil and extant taxa are often unclear (see, e.g., †*Cricetus lophidens*). Lebedev et al. (2018) employed three calibration points within Cricetinae. The first one constrained the origin of *Mesocricetus* to be older than

4.9 Ma, based on †*Mesocricetus primitivus* from Silata (Vasileiadou et al. 2003). However, in this study, †*M. primitivus* is shown to likely not belong to *Mesocricetus* (Fig. 3.2). The next oldest fossil attributed to the genus is *Mesocricetus auratus* from Dursunlu (Güleç et al. 1997; Neumann et al. 2017), which is considerably younger, dated to the latest Early Pleistocene (ca. 0.99–0.78 Ma) (Güleç et al. 2009). Their second calibration point applied a minimum age of 2.7 Ma to the split of *Cricetus* and *Allocricetulus*, supported by three fossils (Lebedev et al. 2018). The first, †*Cricetus runtonensis* from the MN16 of Rebielice-Królewskie, Poland (Fahlbusch 1969), was revised by Pradel (1988), who distinguished between older and younger material (RK-1A and B). †*Cricetus runtonensis* belongs to the younger material and was suggested to be Upper Biharian in age (ca. 0.9–0.5 Ma) (see also Nadachowski 1990). The older material (MN16) was described as *Cricetus* sp. 3. However, this classification is doubtful, given the age of the publication, as several Neogene ‘*Cricetus*’ taxa were later transferred to different fossil genera, such as †*Pseudocricetus* and †*Apocricetus* (Topachevsky and Skorik 1992; Freudenthal et al. 1998). The same issue applies to the second *Cricetus* fossil cited by Lebedev et al. (2018) from Selim-Dzhevar. According to the NOW Community (2023), this record is originally based on a catalog by Borissiak and Belyaeva (1948), which is unavailable to us. Due to the age of the reference, the occurrence requires revision in light of the subsequent transfer of Neogene ‘*Cricetus*’ taxa. The third fossil cited by Lebedev et al. (2018) is *Allocricetulus* from Kiikbay (MN16). In the referenced publication, it was named “*Allocricetus eversmani*” (Tjutkova and Kaipova 1996). Although we cannot rule out the possibility that “*Allocricetus*” and “*eversmani*” are simply typos and that the occurrence refers, in fact, to *Allocricetulus eversmanni*, the large gap to the otherwise oldest known *Allocricetulus* fossils raises additional doubts about this classification. These next-oldest known *Allocricetulus* fossils are from the Middle Pleistocene of Italy, Romania and Russia (ca. 0.424–0.125 Ma) (Rădulescu and Samson 1995; Kowalski 2001; Danukalova et al. 2007; Agadzhanyan et al. 2021). The third calibration point of Lebedev et al. (2018), constrained the divergence of *Cricetulus* to be older than 2.7 Ma based on *Cricetulus* cf. *barabensis* (Transbaikalia, MN16) (Alexeeva and Erbajeva 2008). In the latest description of the Baikalian fauna, this Pliocene fossil was reassigned to *Cricetulus* sp. (Erbajeva et al. 2021). This attribution is problematic, given that *Cricetulus sensu lato* is polyphyletic (including *Urocricetus* and *Nothocricetulus*). The oldest fossil attributed to *C. barabensis*, known to us, is magnetostratigraphically dated to the Early Pleistocene (ca. 2.595–2.14 Ma) (Opdyke et al. 2013; Wu and Flynn 2017). In summary, all three calibration points proposed by Lebedev et al. (2018), the latter two of which were also used by Pan et al. (2025),

seem to be too old. Although the discrepancy is smaller in the third one, their use may have led to an overestimation of divergence dates.

As our estimated divergence ages are substantially younger, they do not produce long ghost lineages, as in previous studies (see, e.g., for *Phodopus* in Neumann et al. 2006). However, in some cases, the oldest fossils, attributed to extant species, predate the respective divergence ages, as illustrated by the stratigraphic ranges depicted in Figure 3.2 (see also Additional file 1 for the oldest fossil occurrences of each taxon). While the difference is small in most cases, it is considerable in the case of *Cricetulus*, for which our estimated divergence date is evidently too young, as there are several fossils from the Early Pleistocene of China and Russia, attributed to *C. barabensis* (Cheng et al. 1997; Zheng et al. 1997; Cai et al. 2013; Wu and Flynn 2017; Kazansky et al. 2022; Shchetnikov et al. 2023) and to *C. longicaudatus* (Zheng 1984; Cheng et al. 1997). The reason of the underestimation in this case and in general may partly be that not all extant species (e.g., *Cricetulus sokolovi*) or fossil species (e.g., taxa potentially assignable to *Cricetus*, see Horáček and Lebedová (2022)) could be included in the analysis. Accounting for the whole known stratigraphic ranges (see Stadler et al. 2018) of both fossil and extant species might likewise result in older estimates. Another possible explanation for older divergence ages within crown hamsters is related to the relatively low support of their monophyly (Fig. 3.2, clade I, PP = 0.52). If additional, older taxa are included in the crown cricetine clade, the inferred divergence ages will also turn out to be older. In summary, most previous divergence estimates are not fully independent and therefore tend to confirm each other. While many of these estimates appear too old compared with the fossil record, our own estimates seem to be too young in the case of *Cricetulus*.

Biogeographical history of cricetines

A recent study on carnivores demonstrated the importance of including fossil taxa when inferring the geographical origin of extant clades (Faurby et al. 2024), an approach that we applied here to hamsters for the first time. While the most probable area of origin of all cricetines is Central to Eastern Europe (Fig. 3.3), the uncertainty is high in the oldest estimates (Additional file 5). Nonetheless, the estimate fits with the oldest members of the group, previous suggestions and reconstructions, as well as proposed ancestors of the Cricetinae (Fahlbusch 1969; Heissig 1995; Kälén 1999; Fejfar et al. 2011; López-Antoñanzas et al. 2024).

From this Central to Eastern Europe area, the cricetines dispersed into South-Western Europe several times. The first arrivals were †*Cricetulodon* and †*Rotundomys* around 12.2 Ma. Apart from †*Allocricetus ehiki* and extant hamsters, the latest species to invade South-Western Europe was †*Neocricetodon polonicus* around 5.5 Ma. All other dispersals into this area (†*Apocricetus*, †*Cricetulodon meini* and †*C. lucentensis*, †*Neocricetodon*, †*Tragomys*) likely occurred in several waves during the Tortonian (11.63–7.246 Ma) (Fig. 3.5a). Globally, the Tortonian, is characterized by increasingly open environments (Steinhorsdottir et al. 2021). The precise timing of habitat opening in Central and Western Europe has however long been debated (see Strömberg et al. 2007). According to climate reconstructions, the spread of open vegetation occurred later in time, especially in Central Europe (Bruch et al. 2011). The Iberian Peninsula, however, already had a more open and drier landscape during the Tortonian (Pound et al. 2011), which coincides with the dispersal of hamsters. Given that today's hamsters are mostly found in open environments, such as steppes, grasslands or semi-deserts (Pardiñas et al. 2017a; Kryštufek and Shenbrot 2025), a connection between the expansion of such habitats and hamster diversification and dispersal seems plausible and has been suggested before (Neumann et al. 2006).

While the first expansions into Eastern Central Asia of †*Colloides* and †*Nannocricetus primitivus* (ca. 12.2–11.4 Ma) remain elusive, due to the high uncertainty in reconstructed ancestral geographic ranges (Additional file 5), the following two waves of dispersal eastwards (during 10.2 to 6.7 Ma), involving several †*Neocricetodon* and †*Allocricetus* taxa (Fig. 3.2, clades C and D), appear to be connected to prior dispersals into South-Western Europe and South-Western Asia (Fig. 3.5a). At the same time (ca. 10–7 Ma), palaeobotanical evidence indicates increasingly open vegetation with spreading grasslands in Greece and Anatolia (Strömberg 2011). Additionally, an expanding open savannah connecting Europe and Eastern Asia from 9 to 6 Ma has been proposed (Kaya et al. 2018), potentially facilitating eastward dispersal of cricetines (but see Denk et al. 2018). Several groups of large mammal are known to have expanded towards the East during the Tortonian as well (Eronen et al. 2009; Kostopoulos 2009; Ataabadi et al. 2013). Some of these taxa, including e.g., Carnivora or Proboscidea, distributed additionally into Africa (Haile-Selassie et al. 2009), in contrast to the hamsters, that remained limited to Europe and Asia.

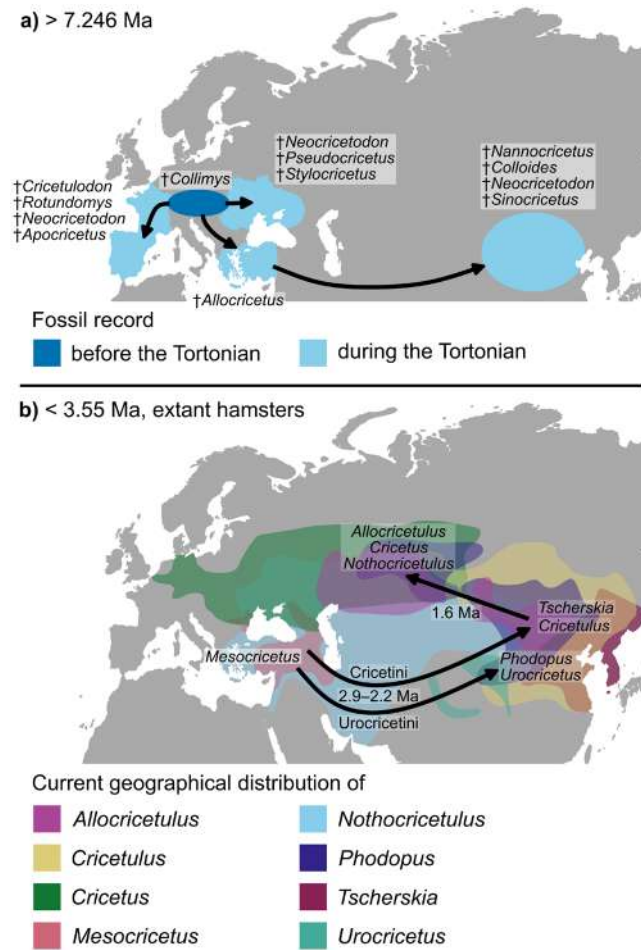


Figure 3.5. Main dispersals **a)** of fossil cricetines before and during the Tortonian, and **b)** of extant cricetines. Arrows in **a)** represent one or multiple dispersal waves during the Tortonian, colored areas and selected genera represent the fossil record. Arrows in **b)** represent main dispersal events, colored areas show today's geographical distribution of extant genera with their inferred geographical origin indicated.

Around 8.9 to 8 Ma, the lineage leading to the extant hamsters moved toward the area of Eastern Europe and South-Western Asia, which is also inferred as the most likely area for the origination of extant hamsters. Given the limited fossil and extant diversity of hamsters documented from Iraq to Pakistan, we focus primarily on the region around the Eastern Mediterranean and the Caucasus, close to the distribution of *Mesocricetus* today. Two main limitations affect this estimation: firstly, the uncertainty of the reconstructed ancestral ranges themselves, especially at the node leading to extant hamsters (Additional file 5), and secondly, the phylogenetic uncertainty. Therefore, the nodes leading to all potential sister groups of extant hamsters, i.e., clades G, H, †*Cricetus lophidens* and †*Allocricetulus ehiki*, should also be considered. Here, either an origin in South-Eastern Europe to South-Western Asia is reconstructed, or a broader

range including this area. Apart from †*Apocricetus*, which is restricted to South-Western Europe, the region around the Eastern Mediterranean and the Caucasus also fits the localities of the respective fossil taxa, that are closest to potential ancestors of crown hamsters. †*Hypsocricetus*, †*Cricetus lophidens*, and †*Mesocricetus primitivus* were all described from Greece (de Bruijn et al. 1970; Daxner-Höck 1992). †*Odessamys* is distributed in the area of Central and Eastern Europe but close to the Eastern Mediterranean and Caucasus, at the Black Sea in Southern Moldova, Southern Ukraine, and Eastern Romania (Topachevsky and Skorik 1992; Pevzner et al. 1996; Nesin and Nadachowski 2001; Nadachowski et al. 2006). Lastly, †*A. ehiki* is more widely distributed but also around the Black Sea and the Eastern Mediterranean (Hordijk and de Bruijn 2009; Krokmal and Rekovets 2010). Except for †*A. ehiki*, all these other taxa diverged in the Late Miocene, when steppe habitats were present specifically along the Black and Caspian Seas (Hurka et al. 2019).

Consequently, the Mesocricetini remained in this region, after the first split within the crown hamsters around 3.55 Ma, while the two lineages, leading to the Urocrinetini and Cricetini, subsequently invaded East Central Asia (Fig. 3.5b). The lineage leading to the Urocrinetini expanded around 2.93 Ma, whereas the lineage leading to the Cricetini dispersed between 2.93 Ma and 2.18 Ma. This interval was characterized by increased aridification in Central Asia (Hurka et al. 2019), consistent with the current distribution of *Phodopus* in deserts and semi-deserts (Ross 1994, 1995). According to our analysis, all taxa within the Urocrinetini, i.e., *Phodopus* and *Urocrinetus*, subsequently diverged in East Central Asia, confirming previous suggestions of an eastern origin (Lebedev et al. 2018). Following the latter lineage, all genera within the Cricetini diverged between ca.1.9 and 1.17 Ma, during a strong cooling event that ended after 1.2 Ma, when the so-called Mid-Pleistocene transition began (Snyder 2016). During that time, glacial cycles shifted from 41 kyr to 100 kyr periods, which resulted in intensified glaciations and additional remarkable climatic changes (Herbert 2023).

Our analysis places the origin of *Tscherskia* in Eastern Asia, although a European origin of *Tscherskia* was recently proposed based on the transfer of European ‘†*Cricetinus*’ species to *Tscherskia* (Xie et al. 2023), a re-assignment not supported by our analysis. Around 1.63 Ma, *Cricetulus* split from the remaining Cricetini, remaining in Eastern Asia, while the other lineage dispersed into Russia and Kazakhstan (Fig. 3.5b), where it subsequently diverged into *Nothocricetulus*, *Allocricetulus*, and *Cricetus* (1.43–1.17 Ma). This coincided with an intensified opening of landscapes during the Early Pleistocene in this region (Popova et al. 2017). Afterwards, *Nothocricetulus*, *Allocricetulus*, and *Cricetus* spread independently into Europe and, in the case of *Nothocricetulus*, also into Eastern Asia, at a time when “prototypes

of periglacial steppes” were found generally in northern Eurasia during cold phases (Hurka et al. 2019). The abundance of fossil species (or subspecies) attributed to *Cricetus* in Europe further suggests subsequent diversification of this genus there (see Hír 1997a).

Conclusion

Due to the partly highly similar dental morphology of different fossil hamster taxa, delimitation of species and genera can be difficult. Our phylogenetic hypothesis allows us to propose several taxonomic changes, most importantly regarding the fossil species that were originally described as †*Cricetinus*. According to our results, †*C. mesolophidos* belongs to †*Neocricetodon*, and †*C. europaeus* to †*Pseudocricetus*, while †*C. koufosi* forms a clade with several species described as †*Allocricetus*. In line with previous studies, we confirm †*Allocricetus* as non-monophyletic and call for erecting of a new genus for the above-mentioned clade that includes †*C. koufosi*. Several additional fossil species do not belong to their originally described genus according to our analysis, namely, †*Apocricetus darderi*, †*Mesocricetus primitivus* and †*Cricetus lophidens*. Furthermore, we confirm that †*Cricetulodon* is non-monophyletic. The type species of the third non-monophyletic genus in our analysis, †*Nannocricetus*, is considered a rogue taxon in our analysis. After accounting for taxonomic changes, we confirm eleven (of 14) genera that are represented by more than one species in our dataset as monophyletic. For the first time, a total-evidence dating analysis is performed on a hamster dataset that includes extant and extinct taxa. We provide estimations of divergence dates, using tip-dating, independently of previous studies and without relying on a priori assumptions about the placement of fossil taxa in the tree. In this way, we infer an origin of crown hamsters in the Pliocene rather than in the Late/Middle Miocene. Probabilistic modeling indicates that the ancestor of the subfamily was located in Central to Eastern Europe, with subsequent dispersal events into Western Europe and Eastern Central Asia. The crown hamsters are inferred to have originated in the region surrounding the Black Sea and the Eastern Mediterranean.

Abbreviations

ASDSF	Average standard deviation of split frequencies
BAYAREALIKE	Likelihood version of BayArea
BIC	Bayesian information criterion
BioGeoBEARS	Biogeography with Bayesian (and likelihood) evolutionary analysis with R scripts
DEC	Dispersal-extinction-cladogenesis
DIVALIKE	Likelihood version of the dispersal-vicariance analysis
ESS	Estimated sample size
FBD	Fossilized birth-death tree model
+G	Including the discrete-gamma model for among site rate variation
GTR	Generalized time reversible substitution model
HKY	Hasegawa-Kishino-Yano substitution model
IGR	Independent gamma rate uncorrelated clock model
ILN	Independent lognormal uncorrelated clock model
+J	Including jump dispersals
m1, m2, m3	First, second, and third lower molars
M1, M2, M3	First, second, and third upper molars
MCMCMC	Metropolis-Coupled Markov chain Monte Carlo
Mkv	Markov model conditioned on variable characters
PSRF	Potential scale-reduction factor
TK02	Continuous autocorrelated clock model
WN	White noise uncorrelated clock model

Declarations**Availability of data and materials**

The data generated or analyzed during this study are included in this published article and its Additional files and in the Supplementary data freely available at Harvard's Dataverse online repository (Dirnberger et al. 2025a). Link for review purposes:

<https://dataverse.harvard.edu/previewurl.xhtml?token=e2aaaa9c-8658-4c86-adb4-9448f1bbf775>

In the context of this thesis, the Additional files of this study are available under:

<https://figshare.com/s/6c91828664b2db2fffa7>

Competing interests

The authors declare that they have no competing interests

Authors' contributions

Conceptualization: MD, RLA, PPC. Methodology: MD, RLA, PPC, TRS. Formal analysis: MD, TRS. Investigation: MD, RLA, PPC. Visualization: MD. Writing original draft: MD. Writing review and editing: MD, RLA, PPC, TRS. All authors read and approved the final manuscript

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Summary

The topology of the retrieved phylogeny is mostly congruent to the one of the previous chapter and shows even increased support for several clades. By extending the taxon set, the relationships of fossil hamsters to extant ones are explored and several potential closest ancestors are proposed. The here reconstructed relationships confirm several previously proposed taxonomic problems in the group of hamsters and contribute to their solving. In this study the first total-evidence dating was performed on the group of hamsters and the resulting divergence times are substantially younger than previously assumed, based on node-calibrated molecular phylogenies. The biogeographical history inferred in this chapter reveals a European origin of the entire group of hamsters, as well as several dispersal events towards West-Europe and Asia. The origin of crown hamsters is narrowed down to a region around the Eastern Mediterranean, and the Black and Caspian Seas. The here reconstructed phylogeny constitutes a valuable basis for further downstream analyses, as conducted in the next chapter.

Chapter 4

Exploring hamster diversification and diversity with Bayesian and deep learning approaches

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Status: in preparation

Besides the phylogenetic relationships and the geographical origin, also the biotic and abiotic factors that impact the macroevolutionary dynamics of hamsters are uncertain. Based on the reconstructed phylogeny of the previous chapter, as well as on here compiled fossil occurrence data, the paleodiversity, as well as the heterogeneity of diversification rates of hamsters are explored. A robust assessment of these processes is necessary to discuss the reasons for the waxing and waning of the group.

Exploring hamster diversification and diversity with Bayesian and deep learning approaches

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Abstract

The study of macroevolutionary rate patterns, including the identification of rate variation between or within lineages, as well as over time, is essential to detect evolutionary events, such as key innovations, or to assess the impact of biotic or abiotic drivers on the waxing and waning of a group. A vital source of information that provides insights into these processes are fossils. Consequently, there are various approaches to study macroevolutionary rates and paleodiversity based on fossil occurrences and on phylogenies that include extinct species. In a case study of hamsters, this work conducts a series of such analyses, employing a recently reconstructed comprehensive total-evidence phylogeny of the group and an extensive fossil occurrence database specifically compiled for this work. Hereby, a widely used framework to find clade-wide diversification shifts and a recently described diffusion model for within-lineage heterogeneity are applied on the tree, as well as an approach using unsupervised Bayesian neural networks for fossil data, to reconstruct species-specific diversification rates while investigating the impact of several time-and-trait-series. Additionally, the true paleodiversity of hamsters is inferred under consideration of temporal, taxonomic, and spatial biases of the fossil record, based on deep learning models. Our results evidence relatively constant and decreasing speciation rates over time but reveal differences in the inferred extinction rates depending on the applied methodology. The lineage-specific extinction rates demonstrate a certain resistance of hamsters against environmental fluctuations, resulting potentially from several ecological strategies to buffer unfavorable conditions. The speciation rate variation is inferred to depend mainly on geographical distribution, as well as global paleotemperature. The inferred true paleodiversity shows a hump-shaped trajectory through time, with a maximum of up to 35 contemporary hamster species during the Messinian and the Pliocene. The apparent incompleteness of the fossil record of hamsters and the highly temporal and taxonomical variation in fossilization reveal limitations in the applied models and highlights the difficulties to account for the various biases often present in fossil data.

4.1 Introduction

The study of macroevolutionary dynamics, the variations in speciation and extinction rates, their underlying drivers, as well as the resulting biodiversity has long been a focus in biology and paleontology. First insights about these processes in deep time were possible by studying the fossil record (Newell 1952) and methods that analyze directly fossil occurrence data are continuously developed since then (Sepkoski 1984; Raup 1985; Foote 2001; Alroy 2008; Silvestro et al. 2014; Alroy 2015; Starrfelt and Liow 2016; Cooper et al. 2024b). In addition, with the rise of molecular phylogenetics, first attempts have been made to utilize phylogenetic trees to reconstruct diversification rates (Nee et al. 1992, 1994), resulting in a vast number of birth–death models, that rely usually on time-calibrated phylogenies of extant taxa (summarized in Morlon et al. 2024). Fueled by debates about the ability to estimate extinction rates on extant trees (Rabosky 2010, 2016; Beaulieu and O’Meara 2015; Louca and Pennell 2020, 2021; Helmstetter et al. 2022; Kopperud et al. 2023), numerous approaches were developed that incorporate fossil data information. They provide alternative solutions, either by constraining extinction rates, based on the fossil record (see Quintero et al. 2024), or by using trees that include fossil taxa (Mitchell et al. 2019; Chabrol et al. 2025; do Rosario Petrucci et al. 2025; Quintero et al. 2025). Common to all these approaches is a central challenge in the incompleteness and bias of the fossil record, which is impacted by socio-economic, taxonomic, and geological factors (Donoghue et al. 2006; Shaw et al. 2021; Raja et al. 2022; Žliobaitė and Fortelius 2022; Krone et al. 2024). Although, this issue is well known and has been addressed since decades (Raup 1972, 1975), leading to a range of methodological frameworks (summarized in Allen et al. 2025), comparisons across methods remain essential, as varying approaches have been shown to yield substantially different results (Martínez-Gómez et al. 2024).

In this study, we further explore these differences by comparing several complementary diversification methods based on both phylogenetic trees and fossil occurrences using hamsters as a case study. True hamsters (Cricetinae) are mouse to rat-sized rodents with characteristic internal cheek pouches, familiar to the public as popular pets. Their extant diversity is modest, comprising 19 species in 9–10 genera (Kryštufek and Shenbrot 2025; Pan et al. 2025), especially when compared with their close relatives Arvicolinae or Sigmodontinae (Pardiñas et al. 2017a). In contrast, their fossil record is remarkably rich, with ~90 fossil species distributed in more than 20 genera (McKenna and Bell 1997; de Bruijn et al. 2012; Dirnberger et al. 2025b) with thousands of occurrences across the Palearctic. This fossil record, dating back to the Middle Miocene (Fejfar et al. 2011) has been so far used in biogeographic and phylogenetic

studies (Neumann et al. 2006; Lebedev et al. 2018; Dirnberger et al. 2024b, 2025b), as well as for European and Asian biostratigraphy (Tchernov 1998; Hír 2002; Prieto and Rummel 2009a; Casanovas-Vilar et al. 2016; Flynn and Wu 2017). Yet, despite this wealth of data, diversification patterns of hamsters have only been analyzed within broader-scale studies of Rodentia or Muroidea, and solely with extant taxa (Fabre et al. 2012; Schenk et al. 2013; Stepan and Schenk 2017). As a result, the macroevolutionary dynamics that led to the present-day diversity of hamsters remain poorly understood. Previous comparisons of diversification models based on trees and fossil occurrences were often based on fragmented fossil records with high numbers of singletons. Analyzing the hamsters, as a group with a high number of fossil occurrences and a low proportion of singletons will provide further insights into the differences between phylogeny and occurrence-based approaches. Moreover, a robust phylogenetic framework, essential to exploring macroevolutionary dynamics, is available for hamsters, as they have recently been comprehensively studied using a total-evidence approach (Dirnberger et al. 2025). Based on such analyses, the mechanisms behind the reduced extant hamster diversity compared to the wealth of fossil taxa can be studied. Additionally, the impact of biotic or abiotic factors on hamster diversification are still widely unknown, beyond hypotheses linking their evolution to the spread of open landscapes and increased aridification (Neumann et al. 2006; Pardiñas et al. 2017a).

To detect shifts in the hamster diversification, we first applied the Bayesian Analysis of Macroevolutionary Mixtures (BAMM) (Rabosky 2014; Rabosky et al. 2017), which enables the estimation of speciation and extinction rate based on phylogenies incorporating fossil tips (Mitchell et al. 2019). Although BAMM has been widely used, it has faced criticism, referring to an extreme prior sensitivity, errors in the likelihood function and other theoretical concerns (Moore et al. 2016; Meyer and Wiens 2018; Meyer et al. 2018), even though some of these issues have since been addressed (Rabosky et al. 2017; Rabosky 2018; Mitchell et al. 2019). In contrast to discrete clade-wide shifts, recent work has suggested that small, gradual changes along the tree may better explain diversification histories (Donoghue and Sanderson 2015; Morlon et al. 2024; Quintero et al. 2024). To account for such gradual dynamics, we also apply the recently described fossilized birth-death diffusion (FBDD) model, in which speciation and extinction rates change through time under a geometric Brownian (Quintero et al. 2025). In both models, BAMM and FBDD, fossilization is assumed to be constant between taxa. In addition, fossilization is constant through time in BAMM, whereas FBDD allows piece-wise changes between user-defined time bins.

In parallel, we use a state-of-the-art birth-death neural network (BDNN) model based on the widely used PyRate framework (Silvestro et al. 2014, 2019; Hauffe et al. 2024), which estimates speciation, extinction and sampling rates drawing from occurrence data. Incorporating occurrence data allows for each species to be sampled more than once, increasing the precision of the sampling rates that are allowed to vary between species, as well as of the estimated species durations and, consequently, of the extinction rates. In BDNN the contribution of phylogeny can be incorporated as a predictor of diversification, together with other factors such as abiotic or biotic variables (Hauffe et al. 2024).

Regarding estimates of paleodiversity, both FBDD and BDNN can be used to overcome some aspects of sampling biases. FBDD allows estimation of paleodiversity, while considering unsampled taxa (Quintero et al. 2025). Hereby, the FBDD model accounts, as mentioned above, for temporal variation in the preservation rate but not heterogeneity of recovery rates between taxa or across space, all of which are known to affect estimations of biodiversity (Benson et al. 2021). In PyRate, origination and extinction dates of each taxon are approximated, which can be used to produce an estimate of the species richness through time with corrected species durations (Silvestro et al. 2015). This approach does, however, not factor in unsampled lineages, and thus the results must be considered with care, as the “true” species richness in deep time remains underestimated. Therefore, we compare diversity curves obtained from FBDD and BDNN PyRate with an ad hoc protocol based on machine learning: Deep learning Diversity Estimation (DeepDive). DeepDive uses the observed features of the fossil record (e.g., number of sample areas, species richness and occurrences per area and time bin, endemisms, shared taxa, etc.) and simulates thousands of diversification and fossilization scenarios, making sure the observed features of the empirical record are included among the simulations. By training the model with this multitude of simulated fossil records and true diversities, DeepDive provides reliable estimates of deep past biodiversity based on occurrence data even under very complex preservational scenarios (Cooper et al. 2024b). By comparing the results from these complementary methods, we provide a robust assessment of the macroevolutionary dynamics of hamsters while drawing practical conclusions about the applied approaches.

4.2 Material and Methods

Phylogenetic data

For the methods based on phylogenies (BAMM, FBDD), we used the total-evidence phylogeny from Dirnberger et al. (2025b, S6), which represents the most comprehensive hamster

phylogeny to date with 71 species (58 fossil and 13 extant). It includes around 65% of all known extinct and extant species (~90 fossil and 19 extant). Trees have been reconstructed using relaxed-clock Bayesian inference on combined morphological and molecular data and are calibrated by tip-dating.

Occurrence data

The occurrence dataset (Suppl. Mat. S1) was initially based on the NOW (New and Old Worlds) fossil mammal database (The NOW Community 2023), accessed February 2023, including the genera: †*Allocricetus*, †*Apocricetus*, †*Collimys*, †*Cricetulodon*, *Cricetulus*, *Cricetus*, †*Hattomys*, †*Kowalskia* (= †*Neocricetodon*), *Mesocricetus*, †*Pseudocricetus*, and †*Rotundomys*. Each entry was carefully cross-checked against the literature and confirmed by primary sources and additional references. We manually curated the data to account for conflicts and changes in the taxonomy of the group (following Dirnberger et al. 2025, S1). The age of each included occurrence was checked as well, to confirm or modify the record from the NOW database. As numerous fossil taxa and occurrences of Cricetinae are not included in the NOW database, an extensive literature search was conducted (Suppl. Mat. S1 contains 1020 references) to expand our occurrence database and to include all currently known species of Cricetinae. In this process, also occurrences of the remaining Cricetinae genera in NOW were added, namely, †*Bahomys*, †*Gromovia*, †*Hypsocricetus*, †*Nannocricetus*, *Phodopus*, †*Pseudocollimys*, †*Sinocricetus*, and †*Stylocricetus*, as well as genera, that are not in NOW, namely, *Allocricetulus*, †*Amblycricetus*, †*Colloides*, and †*Tragomys*. Following Dirnberger et al. (2025) the subfamily comprises around 109 species, of which 19 are extant (see Kryštufek and Shenbrot 2025). For nine living species, no undoubted fossil occurrences could be found at the species level (*Allocricetulus curtatus*, *Cansumys canus*, *Cricetulus sokolovi*, *Phodopus campbelli*, *P. roborovskii*, *P. sungorus*, *Tscherskia ningshaanensis*, *Urocricetus lama*, *U. kamensis*). A subsequent search of all cricetine genera (May 2025) in the NOW database and the Paleobiology Database (<https://paleobiodb.org/>) resulted in 729 and 348 entries respectively. In contrast, our database includes 2751 entries. Of those, we excluded occurrences that could not be verified in the literature, or had unknown ages or uncertain identifications (e.g., open nomenclature, very limited material, conflicting author identifications). After restricting the data to species-level occurrences, the final curated dataset includes 1951 records.

This process illustrates the risks of relying solely on data extracted from existing databases, as many records are missing, taxonomic synonyms may not be properly accounted for, and some

classifications can be erroneous. Such biases can distort macroevolutionary analyses, leading to inaccurate estimates of diversification rates and evolutionary patterns. Unfortunately, as such exhaustive curation is tedious and very time-consuming, it is rarely performed. Nevertheless, to obtain the most reliable results, it is essential to perform exhaustive searches of both the literature and primary fossil collections, integrating multiple data sources to build a complete and more accurate dataset.

Reconstruction of Diversification, Preservation and Paleodiversity

All input and output files, as well as R and Julia codes, are available in the Supplementary Material. Three of the below-listed methods rely on sampling with Markov chain Monte Carlo (MCMC) methods. In these cases, convergence diagnostics and appropriate burn-in were assessed in Tracer v.1.7.2 (Rambaut et al. 2018) or in R (for BAMM, following Title 2019).

Identifying diversification regimes on phylogenies. BAMM aims to find discrete shifts in macroevolutionary rates between regimes of distinct diversification rates, based on time-calibrated phylogenetic trees. The number of these shifts is also inferred by the model, applying reversible-jump Markov chain Monte Carlo. While previous versions of BAMM (Rabosky 2014; Mitchell and Rabosky 2017) were applicable only to phylogenies of extant taxa, the latest version, BAMM v2.6 called ‘Fossil BAMM’ (Mitchell et al. 2019) can also be used on trees combining extinct and extant taxa. It is therefore applied here, compiled from source code (available at <https://doi.org/10.5061/dryad.50m70>). One of the main differences to the former versions of BAMM is the incorporation of a fossil preservation rate, that includes the processes of fossilization, sampling, identification, etc. In the implementation in BAMM, this preservation rate is constant through time and across lineages.

Following previous studies (Mitchell and Rabosky 2017; Černý et al. 2022), we ran several analyses under different priors on the maximum clade compatibility (‘allcompat’) tree of Dirnberger et al. (2025). We compared time-constant and time-variable speciation rates within diversification regimes (with constant extinction rates) and tested the effect of different priors on the expected number of rate shifts (0.1, 1.0, 10). Additionally, we accounted for uncertainty in phylogeny (topology and divergence times) by repeating the analysis on 20 randomly sampled trees from the posterior tree sample of Dirnberger et al. (2025) under default settings (time-variable speciation rate and 1.0 expected shifts). To be able to perform the inference on these trees, branch lengths of zero resulting from sampled ancestors were manually fixed to 0.001. The rate of fossil preservation, constant through time and among lineages, was estimated

based on the number of fossil occurrences that can be assigned to one of the 71 cricetine taxa in the tree (1616 occurrences, extracted from Suppl. Mat. S1). The R-package BAMMtools v.2.1.12 (Rabosky et al. 2014) was used to choose appropriate additional priors for each tree. The analyses were run for 10 million generations, sampled every 1,000 iterations. After discarding the first 10% of the samples as burn-in, the estimated sample size (ESS) was well above 200 in all runs, a value that is generally assumed to indicate convergence of the posterior. BAMMtools was also used to analyze the output. We calculated Bayes factors to assess the support of different numbers of rate shifts and determined the set of distinct ‘core’ shift configurations that make up 95% of the posterior samples. Under the prior alone, there is a non-zero probability of a shift occurring on each branch, which is why there needs to be a distinction between these non-core shifts and the core shifts that are supported by the data. This can be done by calculating marginal odds ratios. Hereby, the posterior probability of a shift is taken relative to the prior probability. Only if this marginal odds ratio is above a predefined value (here 5, as default in BAMM), the shift is considered a core shift. We additionally calculated the respective rates through time across all models for each analysis on the maximum clade compatibility tree (Suppl. Mat. S2) and the average rates based on the 20 randomly sampled trees (Fig. 4.1, 4.2a).

Inferring gradual diversification variations on phylogenies. In contrast to modelling discrete diversification rate shifts, recently described approaches allow for gradual changes in rates, either at cladogenesis events or anagenetically along branches (Maliot et al. 2019; Quintero et al. 2024). The birth–death diffusion (BDD) model of Quintero et al. (2024) represents the latter case, with speciation (λ) and extinction (μ) rates following each a Geometric Brownian. The drift terms of the Geometric Brownian motions represent long-term trends in speciation (α_λ) and extinction (α_μ), while the diffusion terms (σ_λ and σ_μ) represent heterogeneity in the respective rates. The diversification rates are passed on at cladogenetic events, following the fact that traits that influence diversification are also inherited (Quintero et al., 2024). A major challenge in inferring diversification model parameters is the ‘incompleteness’ of phylogenetic trees due to unsampled lineages. For complex birth-death models, analytic likelihood calculations can become computationally intensive or impossible. A solution for that is Bayesian data augmentation (Tanner and Wong 1987), in which the observed data (here, the reconstructed tree) is augmented with non-observed data (here, unsampled lineages). These augmented ‘complete’ trees enable faster and easier likelihood calculations and allow reconstruction of diversity trajectories through time, incorporating even unsampled lineages (Maliot and Morlon 2022; Quintero et al. 2024). The BDD model was recently expanded to

account for both extant and extinct taxa, yielding the FBDD model (Quintero et al. 2025). Unlike the BDD model, FBDD incorporates a fossil preservation process, which is allowed to change between pre-defined time bins but not among lineages. In empirical trees that include extinct taxa, each fossil taxon is typically represented by only one occurrence, usually as an extinct tip (so-called m fossils following Stadler 2010), or representing a sampled ancestor (so-called k fossils). In some cases, however, a substantial number of additional fossil occurrences of the sampled taxa is known from the fossil record. These occurrences are not incorporated in the tree, and the subsequent undersampling of k fossils can severely bias estimations of preservation and extinction rates (see Beaulieu and O’Meara 2023). To address this, the number of additional k fossils can be added to the number of fossil occurrences already in the tree for the FBDD inference (Quintero et al. 2025), similar to the fossil BAMM analysis.

We performed the analysis with the Julia package Tapestry v.0.4.2 (Quintero and Andréoletti 2025), based on the maximum clade compatibility tree of Dirnberger et al. (2025) and the 20 additional trees from the BAMM analysis to account for phylogenetic uncertainty. The preservation rate was set to change at the boundaries of stratigraphic stages (Cohen et al. 2013; updated 2024/10). The extant sampling proportion is fixed to $13/19 = 0.684$. To obtain the number of additional k fossils, the fossil occurrences of the taxa that are included in the tree were counted for each preservation rate time bin. The number of occurrences that were already part of the tree (one per species) was also counted for each time bin and was deducted from the number of fossil occurrences per time slot. The exact position of these sampled ancestors in the tree is not taken into account in FBD models, as clarified by Beaulieu and O’Meara (2023). The MCMC was run for 3 million generations for each run, sampling every 1000 iterations after a burn-in of 10%. The ESS was above 200 for each parameter in 9 of 21 runs, while for the remaining runs, the ESS of up to three parameters was below 200 but above 100. To calculate speciation, extinction and net diversification rates through time, as well as paleodiversity, we subsampled every 10th sample from each of the 20 augmented tree vectors and combined them (6000 samples).

Identifying macroevolutionary patterns and drivers with occurrence data. Contrasting the previous models, the BDNN model estimates diversification and fossilization rates from fossil occurrences and not phylogenies (Hauffe et al. 2024). The model is based on the framework of PyRate (Silvestro et al. 2019), which combines time-variable birth-death processes with a preservation model to jointly infer preservation rates, origination and extinction times of the fossil lineages, and speciation and extinction rate variation through time. In the BDNN model, diversification rates can vary through time and across lineages. They are modelled as a function

of time and potentially multiple other predictors, like time series, or continuous and categorical traits. Nonlinear effects and interactions between predictors can hereby be considered, through the use of an unsupervised neural network (Hauffe et al. 2024).

The preservation model which is best fitting according to a maximum-likelihood analysis (Silvestro et al. 2019), is the Time-variable Poisson Process (TPP), which incorporates piecewise changes through time. The boundaries are defined to follow stratigraphic ages, as for the FBDD model. Additionally, preservation rate heterogeneity across lineages is modelled, with rates drawn from a gamma distribution (Silvestro et al. 2014). Accordingly, the estimated shape parameter of the gamma distribution (α_ψ) reflects the heterogeneity of the preservation rate across species. Following the suggestions and methodology in Hauffe et al. (2024), we set the speciation and extinction rates to also change at the boundaries of stratigraphic stages (but combining the late Pleistocene and Holocene) and examined the effect of three additional predictors (besides time) on the diversification rates: (i) biogeographic distribution, (ii) phylogenetic relatedness, and (iii) global palaeotemperature. For (i) the biogeographic distribution, we followed the geographic areas of Dirnberger et al. (2025) (Fig. 4.3a) and assigned one or multiple of these areas to each taxon using 5 binary traits. Areas were scored based on the fossil record (Suppl. Mat. S1) and current distributions of extant taxa (following Kryštufek and Shenbrot 2025). The second predictor, (ii) the phylogenetic relatedness of taxa, was quantified by calculating phylogenetic eigenvectors using the R package PVR v.0.3 (Phylogenetic Eigenvectors Regression and Phylogenetic Signal-Representation Curve) (Santos 2018). We aim at incorporating major phylogenetic relatedness. For this purpose, the maximum clade compatibility tree of Dirnberger et al. (2025) was first simplified to a clade level phylogeny, where each clade is represented by one tip. The result is a tree with ten tips (Suppl. Mat. S4), for which we computed the phylogenetic eigenvectors and retained the first two as species traits. The species that were not part of the phylogeny were each associated with one of the clades, based on previous studies and morphological comparisons (detailed in the Supplemental Material). *Bahomys hypsodonta* Chow & Li, 1965 and *Gromovia daamsi* Erbajeva, Alexeeva & Khenzykhenova, 2003 (known by two and a single occurrences) could not be placed into one of the clades with confidence and are therefore excluded from the analysis. The third predictor, (iii) the global palaeotemperature, was calculated with equations from Hansen et al. (2013), based on oxygen isotope data from Westerhold et al. (2020). A total of 98 species with 1948 occurrences were included in the BDNN analysis. The phylogenetic eigenvectors and the paleotemperature were z-transformed before the analysis. As ages of fossil

sites are subject to uncertainties, ten replicate data sets were generated, with ages of sites randomly sampled.

Ten separate BDNN analyses were run based on these replicates, each with 200 million MCMC generations, sampled every 10,000 iterations. After discarding the first 10% of generations of each replicate as burn-in, the ESS was above 200 for all parameters in seven of the ten runs. In the remaining three runs, one or two parameters had an ESS between 100 and 200. We took 100 samples of each run and combined them (1000 samples) for the calculation of mean rates across lineages through time and further postprocessing. The origin and extinction times of each lineage that are estimated during the BDNN inference were used to calculate the number of lineages through time based on the range-through method. It is important to note, however, that this approach does not consider unsampled lineages. Before the different tested predictors were ranked in terms of their importance for rate variation, the evidence for rate variation itself was evaluated. For this purpose, the empirical coefficient of rate variation was calculated, to be compared to the upper 95% quantile of coefficient of rate variation obtained by 100 simulations under constant birth-death processes (following Hauffe et al. 2024). In case of higher rate variation in the empirical data, the importance of the predictors is ranked by combining three methods (see Hauffe et al. 2024; Molnar 2025): the decrease in model likelihood after permutation of the predictor, the marginal posterior probability of rate differences using partial dependence plots (Greenwell et al. 2018) and SHAP (SHapley Additive exPlanations) values, that represent the contribution of each predictor to the estimated rates (Lundberg and Lee 2017). In addition to the importance of single predictors, two-ways interactions were ranked.

DeepDive. To account for the incompleteness of the fossil record and its temporal, taxonomic and spatial biases, when reconstructing the paleodiversity, we applied a recent machine learning approach, called DeepDive (Cooper et al. 2024b). Hereby, a wide array of biodiversity scenarios is simulated under consideration of biogeographical differences. Based on these, incomplete fossil records are generated by degrading the data following a myriad of possible preservational scenarios. With these simulations, deep learning models are trained to reconstruct diversity trajectories from incomplete and biased fossil records. Afterwards, the models are used to estimate the palaeodiversity based on empirical fossil data. To ensure that the simulated datasets encompass the empirical one, an ‘autotune’ option is available, that adjusts the simulations according to the specific empirical data (Cooper et al. 2024a).

We created the configuration file for the DeepDive v.1.0.26 analysis (Cooper et al. 2024b) with the R package DeepDiveR v.0.1.0 (Cooper et al. 2024a). The analysis is based on the complete fossil occurrence data, with each occurrence assigned to one of the five large geographical

areas, also used for the BDNN (see Fig. 4.3a, adjusted from Dirnberger et al. 2025), to account for possible spatial sampling biases. Our binning system for the DeepDive is based on stratigraphic boundaries, with some bins further divided so that bins had a duration of around 1 Myr (following Cooper et al. 2024a). Using larger bins may obscure fine-scale features in the fossil record and reduce the temporal resolution of the resulting deep-learning reconstruction. Age uncertainties of sites are addressed by creating 100 replicates with randomly sampled ages. The analysis is also conditioned on the present diversity of hamsters (19 species, following Kryštufek and Shenbrot 2025).

4.3 Results

Diversification history of hamsters estimated by Fossil BAMM

In BAMM analyses, Bayes factors greater than 20 or less than 1/20 are generally interpreted as strong support for or against models of different numbers of rate shifts (Kass and Raftery 1995; Rabosky et al. 2014; Mitchell and Rabosky 2017). Following this value, none of our analyses, either on the maximum clade compatibility tree or on the randomly sampled trees, resulted in strong support for or against a model without rate shifts, apart from a few cases, in which very high numbers of shifts were rejected (Suppl. Mat. S2). Accordingly, considering the 95% credible sets of core shift configurations of the analyses on the maximum clade compatibility tree, configurations without rate shifts are the most frequent in all analysis (0.77–1). A similar pattern is found in 15 of the 20 analyses based on randomly sampled trees (0.46–0.98). The second most frequent shift configuration in these analyses, is the same as the most frequent one in the remaining five analyses. It includes one rate shift located on a branch leading to the crown hamsters. Compared to the remaining cricetines, crown hamsters show here higher speciation rates and lower extinction rates. The only other shift found in the 95% credible sets is associated with an increase in both speciation and extinction rates in *Rotundomys*. Rates through time, averaged across lineages and models, show differences among the three tested priors on the number of diversification shifts (0.1, 1, 10). While with fewer expected shifts, the curves are mostly flat, the shifts in *Rotundomys* and crown hamsters become more apparent with higher numbers of expected shifts (Suppl. Mat. S2). In the averaged rates across the randomly sampled trees under default BAMM settings (one expected shift), a slight increase in speciation and a decrease in extinction are visible from around 4 to 3 Ma (Fig. 4.1), resulting in an increase in net-diversification (Fig. 4.2a), as a result of the diversification shift connected to the origin of

crown hamsters. The weakly supported shift associated with *Rotundomys*, is reflected in a shortly higher upper limit of the 95% credible interval of extinction and speciation rates between 10 and 9 Ma, coinciding with the divergence of several *Rotundomys* species in a short interval, followed by their extinction. Rates in the oldest time interval must be interpreted carefully, due to the different root ages of the analyzed trees, ranging from 19.49 Ma to 15.48 Ma. The apparent increase in speciation rate at the origin of the group is not present in individual tree analyses but is only a side effect of the averaging, as ‘shorter’ trees result in slightly higher diversification rates. Apart from the weakly supported diversification shifts, speciation is also allowed to vary within the regimes over time resulting in a slight deceleration after the initial uncertainty and before the emergence of crown hamsters (Fig. 4.1a). In general, tree-wide diversification rates remain, however, relatively constant (median [95% CI] tree-wide speciation rate: 0.45 [0.34–0.61], extinction rate: 0.38 [0.28–0.52], net-diversification rate: 0.07 [-0.07–0.22] events lineage⁻¹ Myr⁻¹). The median preservation rate (Fig. 4.2c), constant through time and across lineages is high (9.65 [8.02–12.1] events lineage⁻¹ Myr⁻¹).

Diversification and paleodiversity inferred by FBDD

The reconstructed speciation rate on the maximum clade compatibility tree (Fig. 4.1a) shows elevated values at the backbone tree that decrease within the different extinct clades. As an exception, the lineages within †*Collimys* and †*Rotundomys*, show unchanged or even increasing speciation rates in the Serravallian and middle Tortonian. The heterogeneity of the speciation rate, reflected by the diffusion coefficient of the underlying Brownian motion (σ_λ median [95% CI]: 0.28 [0.2–0.39]) is clearly lower than the heterogeneity of the extinction rate (σ_μ : 0.64 [0.32–1.1]) in the maximum clade compatibility tree. Extinction rates increase along the branches of extinct clades, especially in †*Collimys* and †*Rotundomys*. Accounting for phylogenetic uncertainty, the speciation rate through time across all lineages shows a general trend of decreasing by 12% per Myr (α_λ : -0.12 [-0.26–0.01]) but is slightly increasing around 3.5 Ma, coinciding with the emergence of crown hamsters. In contrast the extinction rate increases strongly through time by 56% per Myr (α_μ : 0.56 [0.18–1.1]). Patterns of extinction rate through time show three main peaks, at the end of the Serravallian (ca. 12 Ma), in the middle of the Tortonian (ca. 10–9 Ma) and in the middle of the Pliocene (ca. 4 Ma). Since the speciation through time profile is relatively smooth, net-diversification largely mirrors extinction (Fig. 4.2a) (just reversed), showing a clearly negative phase during the Pliocene, between 5 and 2.5 Ma.

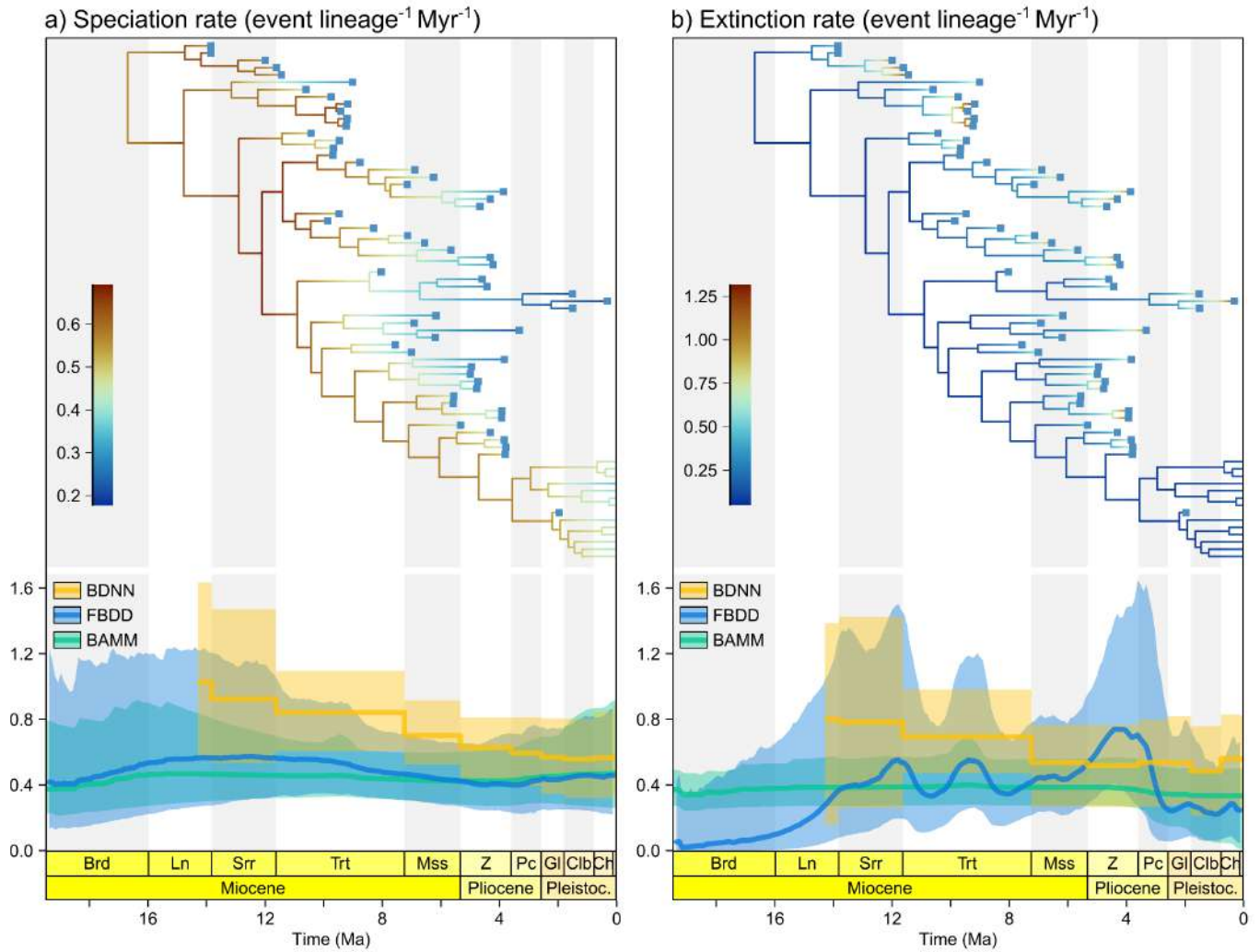


Figure 4.1. Reconstructed a) speciation and b) extinction rates as color gradients on the maximum clade compatibility tree resulting from the fossilized birth-death diffusion (FBDD) model (top) and through time, across lineages (bottom). Rates through time are estimated on several replicate datasets by the birth-death neural network (BDNN) (yellow), FBDD (blue), and the Bayesian Analysis of Macroevolutionary Mixtures (BAMB) (green), shown as median (FBDD, BAMB) or mean (BDNN) with the 95% credible interval (CI) as shaded area. The chronostratigraphic chart follows Cohen et al. (2013; updated 2024/10).

The preservation rate, assumed to be constant across lineages, increases over time (Fig. 4.2c). Before the Late Miocene, preservation rates are relatively low (median rates in stratigraphic stages range between 0.17–0.27 events lineage⁻¹ Myr⁻¹), but increase in the Late Miocene and Pliocene (1.55–3.58 events lineage⁻¹ Myr⁻¹), rise further in the Early to Middle Pleistocene (4.09–35.03 events lineage⁻¹ Myr⁻¹), and reach extremely high values in the Late Pleistocene and Holocene (213.3 and 266.18 events lineage⁻¹ Myr⁻¹).

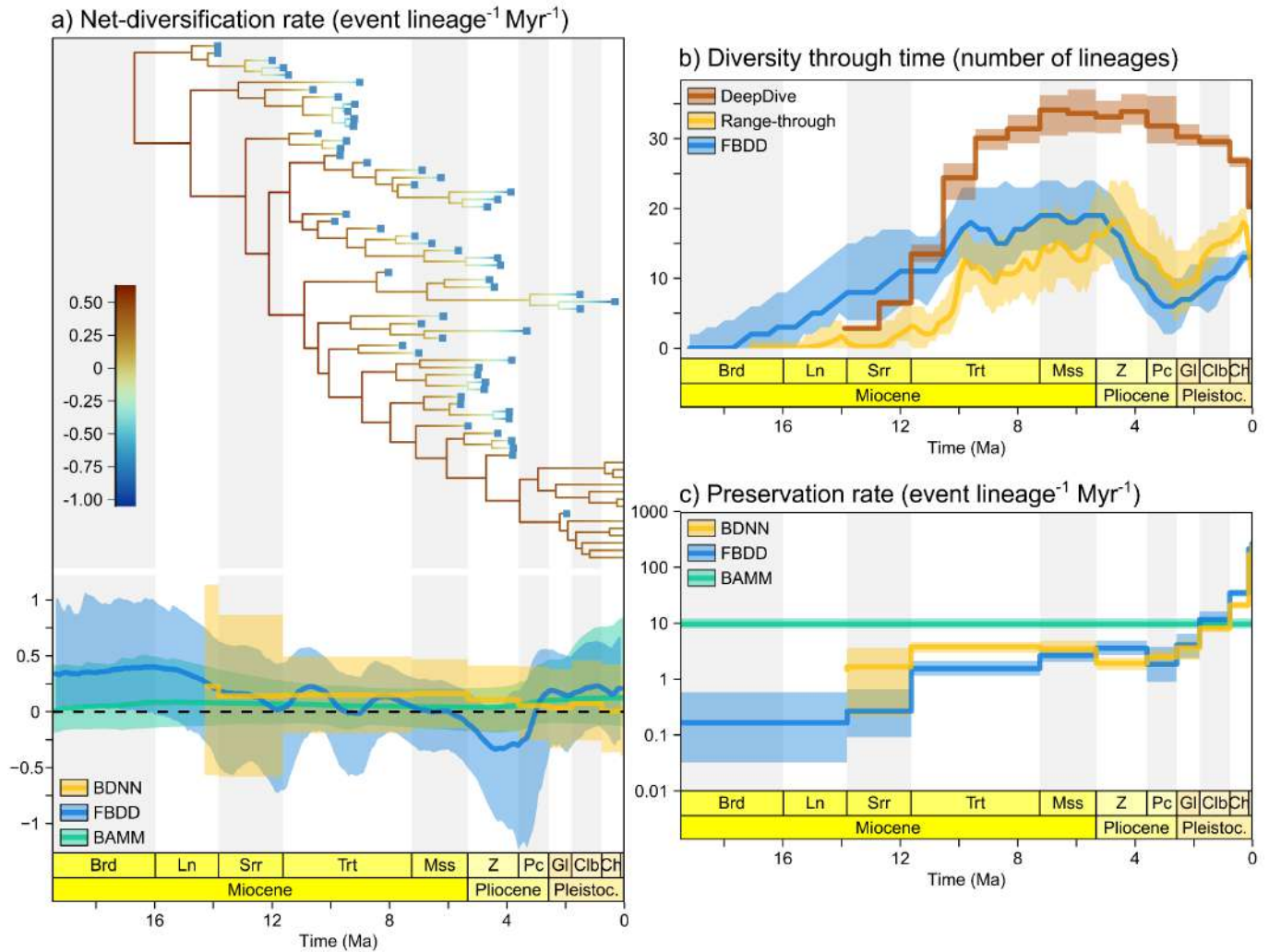


Figure 4.2. a) Reconstructed net-diversification rate as color gradient on the maximum clade compatibility tree resulting from FBDD (top) and through time, across lineages (bottom), b) lineages through time and c) preservation rate through time. Rates through time in a) and c) are estimated on several replicate datasets by BDNN (yellow), FBDD (blue) and BAMM (green), shown as median (FBDD, BAMM) or mean (BDNN) with the 95% CI as shaded area (full range for preservation rate of BDNN). Diversity through time in b) is estimated by FBDD (blue) and DeepDive (brown) and calculated without correcting for unsampled lineages, based on range-through diversity in Pyrate (yellow), shown as mean, with 95% CI for FBDD and with full range for DeepDive and Pyrate. The chronostratigraphic charts follow Cohen et al. (2013; updated 2024/10).

According to the FBDD model, cricetine diversity undergoes a sustained increase between 18 and 10 Ma. After that, the reconstructed diversity fluctuates between 15 and 20 species until the earliest Pliocene, some 5 Ma (Fig. 4.2b). Diversity then declines to ca. 6 species during the Pliocene, before increasing again in the Pleistocene until today.

Diversification and its drivers based on BDNN

The inferred speciation rate (in events lineage⁻¹ Myr⁻¹) decreases through time (Fig. 4.1a) from a mean of 1.03 (95% CI: 0.55–1.63) at the origin of the group to 0.63 (0.42–0.81) in the Early Pliocene with little variation until today (0.57 [0.34–0.89]). The mean extinction rate (Fig. 4.1b) also shows a decreasing trend before the Messinian (from 0.8 [0.16–1.39] to 0.54 [0.27–0.77]) and remains relatively constant until today (0.55 [0.25–0.82]).

Biogeography, phylogeny and paleotemperature capture more variation in speciation than a model where all species share a single speciation regime (coefficients of speciation variation of 0.192 vs. 0.138). However, for extinction this is not the case, suggesting that the candidate predictors are not explaining major differences among species extinction risk (coefficients of speciation variation of 0.260 vs. 0.262). The preservation rate is constant across stratigraphic stages until the Pleistocene (range of means 1.49–3.81 events lineage⁻¹ Myr⁻¹), increases in the Early and Middle Pleistocene (3.63 to 21.14 events lineage⁻¹ Myr⁻¹) and is extremely high in the Late Pleistocene and Holocene (170.167 and 232.13 events lineage⁻¹ Myr⁻¹) (Fig. 4.2c). The inferred heterogeneity of sampling across taxa is extremely high (α_{ψ} mean [95% CI]: 0.24 [0.21–0.28]). The number of lineages through time, counted by the range-through approach, increases from the origin of the group in the Early Miocene to a maximum of ca. 18 species at 4.5 Ma in the Early Pliocene (Fig. 4.2b). Diversity then halves by 2.2 Ma, followed by an increase to ca. 18 species again in the Pleistocene. The final decline towards the recent diversity cannot be considered reliable, as some extant species are not represented by fossil occurrences and are therefore ignored in this analysis.

As there is a slightly higher variation in the speciation rate compared to a constant birth-death process, the impact of the specified predictors on this variation can be analyzed. The geographical distribution of taxa is determined as most important, followed by the first phylogenetic eigenvector and the paleotemperature (Suppl. Mat. S4). The largest and most consistent difference is a ca. 20% lower speciation rate in Eastern Asia compared to the other areas (Fig. 4.3b). In addition, higher global temperatures, especially above 14°C, are associated with an increase in speciation rates (Fig. 4.3c). When evaluating two-ways interactions of predictors, the combination of each predictor with geography (except for time) are ranked as the top three important features to impact the speciation rate. Judging by the partial dependence plot (Fig. 4.3d), the difference between the speciation rate in Eastern Asia compared to the remaining areas increases with higher temperatures.

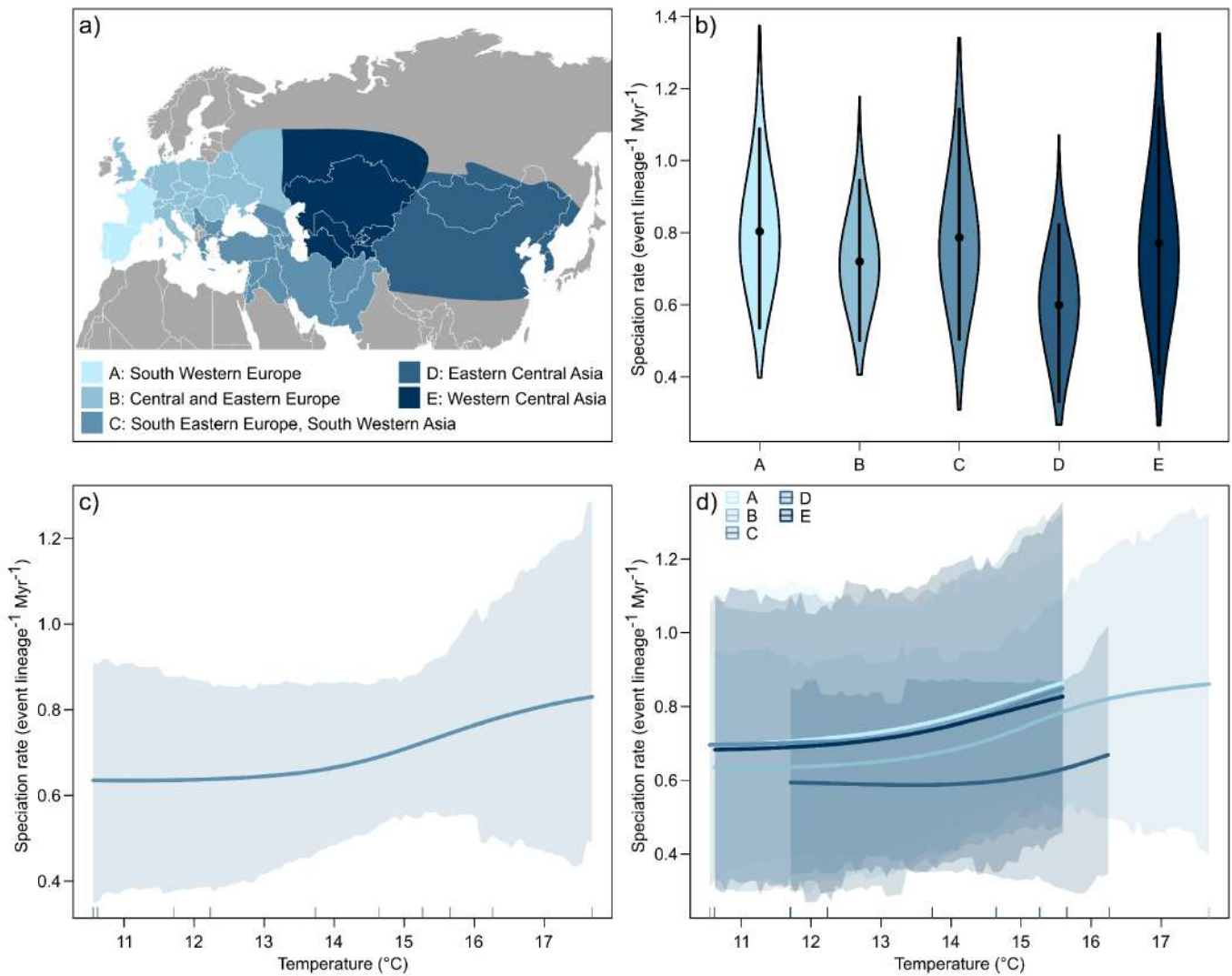


Figure 4.3. a) Geographic areas used for the BDNN analysis and partial dependence plots showing the effect of b) geography, c) temperature, and d) the combination of geography and temperature on the inferred speciation rate. Map adjusted from Dirnberger et al. (2025b).

Paleodiversity estimation under consideration of a biased fossil record

DeepDive is trained on simulated fossil records to reconstruct true paleodiversity. Feature plots show that these simulated datasets encompass the features of the empirical dataset (see Suppl. Mat. S5), ensuring reliable results. Despite the uncertainty in the ages of localities, and shifting sampling and recovery potential across space and time, the confidence intervals around the estimates are never larger than ± 3 species. Reconstructed paleodiversity of hamsters increases steeply from the Middle Miocene to the Messinian, reaching a maximum of around 34 contemporary species (Fig. 4.2b). This high level of diversity remains stable until the Late Pliocene, some 3.6 Ma, when diversity starts to decline down to 26 species in the Middle

Pleistocene. Finally, during the Late Pleistocene and the Holocene an even sharper decline of diversity leads to today's state of 19 known hamster species.

4.4 Discussion

Differences between applied phylogenetic approaches

Inferring clade-wide diversification shifts vs. within lineage variation. Assessing the fit of different diversification models based on phylogenetic trees remains challenging, as classical tools such as Bayes factors are not applicable (May and Rothfels 2023). The currently recommended strategy is to compare different methods, acknowledging the different assumptions and limitations of the models, and to avoid relying on the results of a single approach (Martínez-Gómez et al. 2024). Applying, additionally, fossil-occurrence based models can provide confidence in case of consistent evolutionary patterns. In this study, the methods applied based on phylogenetic trees differ primarily in the type of diversification shifts they aim to detect. BAMM seeks strongly supported, clade-specific shifts, such as those associated with key innovations, but tends therefore to underestimate the true number of rate shifts, especially by neglecting small rate regimes (Rabosky et al. 2017; Mitchell et al. 2019; Černý et al. 2022). In contrast, the FBDD model captures small gradual changes along branches. Such fine-scaled diversification dynamics have been suggested to explain better the evolutionary histories of several bird and mammal groups (Maliot et al. 2019; Ronquist et al. 2021; Quintero et al. 2024). In agreement to these studies, we find for hamsters considerable among-lineage variation in speciation and a similarly imbalanced speciation as described in Quintero et al. (2024): at most nodes, the speciation rate remains high in only one daughter lineage, while the other shows a decline in speciation (Fig. 4.4a). Clade-wide shifts as inferred by BAMM, could also be captured by diffusion models and would result in pattern with rates being more homogeneous in all lineages after such a clade-wide rate shift (see Quintero et al. 2024, fig. S9). Besides, the few possible rate regimes with homogeneous rates are made up by a small number of lineages, which are difficult to detect for BAMM (Rabosky et al. 2017; Meyer and Wiens 2018; Mitchell et al. 2019). Corresponding trends can also be observed in the extinction rate of hamsters, with substantial among-lineage variation and only one daughter lineage maintaining a low extinction rate (Fig. 4.4b). Therefore the evolution of hamsters seem to be not driven by adaptive radiations or key innovations, which would both result in clade-wide diversification shifts (Quintero et al. 2024).

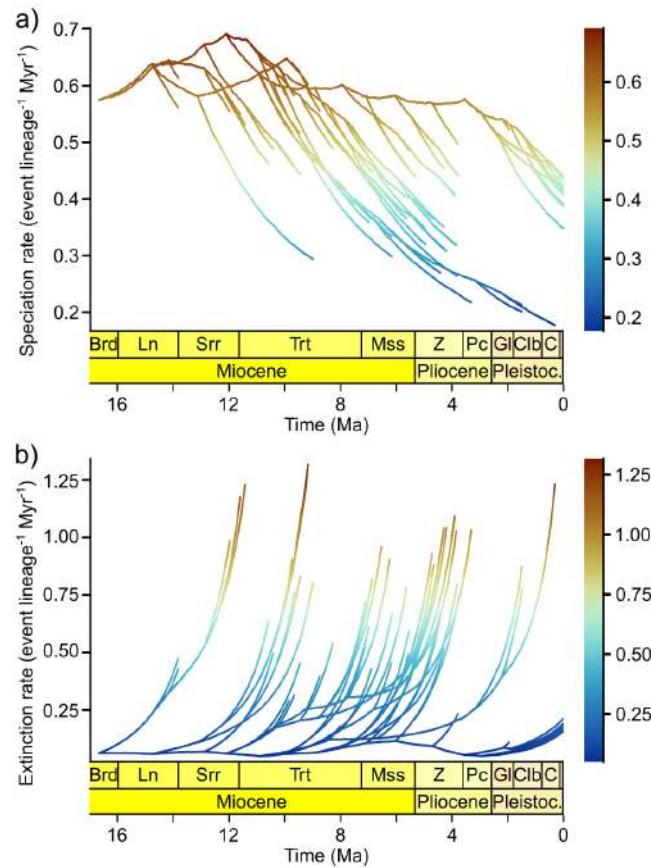


Figure 4.4. a) Speciation rate and b) extinction rate resulting from FBDD along the branches of the maximum clade compatibility tree, as color gradient and on the y-axis. The chronostratigraphic charts follow Cohen et al. (2013; updated 2024/10).

Treatment of sampled ancestors. A further distinction lies in the treatment of sampled ancestors within phylogenies. Both BAMM and FBDD use the total number of fossil occurrences to inform the preservation rate, but in contrast to FBDD, BAMM cannot be applied to trees that include sampled ancestors, typically represented by zero-length branches. To run BAMM on the randomly sampled trees used here, of which some include one or two sampled ancestors, zero-length branches were manually modified, which inflates the number of fossil tips relative to sampled ancestors. Additionally, BAMM and other diversification models are often applied to summary trees with consensus branch lengths, such as the maximum clade compatibility tree used in this study. However, during Bayesian phylogenetic inference, a fossil may be recovered as a sampled ancestor with high probability but not in every single sample. This results in a consensus branch length, that is above zero, effectively transforming that fossil into an extinct tip for downstream analyses (Beaulieu and O’Meara 2023). These alterations may be problematic, as the exact number of sampled ancestors and fossil tips have been shown to be important for accurate diversification rate estimates (Beaulieu and O’Meara 2023; Barido-

Sottani and Morlon 2025). Due to the generally high number of sampled ancestors, especially in datasets with relatively complete fossil records (Parins-Fukuchi and Saulsbury 2025), applying methods that are applicable to trees including zero-length branches such as FBDD, can be advantageous.

Accurately accounting for the preservation rate variation in the hamster dataset

BAMM assumes a constant preservation rate through time, while BDNN and FBDD allows for temporal heterogeneity. Although a constant preservation rate may be overly simplistic in many cases (Wagner and Marcot 2013), BAMM has been demonstrated to have low sensitivity to model violations in this case (Mitchell et al. 2019). In our case, however, preservation rates show unusually strong temporal heterogeneity (Fig. 4.2c), far exceeding values in previous simulation studies (Mitchell et al. 2019; Černý et al. 2022), and with an uncertain impact on diversification rate estimates. For FBDD, low preservation rates (< 0.1 events lineage⁻¹ Myr⁻¹) can lead to underestimated extinction rates (Quintero et al. 2025), but preservation rates inferred here for the hamster dataset are consistently higher than 0.1 across all stratigraphic intervals. Both phylogenetic approaches applied here assume constant preservation rates between taxa. Only the preservation model of BDNN accounts for taxonomical heterogeneity of preservation rates and reconstructs it as extremely high for the hamster dataset (α_ψ mean [95% CI]: 0.24 [0.21–0.28]). This might contribute to the resulting estimates of paleodiversity but also diversification as discussed below.

Diversification rate comparisons

Overall, the inferred speciation rate estimates all show a decelerated speciation over time until the emergence of crown hamsters, after which the trends differ between methods (Fig. 4.1a). The inferred extinction rates are not congruent between methods, with rates from FBDD more variable, compared to decreasing in BDNN and mostly constant in BAMM (Fig. 4.1b). In BAMM, extinction rates are constrained to be constant within diversification regimes, whereas in FBDD and BDNN, extinction is allowed to vary in the same way as speciation. Tree-based methods underestimate speciation and extinction rates, as compared to those obtained from the BDNN analysis (Fig. 4.1).

This finding is expected when comparing rates inferred from stratigraphic data and trees (Cantalapiedra et al. 2015; Law et al. 2018; Silvestro et al. 2018; Černý et al. 2022), and likely

reflects a combination of methodological and sampling differences. The origin of the diversification process inferred by the PyRate framework is typically younger compared to the root ages of the phylogenetic trees analyzed with BAMM and FBDD (see Figs. 1, 2b). In PyRate, the process begins at the estimated time of speciation of the oldest included taxon (Silvestro et al. 2014). For hamsters, this corresponds to *Collimys transversus* (see Prieto and Rummel 2016 for oldest occurrences), which resulted in a mean ‘root’ age of 14.6 Ma inferred by our BDNN PyRate analysis. The difference to the root ages of 19.49 Ma to 15.48 Ma obtained from the 20 randomly sampled total-evidence trees of Dirnberger et al. (2025b), potentially contributes to the higher speciation and extinction rates estimated by BDNN, especially in the early phase of the lineage history. As such differences would affect speciation and extinction equally (Silvestro et al. 2018), the resulting net-diversification rates show substantial overlap among different methods (Fig. 4.2a). Another reason for this gap has been found in the differences in definitions of speciation and extinction rates between stratigraphic and phylogenetic methods, which arise from distinct modes of speciation (Silvestro et al. 2018). Lastly, the differences in sampling between the two types of methods are also reflected in the estimated rates. Occurrence-based analyses typically include more species than phylogenetic datasets, as assembling occurrence data is generally less labor-intensive than scoring morphological characters for inclusion in phylogenies. This is also our case, where the tree has 71 tips and our PyRate analysis included 98 species. The distinct sample sets differ in their species durations and their distribution through time, resulting in different estimated macroevolutionary patterns. For example, for several extant species (four in our case), that are included in the phylogeny, no fossil species-level occurrences may be known and consequently they cannot be included in the PyRate analysis (but see Brée et al. 2022). On the other hand, during the last 5 Myr, BDNN incorporates a larger diversity of extinct taxa into the model than FBDD does (Fig. 4.2b). The result is a higher density of extinction events to inform the model during that interval, allowing BDNN to recover sustained extinction rates of around 0.5 events lineage⁻¹ Myr⁻¹ throughout the Pleistocene, instead of the extinction depression shown by the phylogeny-based methods (Fig. 4.1b). Consequently, the phylogeny-based methods indicate an increase in net-diversification in the last 3.5 Myr associated with the emergence of crown hamsters, whereas this pattern is not supported by BDNN (Fig. 4.2a).

Differences in the inferred paleodiversity and the role of the fossil record heterogeneity and incompleteness

The number of lineages through time estimated using the range-through approach represents a minimum bound to true species richness over time, as it excludes unsampled lineages and does not apply subsampling methods to correct for sampling biases. Accordingly, both FBDD and DeepDive are expected to infer higher species richness. While this is the case for DeepDive, FBDD diversity estimates fall below the range-through counts during the Early Pliocene, as discussed above (Fig. 4.2b).

The average preservation rate across all taxa increases through time (Fig. 4.2c), which leads to fewer assumed unobserved lineages by FBDD. This is reflected in the augmentation process (Quintero et al. 2025). Proposed augmented subtrees, representing unobserved lineages, are only accepted if they do not include a fossilization event (as they would not be unobserved otherwise). With high or very high fossilization rates as in our case (Fig. 4.2c), only few additional lineages are added to the complete tree. Simultaneously, the proportion of known taxa represented in the phylogenies decreases through time in this dataset (see Dirnberger et al. 2025b, additional file 1). For example, our phylogeny includes only four hamster taxa from the Pleistocene (Figs. 4.1 and 4.2). These two opposing trends may explain the underestimated species richness inferred by FBDD after the onset of the Pliocene. The range-through approach has the advantage of including all the total known diversity. Consequently, the inferred species richness of this approach can exceed the FBDD estimate at that time (Fig. 4.2b).

In contrast, the diversity trajectory resulting from the DeepDive approach remains consistently above the range-through minimum (Fig. 4.2b). The main strength of DeepDive lies in its ability to account for taxonomic, temporal and spatial biases in the preservation rate (Cooper et al. 2024b), which is important for the hamster data as shown before. This heterogeneity of preservation also affects the proportion of unsampled lineages. In the case of FBDD, if specific lineages or parts of the tree could show a lower preservation rate, more unobserved lineages would be accepted in the data augmentation process for these lineages. Thus, the underestimation of paleodiversity by FBDD seems to ultimately result from the combination of very high but also substantially heterogeneous preservation rate between lineages, which is not accounted for in this particular model (as in e.g., Barido-Sottani and Morlon 2025). Moreover, the decline in species richness during the Pliocene followed by a Pleistocene rebound, reflected in the fossil record and inferred by FBDD, is not recovered by DeepDive (Fig. 4.2b). This shows that the species sampling is incomplete and even more non-uniform. This is known to be problematic for PyRate analyses, as well, as they assume complete species sampling (Silvestro

et al. 2014; Warnock et al. 2020) but in this case also for FBDD, which can account for unsampled lineages but not all facets of fossil sampling heterogeneity as DeepDive. BAMM has already been criticized for its neglect of rate shifts in unobserved clades (Moore et al. 2016) and assumes equally a constant preservation between lineages.

The discrepancy to the DeepDive results raises concerns about whether models that ignore such important aspects of preservation heterogeneity and species incompleteness are suitable to reconstruct diversification rates based on datasets with an extremely abundant but heterogeneous and apparently considerably incomplete fossil record. Following the previous example, the Pliocene extinction rate increase inferred by FBDD (Fig. 4.1b) seems to be linked to the apparently spurious decline in paleodiversity at that time and is thus itself an artifact of fossil record bias.

In our hamster dataset, a few taxa account for a substantial number of the total occurrences (e.g., *Cricetus cricetus*: 508, *Nothocricetulus migratorius*: 252, *Allocricetus bursae*: 234). This heterogeneity partly reflects temporal variation in preservation rates, as these fossil occurrences are mostly connected to the Pleistocene. Another reason for this bias, is likely the difficulty of identifying and distinguishing closely related fossil hamster species, given the abundance of taxa with partly very similar tooth morphology (de Bruijn et al. 2012; Dirnberger et al. 2025b). Therefore, despite our thorough literature review, some identifications and thus occurrences might not be reliable, as well-known, and widespread hamster taxa (as the ones mentioned above) are possibly more often raised, when fossil faunas are described. In addition, the fossil record is in general well known to be spatially biased, reflecting geological and socio-economic factors (Raja et al. 2022; Krone et al. 2024). Overall, these taxonomic and spatial biases, only accounted for in DeepDive, are probably responsible for the differences in inferred paleodiversity (Fig. 4.2b) and affect also the estimated diversification rates.

Diversification dynamics of hamsters: insights and limitations

Following the discussed limitations of the applied approaches and the differences in their results, the inferred evolutionary patterns of hamsters must be interpreted with caution. Our analyses indicate a general decline in speciation rate and extinction rate, while both rates are relatively constant over time, except for the estimates from FBDD (Fig. 4.1). The absence of major clade-wide diversification shifts in Cricetinae is consistent with earlier studies of molecular phylogenies of extant Muroidea and Rodentia, in which also no diversification shift was found in Cricetinae (Steppan et al. 2004; Fabre et al. 2012; Schenk et al. 2013; Steppan and Schenk 2017). Generally, the diversification of hamsters appears to be not driven by key

innovations, which would have clade-wide effects, but by a more complex combination of several factors, resulting in more numerous and gradual shifts in diversification.

According to our results, neither geographical distribution, global paleotemperature, or phylogenetic relatedness had a major impact on the extinction of hamsters. One explanation for this immutability may lay in the burrowing behavior of hamsters, which has long been suspected to be an important ecological strategy to buffer against environmental changes (Voorhies 1975; Liow et al. 2008). Nonetheless, extinction rates are relatively high in hamsters, compared with other burrowing mammals, which have a generally lower extinction risk compared to non-burrowers (Pinkert et al. 2025). On the other hand, our analyzed predictors explain a slight variation in species-specific speciation rates. Previous studies have suggested the spread of open environments and increasing aridification as potential drivers of hamster diversification (Neumann et al. 2006; Pardiñas et al. 2017a). In our study, geographic distribution of taxa is identified as the most important feature (Fig. 4.3a-b), which could also partly capture the impact from vegetational dynamics. A finer-scale analysis of vegetation changes across space and time would, however, require comprehensive data beyond the scope of this study. In addition, global paleotemperature appears positively correlated with hamster speciation rates (Fig. 4.3c), in line with findings from other clades (see e.g., Condamine et al. 2019; Hauffe et al. 2024; Buffan et al. 2025). This result could stem from two non-exclusive processes. First, intervals with higher temperatures were more productive, being able to sustain a larger diversity of cricetines, as it was also shown for large Neogene mammals (Fritz et al. 2016). Second, we may be also picking up certain signals of diversity-dependent processes, whereby the proliferation of species decays as the ecological and geographical space are filled. The first phylogenetic eigenvector, which captures the basal-most phylogenetic divergences, is also highlighted as another key factor for diversification. It appears that one or multiple unknown traits associated with early cricetine divergences could have influenced speciation rates in hamsters. Identifying such traits and their role in diversification (as in e.g. Cantalapiedra et al. 2021) and assessing the impact of the emergence of closely related groups such as murids and arvicolines (see Kålin 1999) represent promising directions for future studies. Overall, hamsters show relatively constant diversification and diversity following their initial radiation. According to the DeepDive analysis, the group shows a moderate diversity decline around 4 Ma. This decline accelerated around 0.8 Ma, likely stemming from the widespread environmental disruptions of the middle Pleistocene. The late Pleistocene, starting 0.129 Ma, witnessed further species losses (from around 25 to 20 species). This timing and the known impact of climatic fluctuations on suitable habitats and thus the evolution of cricetines and

closely related small mammals (Hír 1997b; von Koenigswald 2007; Ding and Liao 2019; Lebedev et al. 2021; Feoktistova et al. 2023) suggests that environmental forcing stemming from the last glaciations had the main impact on the decline of the group. Nevertheless, the late Pleistocene diversity crisis had a more moderate effect on cricetines, reducing their diversity by approximately 20%, compared to the more severe losses observed in larger mammals (Svenning et al. 2024). This follows the trend of selective impact of Pleistocene extinctions on animals with larger body size (Dirzo et al. 2014) and might be explained in the case of hamster with their ecological strategies of hibernation and burrowing as outlined above.

Conclusion

We conducted two analyses based on phylogenies, that include fossil taxa, as well as two analyses based on fossil occurrence data, one of which also incorporates phylogenetic information, to infer diversification rates in the group of true hamsters, as well as their true paleodiversity. Our results show a speciation rate slightly decreasing through time and affected mainly by biogeography and global paleotemperature. The derived extinction rate on the other hand is considerably different between the applied approaches. We find no evidence for an impact of geographical distribution or global paleotemperature on the extinction risk of hamster, which may be connected to ecological strategies found in hamsters for buffering unfavorable environmental conditions. Corresponding to this hypothesis, the decrease in paleodiversity during the Pleistocene is not as severe as for other mammal groups. An assessment of the violations of the hamster data against the applied models, regarding the incomplete and biased fossil record, calls for caution when using diversification models on similar fossil datasets.

Declarations

Availability of data and materials

The Supplementary Material of this study are available in the context of this thesis under:
<https://figshare.com/s/6c91828664b2db2ffa7>

Competing interests

The authors declare that they have no competing interests

Authors' contributions

Conceptualization: MD, RLA, PPC. Methodology: MD, RLA, PPC, JLC. Formal analysis: MD. Visualization: MD. Writing original draft: MD. Writing review and editing: MD, RLA, PPC, JLC, SR. All authors read and approved the final manuscript

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Summary

In the fourth chapter, further insights into the evolutionary history of hamsters are given, in regards of their macroevolutionary rate dynamics. We found no evidence for clade-wide diversification shifts, which would indicate for example key innovations. The speciation rate appears to also be generally relatively constant through time. Nonetheless, by analyzing specific-specific rates while incorporating several time- and trait-series, we could identify several important drivers for hamster speciation. The derived extinction rate exemplifies the differences that can arise when comparing different approaches. Regarding its drivers, we found no evidence for an influence of several popular candidates, which hints towards a certain resilience in hamsters. The inferred true paleodiversity highlights the difficulty to properly account for the biased and incomplete fossil record in diversification models.

Chapter 5

Discussion

5.1 Building and using a robust phylogeny of extinct and extant hamsters

The main goal of Chapters 2 and 3 was to establish a first phylogenetic hypothesis for extinct hamsters followed by an incorporation of also extant hamsters. Previous attempts working with fossil hamsters were limited in taxonomic scope and relied exclusively on maximum parsimony (Cuenca-Bescós 2003; López-Antoñanzas et al. 2014; Sinitsa and Delinschi 2016; Feroz et al. 2024). The new analyses, I present here, are based on the morphological matrix of López-Antoñanzas and Peláez-Campomanes (2022), which has been expanded by adding additional characters specifically informative for hamsters (and with the exclusion of invariant characters, automatically by MrBayes in Chapter 2 and manually in Chapter 3). To ensure comprehensive taxonomic coverage, I have conducted an extensive data collection campaign. This involved first-hand examination of specimens housed in several major paleontological and zoological collections (ISEM, Montpellier; MNHN, Paris; SNSB-BSPG, Munich; SNSB-ZSM, Munich; UCBL-FSL, Lyon) and a thorough review of the literature.

The comparison of probabilistic and parsimony approaches to phylogenetic inference has been extensively debated, with studies often reaching conflicting conclusions (O'Reilly et al. 2016; Puttick et al. 2017, 2019; Goloboff et al. 2018, 2019; Sansom et al. 2018; Vernygora et al. 2020; Simões et al. 2023). Model-based approaches offer certain advantages, such as the possibility of model selection or quantifying uncertainty in the case of Bayesian inference. Nevertheless, parsimony analyses remain one of the most popular methods for analyzing morphological characters, and are thus particularly often applied in paleontology (e.g., Feroz et al. 2024; Pereyra et al. 2025). Accordingly, in Chapter 2 of this thesis I have applied both maximum parsimony and Bayesian inference and tested the stratigraphic fit of both. The primary focus of this Chapter was not, however, to decide which method is more appropriate for the hamster dataset (see Wright and Lloyd 2020 on potential pitfalls). Instead, the application and comparison of different approaches contributes to the goal of building a reliable phylogenetic

hypothesis. Additionally, I tested different model clock models to further improve our results. The resulting summary trees can be seen in Figures 2 and 3 of Chapter 2.

The use of summary trees, such as the maximum clade credibility (MCC) tree or the maximum a posteriori (MAP) tree, has been shown to be potentially problematic in the context of Bayesian inference, especially with morphological characters (O'Reilly and Donoghue 2018). As an example, highly supported clades (e.g., with posterior probability $PP = 0.95$) can be disregarded in MAP and MCC trees (O'Reilly and Donoghue 2018). The summary trees supported by MrBayes are the 50% majority-rule consensus tree (option 'halfcompat') or the maximum clade compatibility tree (option 'allcompat'). Although the algorithm behind the maximum clade compatibility tree is to my knowledge nowhere explained in detail, the MrBayes command reference states: 'Halfcompat' results in a 50% majority rule tree, 'Allcompat' adds all compatible groups to such a tree.' (Huelsenbeck et al. 2020). Such trees, sometimes also called greedy consensus trees or extended majority consensus trees (Bryant 2003; Jansson et al. 2016), accordingly represent all highly supported clades, in contrast to MAP and MCC trees. Additionally, I identified and removed rogue taxa from the tree set before summarizing to improve the resulting trees (Smith 2022). When reporting and discussing the results, we aimed to carefully address uncertainties in tree topology by referring to clade posterior probabilities, particularly when these values are low. Nonetheless, a detailed visualization of the 'treespace' could be helpful to further explore the differences in the posterior tree samples (see Wright and Lloyd 2020).

Based on the phylogenetic analyses of the morphological matrix extended during this thesis, the monophyly of several genera was confirmed with varying degrees of credibility: †*Collimys* ($PP = 0.76$), †*Rotundomys* ($PP = 1$), †*Pseudocricetus* ($PP = 0.96$), and †*Hattomys* ($PP = 1$). †*Apocricetus* was also recovered as monophyletic, though excluding †*A. plinii* ($PP = 0.98$). Furthermore, the results indicate that †*Neocricetodon* includes †*Cricetulodon complicitens* ($PP = 0.52$). The low posterior probabilities for †*Collimys* and especially †*Neocricetodon* have been increased in the subsequent analysis presented in Chapter 3 (†*Collimys*, $PP = 0.99$; †*Neocricetodon* $PP = 0.88$). Moreover, we found support for the previously proposed non-monophyly of †*Cricetulodon* (López-Antoñanzas et al. 2014) and for †*Kowalskia* as junior synonym of †*Neocricetodon* – a widely debated topic under hamster taxonomists (Daxner-Höck et al. 1996; Freudenthal et al. 1998; Sinitsa and Delinschi 2016; Wu and Flynn 2017; Feroz et al. 2024). Consequently, this work presents a novel phylogenetic hypothesis for extinct hamster genera, which have never been analyzed in a comprehensive phylogenetic study before. Ancestral state reconstruction of morphological dental characters enabled the identification of

synapomorphies for each monophyletic clade, providing a robust morphological basis to resolve taxonomic uncertainties and facilitate future identifications.

Chapter 3 included extant hamsters in the phylogeny and incorporated molecular data from extant taxa to perform total-evidence dating. The inference of phylogenetic relationships among extant hamsters based on dental data is complicated by a general trend of molar simplification, which leads to the loss of structures. As an example, the separation of *Nothocricetulus* from the *Cricetulus* taxa was based on solely molecular data (Neumann et al. 2006; Lebedev et al. 2018). The results of the preliminary analysis based on the comprehensive morphological matrix (see Supplemental Material of Chapter 3, Fig. S5.1) retrieved the monophyly of all extant genera sampled with multiple species (*Phodopus*, *Mesocricetus*, *Cricetulus*), including the separate position of *N. migratorius*. Additionally, the tribe Urocrinetini (*Phodopus* + *Urocrinetus*) is recovered. However, due to the homoplastic nature of simplified molar morphology, the remaining intergeneric relationships differed in part from those in molecular trees. Consequently, molecular data were integrated in the analysis to compensate for the simplified molar morphology of extant hamsters.

The phylogeny in Chapter 3 benefited from an extensive increase in taxonomic sampling, through the addition of 33 species distributed in 17 genera, when compared to Chapter 2. Expanded fossil sampling can improve the accuracy of inferred topologies and divergence times, as well as downstream analyses (Mongiardino Koch et al. 2021; do Rosario Petrucci et al. 2025). Thorough model testing was performed, as in Chapter 2, but with the inclusion of two additional clock models that are only implemented in the development version of MrBayes (Zhang 2022). The resulting phylogeny (Fig. 3.2) helped to resolve several taxonomic questions such as those regarding fossils originally described as †*Cricetinus* (Xie et al. 2023), and confirms the non-monophyly of †*Allocricetus* (Cuenca-Bescós 2003). The analysis presented in this chapter is the first total-evidence dating analysis of extinct and extant hamsters. This approach placed the origin of crown hamsters in the Pliocene, considerably later than previously estimated (e.g., Neumann et al. 2006; Lebedev et al. 2018). These results should, however, be considered with caution due to incomplete sampling of extant species (13/19) and the relatively low posterior probabilities at several nodes leading to the crown group. Nonetheless, we demonstrate that several node calibrations in molecular phylogenetics studies of hamsters are potentially problematic. Moreover, most of these studies ultimately base their calibration points on the results of a few studies (such as Steppan et al. 2004; Neumann et al. 2006), which probably overestimated divergence times. Consequently, the results of the subsequent studies may also be overestimated, which could explain the differences found with our age estimates,

which derive from tip-dating under substantial fossil sampling. I also used the resulting phylogeny to infer ancestral geographic ranges of hamsters (Fig. 3.3). The importance of including fossil taxa in this process has been recently highlighted under a series of simulations and on carnivores (Faurby et al. 2024). The reconstruction I carried out on hamsters suggests a Central to Eastern European origin for hamsters, with the Tortonian identified as a key period for dispersal events into Western Europe and Asia. East Central Asia was recovered as the area of origin for several extant lineages (*Cricetulus*, *Phodopus*, *Tscherskia*, and *Urocricetus*). This result confirms previous suggestions concerning the ancestral area of *Urocricetus* and *Phodopus* (Lebedev et al. 2018) but contrasts with the proposed European origin of *Tscherskia* (Xie et al. 2023). Accounting for phylogenetic uncertainty can be important but also challenging in biogeographic reconstructions and ancestral state inference in general (Duchêne and Lanfear 2015; Magalhaes et al. 2021). In this study, we attempted first to minimize topological uncertainty by removing rogue taxa and then to carefully discuss reconstructed ancestral states considering different possible topologies, such as the different potential sister lineages to crown hamsters. Biogeographic range evolution and phylogenetics can be inferred jointly to directly account for phylogenetic uncertainty (Landis et al. 2021). However, Landis et al. (2021) inferred the position of fossil taxa on the tree based on topological constraints, and biogeographical and biome data rather than from morphological data (see also Ramírez-Barahona 2024). The latter has not been done yet, to my knowledge.

Another concern is the interaction between diversification rates and biogeography (Goldberg et al. 2011). The approach to biogeographic reconstruction employed here, does not incorporate the reciprocal effect of changes in diversification rates and geographical ranges (Matzke 2014). Models that integrate geographical evolution and the diversification process place, however, certain demands on tree size, are limited in the number of regions that can be considered, and are usually developed for extant phylogenies (Goldberg et al. 2011; Landis et al. 2022). An alternative is to study biogeographic range evolution based on fossil occurrence data, with approaches focused on estimating dispersal rates, area-specific extinction, and the impact of biotic and abiotic factors (Silvestro et al. 2016; Hauffe et al. 2022). These methods, however, are limited to comparisons between two regions and are not focused on reconstructing ancestral ranges.

5.2 Reconstructing macroevolutionary rate dynamics of hamsters

The extant diversity of hamsters is very limited, and a comprehensive phylogenetic hypothesis for extinct hamsters has been missing to date. Therefore, regarding their macroevolutionary dynamics, the group has previously been analyzed only as a part of a broader taxonomic group, such as rodents or muroids (Fabre et al. 2012; Schenk et al. 2013; Stepan and Schenk 2017). This is also the case for studies based on fossil occurrences, which have moreover often been limited in their geographical scope (e.g., Van den Hoek Ostende et al. 2020; van Dam et al. 2023).

Accordingly, knowledge about the diversity and variation of diversification within the group of hamsters has been scarce. According to our results, the diversification of hamsters seems to be in general relatively constant through time and across clades (Figs. 4.1, 4.2). The results of the fossilized birth-death diffusion (FBDD) analysis show that variation in diversification within lineages is considerable, while the absence of clade-wide shifts suggests little impact from key-innovations or adaptive radiations (Fig. 4.4) (Quintero et al. 2024). The species-specific extinction rates of hamsters do not depend on changes in temperature, geographical distribution or phylogenetic relatedness. It appears, the group developed strategies to buffer all kinds of unfavorable conditions. Hamsters are for example known to build extensive burrows (Ross 1994; Nechay 2000), and enter deep or short hibernation and/or torpor in winter (Figala et al. 1973; Janský et al. 1984; Waßmer and Wollnik 1997). All these are important strategies for dealing with unexpected environmental changes (Voorhies 1975; Liow et al. 2008; Pinkert et al. 2025). Species-specific speciation rates showed a correlation to our tested predictors, foremost to the geographic distribution (Fig. 4.3), which might be connected to the spread of open vegetation, potentially beneficial for hamsters (Neumann et al. 2006; Pardiñas et al. 2017a). Additionally, a positive correlation of speciation with global paleotemperature is present (Fig. 4.3), possibly as a result of increased resource availability in warm and more productive environments (Clarke and Gaston 2006). The respective partial dependence plot shows a slightly sigmoidal trajectory, which might reflect a certain diversity-dependency with decreasing speciation rates, resulting from competition and ecological limits (Condamine et al. 2019).

A major focus of Chapter 4 was to evaluate the different assumptions about sampling and preservation rate variation across the tested models. Following the results of DeepDive, which accounts for spatial, temporal and taxonomic biases in the fossil record to infer true paleodiversity (Cooper et al. 2024b), the diversity curves inferred by the range-through approach reveal a nonuniformly incomplete fossil record. This is not accounted for in Pyrate,

as it assumes complete sampling for the birth-death process (Silvestro et al. 2014; Warnock et al. 2020). On the other hand, the data augmentation process in FBDD is designed to account for unobserved lineages but the model does not incorporate taxonomic and spatial biases in the preservation rate (Quintero et al. 2025). This might lead to the assumption of nearly complete species sampling (only a few lineages are added in the data augmentation process), while in reality this might not be the case when the DeepDive result is referenced. BAMM has been criticized for not incorporating unobserved rate shifts (Moore et al. 2016) but the impact of this has been suggested to be small (Rabosky et al. 2017; Mitchell et al. 2019; Barido-Sottani et al. 2020b). Nonetheless, BAMM assumes a constant fossil sampling rate not only between taxa but also through time, which is strongly violated by our data. Overall, the comparisons of our models and the paleodiversity inferred by DeepDive exemplify that the biases in the fossil record can lead to biologically meaningful differences in the reconstructed diversification rates, depending on how or if these biases are accounted for. Incorporating heterogeneity in the fossil sampling rate between lineages could be an important extension to the FBDD model in the future.

Another aspect of macroevolutionary analyses refers to which taxonomic level should be employed. All our analyses have been conducted at the species level. Using the genus level as a proxy for species level can be beneficial as it allows the inclusion of occurrences with unclear species identifications. However, the ability of genus-level data to accurately reflect species-level dynamics and correlations is uncertain (Wiese et al. 2016). The use of higher taxonomic levels can especially be valuable, when working on large taxonomic scales or with occurrences rarely identified to the species level, but this is not the case for hamsters. For this group, there are sufficient species-level occurrences while the overall diversity is still manageable. Additionally, the use of “wastebin” genera, which include many distant species, was noted for hamsters (e.g., *Cricetulus* Neumann et al. 2006; Lebedev et al. 2018). This complicates using genus-level data as a proxy for the species level and can also be problematic for data cleaning. As an example, several Neogene fossil taxa have been moved from *Cricetus* to †*Apocricetus* and †*Pseudocricetus* (Topachevsky and Skorik 1992; Freudenthal et al. 1998). If fossil occurrences in older publications are identified to the species level, e.g., “†*Cricetus kormosi*” from Polgárdi, Hungary (Freudenthal and Kordos 1989), the taxonomy can be adjusted for the here used database and the occurrence can be included with its corrected name †*Pseudocricetus kormosi*. If publications, however, can identify only the genus level and note “*Cricetus* sp.”, the occurrence cannot be added to the database, even at genus-level, as it is unclear whether this

specimen refers to †*Apocricetus*, †*Pseudocricetus* or actually *Cricetus*, unless sufficient morphological information is provided in the publication.

This leads to the next topic, the importance of cleaning and extending the occurrence datasets from databases like The Paleobiology Database (PBDB) or the New and Old Worlds database of fossil mammals (NOW). Such online databases have been shown to be partly inaccurate, with regards to duplicates, taxonomic errors or issues with temporal or geographic data (Zizka et al. 2019; Schroeder et al. 2022). Accordingly, guidelines and automatic procedures for cleaning occurrence data have been developed (Zizka et al. 2019; Jones et al. 2025). When scanning PBDB and NOW, inconsistencies or errors in the taxonomy are apparent. As an example, NOW, contains entries for “*Cricetulus trion*”, “*Cricetulus triton*” “*Cricetulus triton varians*”, “*Cricetulus varians*”, and “*Cricetulus rarians*”, which probably all belong to *Tscherskia triton* or †*Tscherskia triton varians* (Xie et al. 2023). Furthermore, several hamster taxa with abundant fossil occurrences are missing from these databases, such as *Allocricetulus eversmanni* in PBDB (see Kowalski 2001; Krokmal and Rekovets 2010). The incompleteness in hamster occurrence data is a considerable concern, which cannot be resolved by data cleaning processes alone. While the overall comprehensiveness of these databases is generally impressive, the confusing taxonomic history of several fossil hamsters and the sheer abundance of specimens present a major challenge. Correcting all inaccuracies and overseeing the expanding of the hamster data is unfeasible for the small teams of volunteers who manage these databases including also millions of occurrences from other groups. Therefore, I compiled an occurrence dataset for hamsters for this thesis. It is originally based on a search for several genera in the NOW database but is now substantially expanded. To compare this new dataset with the widely used online resources, fossil occurrence data for all genera assigned to Cricetinae in this thesis (see Supplemental Material of Chapter 3) have been downloaded from NOW and PBDB. Both datasets have then been adjusted to follow the taxonomy used in Chapter 3. For three occurrences in the PBDB dataset, that showed age uncertainties of over 10 Myr (fossil sites Almenara-Casablanca M, Pertuis, Cucuron) more detailed temporal data were provided (following Ménouret 2014; Minwer-Barakat et al. 2018), but no further data cleaning was performed. The resulting datasets from PBDB and NOW include 43 and 79 species with 250 and 582 fossil occurrences respectively. In contrast, the database I built for this thesis includes 100 species with 1951 occurrences. A limitation is, however, the geographical detail. For several occurrences, no coordinates were published, thus only country-level data are included. This dataset is the most comprehensive compiled for hamsters to date, although a

constant effort is necessary to keep it updated with newly published occurrences and taxonomic adjustments.

Results of macroevolutionary analyses can be expected to vary when using such substantially different datasets. To exemplify this, I performed a PyRate analysis on each dataset (NOW, PBDB and the one built here in Chapter 4). The analyses were run with estimated rate shifts using reversible jump MCMC (Silvestro et al. 2019) under a Time-variable Poisson process (TPP) for preservation rates, using stratigraphic stages boundaries on 5 replicates for each dataset (NOW, PBDB, Chapter 4) to incorporate fossil site age uncertainties. The resulting speciation and extinction rates are shown in Figure 5.1.

The resulting rates show differences in their overall temporal trends, precision, shifts and support for shifts (Fig. 5.1), although proper data cleaning would be necessary for the NOW and PBDB datasets and might result in different rate estimations. This preliminary analysis highlights that the use of different datasets might ultimately result in distinct biological interpretations and lead to either unnecessarily low or erroneously high confidence. This is particularly relevant for inferring true species richness over time, e.g., with DeepDive, as the relatively high number of missing taxa in the NOW and PBDB datasets can be expected to severely impact the results.

Following the assembly of a comprehensive occurrence database and a phylogenetic framework for hamsters, and after this first exploration of macroevolutionary dynamics, further hypothesis-driven analyses can follow to investigate the impact of specific abiotic and biotic factors on the hamster diversification (see e.g., do Rosario Petrucci et al. 2025). Moreover, broadening the taxonomic scope offers promising opportunities for comparative research by including other Cricetidae groups such as the Arvicolinae, the sister group of Cricetinae. Arvicolinae is compared to hamsters an exceptionally diverse group (161 species according to Pardiñas et al. 2017a), even when considering only their Palearctic diversity (128 species according to Kryštufek and Shenbrot 2022). Given their relatively similar geographical and temporal distribution and close phylogenetic position, an intriguing question emerges as to what factors account for the markedly different levels of evolutionary success of these two groups of cricetids.

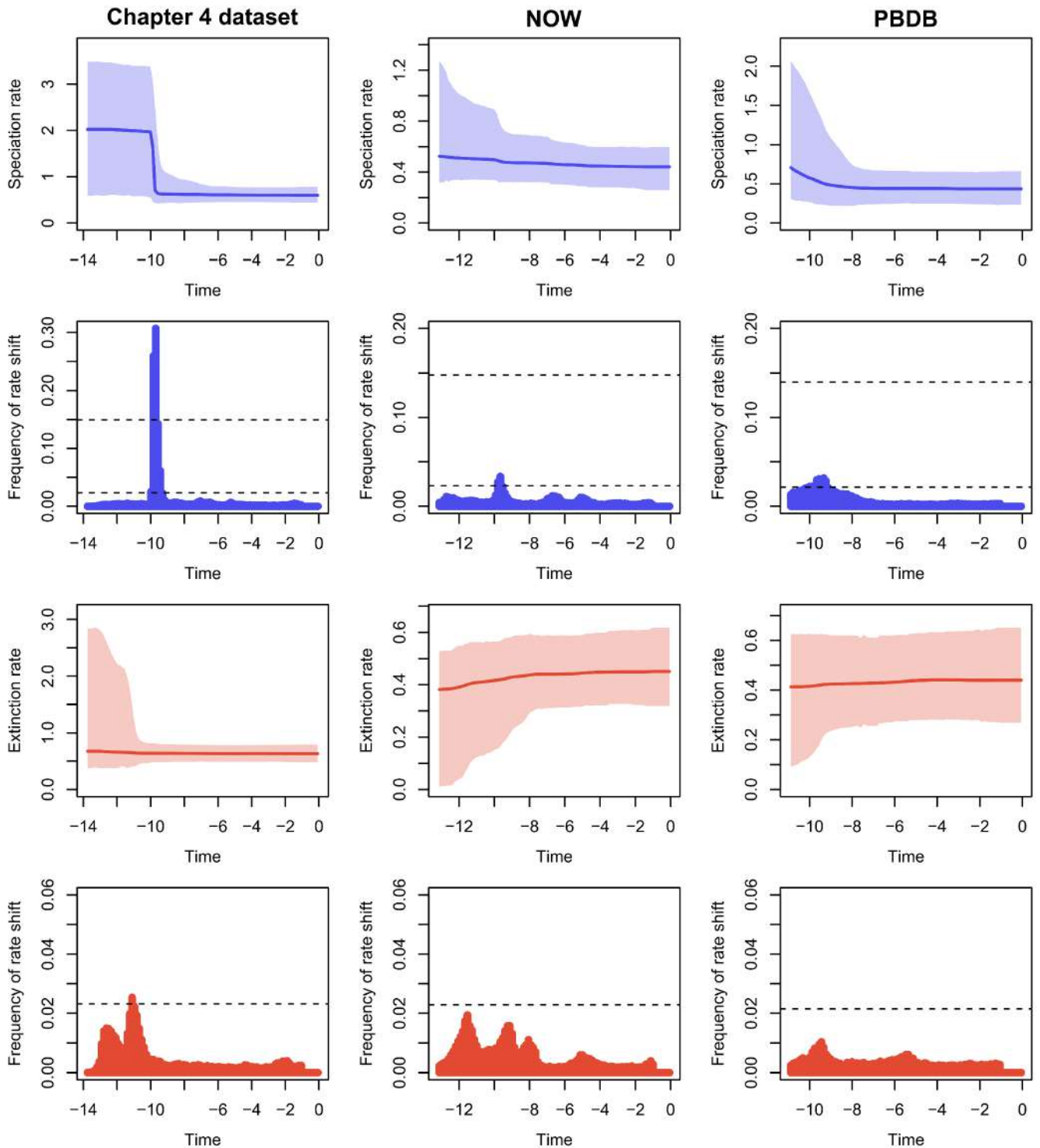


Figure 5.1. Speciation and extinction rates (rows 1 & 3) through time in hamsters retrieved by PyRate based on the datasets of Chapter 4, NOW and PBDB, including 95% credible intervals. The frequency of respective rate shifts is shown below (rows 2 & 4). The dashed lines represent log Bayes factors of 2 (and 6 in the case of speciation) comparing the model including a rate shift to the one without a shift. If the frequency exceeds the line, there is positive (2) or strong (6) evidence for a rate shift at the respective time (Kass and Raftery 1995; Silvestro et al. 2019).

5.3 Challenges in integrating fossils into phylogenies

The incorporation of fossil data into phylogenies has been shown to be beneficial not only for phylogenetic reconstruction itself (Mongiardino Koch and Parry 2020; Mongiardino Koch et al. 2023) but also for phylogenetic comparative methods, such as reconstructing ancestral biogeographical ranges (Faurby et al. 2024) or macroevolutionary rates (Mitchell et al. 2019; do Rosario Petrucci et al. 2025). In the case of hamsters this becomes even clearer, as they are an example of a clade whose extant diversity is considerably smaller than their known extinct richness. The most direct way of incorporating extinct taxa into phylogenies is through the Fossilized Birth-Death (FBD) model (Stadler 2010; Heath et al. 2014), which was used to reconstruct hamster phylogenies in Chapters 2 and 3. The model, as implemented in these chapters assumes constant diversification and fossil sampling rates. While the diversification rates of hamsters appear relatively constant over time (as shown in Chapter 4, except for the extinction rate according to FBDD), the preservation rate does not. When reconstructing phylogenies, applying models that allow diversification and especially preservation rates to vary through time or across lineages (Gavryushkina et al. 2014; Barido-Sottani and Morlon 2025; Chabrol et al. 2025) could lead to changes in tree topology and, in particular, divergence time estimates (see also Barido-Sottani et al. 2023). One of the main challenges of applying the FBD in a total-evidence framework lies in its assumption of specimen-level sampling. In empirical studies, this assumption is frequently violated when using taxon-level data, where each species is represented by a single data point and the uncertainty of the fossil sample represents the stratigraphic range of the taxon (Wright et al. 2022). This is less of a problem, when working with groups for which typically only a single fossil occurrence per taxon is known in the first place (e.g., Dirnberger et al. 2024a). However, in groups like hamsters, some species are represented by a high number of occurrences. Thus, including each fossil species as a single extinct sample is in clear conflict with the FBD sampling assumptions (Wright et al. 2022). To make matters worse, popular fossil occurrence databases like NOW or PBDB are not fully reliable for hamsters as mentioned in the previous section. Therefore, stratigraphic ranges of taxa, such as those used in Chapters 2 and 3, are often the only available data. With the construction of a more complete occurrence dataset, as achieved in Chapter 4, these model violations could be overcome. Single specimens could be incorporated and by accounting for their individual age uncertainties inaccuracy in the results would be further reduced (Barido-Sottani et al. 2019, 2020a). However, as in the case of hamsters, there are often so many fossil specimens that incorporating every single of them directly in the analysis is not feasible. Besides computational and human-based limitations, many specimens lack sufficient

morphological data, which may be unpublished or lost through taphonomic processes. There are ways to still include such fossils without morphological data (Gavryushkina et al. 2014; Heath et al. 2014), which usually requires topological constraints based on taxonomic information (Barido-Sottani et al. 2023; Nikolic et al. 2025). In fossil hamsters, however, several taxonomic uncertainties remain (as shown in Chapters 2 and 3), which could lead to severe misplacement of some fossils. Only a small proportion of such erroneously positioned fossils has been shown to result in less accurate divergence time estimations (Barido-Sottani et al. 2023). Furthermore, the issues related to the large number of specimens remain. A possible solution is to subsample the dataset, either randomly or by including only the oldest known occurrence for each species (Püschel et al. 2020). However, these approaches have been shown to result in divergence dates that are incongruent with the fossil record or overestimated (Matschiner 2019; O'Reilly and Donoghue 2020).

As a consequence of the unreliability of the hamster occurrence data (before Chapter 4), the large number of specimens, and problems with subsampling approaches, in Chapter 2 and 3, stratigraphic species ranges were used for tip-dating, as in several previous empirical studies (e.g., Zhang et al. 2016; Gavryushkina et al. 2017; Turner et al. 2017). In the recent years, an extension the FBD model has been developed, which allows the use of taxon-level data (Stadler et al. 2018). Under consideration of three speciation modes, the assignment of several fossil specimens to one species and thus the incorporation of stratigraphic species ranges is possible in this FBD range model (Stadler et al. 2018). This extension would be the best way to analyze the hamster dataset, resulting in more accurate phylogenies, which could also be used in downstream analyses as in Chapter 4. Unfortunately, however, to my knowledge, no stable, benchmarked and published implementation is available (see also Heckeberg et al. 2025).

Moreover, in reference to the previously mentioned endeavor regarding the incorporation of preservation and diversification rate variation into phylogenetic inference, the available models are still restricted to specimen-level data (Barido-Sottani and Morlon 2025; Chabrol et al. 2025). However, even if these models would be extended to accommodate taxon-level data, the relatively small morphological matrix available for hamsters could be problematic for more complex models, as is also generally the case with inferences based on the Mk model (Puttick et al. 2017; Barido-Sottani et al. 2020a; Luo et al. 2020; Barido-Sottani and Morlon 2025). The number of meaningful discrete characters than can be coded from molars is limited (Simões et al. 2017). Therefore the inclusion of continuous characters (Parins-Fukuchi 2018; Zhang et al. 2024) appears to be a promising future direction for studying hamster phylogenetics. Another option would be the incorporation of fossil occurrences for which no morphological data are

available and no morphological constraint can be justified, by integrating an additional sampling process for these occurrences into FBD inference (Vaughan et al. 2019; Andréoletti et al. 2022). This also related to the above-mentioned problem of uncertain taxonomic information for several hamster species. This occurrence birth-death approach allows the incorporation of temporal information from otherwise unsampled occurrences to inform the macroevolutionary rates during tree reconstruction and to estimate species richness through time based on all known fossils samples. While computational limitations restrict the number of taxa that can be included in such analyses (see also Chabrol et al. 2025), the more manageable size of the hamster group could allow exploring these approaches in the future. This could potentially be combined with topologically constrained taxa (above-mentioned) for which positions in the tree can be assumed. The approach has recently been extended to incorporate within-lineage diversification rate variations (Andréoletti et al. 2025), although without accounting for preservation rate heterogeneity between lineages. This model is however not yet available for full phylogenetic inference. A comparison of such an occurrence birth-death model with the results in Chapter 4 might provide further insights in the paleodiversity of hamsters. Another interesting topic for future research would be a comparison of the models that allow for anagenetic or cladogenetic rate variations (Chabrol et al. 2025; Quintero et al. 2025) and the conclusions we can draw from it for the evolutionary history of hamsters.

5.4 Conclusion

This thesis provides a comprehensive and integrated analysis of the evolutionary history of hamsters (Cricetinae), bridging the gap between their extensive fossil record and their limited extant diversity. The principal conclusions are as follows:

For the first time, a robust and well-supported phylogenetic tree for hamsters has been established that fully integrates both extinct and extant species. This framework, built using both maximum parsimony and Bayesian inference on dental morphology, resolves long-standing taxonomic debates and provides a reliable foundation for all subsequent analyses. The confirmation of monophyletic genera and the clarification of synonymies, such as that of †*Kowalskia* and †*Neocricetodon*, bring much-needed stability to the group's classification.

The first total-evidence dating analysis for hamsters, combining morphological and molecular data, reveals a significantly younger age for the hamster crown group than previous node-

calibrated estimates. The origin of crown hamsters is now placed in the Pliocene, indicating a more recent and rapid radiation than previously assumed.

The biogeographic reconstruction supports a Central and Eastern European origin for the entire hamster group. The evolutionary history of the group is characterized by multiple dispersal events to Western Europe and Asia, with the crown group itself originating in a region around the Eastern Mediterranean, and Black and Caspian Seas.

The macroevolutionary analysis reveals a hump-shaped trajectory of paleodiversity, with a peak of approximately 35 contemporaneous hamster species during the Messinian and Pliocene. Speciation rates were relatively constant but slightly decreasing through time, primarily influenced by geographic distribution and global paleotemperature. Crucially, no strong evidence was found for the impact of these environmental predictors on extinction rates, suggesting a significant degree of resilience in hamsters to unfavorable environmental conditions.

The comparison of different state-of-the-art Bayesian and deep-learning approaches highlights the challenges of accounting for the incompleteness and inherent biases of the fossil record. The discrepancies in estimated extinction rates and paleodiversity between methods underscore the importance of using multiple analytical frameworks to mitigate the limitations of any single one and to draw robust biological inferences.

In conclusion, this thesis fundamentally advances our understanding of hamster evolution by demonstrating their radiation out of Europe, their resilience to environmental change over geological time scales, and the complex macroevolutionary dynamics that have shaped their history. It also provides a critical methodological reflection on the tools used to reconstruct evolutionary history from integrated fossil and molecular data.

Chapter A

Appendix

A.1 Bayesian tip-dated timeline for diversification and major biogeographic events in Muroidea (Rodentia), the largest mammalian radiation

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Not only the phylogenetic relationships within extinct hamsters or the position and closest fossil relatives of extant hamsters are under debate, but also the associations of several other fossil muroid groups to fossil and extant hamsters. In this study the classifications, biogeographic patterns, and divergence times of several extinct and extant muroid taxa are investigated, which also gives phylogenetic context for the group of true hamsters.

RESEARCH ARTICLE

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Bayesian tip-dated timeline for diversification and major biogeographic events in Muroidea (Rodentia), the largest mammalian radiation

Raquel López-Antoñanzas^{1,2*} , Tiago R. Simões³, Fabien L. Condamine¹, Moritz Dirnberger¹ and Pablo Peláez-Campomanes^{2*}

Abstract

Background Extinct organisms provide vital information about the time of origination and biogeography of extant groups. The development of phylogenetic methods to study evolutionary processes through time has revolutionized the field of evolutionary biology and led to an unprecedented expansion of our knowledge of the tree of life. Recent developments applying Bayesian approaches, using fossil taxa as tips to be included alongside their living relatives, have revitalized the use of morphological data in evolutionary tree inferences. Eumuroidea rodents represent the largest group of mammals including more than a quarter of all extant mammals and have a rich fossil record spanning the last ~45 million years. Despite this wealth of data, our current understanding of the classification, major biogeographic patterns, and divergence times for this group comes from molecular phylogenies that use fossils only as a source of node calibrations. However, node calibrations impose several constraints on tree topology and must necessarily make a priori assumptions about the placement of fossil taxa without testing their placement in the tree.

Results We present the first morphological dataset with extensive fossil sampling for Muroidea. By applying Bayesian morphological clocks with tip dating and process-based biogeographic models, we provide a novel hypothesis for muroid relationships and revised divergence times for the clade that incorporates uncertainty in the placement of all fossil species. Even under strong violation of the clock model, we found strong congruence between results for divergence times, providing a robust timeline for muroid diversification. This new timeline was used for biogeographic analyses, which revealed a dynamic scenario mostly explained by dispersal events between and within the Palearctic and North African regions.

Conclusions Our results provide important insights into the evolution of Muroidea rodents and clarify the evolutionary pathways of their main lineages. We exploited the advantage of tip dating Bayesian approaches in morphology-based datasets and provided a classification of the largest superfamily of mammals resulting from robust phylogenetic inference, inferring the biogeographical history, diversification, and divergence times of its major lineages.

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Keywords Bayesian tip dating, Divergence dates, Historical palaeobiogeography, Morphological clock, Muroidea, Rodentia

Background

Establishing an evolutionary timescale is fundamental for tackling a great variety of topics in evolutionary biology, including the reconstruction of patterns of historical biogeography [1], coevolution, and diversification [2]. Without knowledge on the timing of evolutionary events, it is not possible to test hypotheses of causality that invoke extrinsic environmental mechanisms, such as the role of tectonics or climate change in the evolution of the clades [3]. Similarly, it is not possible to determine the rate of evolutionary change in response to such causal factors [4]. The tree of life is pruned by extinction and molecular data cannot be gathered for the vast majority of extinct lineages. Hence, the fossil record once provided the gold standard in attempts to establish evolutionary timescales, which was subsequently complemented by molecular clock approaches for groups with extant representatives [5, 6].

Besides providing fundamental calibration points for critical nodes in divergence time estimates using molecular clocks [3, 7], fossil taxa can now be analyzed either alone—i.e., morphological clocks [8]—or with molecular data in total evidence dating approaches [9, 10]. These advances over the past decade have made it possible to infer the placement of fossil tips without prior assumption of their location in the tree nor employing a priori constraints on tree topology, besides providing more accurate and precise estimate for lineage divergence times across nearly all branches of the tree of life—e.g., [11–16].

Despite the outstanding diversity of rodents among mammals, the richness of their fossil record and their importance to time scaling the molecular trees of living mammals, few attempts have been made to build phylogenies based on morphological data including both extinct and extant rodents [17–19]. Further, no study to date has provided a precise timeline for the evolution of its most speciose clade, Muroidea, which with more than 1695 species (Burgin et al., 2018), is the most diverse group of mammals representing more than a quarter of its total diversity. This group of rodents has a nearly continuous fossil record since the Eocene, which is extremely valuable for informing branch length and, thus, divergence time analyses. These rodents are ubiquitous being distributed in all the geography except for Antarctica. The study of phylogenetic relationships inside the muroids is one of the most complex problems that mammalogists have to tackle. Increasing interest

in this clade has led to some molecular phylogenetic analyses of living Muroidea using mitochondrial and nuclear genes, which have provided some insights into their evolutionary relationships [20–23]. So, huge efforts on molecular studies made over the past two decades, particularly those including broad taxa sampling [22–25], laid the foundation of a subdivision of muroid rodents in twenty subfamilies included in the following six primary monophyletic groups: Platacanthomyidae, Spalacidae, Calomyscidae, Nesomyidae, Cricetidae, and Muridae [23, 24]. According to recent molecular studies, Platacanthomyidae with only two genera and five living species is the most basal muroid clade. It diverges from the remaining muroid families in the middle Eocene (around 40–45 Ma) [23] whereas its most ancient fossil, the sole extinct genus, †*Neocometes* have been recorded from the early Miocene of South Asia and Europe [26, 27]. Spalacidae or mole rats include three extant subfamilies: the Asian Myospalacinae, the African and Asian Rhizomyinae (tribes Tachyoryctini and Rhizomyini), and the Spalacinae, which live in Eastern Europe, Western Asia, and the Middle East. They represent the second earliest diverging clade inside Muroidea, from which they split during the Oligocene [23]. These results are in line with paleontological data, according to which the oldest known crown spalacid, †*Prokanisamys kowalskii* (Rhizomyinae), would come from the Oligocene of Pakistan [28]. However, the first fossil recognized as a true spalacine, †*Pliospalax macovei*, comes from the Pliocene of Romania [29]. Calomyscidae, a poorly known monogeneric family, is distributed in southwestern Asia. Due to plesiomorphic morphological dental characters, this group of rodents has been considered by paleontologists as belonging to a stem group inside Gerbillinae, †*Myocricetodontinae* [30]. However, molecular results point out to a more basal position inside Muroidea, being the sister group of Nesomyidae–Muridae–Cricetidae, from which it diverges circa 20 Ma [23]. Its earliest representatives have been recorded from the Late Miocene of Europe and Turkey [31]. Nesomyidae is divided into six subfamilies Nesomyinae, Delanyomyinae, Mystromyinae, Petromyscinae, Cricetomyinae, and Dendromurinae, all but the Magalasy Nesomyinae live in sub-Saharan Africa and Zanzibar island. Due to their morphological diversity and dental convergence, different species inside this group were classified as belonging to Muridae or Cricetidae [32]. However,

molecular genetic work has evidenced the monophyly of this group [23, 33], which is the sister clade of the most derived muroids (Muridae plus Cricetidae), from which it diverges circa 19 Ma. The oldest known record of nesomyids represented by the extinct species †*Afrocricetodon songhori* and †*Notocricetodon petteri* [34], comes from the Early Miocene of Africa (Kenya, Uganda, and Namibia). These taxa have been placed inside the extinct subfamily †*Afrocricetodontinae*, which has been considered to be part of *Nesomyidae*.

The Muridae, Old World mice and rats, with more than 800 species [35] is the largest family of mammals. Major taxonomic issues concerning the classification of murid rodents resulted from the morphological dental disparity displayed by the teeth of their representatives, so their systematic has been largely discussed for a long time [32, 36–40]. Phylogenetic analysis using molecular data recovered Muridae as distributed in four monophyletic groups, the subfamilies Lophiomyinae, Deomyinae, Gerbillinae, and Murinae, which are generally accepted [22–24]. According to these studies, the most basal murid subfamily is Lophiomyinae followed by Murinae, which is a sister group of Deomyinae plus Gerbillinae. The estimation for the dates of the origin, diversification, and divergence time among and within these subfamilies is still controversial, and molecular results disagree sometimes with paleontological evidences.

The second-most speciose muroid family, with more than 700 living species, corresponds to that of Cricetidae, which brings together New World rats and mice (Sigmodontinae, Neotominae, and Tylomyinae) and Old World hamsters and voles (Cricetinae, Arvicolinae). Molecular studies agree in recovering the sister group relationship of Cricetidae and Muridae but there are still differences in the estimated ages, in which they would have split (e.g., Early Miocene, [23]) versus Late Oligocene [24]. According to molecular results, the diversification of cricetid rodents took place in Late Oligocene times and predates that of murids [24], whereas other authors pointed out that the diversification of cricetids started after that of murids approximately 15 Ma [23]. The issue of estimating divergence ages with accuracy worsens by the inclusion of fossil forms, as there is no consensus regarding the definition of which fossil forms should be included in the family Cricetidae. In fact, it has been a common practice among paleontologists to include inside Cricetidae all fossil forms having a “cricetid” dental pattern and, therefore, to integrate in this group species recorded from the middle Eocene on. However, other researchers consider cricetids exclusively as the most derived forms recorded from the lower Miocene onwards, which probably are more directly related to present-day representatives of this family, an opinion that we share.

All in all, molecular studies dealing on rodents in general and on Muroidea in particular have used fossil taxa merely to indirectly calibrate the divergences (node-dating) between living lineages instead of using the rich fossil record of the group for tip-dated evidence time estimates. Node calibrations are based on the oldest fossil record of a clade, and therefore, they require a prior phylogenetic hypothesis that usually does not exist. Therefore, several different ages have been estimated for the origin and divergence among and within this important group of rodents (e.g., [21, 22, 24, 41]) but no consensus has been reached. Moreover, we still ignore the answer to many important questions concerning the evolutionary origins of the muroid. In fact, we remain ignorant of the phylogenetic position of the fossil taxa with respect to their present-day relatives, which directly impact any node-based fossil calibrations using only molecular data with many deep nodes having poor support. The main reason not to use the fossil species to be included alongside their living relatives in a tip calibration is the lack of morphological data. In fact, the huge number of muroid taxa, their high molar diversity, and morphological convergence of their dental characters have prevented paleontologists and neontologists to build comprehensive phylogenies based on discrete morphological traits including both extinct and extant representatives. It has to be taken in mind that the classical definition of the relationships among fossil rodents is mainly based on dental characters because the teeth are nearly the sole remains preserved in ancient rodents.

Here, we provide the first detailed timeline of muroid evolution and revision of the placement of several of its fossil members by using tip-dated relaxed clock Bayesian inference. Moreover, we use this new timetree to infer their biogeographical history and elucidate the origin and speciation events of the major branches of the muroid tree of life by applying a maximum likelihood inference of geographic range evolution taking into account non-ultrametric dated trees.

Results and discussion

The analyses we present below include the most extensive paleontological database of muroid rodents and constitute the first attempt to build a phylogeny of Muroidea that includes fossils and extant as tips.

Phylogenetics: relationships among major clades

The topology of the tree shows two large sister clades. The backbone tree that summarizes the subfamily relationship is shown in Fig. 1. Within the first major clade, Cricetidae is the earliest diverging clade. This latter is the sister group of a larger clade that splits into two clades, the Megacricetodontinae on one hand and the

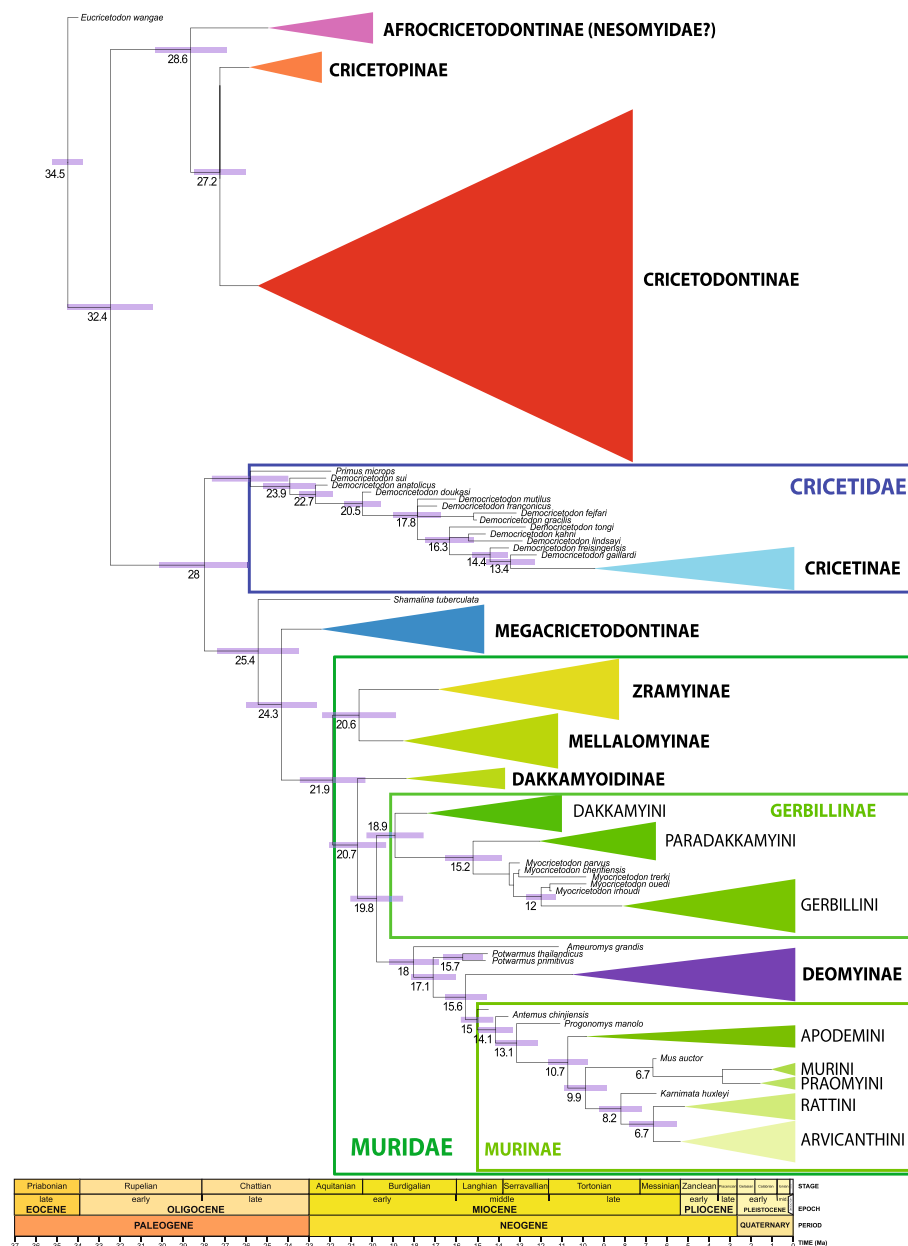


Fig. 1 Summarized phylogeny of Muroidea. Maximum compatible consensus tree summarizing the relationships among subfamilies and tribes inside muroid rodents. Cones represent the sampled diversity of each subfamily and tribe and the depth of each cone corresponds to their most recent common ancestor

Muridae on the other hand. The second clade includes two clades that are sister clades: +Cricetopinae (sensu Maridet and Ni [42]) and +Cricetodontinae. These results evidence that the family Cricetidae, as traditionally conceptualized by paleontologists, that is, considering all cricetid-like rodents as members of this family [42, 43], is paraphyletic and that +Megacricetodontinae and +Cricetodontinae are not part of it and should be excluded. All synapomorphies shared by these clades are listed in Additional File 1: Fig. S1.

Cricetidae

The topology of our tree evidences that democricetodontines are, as suspected by some authors (e.g., [43, 44], closely related to extant hamsters).

According to our results (Figs. 1 and 2), this subfamily originated ~25.8 Ma in China (Figs. 2 and 4). +*Primus microps* appears as the basalmost taxon followed by the earliest deriving +*Democricetodon*, which corresponds to +*Democricetodon sui* from the Early Miocene of the Junggar Basin, China (21.9–21.16 Ma), and represents the oldest

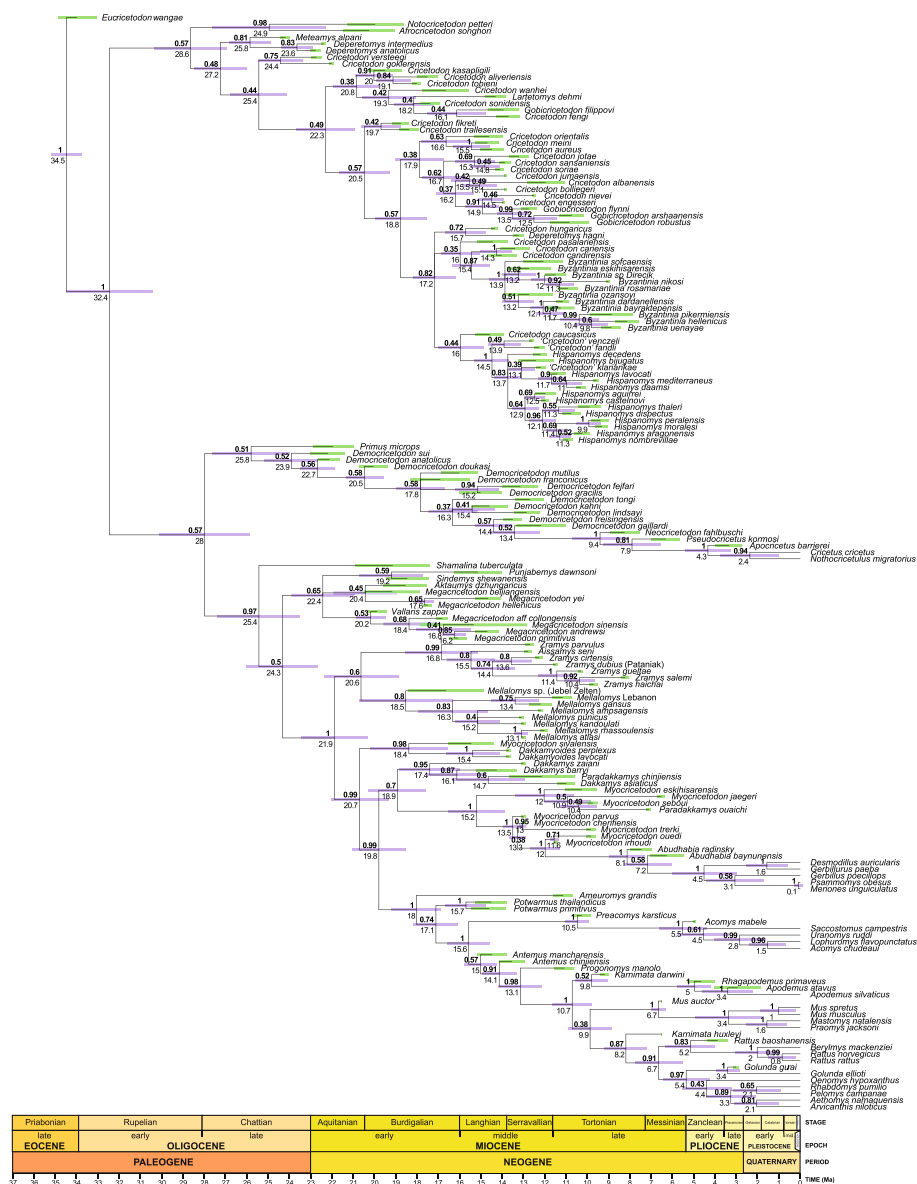


Fig. 2 Relaxed-clock Bayesian inference analysis of the morphological dataset with tip dating using the fossilized birth–death tree model. Maximum compatible consensus tree. Numbers above nodes indicate posterior probabilities, numbers below nodes indicate median estimates for the divergence times, and node bars indicate the 95% highest posterior density for divergence times for internal nodes (magenta) and tips (green)

record of †*Democricetodon* in Asia [45]. Subsequently deriving taxa include the Turkish †*D. anatolicus* and †*D. doukasi*, followed by the remaining species of †*Democricetodon*. The subsequent divergence of †*D. franconicus*, †*D. mutilus*, and then †*D. gracilis* and †*D. feffari* (as sister species) reveals the first entrance of this subfamily of rodents in Europe circa 17.8 Ma, followed by a migration from Europe toward South and West Central Asia ~15.2 Ma (Fig. 2). The origin of extant cricetines, which are here represented by *Cricetus cricetus* and *Nothocricetulus migratorius*, is to be found

in some European species close to †*Democricetodon*: †*D. freisingensis* and †*D. gaillardi*. The basal divergence in the crown clade would have occurred around 13.4 Ma (Fig. 1). This latter clade shares some non-exclusive synapomorphies such as the lack of lingual anteroloph (12 (0→1)) on the M1 and central atoll on the M3 (52 (0→2)), the presence of the posterior spur of the anterocone (anterior ectoloph, (14 (0→1)) on the M1 or a medium length mesoloph ((21 (0→1), 38 (0→1)) on the M1 and M2 (Additional File 1: Fig. S1).

†Megacricetodontinae

We find †Megacricetodontinae as the sister clade to Muridae (Fig. 1). This fact evidences that the former subfamily is closer to murid than to cricetid ancestry, which led this clade outside from the Cricetidae. This result agrees with the previous hypotheses by Flynn [44], in which †Megacricetodontinae is the outgroup to (Murinae, (Gerbillinae, Deomyinae)) and that it could be seen as an earliest deriving Muridae. According to the topology of our tree, †*Shamalina* is the most recent common ancestor of †Megacricetodontinae + Muridae, from which they diverge approximately 24.3 Ma (Fig. 2). Lindsay [46] suggested that †*Shamalina* (Early Miocene, Saudi Arabia) could have given rise to †*Megacricetodon* whereas Wessels [47] included this taxon within the †Myocricetodontinae. Within †Megacricetodontinae, there are three main clades, whose evolutionary relationships cannot be fully resolved. The first one consists of the sister species, †*Punjabemys downsi* and †*Sindemys shewanensis*, from the Early Miocene of Pakistan [47]. Both were considered by Wessels [47] as belonging to †Myocricetodontinae. However, according to our results (Fig. 2), they belong to †megacricetodontines, supporting the ideas of Lindsay [48]. The second clade is composed of a succession of Asian plesiomorphic species that led to †*Megacricetodon hellenicus* from the Early Miocene of Aliveri, Greece [49]. This lineage evidences the first entrance of †*Megacricetodon* into the Aegean–Anatolian region which is set up at ~20.4 Ma. The third clade is constituted by a succession of lineages, the most basal of which is †*Vallaris zappai* from the Early Miocene of Turkey, which splits into two lineages supporting two independent migration events. The first one led to †*Megacricetodon* aff. *col-longensis* (~17.2 Ma), the earliest deriving species of the typical central European †“*Megacricetodon bavaricus* group” [50] and the second one includes the split between †*M. andrewsi* and †*M. primitivus* at ~16 Ma. Interestingly, Oliver and Peláez-Campomanes [51] inferred at least three migration events for early †*Megacricetodon* as the basis for key morphological differences they found between this taxon.

Muridae

Potential candidates have been proposed for the origin of the different subfamilies and tribes inside Muridae. For instance, †*Abudhabia* has been suggested to be good candidate for being the ancestor of Gerbillinae [52], †*Antemus* for Deomyinae [53] or Murinae [52, 54], †*Karnimata* for the tribes Rattini [54] or Arvicanthini–Millardini–Otomyini [52, 55, 56], †*Progonomys* for the tribu Murini [54, 56] and for the tribes Praomyini–Murini–Apodemini [52, 55, 57], and so on.

The topology of our tree as seen above shows that Muridae is the sister group to the †Megacricetodontinae, from which it diverged at ~25.4 Ma (Figs. 1–2). Most of their oldest representatives belong to whole extinct lineages that, according to our results, would warrant subfamily status. So Muridae is divided into two sister main clades that diverge approximately 21.9 Ma. The first of them includes the sister subfamilies †Zramyinae and †Mellalomyinae whereas the second one, which is more speciose, shows six main lineages, the subfamilies †Dakkamyoidinae, Gerbillinae, Deomyinae, and Murinae (Fig. 1). The diversification of †Dakkamyoidinae and Gerbillinae occurred within very short time intervals from each other, <2.5 Ma (Fig. 2) whereas that of Murinae and Deomyinae started much later (15.0 Ma and 10.5 Ma, respectively) (Fig. 1). All these clades show the posterior part of the m3 reduced phylogenetic and they share the synapomorphies of having the entoconid–hypocoid reduced (83(0→1)), extremely reduced (83(1→2)), included in a single cusp (83(1→3)) or even absent (83(2→4)) and the posterosinusid small (85(0→1)) or absent (85(1→2)) (Additional File 1: Fig. S1).

†Zramyinae

The earliest deriving split at ~21.9 Ma within the Muridae separates the sister subfamilies †Zramyinae and †Mellalomyinae from the remaining subfamilies of this diversified group (Figs. 1–2). The systematic placement of †*Zramys* has been for a long time questioned. This genus was first considered as belonging to the Cricetidae [58], then suspected to be a †Myocricetodontinae [59] before being formally reallocated to this group [47]. Nevertheless, the phylogenetic relationships of the taxa belonging to this genus have never been formerly analyzed. The genus †*Aissamys* was coined in an unpublished thesis [60]. It is therefore not available (*nomen ex dissertatione*). The type and only species of the genus, †*Aissamys seni* from Chouf Aïssa, is here considered as †*Zramys* sp. The topology of the tree shows that the polytypic African genus †*Zramys* forms a monophyletic group that belongs to the new subfamily †Zramyinae, whose origin can be inferred at approximately 16.8 Ma (Fig. 2). The †Zramyinae (node 186) share the exclusive synapomorphy of having lingual spur of the anterocone (11→01) on the m1. Additional non-exclusive synapomorphies are the presence of posterior spur of the anterocone (14→01) on the M1, having an elongated M2 (34→01) with a posterior metaloph (39→02) and the presence of an anterior spur on the entoconid on the m1 (92→01) and m2 (93→01) (the latter two, being only shared by †*Notocricetodon petteri*). Moreover, the most derived †Zramyinae share the exclusive synapomorphy of having a transversal protocone spur (23→01) (Additional File 1: Fig. S1).

†Mellalomyinae

The extinct polytypic genus †*Mellalomys* forms a monophyletic group, which splits from †Zramyinae ~20.06 Ma (Figs. 1–2). Its earliest deriving taxon is †*Mellalomys* sp. from the Early Miocene of Jebel Zelten, Libya [61]. One node up this lineage splits to give rise to two clades (Fig. 2). The first one includes all African species of the genus whereas the other one comprises the taxa recorded out of Africa and evidences a migration event from Africa towards the Arabian Peninsula and China ~16.4 Ma.

†Dakkamyoidinae

The †Dakkamyoidinae is an extinct endemic subfamily from Pakistan that splits from more derived murids ~20.7 Ma (Figs. 1–2). Its earliest deriving taxon is †*Myocricetodon sivalensis* from the Middle Miocene of Pakistan [48, 62]. The inclusion of this species into the genus †*Myocricetodon* has been a matter of debate [63, 64]. According to the topology of the tree, †*Myocricetodon sivalensis* should be reallocated into the genus †*Dakkamyoides*. This species is sister species to the clade constituted by the two species of †*Dakkamyoides* (†*D. lavocati*, †*D. perplexus*).

Gerbillinae

According to our results, Gerbillinae diverged from Murinae+Deomyinae at ~19.8 Ma (Figs. 1–2). These three clades share the non-exclusive synapomorphy of lacking the lingual anteroloph on the M2 (31:02) (Additional File 1: Fig. S1). The earliest deriving clade inside the Gerbillinae corresponds to an extinct lineage that we call here tribe †Dakkamyini (Fig. 1). Its earliest most deriving taxon is †*Dakkamys zaiani* from the Middle Miocene of Beni Mellal, Morocco [59]. †*Dakkamys zaiani* is the sister species to the clade composed of the remaining known species of †*Dakkamys* and †*Paradakkamys chinjiensis*, which come from the Middle Miocene of Pakistan (Fig. 2). Our analysis suggests that the origin of this tribe predate Middle Miocene times. According to our results (Fig. 2), †*Paradakkamys chinjiensis* should be reallocated into the genus †*Dakkamys*. The †Dakkamyini is the sister clade of a speciose group from which they diverged approximately 18.9 Ma. This latter is constituted by two sister clades. The first of them that we call here †Paradakkamyini, includes all species of †*Myocricetodon* that have developed an enterostyle on the M1 and M2 and sometimes an ectostylid on the m1. Based on these characteristics, Lindsay [48] excluded these taxa from the genus †*Myocricetodon* and included them together with the genus †*Dakkamys* into the Dendromurinae. However, according to our results, all these species should be reallocated into the genus †*Paradakkamys* in the tribe †Paradakkamyini inside Gerbillinae. Except for its earliest

deriving taxon †“*Myocricetodon*” *eskihisarensis*, which comes from the Middle Miocene of Turkey, the remaining species within this clade are African (†“*M.*” *jaegeri*, †“*M.*” *ouaichi*, †“*M.*” *seboui*). The second group includes the †*Myocricetodon* species that are characterized by lacking enterostyle on the M1 and M2 and more derived gerbillines. Its earliest deriving taxon is †*Myocricetodon parvus* from Beni-Mellal, Morocco. This species is, in turn, sister to two clades. The first one is represented by †*M. trerki* and †*M. cherifiensis* whereas the second one is a larger clade that includes on one hand †*M. irhoudi* and †*M. ouedi* and on the other hand the clade †*Abudhabia radinski* plus more derived gerbils. This latter group includes the living gerbils and it is supposed to be close to the ancestry of all extant members of Gerbillinae. Our results (Figs. 1 and 2) agree with those of Jaeger [59], who suggested the polyphyly of †*Myocricetodon*. As seen above, one of the former lineages considered to be within †*Myocricetodon* is excluded from the genus †*Myocricetodon* and reallocated into the genus †*Paradakkamys*. Our results agree with those pointed out by [64], who considered all species belonging to the genus †*Paradakkamys* and †*Dakkamys* as more closely related to gerbils than to murines. Moreover, the topology of our tree evidence that both †*Dakkamys* and †*Paradakkamys* belong to Gerbillinae. The most derived Gerbillinae (†*Abudhabia radinski* plus more derived gerbils) have sister relationships with †*Myocricetodon irhoudi* as suggested previously by some authors [59, 65, 66]. The origin of the Gerbillinae crown group is inferred at ~4.5 Ma ((median age; 95% highest posterior density [HPD]: 3.08–6.1 Ma). Inside this group, our results show a basal split between (*Desmodillus*, *Gerbillurus*) on one hand and (*Gerbillus* (*Psammomys*, *Meriones*)) on the other hand. These results are in agreement with molecular studies, in which *Desmodillus* and *Gerbillurus* belong to the tribe Taterini (Taterini II sensu Alhajeri et al. [67]), which is in turn the sister group of a clade that includes the tribe Gerbillini (Gerbillini I sensu Alhajeri et al. [67]) with *Psammomys* and *Meriones* being closer to one another than either is to *Gerbillus*. According to our morphological clock, the split between Taterina and Gerbillina (here considered as subtribes) took place between the end of the Miocene and the beginning of the early Pliocene. These results are in line with the fossil record, from which we know that the oldest fossils attributed to modern gerbil genera date from the Early Pliocene (5.1 Ma) of Langebaanweg, South Africa [68].

The species belonging to the tribes †Paradakkamyini and Gerbillini share the synapomorphy of having an extreme degree of reduction on the m3 (76:34) with the exception of †*M. parvus*, in which the m3 is reduced but to a lesser extent (Additional File 1: Fig. S1) as it is in the

representatives of the tribe †Dakkamini. This reduction in the size of the third molars comes together with the reduction on the number of roots from two (on the m3) and three (on the M3) to one (characters 86(0→2) and 89(0→3)). The species included in the clade (†*Myocrice-todon parvus* + more derived species) share the synapomorphy of lacking the entoconid on the m3 (84(1→2)) (Additional File 1: Fig. S1). This synapomorphy is also present in all Deomyinae but the earliest deriving most one (†*Preacomys karsticus*).

Murinae and Deomyinae

Our results agree with molecular studies in recovering the monophyly of Gerbillinae, Murinae, and Deomyinae as well as in evidencing a close relationship between these clades (Fig. 1). However, the phylogenetic position of these clades relative to one another differs from what has been recovered using molecular analyses. Molecular results indicate a sister-group relationship between Murinae and the clade Deomyinae + Gerbillinae [21–23, 69]. However, the topology of our tree shows Gerbillinae as the sister clade to Murinae + Deomyinae. This is a matter of great interest because Deomyinae have been solidly placed inside murines by paleontologists on the basis of the morphology of their molars. Despite numerous attempts to morphologically demonstrate that *Acomys*, a representative of Deomyinae, did not belong to Murinae, the difficulty remained. As Jacobs and Flynn [70] pointed out, *Acomys* if not a murine, represents a very rare case of a nearly perfect convergence in complex dental features among rodents. Our results based on dental morphological data allow separating Deomyinae from Murinae but they place these subfamilies closer to one another than either is to the Gerbillinae.

According to our results (Figs. 1–2), Gerbillinae represents the sister clade of a large group, the earliest deriving taxon of which is †*Ameuromys grandis* from the Late Miocene of Egypt [71]. This species shares with some species of †*Zramys* and †*Mellalomys* the non-exclusive synapomorphy of having three roots, the anterior one of which is divided on the m2 (78(0→1)) (Additional File 1: Fig. S1). †*Ameuromys grandis* splits into two lineages. The smaller one includes the species belonging to †*Potwarmus*, which is the sister group of murines + deomyines, from which it diverges approximately 17.1 Ma. Therefore, our results provide evidence that †*Potwarmus*, which was once considered a stem murine [56, 64] similar to †*Antemus* [72], does not belong to Murinae but to an early dead-end offshoot of the clade leading to Murinae and Deomyinae.

Murinae and Deomyinae share the exclusive synapomorphy of having on the M1 an anterocone divided with its lingual part larger than the labial one (6(2→3)). These

two clades together with *Potwarmus* share the exclusive synapomorphy of having the main cusps of the M1 forming rows (4(1→2)) (Additional File 1: Fig. S1).

Deomyinae is a subfamily of rodents mainly endemic from Africa (except for the genus *Acomys*, which is also found in the Middle East and Crete). The origin of this clade is set up approximately 10.5 Ma. Its earliest deriving member is †*Preacomys karsticus* from the early Late Miocene (circa 10 Ma) of Harasib, Namibia [73], which is the sister species of the remaining taxa of this subfamily. Deomyinae share the exclusive synapomorphy of having the hypocone of the M3 lingual to the protocone (58(3→4)) (Additional File 1: Fig. S1).

Interestingly, Jacobs and Flynn [70] pointed out that if *Acomys* is not a Murinae, it should have diverged from the latter prior to †*Progonomys* or †*Antemus*. Our results are in line with the thoughts of these authors and evidence a split between Deomyinae and Murinae prior to †*Progonomys* and †*Antemus* (see below).

†*Antemus* is known by two Pakistani species, †*Antemus mancharensis*, the oldest one from the late Early Miocene (ca. 15 Ma, [47]), and †*Antemus chinjiensis*, the youngest, (13.8–13.05 Ma [57]). The topology of our tree suggests that the most basal stem murine is †*Antemus mancharensis*, which is not unexpected given the absence of a real t1 in some M1 of this species, which show a low ridge instead ([47] p. 218). †*Antemus chinjiensis* and †*Progonomys manolo* insert sequentially on the stem. Both taxa show a true t1 on all their M1 (novelty already fixed), the development of which already started in †*A. mancharensis* (without reaching fixation in this taxon). The presence of a t1 may have taken place at ~15.02 Ma, age at which we establish the origin of Murinae (Figs. 1 and 2). Our results show agreement with the fossil record, with the oldest murine rodent dating back approximately 15 Ma. According to the topology of our tree, †*Progonomys manolo* leads to two main clades, the first of which with †*Karnimata darwini* as the most basal taxon, leads to the tribe Apodemini, from which it splits at approximately 9.8 Ma. The second clade splits into two clades, the first of which shows as basal †*Mus auctor* and leads to the tribes Praomyini and Murini as sister groups whereas the second one has as basal most taxon †*Karnimata huxleyi* that give rise to the tribes Rattini and Arvicanthini. Interestingly, despite this work focuses on exploring deeper nodes inside Muridae, even a small representation of murines allows recovering the monophyly of this subfamily inside murids. In this way, Murinae shares the synapomorphies of having a t1 but no lingual branch of the anterior cingulum on the M2 (32(1→3)), the presence of the anterolabial cuspid (A1) (96(0→3)) and labial cuspid (C1 or/and C3) on the m2 (102(0→2)). Crown Murinae, which are here represented by all murines but †*Antemus*

and *†Progonomys*, are characterized by the presence of a t1 together with the loss of the lingual branch of the anterior cingulum on the M2 (32(3→4)), by having t3 (33(2→3)) on the M2, a large t1 on the M3 (44(2→4)) and the entoconid and hypoconid merged in a single cusp on the m3 (83:1→3). Our findings also support the monophyly of the tribes analyzed here: Arvicanthini, Rattini, Murini, Praomyini, and Apodemini (synapomorphies are shown in Additional File 1: Fig. S1). Moreover, our analysis identifies a closer relationship between Apodemini and the sister tribes Murini and Praomyini than either is to Arvicanthini, which is in agreement with molecular studies [23, 74] and paleontological data [56]. The basal position of *Golunda* inside the Arvicanthini, followed by the divergence of *Oenomys* has also been recovered in molecular studies [23, 74]. However, we recognize two clearly separate species inside *†Karnimata* that would give rise independently to the tribes Apodemini (*†K. darwini*) on one hand and Arvicanthini and Rattini (*†K. huxleyi*) on the other hand, whereas *†Progonomys* would be a stem murine. Nevertheless, the presence of a t1 together with the already loss of the lingual branch of the anterior cingulum on the M1 exhibited by *†Progonomys* (13(3→4)), a synapomorphy shared by crown murines, leave still room for doubt concerning the exclusion of this genus from the crown group. The origin of Murini and Praomyini is, according to our results, related to another important fossil taxon *†Mus auctor*. According to Kimura et al. [57] and Flynn et al. [56], the origin of Murini and Arvicanthini would be found in *†Progonomys* and *†Karnimata*, respectively.

The inclusion here of a small sample of extinct and extant murine representatives does not allow to fully understand the tempo and mode of evolution of murines, which merits a study on its own. In fact, hundreds of fossils and living species of murine rodents need still to be added in a more comprehensive phylogenetic analysis (in preparation) to finally depict their evolutionary history of what is the most specious subfamily of mammals.

†Cricetodontinae

Our results indicate that Cricetodontinae are not closely related to Cricetidae, Megacricetodontinae, and Muridae (Figs. 1 and 2), so the extinct subfamily Cricetodontinae should be excluded from all these groups. The subfamily Cricetodontinae diverged from the primitive Cricetopinae at approximately 27.24 Ma (Figs. 1 and 2). The earliest deriving clade within the Cricetodontinae (Fig. 2) is represented by two sister species *†Cricetodon versteegi* and *†C. goklerensis* from the Early Miocene of Turkey [75, 76]. This clade is the sister clade to a larger group that includes all remaining species of the subfamily. Its earliest deriving clade has two main branches. The first one

includes a sequence of species of *†Cricetodon* from China giving rise to (*†C. wanhei* (*†C. sonidensis*, *†Mixocricetodon dehmi*, (*†C. fengi* and *†Gobiocricetodon filippovi*))) whereas the second branch includes the *†C. kasapligili* as giving rise to *†C. tobieni* and *†C. aliveriensis* as sister species. The split of these two early deriving clades and the clade comprising more derived species of *†Cricetodon* and the species of *†Byzantinia* and *†Hispanomys* is set up at 22.29 Ma. Inside this clade, the earliest deriving lineage is represented by the Turkish species *†C. fikreti* and *†C. trallesensis* that diverged from the main clade at approximately 20.48 Ma. This latter clade splits into two main branches. The first one comprises (*†C. orientalis* (*C. aureus*, *C. meini*)) as sister species of most of the Middle Miocene European species of *†Cricetodon*. The second main lineage comprises three lineages. The first one is represented by the Middle Miocene European *†C. hungaricus* and *†Deperetomys hagni*. The second one leads to *†Hispanomys* and includes the late Middle Miocene *†Cricetodon* from Central Europe *†Cricetodon caucasicus*, *†C. fandli*, *†C. klariankae*, and *†C. vENZECLI*, which should be considered as *†Hispanomys*. The third one, from which *†Byzantinia* originates, is unresolved, and contains the Middle Miocene *†Cricetodon pasalarensis*, *†Cricetodon cariensis*, and *†C. candirensis*, which would be early representatives of *†Byzantinia*. The clade that includes the most derived species of *†Cricetodon* that lead to all species of *†Hispanomys* and *†Byzantinia* share the synapomorphy (78(0→2)) of having three roots with the posterior one divided on the m2 (except for *†B. sofcaensis*, which is two rooted and the most derived *†B. uenayae*, which is four rooted, 78(2→3)) (Additional File 1: Fig. S1). Some additional synapomorphies consist in the presence of a long longitudinal or oblique backward paracone spur (15(1→2, 3)) and an anterior spur of the metacone (16(0→1)) on the M1 or having an elongated M2 (34(0→1)) (Additional File 1: Fig. S1).

†Cricetopinae Deperetomys/Meteamys clade

On the basis of a cladistics analysis, [42] coined the new subfamily *†Cricetopinae* to include the species *†Paracricetops*, *†Cricetops*, *†Deperetomys* (Early Miocene forms), *†Meteamys*, *†Selenomys*, *†Melissiodon*, *†Mirrabella*, *†Enginia*, *†Muhsinia*, and *†Aralocricetodon*. Two rodents from this analysis (*†Deperetomys* and *†Meteamys*) have been included in ours. The origin of *†Deperetomys* and *†Meteamys* has been widely discussed [43, 45, 76, 77]. Some authors considered that *†Deperetomys* and *†Meteamys* together with *†Cricetodon* and *†Eumyarion* probably derived from the early stock of cricetodontines [76]. Other authors considered them as belonging to a new subfamily, *†Eumyarioninae* [77]. Our results agree with those of [42] in finding *†Early Deperetomys* and

†*Meteamys* as members of a subfamily outside of Cricetodontinae, which they called †Cricetopinae. Interestingly, our results indicate that †*Deperetomys hagni* belongs, in fact, to the genus †*Cricetodon*. The similarities between †*Deperetomys hagni* and some species of †*Cricetodon* have been noticed since the discovery of that species. Fahlbusch [78] named this species as †*Cricetodon sansaniensis hagni* and Mein and Freudenthal [43] coined †*Deperetomys* as a subgenus inside †*Cricetodon* based on the type species †*Cricetodon (Deperetomys) hagni*. In the same line of reasoning, the species †*Cricetodon hungaricus* was named as a subspecies of †*Deperetomys hagni*. Therefore, it is not surprisingly that our results evidence that †*Deperetomys hagni* belongs in fact to †*Cricetodon* and should be reallocated to this genus. As †*Deperetomys hagni* is the type species of the genus, it is plausible that after carrying out a comprehensive analysis of the species belonging to these ancient lineages is necessary to coin a new genus to reallocate all species of †*Deperetomys* but †*D. hagni*.

†*Notocricetodon pelteri* + †*Afrocrisetodon songhori*

†*Notocricetodon* and †*Afrocrisetodon* are the first fossils belonging to the Muroidea that have been recorded in Africa. They date from the early to middle Miocene and have been included in the fossil subfamily †Afrocrisetodontinae by Lavocat [34], who already suspected that they could belong to Nesomyidae. Over the history, various tentative classifications of Nesomyidae, which were considered with different taxonomic rank (family or subfamily, depending on the authors), have been carried out ([79] and references therein) but the high level of dental convergence they show has made very difficult to infer their phylogenetic relationships [80]. Finally, molecular studies have put in evidence that Nesomyinae together with other African groups (Delanymyinae, Mystromyinae, Petromyscinae, Cricetomyinae, Dendromurinae) constitutes the family Nesomyidae, which is basal to the Muridae [23]. However, their relationship with some extinct taxa such as †*Democricetodon* or †*Myocricetodon* is still a matter of discussion ([81] and references therein).

Our results support the monophyly of the sister taxa †*Afrocrisetodon* and †*Neocricetodon* and its basal position with regard to †Cricetodontinae, Cricetidae, and Muridae, which is in agreement with molecular studies [23]. These two taxa are an archaic African lineage that split from †Cricetopinae and †Cricetodontinae approximately 28.6 Ma.

The relationship between these fossil species with extant genera belonging to Nesomyidae are still unknown. In fact, due to the dental convergence of the Nesomyidae, *Saccostomus* branches in our analysis inside the clade Deomyinae. So, many more taxa belonging to extinct and extant members of Nesomyidae as well as additional morphological

characters that complement the dental ones, particularly in that concerning the skull and muscles, need to be added in future phylogenetic analyses to finally depict the relationships between extinct and extant members of this African family of rodents.

Divergence times

Fossil data are fundamental to molecular clock methodology, providing the key means of clock calibration. In order to use fossils as calibration points for molecular clock, we need to know the phylogenetic position of these fossils into a tree and to set up the time, in which the clades to which these fossils belong, originated, and diverged. So, one of the most difficult challenges facing any scientist working in phylogenetic inferences using molecular dataset is the phylogenetic position of the fossils in the tree. In fact, early fossils may be representatives of stem lineages rather than crown groups [23] or belong to a different group, for which they have been used as calibration points. In all these cases, the calibrations will be wrong.

Our analyses herein infer with strong support the placement of several fossil groups never previously tested using Bayesian tip dating approaches. Importantly, we find a strong convergence of divergence time estimates for nearly all nodes within Eumuroidea using multiple relaxed clock models—even when using models (i.e., WN) strongly rejected by our Bayes factors comparisons (Fig. 3, Table 1). This demonstrates that our inferred divergence times are robust to model violation and provide a precise (narrow highest posterior density intervals) timeline of muroid evolution.

We find the origin of the most recent common ancestor (MRCA) of Eumuroidea (sensu [22]) at ~32.45 Ma (median age; 95% highest posterior density (HPD): 30.38–34.45 Ma). This implies that the origin of this large clade (inclusive of all muroids except for Platanthomyidae and Spalacidae) predates the age inferred by molecular clock studies using only node calibrations by at least 7 Ma Schenk et al. [21] and by more than 12 Ma as estimated by Steppan and Schenk [23] and inferred 22% and 37.5% underestimate for the total age of Eumuroidea, respectively.

The topology of our summary tree suggests that neither †Cricetopinae nor †Cricetodontinae are a part of Cricetidae. The split between †Cricetopinae and †Cricetodontinae took place at ~27.2 Ma (median age; 95% highest posterior density (HPD): 26.1, 28.5 Ma). The origin of the most recent common ancestor (MRCA) of †Cricetodontinae is at ~25.43 Ma (median age; 95% highest posterior density (HPD): 24.1, 26.8 Ma) and that of the †Cricetopinae is ~25.8 Ma. The divergence between Cricetidae and



Fig. 3 Density tree contrasting divergence times under distinct clock models. Each tree represents the maximum compatibility tree and median divergence times obtained from each model. Gray bars at nodes indicate range between maximum and minimum divergence times for each model and node values represent the midpoint between these respective maximum and minimum divergence times. IGR independent gamma rates uncorrelated clock, ILN independent lognormal relaxed clock, TK02 continuous autocorrelated clock, WN white noise uncorrelated clock

Table 1 Results of model fitting analyses using the stepping-stone algorithm for Bayes factors. *BF* Bayes factor relative to best-fit model, *mLnL* marginal natural log-likelihood

Clock model	mLnL	BF
IGR	−7329.76	190.34
ILN	−7330.05	190.92
WN	−7396.93	324.68
TK02	−7234.59	0

the clade Muridae+†Megacricetodontinae is at ~28Ma and the origin of the MRCA of Cricetidae, around 25.8 Ma. Hedges and Kumar [82], based on the supertree they built, estimated a much older age (40.2 Ma) for the origin of Cricetidae whereas [21] and [23] a much younger one (17.7 Ma and 14.6 Ma, respectively). According to these latter authors, Cricetidae would have originated after the Muridae. However, our results agree with the fossil record that evidences without any doubt that the origin of cricetids took place much earlier than that of murids.

The inclusion in our analysis of murid fossils now recognized as stem Muridae (†Zramyinae, †Mellalomyinae, †Dakkamyoidinae) evidence the existence of Muridae at least since 21.89 Ma. So, our estimate predates the age inferred by molecular clock studies given by Aghová et al. [52] (20.1 Ma), [21] (20.7 Ma), and [23] (17.4 Ma).

We find the origin of the MRCA of murines at ~15 Ma, one million years older to that obtained for the origin of this group by Schenk et al. [21] (14.2 Ma) and Rowe et al. [83] (14 Ma) but somewhat younger than the age estimated by Aghová et al. [52] (15.9 Ma). The divergence age between †*Progonomys* and the crown group of murines is here set up in circa 13.1 Ma and the origin of the MRCA of the crown murines that leads to all known extant species of murines in 10.7 Ma. The oldest fossil occurrences of †*Antemus* (†*A. mancharensis*), †*Progonomys* (†*P. morganae*), and †*Karnimata* (†*K. feffari*) fall within the time ranges we have estimated by our divergence dates. If we consider †*K. darwini* as the most basal representative of Apodemini, this tribe would have originated around 9.8 Ma, which is consistent with molecular studies [74]. However, the origin of the crown group of this tribe is considerably younger (4.4–5.9 Ma). Our estimates for the timing of the origin of Praomyini and Murini lineages is 1.9–5.1 Ma and that for the origin of the tribes Rattini and Arvicanthini 5.5–7.8 Ma. Our presented divergence time estimates for extant tribes are younger than that inferred by other studies [74, 84] who inferred for instance an approximate age of 8 Ma and 8.4 Ma for the origin of Arvicanthini and circa 8 Ma and 7.6 Ma for the Praomyini. Despite the fact that fossil representatives of modern genera among murines are not reported with

certainty before 5–7 Mya [53] and therefore their evolutionary time frame should not be much older than that, we remain cautious concerning the origin and divergence ages of the most derived clades inside murines until the completion of a more comprehensive analysis.

According to our results, Deomyinae originated at ~10.47 Ma, approximately 2 Myrs before the age estimated by Schenk et al. [21] and 2 and 3 Myrs after that estimated by Steppan and Schenk [23] and Aghová et al. [52], respectively. The earliest fossil Deomyinae showing a modern deomyine dental pattern (†*Preacomys*) has been recovered from Ethiopia and Namibia around 11–9.5 Ma [73, 85].

Gerbillinae is here considered to be the clade (†Dakkamyini (†Paradakkamyini+†Gerbillini)). The origin of its MRCA is at ~18.92 Ma and that of its crown group (†*Abudhabia radinski*+more derived gerbils) at ~8.13 Ma. Previous molecular clock hypotheses have placed the origin of the Gerbillinae either earlier, around 23.7–23.6 Ma ([86] p. SA1, 56) or later, approximately, 10.2 Ma [23]. One of the possible reasons for such disparity could be the inherent problems due to the use of the origin of clades as age constraints. A usual error is to neglect the extinct lineages belonging to these clades and, therefore, considering them younger than what they probably are. For instance, according to the topology of our tree, we could have considered Gerbillinae as the clade including †*Abudhabia radinski* plus more derived gerbils (origin at ~8.13 Ma), or †*Myocricetodon parvus*+more derived gerbils (origin at ~13.29 Ma), or (†Paradakkamyini+(†*Myocricetodon parvus*+more derived gerbils)) (origin ~15.21 Ma) or (†Dakkamyini+(†Paradakkamyini+(†*Myocricetodon parvus*+more derived gerbils))), whose origin is at ~18.92 Ma.

Each of these clades has different origin-time estimates and the results of molecular clock analyses will be impacted depending on which one is taken as calibration point. The accurate choice of these calibration points requires deep paleontological knowledge.

Historical biogeography

Fossil data are vital to infer the biogeographical history of a given group [87]. The results of the dispersal-extinction-extended cladogenesis (DECX) analysis suggest that the origin of the extinct subfamilies †Cricetopinae and †Cricetodontinae is estimated in Anatolia, whereas that of Cricetidae, †Megacricetodontinae, and Muridae includes several regions (Fig. 4).

†Cricetodontinae

Our results show some inter and intra-continental exchanges since their origin in Anatolia some 25.5 Ma. The first dispersal from Anatolia towards Central Asia occurred circa 19.32Ma. This event could have promoted

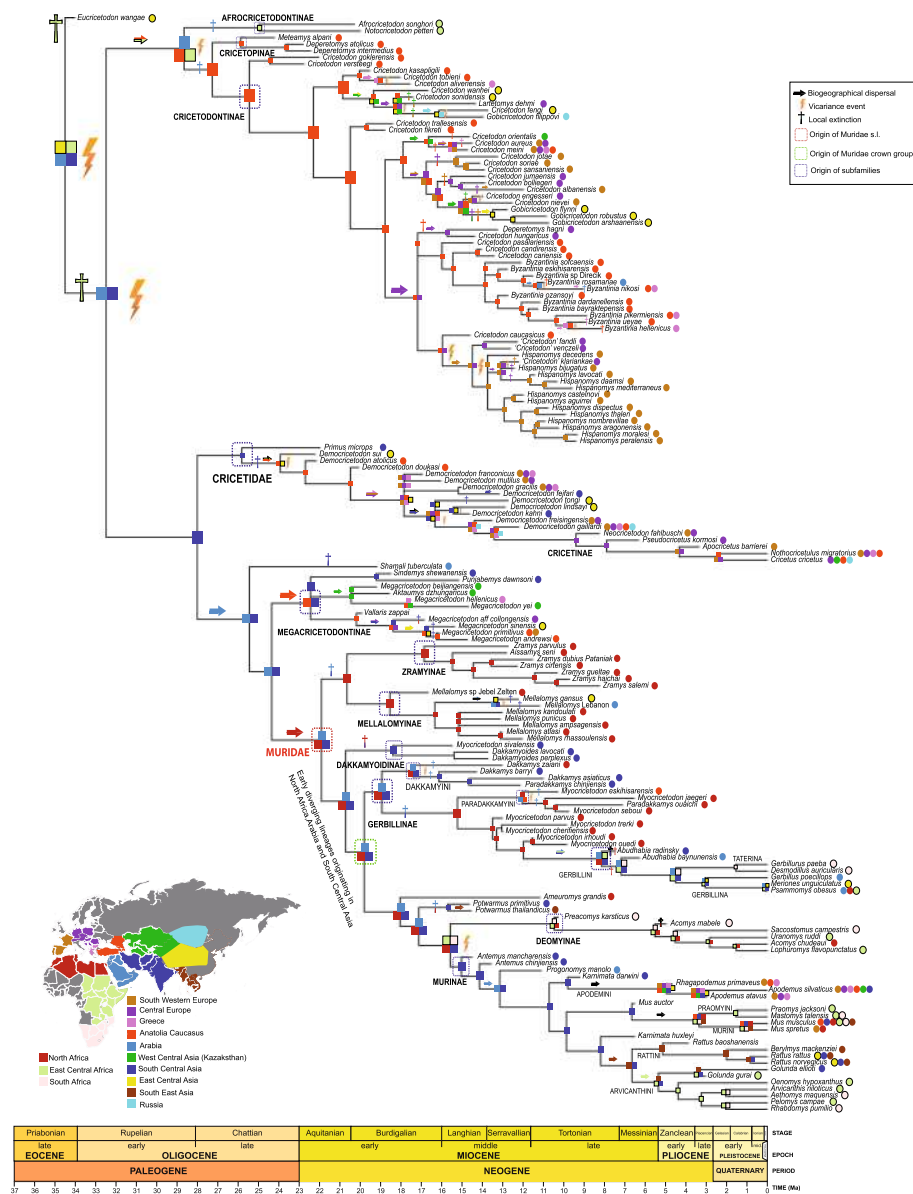


Fig. 4 Historical biogeography of Muroidea. The schematic map shows the 13 biogeographical areas for the DECX analysis. Colored areas on the map correspond to colored squares for each node, representing inferred ancestral range(s) with the DECX model. Colored circles at tips correspond to present-day distributions

the diversification of Chinese plesiomorphic species of †*Cricetodon*. The first migration of †*Cricetodon* from Turkey towards Greece took place in the early Miocene (*ca.* 19.3 Ma). Most European cricetodontines come from an independent Anatolian lineage that dispersed into the central and southwestern part of the continent between 16.74 and 17.89 Ma. Another dispersal event towards western China is inferred between 14.93 Ma and 13.5 Ma. A subsequent dispersal event of †*Cricetodon* from

Anatolia to Central and SW Europe (*ca.* 15.96 Ma) would have led to the origin of the genus †*Hispanomys* but continued evolving in Anatolia as the genus †*Byzantinia*, whose origin has been estimated approximately 15.97 Ma. According to our results, the Late Miocene cricetodontines are not closely related to the Middle Miocene European ones. In fact, they evolved from an Anatolian ancestor that dispersed in Southwestern Europe between 17.89 and 16.74 Ma.

Cricetidae

According to our results, extant cricetines evolved from some lineages belonging to Democricetodontinae as already suspected [88]. So, we here consider democricetodontines as *Cricetidae* sensu stricto and inferred their ancestral origins to be in South Central Asia (Fig. 3) from where they rapidly dispersed into West Central Asia and Anatolia. The DECX analysis indicates a dynamic biogeographical history for cricetids with speciation mainly occurring by dispersal. The first entrance of these rodents from Anatolia into Greece and Central and Southwestern Europe is estimated between 19 and 16.7 Ma. The delayed entrance of this group into Europe could have been the result of the geographical isolation of the Dinarian-Anatolian Island [89]. Later on, these rodents dispersed eastwards from Europe towards South and East Central Asia (17.58–15.28 Ma) but continued evolving in Europe from where extant cricetines are supposed to have originated and then dispersed into Asia and the Middle East.

However, as our sample only includes a limited sample of extinct and extant cricetids, the biogeographical inferences concerning this group should be taken with caution.

†*Megacricetodontinae*

The ancestral area of this subfamily, which is recovered as the sister group of Muridae, comprises a range including the regions of Anatolia and South Central Asia. Its biogeographical history is characterized by no vicariance but some dispersal events. Early Miocene dispersals took place from South Central Asia to West Central Asia (22.4–20.4 Ma) and from there to Greece (20.4 and 17.64 Ma). The first entrance of the genus †*Megacricetodon* into Greece is estimated between 18.17 and 17.31 Ma. Two additional dispersal events in Europe for early species of †*Megacricetodon* have been identified: the first from Asia to Central Europe (20.18–18.37 Ma) and the second from Anatolia to Southwestern Europe (ca. 16.2 Ma). These results are in agreement with a recent study [51], who inferred at least three migration events in Europe for early †*Megacricetodon* forms. As with cricetids, our study only includes a partial representation of megacricetodontines and the results should be interpreted with caution as well.

Muridae

The most likely ancestral geographic origin at the root of Muridae covers South Central Asia, Arabia, and North Africa (ca. 21.9 Ma). Our results also point to a dynamic biogeographical history, in which vicariance events ($n=4$) were less numerous than dispersal ($n=11$). The fact that North Africa may have been an important

ancestral area for the historical biogeography of Muridae was expected. In fact, after the collision of the Afro-Arabian plate with Eurasia [90], the subsequent closure of the Tethys Sea and the emergence of a land bridge between these continents have facilitated dispersals from Asia towards Africa and vice versa. Even though the fossil record brings to light the beginning of a foremost faunal interchange ca. 20–19 Ma, there is evidence that indicates intermittent trans-Thetysan dispersals before this time ([91] and references therein). So, it is plausible that ancestral murids had already spread from Asia and Arabia into North Africa shortly before 22 Ma. The analysis of the three main subfamilies within Muridae (Gerbillinae, Murinae, and Deomyinae) together with the three extinct murid subfamilies (†Zramyinae, †Mellalomyinae, †Dakkamyoidinae), evidence that some subfamilies such as the †Mellalomyinae (18.5 Ma), †Zramyinae (16.8 Ma), Gerbillinae (18.9 Ma), and Deomyinae (10.5 Ma) mostly or completely diversified into Africa. In fact, early murid clades are of African origin. So, †Zramyinae are endemic from Africa where they originated some 16.85 Ma and then diversified. The ancestral origin of the †Mellalomyinae (18.53 Ma) is also estimated in Africa. However, this extinct lineage seems to have dispersed into Arabia and South Asia between this time and 2 My after and later dispersed into East Asia (16.3–13.4 Ma) where †*Mellalomys gansus* has been found. The ancestral area of the †Dakkamyoidinae is recovered in South Central Asia where they evolved and diversified before become extinct in this region. The most likely ancestral geographic origin for the crown group of Muridae is similar with that evidenced for the root of all murids.

The origin of Gerbillinae, which includes the two extinct tribes †Dakkamyini and †Paradakkamyini as well as the tribe Gerbillini, is also inferred in South Central Asia, Arabia, and North Africa as all early-diverged murids and the tribe †Dakkamyini. DECX analysis points out to a North African origin for all gerbillines, except for †Dakkamyini that had a broader ancestral range, from where they dispersed to Anatolia (15.2–12.0 Ma), and to Arabia and South Central Asia (12.0–8.13 Ma) to give rise to the Gerbillini.

The analysis recovers an ancestral area for †*Potwarmus* that includes South Central Asia, Arabia, and North Africa. The oldest record of this taxon comes from Pakistan (locality 592) from where it is known since 16 Ma [56]. †*Potwarmus* dispersed eastwards about 15.7 Ma and enter into South Central Asia where it has been recorded in Thailand [92]. At the same time, this taxon dispersed westwards through Arabia entering into North Africa. In fact, remains of †*Potwarmus* have been found in early Middle Miocene deposits et al.-Jadidah, Arabia (12–14 Ma; [64]) and Jebel Zelten, Libya (14–15 Ma; [61]).

Deomyinae, mostly endemic from Africa, are supposed to have lived and diversified in this continent. A dispersal event from North Africa towards its South and East parts took place between 15.57 and 10.47 Ma.

Murinae, also known as “true murids” originated in South Central Asia *ca.* 15.6 Ma. The oldest known murine rodents have been recorded in the Potwar Plateau (Pakistan) where they had a remarkable diversity through time [56]. Primitive murine rodents seem to have endemically evolved at least during 2 Ma in the Indian subcontinent, which was rather isolated by high mountains to the north, large rivers on the east, and desertic areas on the west [56]. The exodus of †*Progonomys* out of South Central Asia could have been bolstered by the sea level drop that took place some 11.6 Ma [93]. †*Progonomys*, dispersed out of Pakistan between 13.1 Ma and 10.5 Ma, date in which this taxon is already recorded in Turkey and Lebanon [94]. In fact, the arrival of †*Progonomys* into the Arabian Peninsula is possibly related with the final closure of the eastern Tethys and the end of the marine connection between the Mediterranean Sea and the Indian Ocean along northern Arabia [95] and see discussion in López-Antoñanzas et al. [94].

Our results indicate an important wave of dispersal for Muridae that seems to have been at the origin and diversification of all important subfamilies. In fact, early Miocene ancestors of murids entered in Africa, colonizing a new environment. Interestingly, all early diverging lineages that gave rise to the main murid clades share a common biogeographical range that covers North Africa, Arabia, and South Central Asia. So, the collision between Afro-Arabia and Eurasia likely played an important role in the origin and diversification of murids, the largest family of mammals.

During the Miocene (23.03–5.3 Ma), plate tectonics drove sea level changes and orogeny, all of which significantly altered the climate and marked the start of the late Cenozoic cold house mode and the establishment of modern terrestrial biomes [96–98]. Until the beginning of the mid-Miocene Climatic Optimum (17–15 Ma), eustatic sea-level changes indicate the presence of a moderate to large ice sheet in Antarctica [99]. The existing land bridge between Africa and Eurasia together with some periods of sea-level drop could have favored the entrance of the murid ancestors from South Asia into North Africa through the Levant. During the mid-Miocene Climatic Transition (~15–13 Ma), three events of cooling and sea-level fall have also been recorded at 14.8 (Mi2a), 13.8 (Mi3), and 12.8 Ma (Mi4) that culminated with the establishment of the permanent East Antarctic ice sheet [99]. These sea-level low-stand periods could have opened terrestrial routes that could have allowed the migration of the ancestors of †*Hispanomys* from Anatolia into Europe

(Mi2a) and the eastwards dispersals (approximately 14–13.5 Ma) of some species of †*Mellalomys*, †*Democricetodon*, and † “*Cricetodon*” into East Central Asia (Mi3). These dispersals could have been triggered by the well-documented [99] contraction of the tropics in East Asia due to an increase in monsoon seasonality, coincident with the establishment of a quasi-permanent East Antarctic ice sheet [99]. The entrance of the ancestors of Gerbillini from North Africa to Arabia and from there to South Central Asia (*c.* 8.1 Ma) could have also been triggered by the global drying and cooling climate occurring at this time [97–100]. Likewise, the origin and diversification of the African subfamily Deomyinae is linked to a dispersal event from South Central Asia that took place in the Late Miocene *ca.* 10.5 Ma. The exodus of the ancestor of this subfamily out of Asia could have been favored by the Serravallian-Tortonian sea level fall that has been documented worldwide [101].

Conclusions

Our results evidence that neither †Cricetopinae nor †Cricetodontinae can be considered as belonging to the Cricetidae, the latter being related to Democricetodontinae. The split between these two extinct clades took place at approximately 27.2 Ma. In this work, we provide unprecedented agreement between morphological and molecular data concerning the monophyly of the Deomyinae and Gerbillinae as distinct clades within Murinae. For the first time, relaxed clock approaches using tip dating (which can account for uncertainty in the placement of fossil taxa) were used to estimate divergence times for the major groups of muroids. Even under strong violations of the best-fitting clock model, we find low levels of uncertainty for divergence times, thus providing an accurate timeline for muroid evolution. The origin of the Muridae (~21.9 Ma) is estimated with greater precision thanks to the inclusion of fossils now recognized as stem Muridae (†Zramyinae, †Mellalomyiinae, †Dakkamyoidinae). Early diverging murid lineages originated during the Early Miocene in low latitudes (North Africa, Arabia, and South Central Asia). The ancestor of murids dispersed from South Central Asia to North Africa before 21.9 Ma. The collision between these two continents seems to have played an important role in the origin and diversification of this family of rodents. The origin of the clade that will lead to all extant murids is estimated at ~19.8 Ma and that of murines at ~15 Ma. Our results show a dynamic biogeographical history for Muroidea, in which dispersal events seem to have been more numerous than vicariance. Eastwards early Middle Miocene long-distance dispersals towards East Asia could have been favored by the increase in seasonality during the Middle Miocene Climatic Transition

(MMCT) that resulted in a contraction of the tropics in this part of the continent.

Methods

Material

The systematic study presented below is based on the examination of original specimens and casts and data from the literature listed in Additional file 1: Table S1. First, second, and third lower molars are designated as m1, m2, and m3, respectively, and first, second, and third upper molars as M1, M2, and M3, respectively.

Morphological dataset

The dataset presented here represents a major expansion of the morphological data matrix for cricetodontines [16]. The inclusion of Muridae has required the addition of new characters and state of characters. It has been expanded from 77 to 166 taxa and from 82 to 108 morphological characters, thus representing an increase to more than double in taxonomic sampling and coverage across all major lineages of Muroidea. The data matrix (Additional file 1: File S1) has been built using Mesquite 3.0 [102] and the morphological characters are listed in it.

Character evolution model

We used the Markov model Mk(v) of character evolution [103] for morphological characters, implementing assortment bias correction for the inclusion of variable characters only. We allowed for among character rate variation by using a gamma distribution with four rate categories and with shape (alpha) sampled from an exponential distribution with mean = 1.0.

Clock models

We assessed the fit of various clock models to our data, including a new suit of clock models available in the developer's version of MrBayes (future MrBayes 3.2.8 release) [10] compiled from source code available at (<https://github.com/NBISweden/MrBayes>). These models include the continuous autocorrelated clock model (TK02) [104] and three uncorrelated clock models: the white noise model (former IGR model in previous versions of MrBayes) [105] and the new independent gamma rate (IGR) and independent log-normal clock (ILN) models [106] (Additional File 2: files S1-S8). These different models represent radically different interpretations of how and where in the tree evolution rates are allowed to change across lineages, which can substantially impact divergence time estimates using morphological and/or molecular data [13, 106].

We tested model fit using the stepping-stone sampling strategy to assess the marginal model likelihoods [107] and calculated Bayes Factors (BF)—30 steps (+ 5 as burn in) between for 250 million generations. Using standard

significance thresholds [108], we found a strong support (BF > 5) for the autocorrelated clock model (TK02) relative to the IGR and ILN models (BFs $\approx$ 8) and a very strong preference for the TK02 model relative to the WN model (BF = 44.8) (Table 1).

The mean and standard deviation for the probability distribution defining the prior on the clock rate were given informative values based on previous non-clock analysis, following Ronquist et al. [109] and subsequent studies [11, 13, 110]—the median value for tree height in substitutions from posterior trees divided by the age of the tree based on the median of the distribution for the root prior (16.6429/37.6 = 0.4426). The mean of the lognormal distribution was given the value based on the non-clock tree estimate in natural log scale: $\ln(0.4426) = -0.8150$. Finally, we chose a broad standard deviation around the mean ($\sigma = 1.5$).

Tree model and age calibrations

The age of the root has been set with a soft lower bound. The minimum age of the root corresponds to the oldest age for the oldest fossil belonging to *Eucrietodon* (34 Ma) and the maximum root has been estimated at 41.2 Ma (Middle/Late Eocene boundary), which is the maximum soft age for the Eucrietodontinae clade. We used an offset exponential distribution for the prior on-root age, which gives a higher sampling probability for values closer to the minimum age and a relatively low (but non-zero) probability of age values higher than 41.2 Ma.

All fossil calibrations (apart from the root node) were based on tip dates only, which account for the uncertainty in the placement of fossil taxa and avoids the issue of assigning maximum age constraints that are necessary for node-based age calibrations [109]. Additionally, we avoided recently detected strong biases that can be introduced by point age calibrations on the age of the fossils [111] by using a uniform prior distribution on the age range of the stratigraphic occurrence of the oldest fossils for each species (Additional File 2). We implemented the FBD process for the tree model with a flat prior (uniform distribution: $\text{samples} \in U[0, 1]$) on the parameters of relative extinction (= turnover) and probability of sampling fossils, whereas the net diversification rate was sampled from an exponential distribution with mean = 1.0.

Further, we performed a dense taxonomic sampling of closely related fossil lineages within a relatively short geological time span, which meets the conditions in which inferring direct ancestor-descendent relationships should be expected, and so we modeled our tree prior accordingly by allowing fossils to be either fully extinct tips or sampled ancestors [112, 113].

MCMC and diagnostic parameters

All analyses were conducted using the developer's version of MrBayes [10] available at (<https://github.com/>

NBISweden/MrBayes) using the Della computer cluster at Princeton University. Convergence of independent runs was assessed using: average standard deviation of split frequencies (ASDSF ≈ 0.01), potential scale reduction factors (PSRF ≈ 1 for all parameters), and effective sample size (ESS) for each parameter greater than 200 and analyzed using Tracer 1.7.1 [114] and the R package RWTY (R We There Yet) [115] (see Additional File 3: Fig. S1).

Stochastic character mapping

Stochastic character mapping was performed using the Stochastic character mapping of discrete traits on phylogenies (SIMMAP) approach [116] as implemented in the *make.simmap* function of the R package phytools 2.1.1 [117]. The R script is provided in Additional File 3: File S1. For each character, three different models were tested, concerning the rate of changes between states, estimated by Bayesian Markov chain Monte Carlo (MCMC): equal-rates model (ER, in which all rates are equal), symmetrical model (SYM where rates for symmetrical transition are the same), or the all-rates-different model (ARD, in which each rate can take on a potentially different value) [118]. The best-fit model was then chosen based on the Akaike information criterion and used for the stochastic character mapping. The resulting posterior probabilities of each state are represented at the nodes of the trees. Most the synapomorphies mentioned in the text does not show a large amount of uncertainty in their probability of having a given character state. However, for the few ones, which show a fair amount of uncertainty, we have taken a threshold value of 50%, above which, we consider that the node has a given character state. For some characters (marked as ER), symmetrical rate changes were favored but resulted in misbehaving analyses. In these cases, the equal rates option was chosen instead (Additional File 3: Table S1).

Inference of historical biogeography

We estimated the ancestral ranges of origin and geographic range evolution for muroids using the maximum likelihood (ML) approach of dispersal-extinction-cladogenesis model (DEC [119]) as implemented in the *DEC eExtended* model (DECX [120]). DECX can infer the ancestral ranges using the geographic ranges of the sampled species taking into account non-ultrametric trees that include fossils as species, by modeling anagenetic events, classical vicariance, and cladogenetic events [119]. The most likely ancestral states at each node and the geographic range evolution of Muroidea have been estimated using a time-calibrated tree, a species distribution matrix, and connectivity matrices, representing the palaeogeographic history. As the input data, we have used our resulting time-calibrated Bayesian majority-rule

consensus tree, the distribution of each species for a set of geographic areas, and a geographic model that is represented by connectivity and dispersal scalar matrices spanning the evolutionary history of the group. The geographic distribution of the muroid species included in our study has been categorized by coding the presence or the absence of each species in 13 areas as follows: (1) South Western Europe (SWE); (2) Central Europe (CE); (3) Greece (GR); (4) Anatolia Caucasus (AC); (5) West Central Asia (WCA); (6) Arabia (AR); (7) South Central Asia (SCA); (8) East Central Asia (ECA); (9) North Africa (NA); (10) East Africa (EA); (11) South Africa (SA); (12) Russia (RU); and (13) South East Asia (SEA). The geographic model has been built using connectivity matrices [118] that take into account paleogeography and the possibility for a species to colonize a new area, with disconnection between areas coded as 0 and connection between areas coded as 1 (Additional File 3: Table S2).

Abbreviations

ARD	All-rates-different model
ASDSF	Average standard deviation of split frequencies
BF	Bayes factors
DEC	Dispersal-extinction-cladogenesis
DECX	Dispersal-extinction-eExtended cladogenesis
ESS	Effective sample size
ER	Equal-rates model
IGR	New independent gamma rate uncorrelated clock model
ILN	Independent lognormal uncorrelated clock model
m1, m2, m3	First, second, and third lower molars
M1, M2, M3	First, second, and third upper molars
MCMC	Markov chain Monte Carlo
MK (v)	Markov model that conditions on variable characters
ML	Maximum likelihood
MMCT	Middle Miocene climatic transition
MRCA	Most recent common ancestor
PSRF	Potential scale reduction factors
RWTY	R We There Yet
SIMMAP	Stochastic character mapping of discrete traits on phylogenies
SYM	Symmetrical model
TK02	Continuous autocorrelated clock model
WN	White noise uncorrelated clock model
†	Whole extinct clade

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12915-024-02053-2>.

Additional file 1: Figure S1, stochastic mapping of all synapomorphies. Pie charts at nodes indicate the relative Bayesian posterior probability of each character state. Duplicated characters are those for which the mapping with the equal rates (ER) model was added as well. Characters missing are those that presented some non-applicable states, for which the analysis did not work. File S1, morphological data matrix in Nexus format used in this study. File S2, list of morphological characters used in the analyses. Table S1, table with full taxonomic sampling, including specimen accession numbers and ages.

Additional 2: Files S1–S8. All input (S1–S4) and output (S5–S8) files from relaxed clock Bayesian inference analyses, including all necessary code to reproduce the results.

Additional file 3: Figure S2, convergence diagnostics from Bayesian inferences using the R package RWTY [115]. Script S1, R Script used in this work to carry out the stochastic character mapping performed using the

SIMMAP approach [116] as implemented in the `make.simmap` function of the R package `phytools` 2.1.1 [117]. Table S2, table showing the characters for which the equal rate model has been chosen despite not being the best fitting model. Table S3, geographic distribution of the muroid species included in this study and categorized by coding the presence or the absence of each species in 13 areas. The geographic model has been built using connectivity matrices [118].

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Authors' contributions

Conceptualization: RLA, PPC. Methodology: RLA, PPC, TRS, FC. Formal analysis: RLA, PPC, TRS, FC, MD. Investigation: RLA, PPC. Visualization: RLA, PPC, TRS. Writing original draft: RLA. Writing review and editing: all authors. All authors read and approved the final manuscript.

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Data availability

The authors confirm that the data supporting the findings of this study are available within the article and its Additional Files.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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A.2 French Summary

Chapitre 1. Introduction

Avec plus de 700 espèces vivantes, les Cricetidae constituent la deuxième famille la plus riche en espèces au sein des Muroidea. Selon des études moléculaires (Musser et Carleton 2005; Neumann et al. 2006; Steppan et Schenk 2017), elle comprend les sous-familles suivantes : les rats et souris du Nouveau Monde (Sigmodontinae, Neotominae et Tylomyinae), le groupe des campagnols, rats musqués et lemmings (Arvicolinae), et les hamsters de l'Ancien Monde (Cricetinae). Les Cricetinae (communément appelés hamsters) sont un groupe de rongeurs de la taille d'une souris ou d'un rat, dotés de poches jugales et d'une queue courte, qui comprend aujourd'hui 19 espèces réparties en neuf ou dix genres. Ils vivent actuellement dans le domaine paléarctique, principalement dans les steppes et les prairies, mais aussi dans les zones désertiques et les environnements urbains (Pardiñas et al. 2017a). Les fossiles de ce groupe remontent au Miocène moyen (Kälin 1999; Prieto et Rummel 2016) et comprennent environ 90 espèces fossiles connues et des milliers de découvertes fossiles. Les tentatives de reconstruction des phylogénies des hamsters fossiles ont été limitées en termes d'échantillonnage taxonomique, ne comprenant qu'un ou deux genres, et s'appuyant exclusivement sur des approches de parcimonie maximale (Cuenca-Bescós 2003; López-Antoñanzas et al. 2014; Sinitsa et Delinschi 2016; Feroz et al. 2024). De plus, les dates de divergence des hamsters n'ont jusqu'à présent été estimées qu'à partir des phylogénies moléculaires des taxons existants, les données fossiles n'étant utilisées que comme points de calibrage (Neumann et al. 2006; Steppan et Schenk 2017; Lebedev et al. 2018; Jiang et al. 2024; Pan et al. 2025). En conséquence, les relations phylogénétiques entre les genres fossiles de hamsters, le lien entre les hamsters actuels et leurs ancêtres fossiles, ainsi que la chronologie et l'origine géographique des principaux groupes de hamsters restent largement inconnus. Les chapitres 2 et 3 de cette thèse ont donc pour objectif principal d'aborder ces questions. En outre, les schémas macroévolutifs qui ont façonné le groupe des hamsters sont tout aussi flous, mis à part une influence possible non précisée de l'aridification et de l'ouverture de la végétation (Neumann et al. 2006; Pardiñas et al. 2017). Par conséquent, l'objectif du chapitre 4 est d'explorer les modèles et les tendances de diversification au sein des Cricetinae et d'évaluer l'impact de facteurs potentiels tels que la paléotempérature ou la biogéographie. En comparant plusieurs méthodologies de pointe qui ont rarement été appliquées à des ensembles de données empiriques jusqu'à présent, nous pouvons également tirer des conclusions pratiques sur ces méthodologies lorsqu'elles sont utilisées avec des groupes qui présentent des archives fossiles relativement abondantes.

Chapitre 2. Relations phylogénétiques des hamsters du Néogène (Mammalia, Rodentia, Cricetinae) révélées par inférence Bayésienne et parcimonie maximale

Dans ce chapitre, les relations entre sept genres importants de hamsters du Néogène ont été explorées en comparant deux approches phylogénétiques, à savoir la parcimonie maximale et l'inférence bayésienne, après des tests approfondis des modèles. De plus, les taxons aberrants ont été identifiés et exclus des arbres de consensus afin d'améliorer la résolution et le soutien des clades. Enfin, l'adéquation stratigraphique des phylogénies obtenues a été évaluée. Sur la base des arbres obtenus, des synapomorphies ont été définies pour les genres qui sont récupérés comme monophylétiques après reconstruction des caractères ancestraux. Les phylogénies obtenues par inférence bayésienne (Fig. A.2.1) et par parcimonie maximale sont largement concordantes dans leur topologie, ce qui aboutit à une hypothèse phylogénétique robuste concernant plusieurs hamsters du Néogène. Sur la base de ces résultats, plusieurs questions taxonomiques et systématiques persistantes concernant le groupe des hamsters, par exemple concernant la synonymie de †*Kowalskia* et †*Neocricetodon*, sont clarifiées grâce à la classification de groupes monophylétiques bien définis. Cette étude est la première à utiliser une approche de « tip-dating » dans le cas des hamsters, fournissant de nouvelles estimations des dates de divergence pour les clades éteints inclus.

Chapitre 3. La phylogénie totale révèle la radiation récente du groupe couronne et l'histoire biogéographique des hamsters

Afin de répondre à la question de la position phylogénétique des hamsters actuels, ce chapitre, s'appuyant sur le cadre présenté dans le chapitre précédent, élargit considérablement l'ensemble des taxons en incluant les hamsters asiatiques et les hamsters fossiles du Quaternaire, ainsi que les hamsters actuels. De plus, les données moléculaires des hamsters actuels sont intégrées afin de réaliser la première analyse phylogénétique d'évidence totale des hamsters. La phylogénie reconstruite est ensuite utilisée pour contribuer aux questions taxonomiques concernant les hamsters éteints, comparer les estimations du temps de divergence avec les études précédentes et explorer l'histoire biogéographique des hamsters fossiles et actuels. La topologie de la phylogénie obtenue (Fig. A.2.2) est en grande partie conforme à celle du chapitre précédent et montre un soutien encore plus important pour plusieurs clades, tels que †*Collimys* et †*Neocricetodon*. En élargissant l'ensemble des taxons, les relations entre les hamsters fossiles et les hamsters actuels sont explorées et plusieurs ancêtres potentiels les plus proches sont proposés.

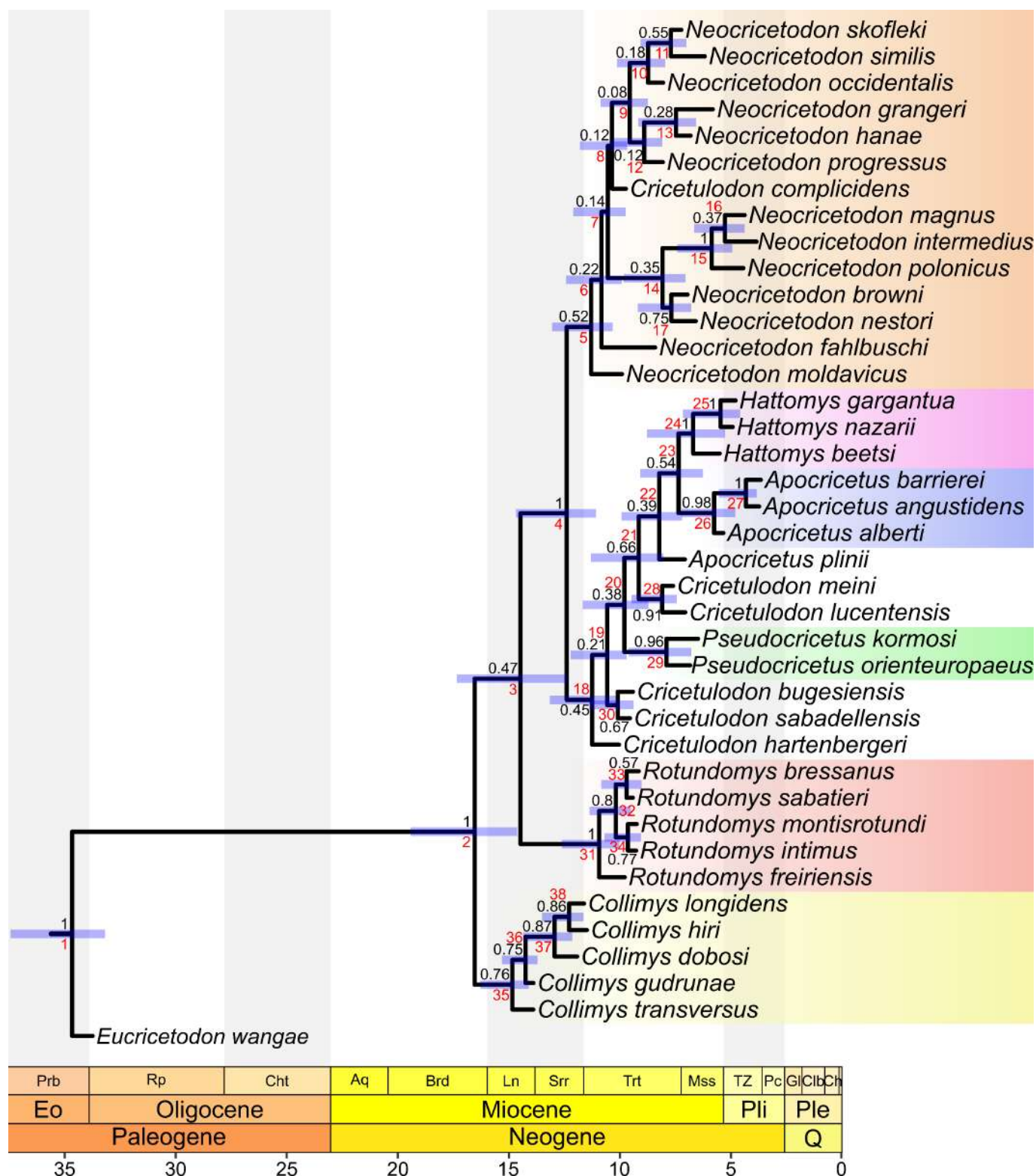


Figure A.2.1 Arbre de compatibilité maximale des clades issu de l'analyse Bayésienne IGR à horloge relaxée calibrée dans le temps. Les probabilités a posteriori des clades sont indiquées en noir au niveau des nœuds respectifs, les numéros des nœuds sont indiqués en rouge, les barres des nœuds indiquent la densité a posteriori la plus élevée à 95 % pour les temps de divergence. L'échelle est exprimée en Ma, le tableau chronostratigraphique suit Cohen et al. (2013; mis à jour en 2024/10).

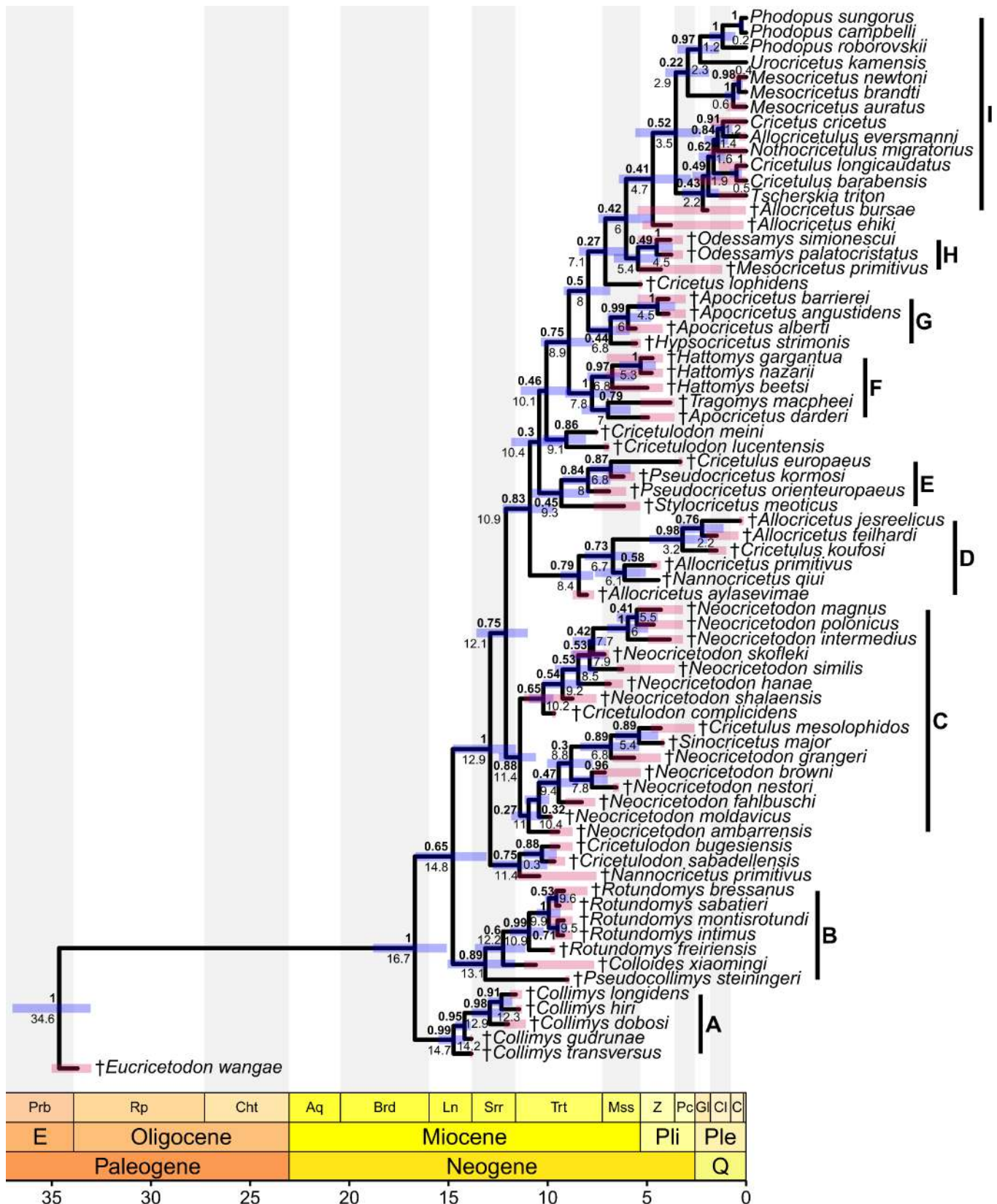


Figure A.2.2. Arbre de compatibilité maximale des clades issu de l'analyse Bayésienne WN à horloge relaxée calibrée par les extrémités, combinant des données moléculaires et morphologiques. Les probabilités a posteriori (en gras) et les âges médians des clades sont indiqués aux nœuds respectifs, les barres bleues indiquent la densité a posteriori la plus élevée à 95 % pour les temps de divergence, les barres rouges indiquent la gamme stratigraphique du taxon respectif. L'échelle est en Ma, le tableau chronostratigraphique suit Cohen et al. (2013; mis à jour en 2024/10).

Les temps de divergence qui en résultent sont nettement plus récents que les estimations précédentes calibrées par nœuds, selon lesquelles les hamsters couronnés auraient apparu au Pléistocène. L'histoire biogéographique déduite dans ce chapitre révèle une origine européenne de l'ensemble du groupe des hamsters, ainsi que plusieurs événements de dispersion vers l'Europe occidentale et l'Asie. L'origine des hamsters couronnés est réduite à une région située autour de la Méditerranée orientale, de la mer Noire et de la mer Caspienne.

Chapitre 4. Exploration de la diversification et de la diversité des hamsters à l'aide d'approches bayésiennes et d'apprentissage profond

Le quatrième chapitre fournit des informations supplémentaires sur l'histoire évolutive des hamsters, en ce qui concerne la dynamique de leur taux de macroévolution. Sur la base de la phylogénie reconstruite au chapitre précédent, ainsi que des données compilées ici sur la présence de fossiles, la paléodiversité et l'hétérogénéité des taux de diversification des hamsters sont explorées. Une évaluation rigoureuse de ces processus est nécessaire pour discuter des raisons de l'essor et du déclin de ce groupe. Nous n'avons trouvé aucune preuve de changements de diversification à l'échelle du clade, qui indiqueraient par exemple des innovations clés. Nos résultats montrent des taux de spéciation relativement constants et décroissants au fil du temps, mais révèlent des différences dans les taux d'extinction déduits en fonction de la méthodologie appliquée (Fig. A.2.3). Les taux d'extinction spécifiques à chaque lignée démontrent une certaine résistance des hamsters aux fluctuations environnementales, résultant potentiellement de plusieurs stratégies écologiques visant à amortir les conditions défavorables. La variation du taux de spéciation dépendrait principalement de la répartition géographique, ainsi que de la paléotempérature mondiale. La paléodiversité réelle déduite montre une trajectoire en forme de bosse à travers le temps, avec un maximum de 35 espèces contemporaines de hamsters pendant le Messinien et le Pliocène. Le caractère apparemment incomplet des archives fossiles des hamsters et la variation temporelle et taxonomique élevée de la fossilisation révèlent les limites des modèles appliqués et soulignent les difficultés à tenir compte des divers biais souvent présents dans les données fossiles.

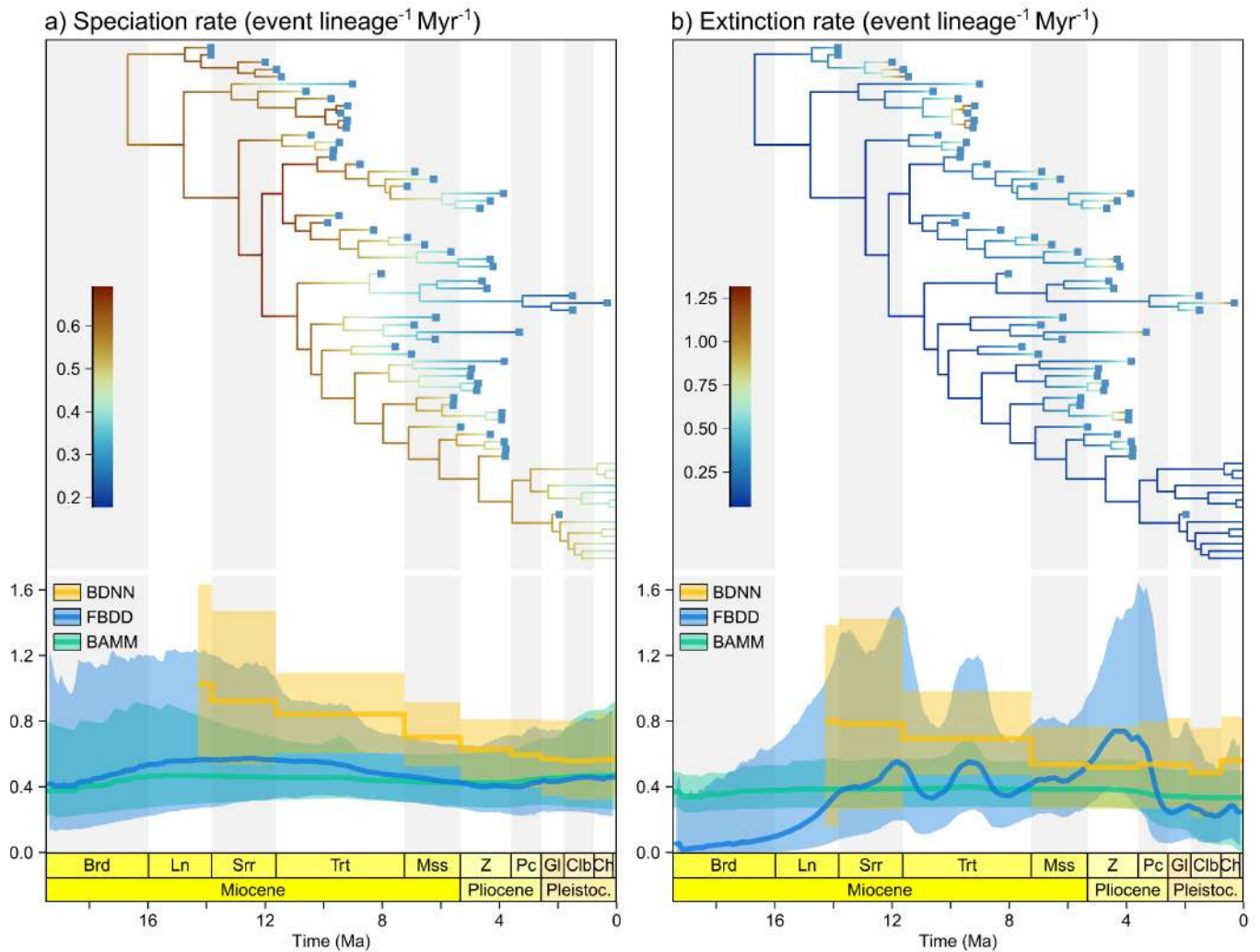


Figure 4.1. Reconstitution des taux de spéciation (a) et d'extinction (b) sous forme de dégradés de couleurs sur l'arbre de compatibilité maximale des clades résultant du modèle de diffusion fossilisée naissance-mort (FBDD) (en haut) et à travers le temps, pour toutes les lignées (en bas). Les taux au fil du temps sont estimés à partir de plusieurs ensembles de données répliqués par le réseau neuronal naissance-mort (BDNN) (jaune), le FBDD (bleu) et l'analyse bayésienne des mélanges macroévolutifs (BAMM) (vert), représentés sous forme de médiane (FBDD, BAMM) ou de moyenne (BDNN) avec un intervalle de crédibilité (IC) de 95 % sous forme de zone ombrée. Le tableau chronostratigraphique suit Cohen et al. (2013 ; mis à jour en 2024/10).

Chapitre 5. Discussion

Dans le chapitre 5, les résultats des chapitres précédents sont discutés en mettant l'accent sur les défis qui peuvent se poser lors de l'intégration d'espèces fossiles dans les analyses phylogénétiques et macroévolutives. Un échantillonnage fossile exhaustif est important pour de telles analyses (Mongiardino Koch et al. 2021; do Rosario Petrucci et al. 2025), c'est pourquoi j'ai mené une vaste campagne de collecte de données, comprenant l'examen direct de

spécimens conservés dans plusieurs grandes collections paléontologiques et zoologiques et une revue approfondie de la littérature. L'absence supposée de changements discrets de diversification à l'échelle du clade chez les hamsters dans cette thèse est conforme aux études précédentes qui ont analysé des groupes taxonomiques plus larges incluant les hamsters (Fabre et al. 2012; Steppan et Schenk 2017). Plusieurs idées pour de futures analyses sur le groupe des hamsters sont proposées, telles que l'estimation conjointe du processus de diversification et de l'évolution biogéographique (Goldberg et al. 2011), l'intégration directe des données sur la présence de fossiles dans l'inférence phylogénétique (Andréoletti et al. 2022), ou la prise en compte de la diversification et de la variation du taux de préservation lors de la reconstruction phylogénétique (Barido-Sottani et Morlon 2025; Chabrol et al. 2025). Les résultats indiquent également que les bases de données en ligne sur les occurrences doivent être évaluées avec soin lorsqu'elles sont utilisées pour des analyses macroévolutives, et révèlent les limites de plusieurs approches actuelles (Mitchell et al. 2019; Silvestro et al. 2019; Quintero et al. 2025) pour tenir compte du caractère incomplet et biaisé des archives fossiles (Warnock et al. 2020). Enfin, j'ai souligné les défis qui peuvent se poser lorsque des taxons fossiles sont intégrés dans des cadres phylogénétiques, et j'ai mis l'accent ici sur le conflit entre le modèle fossile naissance-mort (Stadler 2010; Heath et al. 2014), qui est basé sur des spécimens fossiles, et les données empiriques typiques (Wright et al. 2022). Une approche prometteuse pour surmonter certaines de ces limites à l'avenir pourrait être l'extension décrite du modèle afin d'intégrer des données au niveau des taxons (Stadler et al. 2018).

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Supplementary Material

The Supplementary Material of Chapter 2 is published online together with the article at <https://dx.doi.org/10.7717/peerj.18440>

The Additional files of Chapter 3 are, in the context of this thesis, available online at: <https://figshare.com/s/6c91828664b2db2fffa7>

The Additional files 1 and 5 are also included below, directly in this thesis.

The Supplementary Material of Chapter 3 is available online for review purposes at <https://dataverse.harvard.edu/previewurl.xhtml?token=e2aaaa9c-8658-4c86-adb4-9448f1bbf775>

The Supplementary Material of Chapter 4 is, in the context of this thesis, available online at: <https://figshare.com/s/6c91828664b2db2fffa7>

The Supplementary Material of Chapter A.1 is published online together with the article at <https://doi.org/10.1186/s12915-024-02053-2>

Additional file 1 of Chapter 3**Total-evidence phylogeny reveals recent crown group radiation and biogeographical history of hamsters**

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Additional information about the studied taxa and material

Overview. In total, Neogene and Quaternary combined, including all true hamsters, there are 28 genera including around 109 species, of which 82 (75 %) were included in the phylogenetic reconstruction of this study. All genera are represented by at least one species in this analysis, except the monospecific genera, †*Amblycricetus* Zheng, 1993, †*Bahomys* Chow & Li, 1965, *Cansumys* Allen, 1928, and †*Gromovia* Erbajeva, Alexeeva & Khenzykhenova, 2003, due to limited available material or descriptions.

Neogene taxa. There are 15 genera from the Neogene that are clearly associated with the Cricetinae. There is one additional genus, *Aepyocricetus* Li, Stidham, Ni & Li, 2017 that is possibly a member of the Cricetinae but the unique morphology of the genus impedes a confident association to the hamsters (Li et al., 2017). Out of the 65 species identified within these 15 genera, 55 (85%) could be added to the matrix, including at least one member of each genus (Tab. S1.1). The remaining species were either too limited in their material, with several molars not known, or no material or sufficient description was available to the authors and therefore no meaningful number of characters could be coded. The information given here for *Apocricetus*, *Collimys*, *Cricetulodon*, *Hattomys*, *Neocricetodon*, *Pseudocricetus*, and *Rotundomys* follows in most parts Dirnberger et al. (2024).

Table S1.1. Total number of accepted species and number of species included in the phylogenetic analysis of this study of all 15 extinct cricetine genera from the Neogene.

Genus	Accepted species	Species included here
<i>Apocricetus</i>	5	5 (100 %)
<i>Collimys</i>	7	5 (71 %)
<i>Colloides</i>	1	1 (100 %)
<i>Cricetulodon</i>	7	6 (86 %)
<i>Hattomys</i>	3	3 (100 %)
<i>Hypsocricetus</i>	1	1 (100 %)
<i>Nannocricetus</i>	3	3 (100 %)
<i>Neocricetodon</i>	20	15 (75 %)
<i>Odessamys</i>	2	2 (100 %)
<i>Pseudocollimys</i>	1	1 (100 %)
<i>Pseudocricetus</i>	4	3 (75 %)
<i>Rotundomys</i>	6	5 (83 %)
<i>Sinocricetus</i>	3	3 (100 %)
<i>Stylocricetus</i>	1	1 (100 %)
<i>Tragomys</i>	1	1 (100 %)
Total	65	55 (85 %)

Quaternary taxa. There are 13 genera from the Quaternary that are clearly associated with the Cricetinae, including 25 extinct and 19 extant species (following Kryštufek & Shenbrot, 2025), although there are ongoing discussions, e.g., the status of ‘*Cricetulus griseus*’ Milne-Edwards, 1867 (Ivanova et al., 2022; Lebedev et al., 2018), which is here treated as a subspecies of *Cricetulus barabensis*. Out of the total 44 species, 27 (61 %) could be added to the matrix (Tab. S1.2), including, 13 (68 %) extant species, whose given type localities and distributions follow Musser and Carleton (2005) and Kryštufek and Shenbrot (2025), as well as references therein. The information given here for *Cricetus cricetus* and *Nothocricetulus migratorius* is compiled in most parts in Dirnberger et al. (2024).

Table S1.2. Total number of accepted species and number of species included in the phylogenetic analysis of this study of all 13 cricetine genera from the Quaternary.

Genus	Accepted species	Species included here
† <i>Allocricetus</i>	9	7 (78 %)
<i>Allocricetulus</i>	2	1 (50 %)
† <i>Amblycricetus</i>	1	0
† <i>Bahomys</i>	1	0
<i>Cansumys</i>	1	0
<i>Cricetulus</i>	9	7 (78 %)
<i>Cricetus</i>	6	2 (33 %)
† <i>Gromovia</i>	1	0
<i>Mesocricetus</i>	6	4 (67 %)
<i>Nothocricetulus</i>	1	1 (100 %)
<i>Phodopus</i>	3	3 (100 %)
<i>Tscherskia</i>	2	1 (50 %)
<i>Urocricetus</i>	2	1 (50 %)
Total	44	27 (61 %)

Sampled taxa, including material and references studied for coding

Neogene taxa

Apocricetus Freudenthal, Mein & Martín Suárez, 1998

All 5 accepted species are included in the analysis:

Apocricetus alberti Freudenthal, Mein & Martín Suárez, 1998

Geographical distribution of occurrences: Spain

Type locality: Crevillente 6, Spain

Holotype: Department Aardwetenschappen, University Utrecht, CR6 111, m1 (figured in de Bruijn et al. (1975), pl. 3, fig. 4, as *Cricetus* cf. *kormosi*)

Age range [Max–Min (Mean)]: 6.56–4.2 Ma (5.38 Ma)

Oldest occurrences: Arquillo 1, La Gloria 5, Las Casiones, Las Casiones superior, Masada del Valle 7, Purcal 23, Purcal 24A, Purcal 25, Purcal 25A, Valdecebro 3, Valdecebro 6, Villastar (van Dam et al., 2023)

Youngest occurrence: Alcoy-N (van Dam et al., 2023)

Anatomical references: (de Bruijn et al., 1975; Freudenthal et al., 1991, 1998; García-Alix et al., 2008; Mansino et al., 2014; Minwer-Barakat et al., 2009; Piñero & Agustí, 2019)

Apocricetus angustidens (Depéret, 1890)

Geographical distribution of occurrences: France and Spain

Type locality: Serrat d'en Vacquer, France

Holotype: Musée Guimet d'Histoire Naturelle, Lyon (Donnezan coll.), Pp 75, mandible (figured in Depéret (1890-1895), pl. 4, figs. 26, 26a, as *Cricetus angustidens*)

Age range: 4.27–3.03 Ma (3.65 Ma)

Oldest occurrences: Le Soler, Mont-Hélène, Nîmes, Sète, Serrat d'en Vacquer, Villeneuve de la Raho: MN 15 (Fejfar, 1970; Freudenthal et al., 1998)

Youngest occurrences: Puimoisson II, Lo Fournas 13: MN 16a (Biquand et al., 1990; Guérin et al., 1970)

Problematic localities: Port-la-Nouvelle, Castelnou 1: mix of different ages (Freudenthal et al., 1998)

Observed material: Faculty of Sciences, University Claude Bernard, Lyon (FSL): casts of type material. Bavarian State Collection for Palaeontology and Geology, Munich (SNSB-BSPG) 1967 XIV 19–24: m1, m2, m3, M1, M2, M3 from Nîmes

Anatomical references: (Aguilar et al., 1986; Freudenthal et al., 1998; Hugueney & Mein, 1966; Martín Suárez, 1988)

Remark: According to Freudenthal et al. (1998), there is *Apocricetus angustidens* in the Spanish locality Vilafant. This locality was related to MN 14 (Gibert et al., 1980), however on the basis that the species is *A. barrierei* (see also Agustí, 1982). As the species identification is not clear, the locality is not included here.

Apocricetus barrierei (Mein & Michaux, 1970)

Geographical distribution of occurrences: France and Spain

Type locality: Chabrier, France

Holotype: FSL 65356, M1 (figured in Mein and Michaux (1970), pl. 1, fig. 1, as *Cricetus barrierei*)

Age range: 5.47–3.04 Ma (4.255 Ma)

Oldest occurrences: Celleneuve, Chabrier, Hautimagne, Vendargues: MN 14

Youngest occurrence: Huéscar 3 (van Dam et al., 2023)

Problematic localities: Castelnou 1, 3: mix of different ages (Freudenthal et al., 1998);
Almudena 1D: age unknown (Freudenthal et al., 1998; Ruiz-Sánchez et al., 2014)

Observed material: FSL 65357, 65358, 65359: casts of type material: m1, m3, M3; cast of M3
and m2 from Hautimagne

Anatomical references: (Aguilar et al., 1991; Ambert et al., 1998; Freudenthal et al., 1998;
García-Alix et al., 2008; Mansino et al., 2015; Martín Suárez, 1988; Mein & Michaux, 1970;
Piñero & Agustí, 2020; Piñero et al., 2017; Ruiz Bustos et al., 1984; Ruiz-Sánchez et al.,
2014)

Remark: FSL 65359 is from Hautimagne according to Mein and Michaux (1970) but from
Chabrier (Type locality) according to the specimen label.

Apocricetus darderi Torres-Roig, Agustí, Bover & Alcover 2019

Single occurrence/type locality: Na Burgesa-1, Spain

Holotype: Institut Mediterrani d'Estudis Avançats, Esporles, IMEDEA 94726, mandible
(figured in Torres-Roig et al. (2019), figs. 5(A), 7(A), 7(B))

Age: 5.333–3.6 Ma (4.467 Ma): Zanclean (Torres-Roig et al., 2019)

Anatomical reference: (Torres-Roig et al., 2019)

Apocricetus plinii (Freudenthal, Lacombe & Martín Suárez, 1991)

Geographical distribution of occurrences: France and Spain

Type locality: Crevillente 15, Spain

Holotype: Naturalis Biodiversity Center, Leiden, RGM 402 10, M1 (figured in Freudenthal et
al. (1991), pl. 4, fig. 1, as *Neocricetodon plinii*)

Age range: 7.67–6.27 Ma (6.97 Ma)

Oldest occurrence: Crevillente 5A (van Dam et al., 2023)

Youngest occurrence: Otura 3 (van Dam et al., 2023)

Problematic locality: Castelnou 1: mix of different ages (Freudenthal et al., 1998)

Anatomical references: (Freudenthal et al., 1991, 1998)

***Collimys* Daxner-Höck, 1972**

Of the 7 total species, 2 were not included in the analysis: *Collimys caucasicus* Tesakov, 2017 (in Tesakov et al., 2017) and *C. primus* Daxner-Höck, 1972, due to a limited amount of found specimens. The 5 species that are included are the following:

***Collimys dobosi* Hír, 2005**

Geographical distribution of occurrences: Germany and Hungary

Type locality: Felsötárkány 3/2, Hungary

Holotype: Municipal Museum of Pásztó, Nógrád County Museums, No. 2003.79, M1 (figured in Hír (2005), fig. 1)

Age range: 12.2–11.1 Ma (11.6 Ma)

Oldest occurrence: Felsötárkány 2/3 (Hír, 2006; Hír et al., 2016, 2017)

Youngest occurrence: Felsötárkány 3/10 (Hír, 2006; Hír et al., 2024)

Observed material: SNSB-BSPG 1979 XVIII 1–81: 81 isolated molars from Hillenlohe

Anatomical references: (Hír, 2004, 2005, 2006; Hír & Kókay, 2010; Prieto & Rummel, 2009a)

***Collimys gudrunae* Prieto & Rummel, 2009b**

Single occurrence/type locality: Petersbuch 31, Germany

Holotype: Naturmuseum Augsburg, NMA 2007/1/2017, M1 (figured in Prieto and Rummel (2009b), fig. 4 b1, b2)

Age: 13.9–13.8 Ma (13.85 Ma) (Prieto & Rummel, 2009b, 2016)

Anatomical reference: (Prieto & Rummel, 2009b)

***Collimys hiri* Prieto & Rummel, 2009a**

Geographical distribution of occurrences: Germany

Type locality: Hammerschmiede 1, Germany

Holotype: SNSB- BSPG 1973-XIX-182, m1 (figured in Prieto and Rummel (2009a), fig. 2a)

Age range: 11.6–11.3 Ma (11.45 Ma)

Oldest occurrence: Hammerschmiede 1 (Prieto & Rummel, 2009a, 2016)

Youngest occurrence: Hammerschmiede 3 (Prieto & Rummel, 2009a, 2016)

Observed material: SNSB-BSPG 1973 XIX 182–185, 265, 266: type material: m1, m2, m3, M1, M2, M3; SNSB-BSPG 1980 XXVIII 1–58: 58 isolated molars from Hammerschmiede 3

Anatomical reference: (Prieto & Rummel, 2009a)

Remark: The dating of the Hammerschmiede locality is based on the levels 4 and 5. Hammerschmiede 1 can be correlated to Hammerschmiede 5 (Kirscher et al., 2016). As a more precise dating of Hammerschmiede 1 and 3 is not available, the range for the locality given in Hír et al. (2016) and Prieto and Rummel (2016) is taken.

***Collimys longidens* Kälin & Engesser, 2001**

Single occurrence/type locality: Nebelbergweg, Switzerland

Holotype: Natural History Museum Basel (NMB), Nbw 92, m1 (figured in Kälin and Engesser (2001), fig. 37b)

Age: 11.9–11.3 Ma (11.6 Ma)

Anatomical reference: (Kälin & Engesser, 2001)

Remark: According to Kälin and Engesser (2001), the locality Nebelbergweg is related to early MN 9 but according to Hír and Kókey (2010), it can also be late MN 7/8. The relative age of Nebelbergweg compared to Hammerschmiede is uncertain (Prieto et al., 2014; Prieto & Rummel, 2009a). Because of similarities of both faunas a similar or, following Prieto and Rummel (2016), a slightly older age of Nebelbergweg is assumed.

***Collimys transversus* Heissig, 1995**

Single occurrence/type locality: Steinheim am Albuch, Germany

Holotype: NMB, Sth. 106 H1, M1 (figured in Heissig (1995), pl. 1, fig. 7 and in Kälin and Engesser (2001), fig. 36e)

Age: 13.9–13.8 (13.85 Ma)

Anatomical references: (Heissig, 1995; Kälin & Engesser, 2001)

Remark: *Collimys transversus* is also reported from Belchatow A, Poland (MN 9) (Garapich, 2002). The material is however low in number and is referred to as *Collimys* sp. by other authors (Prieto et al., 2014).

***Colloides* Qiu & Li, 2016**

The only species of the genus is included in the analysis. There is a second species found at Halajin Hushu, China, delimited by its smaller size, but it is not named nor formally

described (Qiu et al., 2021). The status as a genus is doubted by Tesakov et al. (2017), who suggest a possible attribution to *Collimys*.

***Colloides xiaomingi* Qiu & Li, 2016**

Geographical distribution of occurrences: China

Type locality: Balunhalagen, China

Holotype: IVPP V19840, m1 (figured in Qiu and Li (2016), fig. 174G)

Age range: 11.2–7.65 Ma (9.425 Ma)

Oldest occurrence: Amuwusu: MN 9 (Qiu & Li, 2016; Qiu et al., 2021)

Youngest occurrences: Shala: MN 11 (Qiu & Li, 2016; Qiu et al., 2021)

Problematic locality: Balunhalagen: mix of different ages (Qiu & Li, 2016)

Anatomical references: (Qiu & Li, 2016; Qiu et al., 2021)

***Cricetulodon* Hartenberger, 1965**

Of the 7 total species, 1 was not included in the analysis: *Cricetulodon cretensis* (de Bruijn & Meulenkamp, 1972) (see de Bruijn et al., 2012), due to a limited number of specimens found. The 6 species that are included are the following:

***Cricetulodon bugesiensis* Freudenthal, Mein & Martín Suárez, 1998**

Geographical distribution of occurrences: France

Type locality: Soblay, France

Holotype: FSL 65897, m1 (figured in Freudenthal et al. (1998), pl. 1, fig. 1)

Age range: 9.9–8.75 Ma (9.325 Ma)

Oldest occurrence: Douvre: MN 10 (Freudenthal et al., 1998; Mein, 1999)

Youngest occurrences: Dionay, Lo Fournas 16-M: MN 10 (Aguilar, Lazzari, et al., 2007; Freudenthal et al., 1998; Mein, 1999)

Observed Material: FSL 65897: holotype m1; isolated molars from Soblay

Anatomical reference: (Freudenthal et al., 1998)

Remark: Aguilar, Lazzari, et al. (2007) assumed MN 11 and an age of ca. 9.5 Ma for the locality Lo Fournas 16-M, which is considerably older than usual assumptions for the range of this zone, with a maximum of 8.9 or 8.75 Ma (Hilgen et al., 2012; van Dam et al., 2023). We assume an association to MN 10, equal to the other French localities containing *Cricetulodon bugesiensis* (Mein, 1999).

Cricetulodon complicitens Topachevsky & Skorik, 1992

Single occurrence/type locality: Gritsev, Ukraine

Holotype: Schmalhausen Institute of Zoology of National Academy of Sciences of Ukraine, (NASU-IZAN) No. 46, maxilla (figured in Topachevsky and Skorik (1992), fig. 4a)

Age: 9.78–9.6 Ma (9.69 Ma) (Topachevsky & Skorik, 1992)

Anatomical reference: (Topachevsky & Skorik, 1992)

Remarks: The species shows associations to *Neocricetodon* and should possibly be transferred to that genus (Dirnberger et al., 2024).

Gritsev is Bessarabien in age, younger than Buzhor 1 and Kalfa and with reversed palaeomagnetism (Pevzner & Vangengeim, 1993; Vangengeim et al., 2006), therefore it is probably correlated with C4Ar.2r or C4Ar.3r: lower border: 9.78 Ma (Ogg, 2020; see Vasiliev et al., 2011), the upper border follows the upper border of the Bessarabian: 9.6 Ma (Palcu et al., 2021).

Cricetulodon hartenbergeri (Freudenthal, 1967)

Geographical distribution of occurrences: Spain and Turkey

Type locality: Pedregueras 2C, Spain

Holotype: Institut de Paleontologia, Sabadell, PEC 585, m1 (figured in Freudenthal (1967), pl. 1, fig. 14, as *Rotundomys hartenbergeri* and in Erten et al. (2014), fig. 11, 4)

Age range: 10.6–9.49 Ma (10.045 Ma)

Oldest occurrence: Pedregueras 2A (van Dam et al., 2014)

Youngest occurrence: Puente Minero 10 (van Dam et al., 2006)

Observed Material: FSL: isolated molars from Pedregueras 2C. SNSB-BSPG1966 III 91–102: 2 m1, 2 m2, 2 m3, 2 M1, 2 M2 (broken), 2 M3 from Pedregueras 2C

Anatomical references: (Agustí et al., 1979; Erten et al., 2014; Freudenthal, 1967; Freudenthal et al., 1998; Pineda-Muñoz et al., 2011; Şen, 1991)

Remark: *Cricetulodon hartenbergeri* was reported from Hostalets de Pierola Inferior (see Alba et al., 2006), which would be an even older location than Pedregueras 2A. The material was, however, recently assigned to another species, *Democricetodon nemoralis*, which is why this locality is not included here (see SOM S1 in Alba et al., 2022).

Cricetulodon lucentensis (Freudenthal, Lacomba & Martín Suárez, 1991)

Geographical distribution of occurrences: Spain

Type locality: Crevillente 17, Spain

Holotype: Naturalis Biodiversity Center, Leiden, RGM 404 67, m1 (figured in Freudenthal, et al. (1991), pl. 5, fig. 1, as *Neocricetodon lucentensis*)

Age range: 7.17–6.89 Ma (7.03 Ma)

Oldest occurrence: Crevillente 5 (van Dam et al., 2023)

Youngest occurrence: Crevillente 17 (van Dam et al., 2023)

Anatomical references: (Freudenthal et al., 1991, 1998)

Cricetulodon meini (Agustí, 1986)

Single occurrence/type locality: Casa del Acero, Spain

Holotype: Institut de Paleontologia, Sabadell, FCA-237, M1 (figured in Agustí (1986), pl. 2, fig. 8, as *Kowalskia meini*)

Age: 7.58–7.53 Ma (7.555 Ma) (van Dam et al., 2023)

Anatomical references: (Agustí, 1986; Freudenthal et al., 1998)

Remark: The material from Maramena, Greece described as *Allocricetus* cf. *ehiki* (by Daxner-Höck, 1992), shows similarities to *Cricetulodon meini* (Cuenca-Bescós, 2003). With an age of around 5.5 Ma (Hordijk & de Bruijn, 2009) this would be a considerably younger occurrence but due to the uncertainties regarding the taxonomic identification, it is not considered here.

Cricetulodon sabadellensis Hartenberger, 1965

Geographical distribution of occurrences: Spain

Type locality: Can Llobateres, Spain

Holotype: Institut de Paleontologia, Sabadell, CL 1392, maxilla (figured in Hartenberger (1965), pl. 10, fig. c)

Age range: 9.98–9.12 Ma (9.55 Ma)

Oldest occurrences: Can Casablanques 1B, Can Coromines 2, Can Pallars de Llobateres (Casanovas-Vilar et al., 2016; Checa Soler & Rius Font, 2003; Garcés et al., 1996)

Youngest occurrence: Torrent de Febulines M (Casanovas-Vilar et al., 2014; Freudenthal et al., 1998)

Observed specimens: University Montpellier: type material

Anatomical references: (Freudenthal et al., 1998; Hartenberger, 1965)

Remarks: Can Coromines 2 is associated to chron C5n.1r (Agustí et al., 1997), Can Pallars de Llobateres belongs to the *Cricetulodon hartenbergeri* – *Progonomys hispanicus* interval subzone (Casanovas-Vilar et al., 2016). The lower boundaries are shared (9.98 Ma). Can

Casablanques 1B is not precisely dated but can be assumed to fall in the same range. According to Casanovas-Vilar et al. (2016) the *Cricetulodon* species in Autopista de Rubí-Terrassa 8 and 3B, Spain are *C. cf. hartenbergeri* and *C. hartenbergeri*. But according to López-Antoñanzas et al. (2014), it is *C. sabadellensis*, which would make Autopista de Rubí-Terrassa 8 the oldest occurrence but due to the uncertainties regarding the taxonomic identification, it is not considered here.

***Hattomys* Freudenthal, 1985**

All 3 accepted species are included in the analysis:

***Hattomys beetsi* Freudenthal, 1985**

Geographical distribution of occurrences: Italy

Type locality: Biancone 1, Italy

Holotype: Naturalis Biodiversity Center, Leiden, RGM 263 775, mandible (figured in Freudenthal (1985), pl. 1, fig. 1)

Age range: 7.02–4.19 Ma (5.605Ma)

Oldest occurrence: Biancone 1: MN 13 (Freudenthal, 1985; Kälin, 1999; Savorelli, 2013)

Youngest occurrence: Trefossi 1 or Chiro 19: MN 13/14 (Freudenthal, 1985; Savorelli, 2013)

Anatomical references: (Freudenthal, 1985; Savorelli, 2013)

Remark: For all localities of *Hattomys*, an age of MN 13/14 is used (Freudenthal & Martín Suárez, 2006, 2010; Kälin, 1999).

***Hattomys gargantua* Freudenthal, 1985**

Geographical distribution of occurrences: Italy

Type locality: Chiro 2N, Italy: MN 13/14? (Kälin, 1999)

Holotype: Naturalis Biodiversity Center, Leiden, RGM 263 55, cranium with maxilla (figured in Freudenthal (1985), pl. 3, fig. 3 and pl. 5, fig. 10–11)

Age range: 7.02–4.19 Ma (5.605Ma)

Observed material: SNSB-BSPG2004 IX 4: mandible with m1, m2, m3 from Monte Gargano, Fissure GP XI

Anatomical references: (Freudenthal, 1985; Savorelli, 2013)

***Hattomys nazarii* Freudenthal, 1985**

Geographical distribution of occurrences: Italy

Type locality: Nazario 2B, Italy: MN 13/14? (Kälin, 1999)

Holotype: Naturalis Biodiversity Center, Leiden, RGM 263 574, m1 (figured in Freudenthal (1985), pl. 1, fig. 18)

Age range: 7.02–4.19 Ma (5.605 Ma)

Anatomical references: (Freudenthal, 1985; Savorelli, 2013)

***Hypsocricetus* Daxner-Höck, 1992**

The only species of the genus is included in the analysis:

***Hypsocricetus strimonis* Daxner-Höck, 1992**

Single occurrence/type locality: Maramena, Greece

Holotype: Department Aardwetenschappen, University Utrecht, MAA1 1812, maxilla (figured in Daxner-Höck (1992), fig. 20, 1a, b)

Age: 5.8–5.3 Ma (5.55 Ma) (Hordijk & de Bruijn, 2009)

Anatomical references: (Daxner-Höck, 1992; Vasileiadou & Sylvestrou, 2022)

***Nannocricetus* Schaub, 1934**

There are 3 species described as *Nannocricetus*. A fourth one, ‘*Nannocricetus wuae*’ Zhang, Wang, Liu & Liu, 2011 is considered to belong to *Democricetodon*, due to several ‘primitive’ morphological traits (Li et al., 2017; Qiu & Li, 2016). The material is relatively scarce, as there is e.g., no m3 and only a single partly broken M3 (Z.-Q. Zhang et al., 2011). The remaining 3 species are included here:

***Nannocricetus mongolicus* Schaub, 1934**

Geographical distribution of occurrences: China

Type locality: Ertemte 1, China (Wu, 1991)

Holotype: mandible (figured in Schaub (1934), pl.-fig. 3) (as cited in Wu, 1991)

Age range: 7.14–2.58 Ma (4.86 Ma)

Oldest occurrence: Baode-JJG02 (Kaakinen et al., 2013)

Youngest occurrences: several occurrences in the Nihewan basin (Cai et al., 2013; see also Liu et al., 2021)

Problematic localities: Bilutu and Balunhalagen: mix of different ages (Qiu & Li, 2016)

Anatomical references: (Li, 2010; Li et al., 2014; Qiu & Li, 2016; Qiu & Storch, 2000; Wu, 1991)

Nannocricetus primitivus Zhang, Zheng & Liu, 2008

Geographical distribution of occurrences: China and Mongolia

Type locality: Lantian, Loc. 12, China

Holotype: IVPP V15700, maxilla (figured in Zhang, Zheng and Liu (2008), fig. 1A)

Age range: 11.6–7.537 Ma (9.569 Ma)

Oldest occurrence: Yihachi (Qiu et al., 2023; R. Zhang et al., 2021)

Youngest occurrence: Baogedawula (Sun et al., 2018)

Problematic localities: Bilutu and Balunhalagen: mix of different ages (Qiu & Li, 2016)

Anatomical references: (Daxner-Höck et al., 2022; Li, 2010; Maridet et al., 2014; Qiu & Li, 2008, 2016; Qiu et al., 2023; Z.-Q. Zhang et al., 2008)

Remark: The material of the single occurrence outside of China in Builstyn Khudag, Mongolia was first assigned to an affine form, *Nannocricetus* aff. *primitivus*, due to a ‘higher advanced’ morphology (Maridet et al., 2014). Recent studies assign the fossil however directly to *N. primitivus*, based on additional material of the species found in other Late Miocene localities (Daxner-Höck et al., 2022).

Nannocricetus qiui Li, Stidham, Ni & Li, 2017

Single occurrence/type locality: Tibet Zanda ZD1001, China

Holotype: IVPP V 23220, M1 (figured in Li et al. (2017), fig. 1A)

Age: 4.42 Ma (Li et al., 2017; X. Wang et al., 2023)

Anatomical reference: (Li et al., 2017)

Neocricetodon Schaub, 1934

There are 31 species that have been associated with *Neocricetodon* at times (Daxner-Höck et al., 1996; Freudenthal et al., 1998; Qiu & Li, 2016; Sinita & Delinschi, 2016). We follow Freudenthal et al. (1998) in the question of synonymy of *Kowalskia* and *Neocricetodon* (see Dirnberger et al., 2024) and for the European species but add *N. progressus* (see Sinita, 2012). For the Chinese species, Qiu and Li (2016) is followed (note that in the English translation only, *N. similis* is missing in their list (Qiu & Li, 2016, pp. 633-634)). Differing from their list, we include *N. grangeri* but exclude *N. zhengi* (Qiu & Storch, 2000) and *N. neimengensis* (Wu, 1991) (see Sinita & Delinschi, 2016). Therefore, there are 20 species

included in the genus. Five of these species were not included in the analysis: *N. lavocati* (Hugueney & Mein, 1965), *N. lii* (Zheng, 1993), *N. schaubi* (Kretzoi, 1951), *N. seseae* Aguilar, Calvet & Michaux, 1995, *N. yinanensis* (Zheng, 1984a). Very recently one additional species of the genus was described as ‘*Kowalskia weni*’ Feroz, Li, Fazal, Qiu & Ni, 2025. Due to the recency, the species could not be incorporated in this study. The 15 species that are included in the analysis are the following:

Neocricetodon ambarrensis Freudenthal, Mein & Martín Suárez, 1998

Geographical distribution of occurrences: France and Spain

Type locality: Ambérieu 2C, France

Holotype: FSL 65907, m1 (figured in Freudenthal, Mein and Martín Suárez (1998), pl. 3, fig. 1)

Age range: 9.9–8.75 Ma (9.325 Ma)

Oldest occurrence: Ambérieu 1: MN 10 (Freudenthal et al., 1998; Mein, 1999)

Youngest occurrence: Ambérieu 2A: MN 10 (Freudenthal et al., 1998; Mein, 1999)

Anatomical references: (Freudenthal et al., 1998; Sinitsa & Delinschi, 2016)

Neocricetodon browni (Daxner-Höck, 1992)

Geographical distribution of occurrences: Greece and Moldova

Type locality: Maramena, Greece

Holotype: Department Aardwetenschappen, University Utrecht, MAA1 1343, M1 (figured in Daxner-Höck (1992), fig. 18/3, as *Kowalskia browni*)

Age range: 7.68–5.3 Ma (6.49Ma)

Oldest occurrence: Chimishlija (Cimislija): lower border of MN 12 (Lungu & Delinschi, 2008)

Youngest occurrence: Maramena (Daxner-Höck, 1992; Hordijk & de Bruijn, 2009)

Anatomical references: (Daxner-Höck, 1992; Sinitsa & Delinschi, 2016; Vasileiadou & Sylvestrou, 2022)

Remark: The material from Maramena is mentioned as ‘*Kowalskia fahlbuschi*’ in van Kolfschoten and van der Meulen (1986) and in de Bruijn (1989).

Neocricetodon fahlbuschi (Bachmayer & Wilson, 1970)

Geographical distribution of occurrences: Austria, Hungary, and Spain

Type locality: Kohfidisch, Austria

Holotype: Museum of Natural History, Vienna (NHM), No. 1970/1393, maxilla (figured in Bachmayer and Wilson (1970), pl. 11, fig. 57 and in Bachmayer and Wilson (1980), pl. 2, fig. 9, as *Kowalskia fahlbuschi*)

Age range: 9.105–7.6 Ma (8.353 Ma)

Oldest occurrence: Sümeg (Kordos, 1987)

Youngest occurrence: Ronda Oest de Sabadell A1 (Casanovas-Vilar et al., 2016)

Observed material: SNSB-BSPG1972 XXII 1–4: 2 m2, m3, M2 from Kohfidisch

Anatomical references: (Bachmayer & Wilson, 1970, 1978, 1980; Daxner-Höck, 1972, 1992; Daxner-Höck & Höck, 2015; Freudenthal et al., 1998; Kordos, 1987; Sinita & Delinschi, 2016)

Remark: The type material is numbered KO 137 1-157 according to Freudenthal et al. (1998). Different ages of the locality Sümeg are given in several papers: MN 10 (e.g., Kordos, 1987), MN 11 (e.g., Hír & Kókay, 2010) or MN 12 (e.g., Fejfar et al., 1997). Bernor et al. (2003) give an absolute age of 9.7–9.0 Ma. Here, Angelone and Čermák (2015) are followed, who gave an overview of the discussions in the past and placed the locality between MN 10 and MN 11. Following their fig. 1, the age is within chron C4An or C4r (9.105–8.125 Ma, following Ogg (2020)).

***Neocricetodon grangeri* (Young, 1927)**

Geographical distribution of occurrences: China

Type locality: Chia yu Tsun, Yushe-YS161, China (following Wu & Flynn, 2017)

Holotype: Uppsala University (Lagrelus coll.), Yu She 173, mandibles, cranium, postcranial fragments (figured in Daxner-Höck et al. (1996), fig. 16.1 and in Wu and Flynn (2017), fig. 10.2)

Age range: 6.727–4.3 Ma (5.514 Ma) (Wu & Flynn, 2017)

Oldest occurrences: Yushe-YS139, 142 (Flynn & Wu, 2017; Tedford et al., 2013)

Youngest occurrences: Yushe-YS4, 97 (Opdyke et al., 2013; Wu & Flynn, 2017)

Anatomical references: (Daxner-Höck et al., 1996; Sinita & Delinschi, 2016; Wu & Flynn, 2017)

Remark: According to Opdyke et al. (2013), fig. 4.8, the locality Yushe-YS97 is in the Mazegou Formation correlated with C2An.3n: 3.596–3.33 Ma (Ogg, 2020). This age is considerably younger than the remaining localities of *N. grangeri*. Therefore, Tedford et al. (2013), fig. 3.4 and Wu and Flynn (2017), fig. 10.6 are followed, according to whom Yushe-YS97 is close in age to Yushe-YS4. That way, the age of Yushe-YS97, 4.3 Ma, as mentioned in Wu and Flynn (2017), is congruent with the age of Yushe-YS4, as determined by Opdyke et al. (2013), correlated with the base of C3n.1n: 4.3 Ma (Ogg, 2020).

***Neocricetodon hanae* (Qiu, 1995)**

Geographical distribution of occurrences: China

Type locality: Lufeng-shihuiba, China

Holotype: Institute of Vertebrate Paleontology and Paleoanthropology of the Chinese Academy of Sciences, Beijing (IVPP), V10843, M1 (figured in Qiu (1995), pl. 1, fig. 1, as *Kowalskia hanae*)

Age range: 7.15–6.2 Ma (6.675 Ma)

Occurrences: Leilao 9903–9906 (Dong & Qi, 2013; Ni & Qiu, 2002): 7.15–7.10 Ma; Lufeng-shihuiba (Dong & Qi, 2013; Qiu, 1995): ca. 6.9–6.2 Ma

Anatomical references: (Qiu, 1995; Sinita & Delinschi, 2016)

***Neocricetodon intermedius* (Fejfar, 1970)**

Single occurrence/type locality: Ivanovce, Slovakia

Holotype: SÚÚg, OF ‘Geologische Zentralanstalt’, Prague, No. 652376, mandible (figured in Fejfar (1970), fig. 1a, as *Kowalskia intermedia*)

Age: 4.95–3.18 Ma (4.065 Ma) MN 15

Observed material: SNSB-BSPG 73272, 65125, 65128, 652382, -13, -92, -50, -88, -77, -93: casts of 9 isolated molars and a maxilla of ‘collection Prague’ from Ivanovce

Anatomical references: (Fejfar, 1970; Sinita & Delinschi, 2016)

Remarks: De Bruijn (1989) and Koufos (2006) mention ‘*Kowalskia intermedia*’ in Lefkon, Greece. There is however no other reference mentioning this occurrence and between Lefkon (MN 10/11) and the type locality Ivanovce (MN 15) is such a large gap, that Lefkon is not included, here.

The locality Ivanovce is mentioned in several papers, a more precise age of the locality than MN 15 was however not found (Fejfar, 1970, 2001; Fejfar et al., 2016; Fejfar & Sabol, 2004; Kordos, 1987; Sabol et al., 2004). The boundaries of MN 15 follow Hordijk and de Bruijn (2009) and van Dam et al. (2023).

***Neocricetodon magnus* (Fahlbusch, 1969)**

Geographical distribution of occurrences: Hungary, Italy and Poland

Type locality: Podlesice, Poland

Holotype: Institute of Systematic and Experimental Zoology, Polish Academy of Sciences, Cracow, MF/823/1, m1 (figured in Fahlbusch (1969), pl. 17, fig. 1, as *Kowalskia magna*)

Age range: 5.47–3.18 Ma (4.325 Ma)

Oldest occurrence: Osztramos 1b, c, e: MN 14 (Jánossy & Kordos, 1977)

Youngest occurrence: Osztramos 10: MN 15 (Jánossy & Kordos, 1977)

Anatomical references: (Colombero et al., 2014; Fahlbusch, 1969; Jánossy, 1972; Kordos, 1987; Pradel, 1988)

Remark: The occurrence of *N. magnus* in the latest Pliocene localities Zalesiaki 1B and Zamkowa Dolna Cave B is based on a single m1 and on a single m3, respectively (Pradel, 1988). Due to the limited material, these occurrences are not considered here.

Neocricetodon moldavicus (Lungu, 1981)

Geographical distribution of occurrences: Moldova

Type locality: Kalfa, Moldova

Holotype: Department of Geography, Tiraspol State University, Kishinev, TSU Caf-2425, mandible (figured in Lungu (1981), pl. 15, figs. 1a, b, as *Kowalskia moldavica* and in Sinitsa and Delinschi (2016), figs. 2.2, 4.2, 4.1)

Age range: 9.937–9.786 Ma (9.862 Ma)

Occurrences: Buzhor 1, Kalfa (Lungu, 1981; Sinitsa & Delinschi, 2016)

Observed material: FSL: TSU Caf-2425: casts of holotype: m1–3, M1–3

Anatomical references: (Lungu, 1981; Sinitsa & Delinschi, 2016)

Remarks: Buzhor 1 and Kalfa are associated with the Bessarabian, probably correlated with C5n1n: 9.937–9.786 Ma (Ogg, 2020; see Vasiliev et al., 2011)

Neocricetodon moldavicus is mentioned from two additional localities in Moldova, Hirova (Girovo) and Veverița 1 (Veveritsa 1) (Lungu & Rzebik-Kowalska, 2011). This material does, however, not resemble *N. moldavicus* or any other species of *Neocricetodon*, according to Sinitsa and Delinschi (2016).

Neocricetodon nestori (Engesser, 1989)

Single occurrence/type locality: Baccinello V3, Italy

Holotype: NMB, Bac. 242, M2 (figured in Engesser (1989), fig. 16A, as *Kowalskia nestori*)

Age: 6.7–6.4 Ma (6.55 Ma) (Engesser, 1989; Rook, 2016)

Anatomical references: (Engesser, 1989; Sinitsa & Delinschi, 2016)

Remark: The type locality is given as ‘Podere Santa Croce 1’ (Engesser, 1989; Kálin, 1999), sometimes just the faunal horizon ‘Baccinello V3’ is given (Colombero et al., 2014; Rook et al., 1996). The age range follows Rook (2016), where only the horizon is given.

Neocricetodon occidentalis (Aguilar, 1982)

Geographical distribution of occurrences: Spain

Type locality: Crevillente 2, Spain

Holotype: Department Aardwetenschappen, University Utrecht, CR2 63, M1 (figured in de Bruijn et al. (1975), pl. 4, fig. 10, as *Kowalskia fahlbuschi*)

Age range: 8.51–7.62 Ma (8.065 Ma)

Oldest occurrence: Masada Ruea 2 (van Dam et al., 2023)

Youngest occurrence: Concud Barranco de las Calaveras (van Dam et al., 2023)

Anatomical references: (de Bruijn et al., 1975; Freudenthal et al., 1991, 1998; Sinita & Delinschi, 2016; van de Weerd, 1976)

Remark: Freudenthal et al. (1998), also mentioned *Neocricetodon occidentalis* in the considerably older locality Masia del Barbo 2B but expressed reservations due to a limited number of specimens.

Neocricetodon polonicus (Fahlbusch, 1969)

Geographical distribution of occurrences: France, Hungary, Poland, and Ukraine

Type locality: Podlesice, Poland

Holotype: Institute of Systematic and Experimental Zoology, Polish Academy of Sciences, Cracow, MF/822/1, mandible (figured in Fahlbusch (1969), pl. 16, fig. 1a–c, as *Kowalskia polonica*)

Age range: 5.47–3.18 Ma (4.325 Ma)

Oldest occurrences: Hauterives: MN 14 (Freudenthal et al., 1998); Osztramos 1b, c, e: MN 14 (Jánossy, 1972; Jánossy & Kordos, 1977; Kordos, 1987); Podlesice: MN 14 (Fahlbusch, 1969; Fejfar et al., 2011; Freudenthal et al., 1998; Nadachowski, 1989)

Youngest occurrence: Osztramos 10: MN 15 (Jánossy & Kordos, 1977)

Observed material: SNSB-BSPG1976 I 2–7: m1–m3, M1–M3 from Podlesice

Anatomical references: (Fahlbusch, 1969; Freudenthal et al., 1998; Jánossy, 1972; Kordos, 1987; Pradel, 1988; Sinita & Delinschi, 2016)

Remarks: The occurrence of *N. polonicus* in the latest Pliocene to Pleistocene localities Zamkowa Dolna Cave B and Rebielice Królewskie 1A is based on one m2 and m3 and on a single m2, respectively (Pradel, 1988). Due to the limited material, these occurrences are not considered here.

There are also 6 additional occurrences in China of *N. neimengensis* and *N. zhengi*, that are

synonyms of *N. polonicus* according to Sinita and Delinschi (2016). They are not considered here. If they are included, Ertemte might be the oldest occurrence (MN 12) (Wu, 1991).

Neocricetodon progressus (Topachevsky & Skorik, 1992)

Geographical distribution of occurrences: Romania and Ukraine

Type locality: Novo-Elizavetovka 2, Ukraine (see Sinita, 2012)

Holotype: NASU-IZAN No. 47, maxilla (figured in Topachevsky and Skorik (1992), fig. 19, as *Kowalskia progressa*)

Age range: 8.5–7.65 Ma (8.075 Ma)

Occurrences: Dolhești-1: MN 11 (Badea et al., 2025); Palievo (Sinita, 2012; Sinita & Nesin, 2018); Maloye, Novo-Elizavetovka 2, Otradovo: upper MN 11 (Nesin & Storch, 2004; Sinita, 2012; Sinita & Nesin, 2018)

Anatomical references: (Sinita, 2012; Sinita & Delinschi, 2016; Topachevsky & Skorik, 1992)

Neocricetodon shalaensis (Qiu & Li, 2016)

Geographical distribution of occurrences: China

Type locality: Shala, China

Holotype: IVPP V19859, M1 (figured in Qiu and Li (2016), fig. 182A, as *Kowalskia shalaensis*)

Age range: 11.2–7.537 Ma (9.368 Ma)

Oldest occurrence: Halajin Hushu: MN9-10 (Qiu et al., 2021)

Youngest occurrence: Baogedawula (Sun et al., 2018)

Anatomical references: (Qiu & Li, 2016; Qiu et al., 2021)

Neocricetodon similis (Wu, 1991)

Geographical distribution of occurrences: China

Type locality: Ertemte 2, China

Holotype: IVPP V8723.50, M1 (figured in Wu (1991), pl. 2, fig. 22, as *Kowalskia similis*)

Age range: 6.5–3.6 Ma (5.05 Ma)

Oldest occurrence: Ertemte 2 (Qiu & Li, 2016; Qiu et al., 2013; Wu, 1991)

Youngest occurrence: Daodi_Laowogou-2 (Cai et al., 2013)

Anatomical references: (Sinitsa & Delinschi, 2016; Wu, 1991)

Remark: Qiu and Li (2016) ‘arbitrarily’ assign five molars from Balunhalagen, China and four molars from Bilutu, China to *Neocricetodon similis* (*‘Kowalskia similis’*), due to a close geographic position to Ertemte and Harr Obo. Especially Balunhalagen (up to 11 Ma) is however considerably older than Ertemte and Harr Obo and due to poor material and an arbitrary assignment (Qiu & Li, 2016), these localities are not considered here.

***Neocricetodon skofleki* (Kordos, 1987)**

Geographical distribution of occurrences: Austria, France, Hungary and Slovakia

Type locality: Tardosbánya, Hungary

Holotype: Paleontological Department, Hungarian Natural History Museum, Budapest, V.86.2, mandible (figured in Kordos (1987), pl. 1, fig. 1, as *Karstocricetus skofleki*)

Age range: 8.75–6.9 Ma (8.2 Ma)

Oldest occurrence: Ambérieu 3: MN 11 (Freudenthal et al., 1998; Mein, 1999)

Youngest occurrences: Tardosbánya, Šalgovce: MN 12 (Kordos, 1987; Sabol et al., 2021)

Observed material: FSL: casts of M1–3, m1–3 from Tardosbánya

Anatomical references: (Daxner-Höck, 1972; Daxner-Höck & Höck, 2015; Freudenthal et al., 1998; Kordos, 1987; Sinitsa & Delinschi, 2016)

Remark: The material of Dionay and Ambérieu 2C, is only associated to *N. skofleki* with doubt in Freudenthal et al. (1998), while the material of Ambérieu 1 and 2A is not mentioned at all.

***Odessamys* Topachevsky & Skorik, 1992**

‘Cricetulus simionescui’ Schaub, 1931, described from Berești, Romania, was assigned to the newly erected subgenus *Moldavimus* Samson & Radulescu, 1973 and later to the new genus *Odessamys* Topachevsky & Skorik, 1992. Recently, Crespo et al. (2023), re-examined the material and transferred it to *Allocricetus* indet. The type species *O. palatocristatus* and *O. simionescui* are both included in this study (following Topachevsky & Skorik, 1992):

***Odessamys palatocristatus* Topachevsky & Skorik, 1992**

Geographical distribution of occurrences: Moldova and Ukraine

Type locality: Odessa Catacombs, Ukraine

Holotype: NASU-IZAN No. 24, maxilla (figured in Topachevsky and Skorik (1992), fig. 53a)

Age range: 4.27–3.18 Ma (3.725 Ma), Tătărești: MN 15 (Nadachowski et al., 2006)

Anatomical reference: (Topachevsky & Skorik, 1992)

Odessamys simionescui (Schaub, 1931)

Geographical distribution of occurrences: Moldova, Romania and Russia

Type locality: Berești, Romania

Lectotype: mandible without collection number, stored either in the Laboratory of Palaeontology, Faculty of Geology and Geophysics, University of Bucharest (see Samson & Rădulescu, 1973) or in the ‘Emil Racoviță’ Institute of Speleology, Romanian Academy, Bucharest (see Crespo et al., 2023) (figured in Schaub (1931), fig. 1, as *Cricetulus simionescui*)

Age range: 5.47–3.18 Ma (4.325Ma), Berești: MN 14–MN 15 (Crespo et al., 2023)

Anatomical reference: (Crespo et al., 2023; Samson & Rădulescu, 1973; Schaub, 1931; Topachevsky & Skorik, 1992)

Pseudocollimys Daxner-Höck, 2004

The status as a genus is doubted by Tesakov et al. (2017). The only species of the genus is included in the analysis.

Pseudocollimys steiningeri Daxner-Höck, 2004

Single occurrence/type locality: Schernham bei Haag, Austria

Holotype: NHM 2003z0043/0001, M1 (figured in Daxner-Höck (2004), pl. 1, fig. 1a-c)

Age: 9.1–8.9 Ma (9 Ma) (Daxner-Höck et al., 2016)

Anatomical references: (Daxner-Höck, 2004; Daxner-Höck & Höck, 2015)

Pseudocricetus Topachevsky & Skorik, 1992

Of the 4 total species, one was not included in the analysis: *Pseudocricetus antiquus* Topachevsky & Skorik, 1992, due to a limited amount of found specimens. The 3 species that are included are the following:

Pseudocricetus kormosi (Schaub, 1930)

Geographical distribution of occurrences: Ukraine and Hungary

Type locality: Polgardi 2, Hungary

Lectotype: Palaeovertebrate Collection of the Hungarian Geological Institute, Budapest, Ob. 4185, mandible (figured in Freudenthal and Kordos (1989), fig. 8, 1, as *Cricetus kormosi*)

Age range: 7.02–5.6 Ma (6.31 Ma)

Occurrences: Polgardi 2 (Freudenthal & Kordos, 1989); Andreevka (Nesin & Storch, 2004; Sinita, 2010); Novoukrainka 1, Orekhova, (Nesin & Storch, 2004; Sinita, 2010; Topachevsky & Skorik, 1992); Odessa Pontian Lectostratotype (Nesin & Storch, 2004; Topachevsky & Skorik, 1992); Nerubayskoe/NATI 2 (Sinita & Delinschi, 2022)

Observed Material: FSL: casts of type material: Ob. 4185 M1–3 and M1–2, M1; originals from NMB & Paleontological Department, Hungarian Natural History Museum, Budapest. SNSB-BSPG 1979 I 154: casts of type material: 3 maxilla fragments with M1, M2 and 3 mandible fragments with m1, m2, m3 of the Hungarian Natural History Museum, Budapest

Anatomical references: (de Bruijn et al., 1975; Freudenthal & Kordos, 1989; Sinita, 2010; Topachevsky & Skorik, 1992)

Remark: Following Nesin and Storch (2004) all the Ukrainian localities belong to MN 13 and the upper Maeotian or Pontian. Therefore, the lower border of MN 13 of van Dam et al. (2023), and the upper border of the Pontian is taken (following Krijgsman et al., 2010; Vasiliev et al., 2011). The Hungarian locality Polgardi 2 (MN 13, Pontian) is assumed to fall in this range as well (Freudenthal & Kordos, 1989).

***Pseudocricetus orienteuropaeus* Topachevsky & Skorik, 1992**

Geographical distribution of occurrences: Ukraine

Type locality: Protopovka 3, Ukraine (see Sinita, 2010)

Holotype: NASU-IZAN No. 50, mandible (possibly figured in Topachevsky and Skorik (1992), fig. 34a or 36, there is no collection number given for the figured mandibles)

Age range: 7.68–6.04 Ma (6.86 Ma)

Occurrences: Belka 2; Egorovka 1, 2; Kubanka 2; Novo-Elizavetovka 3; Protopopovka 3 (Nesin & Storch, 2004; Rosina & Sinita, 2014; Sinita, 2010; Sinita & Nesin, 2018)

Anatomical references: (Sinita, 2010; Topachevsky & Skorik, 1992)

Remarks: All the localities belong to MN 12 and the Maeotian. Therefore, the lower border of MN 12 of van Dam et al. (2023), and the upper border of the Maeotian is taken (following Krijgsman et al., 2010; Vasiliev et al., 2011). As there is disagreement about the border of MN 12 to MN 13 in the Maeotian, the here used absolute ages overlap.

Pseudocricetus polgardiensis (Freudenthal & Kordos, 1989)

Geographical distribution of occurrences: Hungary

Type locality: Polgardi 4, Hungary

Holotype: Palaeovertebrate Collection of the Hungarian Geological Institute, Budapest, V. 14033 (Vt. 126), mandible (figured in Freudenthal and Kordos (1989), figs. 1, 1; 5, as *Cricetus polgardiensis*)

Age range: 7.02–5.6 Ma (6.31 Ma)

Occurrences: Polgardi 4 (Freudenthal & Kordos, 1989); Polgardi 3 (Freudenthal & Kordos, 1989; Kordos, 2022; Mihály, 1975)

Anatomical reference: (Freudenthal & Kordos, 1989)

Remark: The age is assumed to be MN 13/Pontian (Freudenthal & Kordos, 1989). The age of Polgardi 2 is followed (see remark of *P. orienteuropaeus*).

***Rotundomys* Mein, 1965**

Of the 6 total species, 1 was not included in the analysis: *Rotundomys mundi* Calvo, Elizaga, López-Martínez, Robles & Usera, 1979. The 5 species that are included are the following:

Rotundomys bressanus Mein, 1975

Geographical distribution of occurrences: France and Spain

Type locality: Soblay, France

Holotype: FSL 65443, M1

Age range: 9.9–8 Ma (8.95 Ma)

Oldest occurrence: Soblay: MN 10 (Mein, 1999)

Youngest occurrence: Bernardière: Early MN 11 (Mein, 1999)

Observed material: FSL: casts of type material

Anatomical references: (Fejfar et al., 2011; Freudenthal et al., 1998; López-Antoñanzas et al., 2014; Mein, 1975; Topachevsky & Skorik, 1992)

Remark: A locality ‘Santa Margarida’ is mentioned with an occurrence of *R. bressanus* with no further comment in Freudenthal et al. (1998). There is a locality with that name from Portugal mentioned in Antunes and Mein (1989), with an age of ca. 6 Ma, which is younger than all other localities of *R. bressanus*. In NOW there is a locality ‘Santa Margarida’ in Spain, associated with MN 5, which is too old for *R. bressanus*. For neither of the two localities, *R. bressanus* is mentioned in the respective references.

Rotundomys freiriensis Antunes & Mein, 1979

Geographical distribution of occurrences: Portugal and Spain

Type locality: Freiria do Rio Manor, Portugal

Holotype: Department of Earth Sciences, NOVA School of Science and Technology, NOVA University Lisbon, M1 (figured in Antunes and Mein (1979), pl. 1, fig. a, i)

Age range: 9.87–9.62 Ma (9.745 Ma)

Occurrences: Asseiceira (Antunes et al., 1992); Freiria do Rio Manor (Antunes & Mein, 1979); Autopista de Rubí-Terrassa 7C, 11 (Casanovas-Vilar et al., 2016)

Observed material: FSL: casts of type material

Anatomical references: (Antunes & Mein, 1979; Antunes et al., 1992; Freudenthal et al., 1998; López-Antoñanzas et al., 2014)

Remark: The localities Autopista de Rubí-Terrassa 7C and 11 are associated to the *Cricetulodon sabadellensis* + *Progonomys hispanicus* concurrent range subzone (= 9.73–9.65 Ma) of Casanovas-Vilar et al. (2016), which falls within the range of the J1 subzone (= 9.87–9.62 Ma) of van Dam et al. (2006), that Asseiceira and Freiria do Rio Manor are correlated to (Mein, 2000).

Rotundomys intimus López-Antoñanzas, Peláez-Campomanes & Álvarez-Sierra, 2014

Geographical distribution of occurrences: Spain

Type locality: Cerro de los Batallones 5, Spain

Holotype: National Museum of Natural Sciences, Madrid, BAT5'10-07, mandible (figured in López-Antoñanzas, Peláez-Campomanes and Álvarez-Sierra (2014), fig. 3A)

Age range: 9.71–8.75 Ma (9.23 Ma)

Oldest occurrence: Cerro de los Batallones 10 (López-Antoñanzas et al., 2014, 2010; Martín-Perea et al., 2021)

Youngest occurrence: Cerro de los Batallones 3 (López-Antoñanzas et al., 2014, 2010; Martín-Perea et al., 2021)

Anatomical reference: (López-Antoñanzas et al., 2014)

Remark: The Cerro de los Batallones system is placed in the local subzones J2 or J3 (López-Antoñanzas et al., 2010), so this range is taken from van Dam et al. (2006).

Rotundomys montisrotundi (Schaub, 1944)

Geographical distribution of occurrences: France and Spain

Type locality: Montredon, France

Holotype: NMB, A Mo 849, m1 (figured in Mein (1965), fig. 1a and in Fejfar et al. (2011), fig. 2(19a))

Age range: 9.9–8.75 Ma (9.325 Ma)

Occurrences: Ampudia 3; Camí de Can Tarumbot 2, 3; Can Casablanques; Can Casablanques 2; Chabeuil - Les Bourbons; La Tarumba 1; Lo Fournas 6, 7, 1993; Montredon; Racor: MN 10 (Casanovas-Vilar et al., 2016; Guerra-Merchán et al., 2001; López Martínez et al., 1986; Mein, 1999)

Anatomical references: (Aguilar, 1982; Aguilar, Michaux, et al., 2007; Fejfar et al., 2011; Freudenthal et al., 1998; Guerra-Merchán et al., 2001; López-Antoñanzas et al., 2014; Mein, 1965)

Rotundomys sabatieri Aguilar, Michaux & Lazzari, 2007

Geographical distribution of occurrences: France

Type locality: Lo Fournas 16-M, France

Holotype: Faculty of Science, University of Montpellier, Fou 16-M no 395, m1 (figured in Aguilar, Michaux, et al. (2007), pl. 2, fig. 2)

Age range: 9.9–8.75 Ma (9.325 Ma)

Oldest occurrence: Lo Fournas 6 (Aguilar, Lazzari, et al., 2007; Aguilar, Michaux, et al., 2007)

Youngest occurrence: Lo Fournas 16-M (Aguilar, Lazzari, et al., 2007; Aguilar, Michaux, et al., 2007)

Anatomical references: (Aguilar, Michaux, et al., 2007; López-Antoñanzas et al., 2014)

Remark: Both localities are considered MN 10, which fits with the age given for Lo Fournas 16-M (± 9.5 Ma) in Aguilar et al. (2007).

Sinocricetus Schaub, 1930

There are 3 species, that are undoubtedly members of *Sinocricetus*. Additionally, ‘*Kowalskia complicidens*’ Topachevsky & Skorik, 1992 was excluded from *Neocricetodon* (or ‘*Kowalskia*’) by Daxner-Höck et al. (1996) and tentatively transferred to *Sinocricetus* by Sinitsa and Delinschi (2016), as was ‘*Kowalskia dalinica*’ Wang, 1988 (*Sinocricetus dalinicus*) by Xie et al. (2021), which is however only known from a single m1. The 3 species, certain to be *Sinocricetus*, are included here:

***Sinocricetus major* Li, 2010**

Geographical distribution of occurrences: China

Type locality: Gaotege DB02, China

Holotype: IVPP V17022, m2 (figured in Li (2010), fig. 1G)

Age range: 4.3–4.1 Ma (4.2 Ma)

Oldest occurrence: Gaotege DB02 (Li, 2010; Y. Xu et al., 2007)

Youngest occurrence: Gaotege DB03-1 (Li, 2010; Y. Xu et al., 2007)

Anatomical reference: (Li, 2010)

***Sinocricetus progressus* Qiu & Storch, 2000**

Geographical distribution of occurrences: China

Type locality: Bilike, China

Holotype: IVPP V11914, maxilla (figured in Qiu and Storch (2000), pl. 8, fig. 1)

Age range: 5.333–2.58 Ma (3.957 Ma)

Oldest occurrences: Daodi_Laowogou-2 (Cai et al., 2013)

Youngest occurrences: several occurrences in the Nihewan basin (Cai et al., 2013; see also Liu et al., 2021)

Problematic locality: Bilutu: mix of different ages (Qiu & Li, 2016)

Anatomical references: (Li, 2010; Qiu & Li, 2016; Qiu & Storch, 2000)

Remark: *Sinocricetus zdanskyi* from Daodi, Laowogou (see Z.-Q. Zhang et al., 2003) is considered *S. progressus* (Cai et al., 2013; Li et al., 2008).

***Sinocricetus zdanskyi* Schaub, 1930**

Geographical distribution of occurrences: China

Type locality: Ertemte 1, China (Wu, 1991)

Lectotype: mandible (figured in Schaub (1934), fig. 30) (as cited in Wu, 1991)

Age range: 8.75–4.63 Ma (6.69 Ma)

Oldest occurrence: Shala: MN 11

Youngest occurrences: Lingtai Wenwanggou(93002)-3, Xiaoshigou-2 (Zheng & Zhang, 2001)

Problematic localities: Bilutu and Balunhalagen: mix of different ages (Qiu & Li, 2016)

Anatomical references: (Li, 2010; Schaub, 1930; Schlosser, 1924; Qiu & Li, 2016; Wu, 1991; L.-M. Zhang et al., 2021)

Remark: According to Li (2010), *Sinocricetus zdanskyi*, that has been mentioned in the Zones IV and V of Lingtai, Gansu Province, is actually *S. progressus*. In the descriptions of these localities, however, *S. zdanskyi* is not mentioned in Zone IV or V at all (Z.-Q. Zhang & Zheng, 2000, 2001; Zheng & Zhang, 2000, 2001).

***Stylocricetus* Topachevsky & Skorik, 1992**

The only species of the genus is included in the analysis:

***Stylocricetus meoticus* Topachevsky & Skorik, 1992**

Geographical distribution of occurrences: Kazakhstan, Ukraine and Russia

Type locality: Cherevichnoe 3, Ukraine

Holotype: NASU-IZAN No. 51, maxilla (possibly figured in Topachevsky and Skorik (1992), fig. 51, there is no collection number given for the figured maxilla)

Age range: 7.68–5.33 Ma (4.55 Ma)

Oldest occurrences: Cherevichnoe 3, Novo-Elizavetovka 3, Protopopovka 3 (Topachevsky & Skorik, 1992)

Youngest occurrences: Selety 1A: MN 13 (Maul et al., 2017)

Anatomical reference: (Topachevsky & Skorik, 1992)

Remark: For the age of Cherevichnoe 3, Novo-Elizavetovka 3, Protopopovka 3, see the remark of *Pseudocricetus orienteuropaeus*.

***Tragomys* Agustí, Bover & Alcover, 2012**

The only species of the genus is included in the analysis:

***Tragomys macpheeii* Agustí, Bover & Alcover, 2012**

Single occurrence/type locality: Caló den Rafelino, Spain

Holotype: Institut Mediterrani d'Estudis Avançats in Esporles, IMEDEA 90614, ml (figured in Agustí et al. (2012), fig. 2J)

Age: 5.333–3.6 Ma (4.467 Ma): early Pliocene (Agustí et al., 2012; Bover et al., 2010; Torres-Roig et al., 2019)

Anatomical references: (Agustí et al., 2012; Torres-Roig et al., 2019)

Quaternary taxa***Allocricetulus* Argyropulo, 1933**

There are 2 species, 1 of them, *Allocricetulus curtatus* (Allen, 1925), is not included here. Its status as a species is currently under debate (Pan et al., 2025). The type species is included here:

***Allocricetulus eversmanni* (Brandt, 1859)**

Geographical distribution of fossil occurrences: Moldova, Romania, Russia, and Ukraine

Type locality: near Orenburg, Russia

Distribution: Southern Russia, Northern and Eastern Kazakhstan, North-Western China

Oldest fossil occurrence: Gura Dobrogei 4-2, Romania: 0.374–0.125 Ma: Saalian (Rădulescu & Samson, 1995)

Observed material: French National Museum of Natural History, Paris (MNHN) 2024-1556–1558: 3 skulls with mandibles from Kokehada, Xinjiang, China; 1958-748: skull with mandible from Russia; 1961-711: skull from Russia

Anatomical references: (Kryštufek & Shenbrot, 2025; Lebedev et al., 2018; Topachevsky & Skorik, 1992)

Remarks: *Allocricetulus* and †*Allocricetus* has been synonymised by some authors (see Kryštufek & Shenbrot, 2025). There is ‘*Allocricetus eversmanni*’ mentioned in Cai et al. (2013) and in Tjutkova and Kaipova (1996). As these fossils are distinctively older than any other known *A. eversmanni* fossils, they are treated as †*Allocricetus* indet. and are not included here.

In Markova et al. (2019), there are different combinations mentioned (‘*Allocricetus eversmanni*’, ‘*Allocricetulus eversmanni*’ and *Allocricetulus eversmanni*), while Krokhmal and Rekovets (2010) list for the same locations ‘*Cricetulus eversmanni*’ (and once probably by mistake ‘*Cricetus eversmanni*’).

Uncertainties about †*Allocricetus* vs *Allocricetulus* can also be seen in Serdyuk and Zenin (2014, text vs. fig. 1, 2016, tab. 2 vs. fig. 3).

†*Allocricetus* Schaub, 1930

The status of *Allocricetus* as a valid genus is doubted by several authors (see e.g., van den Hoek Ostende et al., 2015), due to the similarity of their molar morphologies to *Cricetulus*, that can only be separated by proportions of morphotypes (Hír, 1993b). Cuenca-Bescós (2003) kept the two genera separated, based on their phylogenetic analysis. According to this analysis, the genus should be split up, with the type species *A. bursae* on the one side and a new genus including *A. ehiki*, *A. correzensis* (Chaline, 1972) (as cited in Cuenca-Bescós, 2003), *A. croaticus* Paunović & Rabeder, 1996 and *A. anterolophidens* on the other side. *Allocricetus jesreelicus* and *A. teilhardi* should be transferred to *Cricetulus*. Two additional

species, *A. aylasevima* and *A. primitivus*, were described later and are included here, as well. We included the following 7 of the 9 mentioned species, excluding *A. correzensis* and *A. croaticus*:

†*Allocricetus aylasevima* Ünay, de Bruijn & Suata-Alpaslan 2006

Single occurrence/type locality: Corakyerler, Turkey

Holotype: Department Aardwetenschappen, University Utrecht, no. 54, M2 (figured in de Ünay et al. (2006), pl. 1, fig. 19)

Age: 8.75–7.65 Ma (8.2 Ma)

Anatomical reference: (Ünay et al., 2006)

†*Allocricetus anterolophidens* Topachevsky & Skorik, 1992

Geographical distribution of occurrences: Moldova and Ukraine

Type locality: Zhevakhova Gora ‘lower level’, Ukraine (= Zhevakhova Gora 11, following Kowalski (2001))

Holotype: NASU-IZAN No. 31, M2 (possibly figured in Topachevsky and Skorik (1992), fig. 67d or e, there is no collection number given for the figured M2s)

Age range: 4.27–2.54 Ma (3.405 Ma)

Oldest occurrence: Luceşti: MN 15–MN 16a (Nadachowski et al., 2006)

Youngest occurrences: Cherevichnoe 2, Zhevakhova Gora 15: MN 16 (Nesin & Nadachowski, 2001; Topachevsky & Skorik, 1992)

Anatomical references: (Cuenca-Bescós, 2003; Topachevsky & Skorik, 1992)

†*Allocricetus bursae* Schaub, 1930

Geographical distribution of occurrences: Austria, Bulgaria, China, Croatia, Czech Republic, France, Georgia, Germany, Greece, Hungary, Israel, Italy, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Switzerland, Turkey, and Ukraine

Type locality: Fortyogóberg bei Brassó, Romania = Fortyogó or Şprengi hill near Braşov

Holotype: Schaub (1930) did not assign a holotype for *Allocricetus bursae*. There was however one maxilla figured by him (Schaub, 1930, fig. 13), that is usually taken as holotype (according to Agustí et al., 2022).

Age range: 5.47–0.0056 Ma (2.738 Ma)

Oldest occurrence: Igdeli: MN 14 (Suata-Alpaslan, 2010; Suata-Alpaslan et al., 2010)

Youngest occurrences: Buena Pinta Cave 1 (Laplana et al., 2016; López-García, 2008)

Observed material: FSL: 3 mandibles with m1–3 from Villány

Anatomical references: (Aguilar et al., 2001; Agustí et al., 2022; Cuenca-Bescós, 2003; Cuenca-Bescós et al., 1997; Fahlbusch, 1969; Hír, 1992a, 1993b; Hír & Venczel, 1991; López Martínez, 1980; Malez & Rabeder, 1984; Paunović & Rabeder, 1996; Popov, 1988, 1989, 2004, 2017; Pradel, 1988; Schaub, 1930; Sesé, 1989; Sesé, Panera, et al., 2011; Sesé, Rubio-Jara, et al., 2011; Sesé & Ruiz Bustos, 1992; Sesé et al., 2000; Suata-Alpaslan et al., 2010; Topachevsky & Skorik, 1992; Wang Y. et al., 2022)

Remark: Schaub (1930) did also not assign a type locality but the figured maxilla, that is usually taken as holotype, and the only figured molars are from Braşov. (Schaub, 1930, pl. 1, fig. 9).

†*Allocricetus ehiki* Schaub, 1930

Geographical distribution of occurrences: Bulgaria, China, Czech Republic, Greece, Hungary, Italy, Moldova, Poland, Russia, Slovakia, and Ukraine

Type locality: Villány, Kalkberg, Hungary (= Villány 3 according to Horáček and Lebedová (2022) and Kowalski (2001), = Villány (? 5) according to Hordijk and de Bruijn (2009))

Holotype: Schaub (1930) did not assign a holotype for *Allocricetus ehiki*. There is however one lower tooththrow figured by him from Villány (Schaub, 1930, pl. 1, fig. 10).

Age range: 5.23–0.13 Ma (2.68 Ma)

Oldest occurrence: Komanos 1 low (Hordijk & de Bruijn, 2009; Koufos & Vasileiadou, 2015)

Youngest occurrences: Jinpendong (Jin et al., 2004; Q.-Q. Xu et al., 2009)

Observed material: FSL: 3 mandibles with m1–3 from Villány, 1 mandible with m1–2

Anatomical references: (Cuenca-Bescós, 2003; Fahlbusch, 1969; Hír, 1993a, 1993b; Pazonyi et al., 2016; Popov, 2004; Pradel, 1988; Schaub, 1930; Topachevsky & Skorik, 1992; Zheng, 1984b)

Remark: Schaub (1930) did also not assign a type locality, but the only figured molars are from Villány (Schaub, 1930, pl. 1, fig. 9), which was also the locality with the most material of the species, known to him.

†*Allocricetus jesreelicus* Bate, 1943

Geographical distribution of occurrences: Israel

Type locality: Tabun Cave F, Israel

Holotype: NHMUK M 16006, skull with dentition (figured in Bate (1943), figs. 4, 5a, b)

Age range: 0.361–0.103 Ma (0.232 Ma)

Oldest occurrence: Oumm Qatafa Cave (Tchernov, 1968)

Youngest occurrence: Hayonim E (Tchernov, 1998)

Observed material: FSL: 3 mandibles with m1–3 from Oumm Qatafa Cave

Anatomical references: (Bate, 1943; Cuenca-Bescós, 2003; Tchernov, 1968)

Remark: According to Cuenca-Bescós et al. (2003), the species should be transferred to *Cricetulus*.

†*Allocricetus primitivus* Wu & Flynn, 2017

Geographical distribution of occurrences: China

Type locality: Yushe-YS4, China

Holotype: IVPP V9895.1, maxilla (figured in Wu and Flynn (2017), fig. 10.4d)

Age range: 4.799–4.3 Ma (4.55 Ma)

Oldest occurrences: Yushe-YS39, 50 (Opdyke et al., 2013; Wu & Flynn, 2017)

Youngest occurrences: Yushe-YS4, 43 (Opdyke et al., 2013; Wu & Flynn, 2017)

Anatomical references: (Wang Y. et al., 2022; Wu & Flynn, 2017)

Remark: According to Opdyke et al. (2013), Yushe-YS97 would be the youngest occurrence of *A. primitivus* but see the remark of †*Neocricetodon grangeri*.

†*Allocricetus teilhardi* Zheng, 1984b

Geographical distribution of occurrences: China

Type locality: Zhoukoudian 13, China

Holotype: IVPP RV410012, posterior part of skull (figured in Teilhard de Chardin and Pei (1941), fig. 38A) (as cited in Zheng, 1984b)

Age range: 2–0.374 Ma (1,187 Ma)

Oldest occurrences: Fanchang_Renzidong (Dong et al., 2020; Xie et al., 2023)

Youngest occurrences: Zhoukoudian 1 (Shen et al., 2009; Zheng, 1984b)

Anatomical references: (Cuenca-Bescós, 2003; Wang Y. et al., 2022; Zheng, 1984b)

Remark: According to Cuenca-Bescós et al. (2003), the species should be transferred to *Cricetulus*.

***Cricetulus* Milne-Edwards, 1867**

Musser and Carleton (2005) list 6 extant species of that genus. Molecular studies suggest however, a non-monophyly of *Cricetulus* in that broad sense (Lebedev et al., 2018; Neumann et al., 2006; Steppan & Schenk, 2017). Consequently, the genus is split up (see *Nothocricetulus* and *Urocrinetus*) and is assumed to only include 3 extant species, of which 1 is not included here, *Cricetulus sokolovi* Orlov & Malygin, 1988. Regarding the extinct taxa, 7 species have been referred to ‘*Cricetinus*’ Zdansky, 1928, which is synonymized with *Cricetulus* by several authors (see e.g., McKenna & Bell, 1997; Musser & Carleton, 2005). Recently, the type species ‘*Cricetinus varians*’ Zdansky, 1928, has been identified as a subspecies of *Tscherskia triton* (see Xie et al., 2023). The remaining 6 species have been tentatively assigned to different genera (Xie et al., 2023) but for now, they are referred to as *Cricetulus*, here. Of these 6 species, 1 is not included here, †*Cricetulus janossyi* (Hír, 1996a), as no M1 is known to date. In total, the following 7 (of 9) *Cricetulus* species are included in the phylogenetic reconstruction:

***Cricetulus barabensis* (Pallas, 1773)**

Geographical distribution of fossil occurrences: China and Russia

Type locality: South-Western Siberia, Russia

Distribution: Southern Siberia, Mongolia, North-Eastern China, North-Eastern Kazakhstan, North Korea

Oldest fossil occurrence: Yushe-YS120, China: 2.595–2.14 Ma: C2r.2r (Opdyke et al., 2013; Wu & Flynn, 2017)

Observed material: SNSB-BSPG1973 I 211: skull with mandible from an unknown locality. MNHN 1867-152: skull with mandible from Shuen hoa fou, China; 1868-1384: skull with single broken hemimandible, from Siberia, Russia; 1962-2345: skull with mandible from Peking, China.

Anatomical references: (Kryštufek & Shenbrot, 2025; Lebedev et al., 2018; Wu & Flynn, 2017; Xie & Li, 2016; Xie et al., 2023)

***Cricetulus longicaudatus* (Milne-Edwards, 1871)**

Geographical distribution of fossil occurrences: China

Type locality: Northern Shanxi, near Saratsi, China

Distribution: Russia, East Kazakhstan, North-Western China, Mongolia, Tibet

Oldest fossil occurrence: Zhoukoudian 9, China: 1.376–0.774 Ma: late third of the Early Pleistocene (Cheng et al., 1997; Zheng, 1984b)

Observed material: MNHN 2024-1312–1315: 4 skulls with mandibles from Xiji, Ningxia, China; 2024-1368–1315: 4 skulls with mandibles from; –1315: 4 skulls with mandibles from

Anatomical references: (Argyropulo, 1933; Kryštufek & Shenbrot, 2025; Lebedev et al., 2018; Xie & Li, 2016; Xie et al., 2023; Zheng, 1984b)

†*Cricetulus beremendensis* (Hír, 1994a)

Geographical distribution of occurrences: Hungary

Type locality: Beremend 15, Hungary

Holotype: Paleontological Department, Hungarian Natural History Museum, Budapest, no number given, mandible (figured in Hír (1994), figs. 1, 3 as *Cricetinus beremendensis*)

Age range: 3.4–2.7 Ma (3.05 Ma)

Oldest occurrence: Csarnota 4 (Hír, 1994, 1996b)

Youngest occurrence: Beremend 15 (Hír, 1994; Pazonyi, 2011)

Anatomical references: (Hír, 1994, 1996b, 2002; Koufos et al., 2001; Xie et al., 2023)

Remark: According to Xie et al. (2023), the species should be transferred to †*Allocricetus*.

†*Cricetulus europaeus* (Kretzoi, 1959)

Single occurrence/type locality: Csarnota 2

Holotype: Kretzoi (1959) did not formally assign a holotype for this species. He did however specifically address the m1 among the three specimens available to him (M1, M2, m1), which might be the reason, Kordos (1987) calls it the holotype. Hír (1994) lists all three molars as holotype, instead. The M2 is however later identified as †*C. janossyi* (Hír, 1996a), leaving the syntypes: Palaeovertebrate Collection of the Hungarian Geological Institute, Budapest, V. 12769, M1 and m1 (figured in Kretzoi (1962), pl. 4, figs. 1 and 2, in Kordos (1987), figs. 1:10 and 5:6, and in Hír (1994), figs. 6 and 11; as *Cricetinus europaeus*).

Age: 3.4–3.2 Ma (3.3 Ma) (Marciszak et al., 2023)

Anatomical reference: (Hír, 1994, 1996b, 1996a, 2002; Kordos, 1987; Koufos et al., 2001; Kretzoi, 1959, 1962; Xie et al., 2023)

Remark: According to Xie et al. (2021, 2023), the species should be transferred to *Tscherskia*.

†*Cricetulus gritzai* (Topachevsky & Skorik, 1992)

Single occurrence/type locality: Odessa Catacombs, Ukraine

Holotype: NASU-IZAN No. 35, maxilla (figured in Topachevsky and Skorik (1992), fig. 77, as *Cricetinus gritzai*)

Age: 4–3.5 Ma (3.75 Ma) (Tesakov, 2021)

Anatomical reference: (Koufos et al., 2001; Topachevsky & Skorik, 1992; Xie et al., 2023)

Remarks: The measurement of the M3 given by Topachevsky and Skorik (1992) is not taken for the morphological matrix. Its length is larger than in the M2, disagreeing with their diagnosis of ‘*Cricetinus*’. As there is only one M3, this might be a statistical error.

According to Xie et al. (2021, 2023), the species should be transferred to *Tscherskia*.

The species is also mentioned as ‘*C. grizai*’ (The NOW Community, 2023), ‘*C. grizaii*’ (Tesakov, 2021) and ‘*C. gritzaii*’ (Vasileiadou & Sylvestrou, 2022).

There is an additional locality called ‘Angangxi_Daxingtun’, listed for this species in The Now Community (2023). The given reference for this occurrence mentions only *C. cf. griseus* however (reference erroneously given in The Now Community (2023) as: ‘Huang & Zhang (1984). Discovery of paleolithic artifacts at Angangxi, Qiqihaer, Heilongjiang. Acta Geologica Sinica 3 (3)’; see instead: (Huang et al., 1984) in the reference list).

†*Cricetulus koufosi* (Koliadimou, 1996)

Geographical distribution of occurrences: Greece

Type locality: Ravin Vulgarakis, Greece (Vasileiadou & Sylvestrou, 2022)

Holotype: Geology-Paleontology Museum of the School of Geology of the Aristotle University of Thessaloniki, RVL289, mandible (figured in Koliadimou (1996), fig. 4.38a) (as cited in Vasileiadou & Sylvestrou, 2022)

Age range: 1.9–1 Ma (1.45 Ma)

Oldest occurrence: Marathoussa (Doukas & Papayianni, 2016; Koufos et al., 2001)

Youngest occurrence: Ravin Vulgarakis (Doukas & Papayianni, 2016; Koufos et al., 2001)

Anatomical references: (Koufos et al., 2001; Vasileiadou & Sylvestrou, 2022; Xie et al., 2023)

Remark: According to Xie et al. (2021, 2023), the species should be transferred to *Tscherskia*.

†*Cricetulus mesolophidos* (Wu & Flynn, 2017)

Geographical distribution of occurrences: China

Type locality: Yushe-YS97, China

Holotype: IVPP V9885.1, m1 (figured in Wu and Flynn (2017), fig. 10.3i, as *Cricetinus mesolophidos*)

Age range: 4.799–2.6 Ma (3.7 Ma)

Oldest occurrence: Yushe-YS50 (Opdyke et al., 2013; Wu & Flynn, 2017)

Youngest occurrences: Lingtai_Wenwanggou(93001)-4, Lingtai_Xiaoshigou-4 (Zheng & Zhang, 2001)

Anatomical references: (Wu & Flynn, 2017; Xie et al., 2021, 2023)

Remarks: According to Xie et al. (2021, 2023), the species should be transferred to †*Neocricetodon*.

For a comment on the age of Yushe-YS97, see the remark of †*Neocricetodon grangeri*. According to Wu and Flynn (2017), the locality Yushe-YS59 should not be taken into account in the stratigraphic range of this taxon, as the specimen might be recent. The age of the locality is uncertain. According to Opdyke et al. (2013), fig. 4.8, it is in the Mahui Formation but according to Wu and Flynn (2017), it is in the Taoyang Member of the Gaozhuang Formation. Either way, this age would be considerably older than the remaining localities of †*C. mesolophidos*. On the other hand, following Tedford et al. (2013), fig. 3.4, Yushe-YS59 is close in age to Yushe-YS58. Yushe-YS58 is correlated to C3n.2n: 4.631–4.493 Ma, which fits to the age range of †*C. mesolophidos* and is the assumed age of Yushe-YS59, here.

Cricetus Leske, 1779

The only extant species, the type species *Cricetus cricetus* is included in the analysis. Regarding extinct species, there is †*C. lophidens* from the Late Miocene or Early Pliocene. Other species from the Neogene, that have been previously assigned to *Cricetus*, were however transferred to different genera (†*Apocricetus* and †*Pseudocricetus*, Freudenthal et al., 1998). Additionally, there are 4 taxa from the Pleistocene, †*C. major* (Woldřich, 1880), †*C. nanus* (Schaub, 1930), †*C. praeglacialis* (Schaub, 1930) and †*C. runtonensis* (Newton, 1909). The status of these taxa has been discussed extensively over the last century (see summaries in Hír, 1997; Kryštufek et al., 2020). The main questions regard the species or subspecies level of the 4 taxa (Hír, 1997; Horáček & Lebedová, 2022; Sabol & Zervanová, 2014) and the status of the 2 largest forms, †*C. runtonensis* and †*C. major* (Fahlbusch, 1976; Maul & Parfitt, 2010; Pradel, 1985). Of the 6 possible total species, the two certain ones *C. cricetus* and †*C. lophidens* are included in this analysis, while the latter might belong to a different genus (see, de Bruijn et al., 2013):

Cricetus cricetus (Linnaeus, 1758)

Geographical distribution of fossil occurrences: Armenia, Austria, Belarus, Belgium, Bulgaria, Croatia, Czech Republic, France, Georgia, Germany, Hungary, Israel, Italy, Moldova, Montenegro, Poland, Romania, Russia, Serbia, Slovakia, Slovenia, Switzerland, Turkey, and Ukraine

Type locality: Germany

Extant distribution: Central and Eastern Europe, Western Siberia, Northern Kazakhstan, North-Western China

Oldest fossil occurrence: Salcia 1, 2, Moldova: 1.785–1.585 Ma: MIS 62–41 (Krokhmal & Rekovets, 2010; Krokhmal et al., 2021)

Observed material: SNSB-BSPG 1991 IV 1–95: 95 skulls with mandibles from Rhine Hesse, Germany (‘Raum Alsheim, Wintersheim, Dorn Dürkheim’). Bavarian State Collection for

Zoology, Munich (SNSB-ZSM) 1988/0061, 0068, 0069, 0072, 0076, 0080, 0095, 0107, 0108, 0111, 0120–0122, 0125, 0129, 0131–0133, 0140, 0141, 0145: 21 skulls with mandibles from the same location.

Anatomical references: (Fahlbusch, 1976; Hir, 1997; Kryštufek & Shenbrot, 2025; Pradel, 1985)

†*Cricetus lophidens* de Bruijn, Dawson & Mein, 1970

Single occurrence/type locality: Maritsa 1, Greece

Holotype: Department Aardwetenschappen, University Utrecht, MA-16, M1 (figured in de Bruijn et al. (1970), pl. 4, fig. 3 and in Vasileiadou and Sylvestrou (2022), fig. 22a)

Age: 5.4–5.23 Ma (5.315 Ma)

Observed material: FSL: 2 M1, M2, 2 M3, m1, 3 m2, m3 from Maritsa

Anatomical reference: (de Bruijn et al., 1970, 2013)

Remark: The assignment to *Cricetus* is apparently doubted by de Bruijn et al. (2013), as they put the genus in quotation marks (“*Cricetus*” *lophidens*)

The age of Maritsa 1 is determined as either Late Miocene, MN 13 (see e.g. Hordijk & de Bruijn, 2009; Vasileiadou & Sylvestrou, 2022) or Early Pliocene, MN 14 (see e.g. Koufos, 2006; Şen et al., 2017) (see also Čermák et al., 2019; Suata-Alpaslan, 2010; van der Meulen & van Kolfschoten, 1986). For absolute ages, Koufos and Vasileiadou (2015) are followed, who suggest an age of 5.4–5.23 Ma for Turolian/Ruscinian localities.

***Mesocricetus* Nehring, 1898b**

Of the 4 extant species, only *Mesocricetus raddei* (Nehring, 1894) is not included in this analysis. Regarding the extinct species, there are 2, †*M. rathgeberi* Pieper, 1984, which is however only known from one skull fragment including its upper tooth row (Pieper, 1984) and †*M. primitivus*, which is included here. Another possible species is ‘*M. aramaeus*’ Bate, 1943, which might be a synonym of *M. auratus* (see Tchernov, 1975, vs. 1996, 1998). In Erbajeva and Alexeeva (2013), ‘*Mesocricetus* cf. *primaevus*’ is mentioned (see also The NOW Community, 2023), there is however no other mentioning of this species known to the authors. For the same locality, ‘*Mesocricetus* sp.’ is listed in a more recent publication (Lopatin, 2020). A similar case is ‘*Mesocricetus armatus*’, mentioned in Turnbull (1975) (see Kryštufek & Shenbrot, 2025; Turnbull & Reed, 1974).

***Mesocricetus auratus* (Waterhouse, 1839)**

Geographical distribution of fossil occurrences: Israel, Turkey

Type locality: Aleppo, Syria

Extant distribution: around Aleppo, Syria, South-Eastern Turkey

Oldest fossil occurrence: Dursunlu, Turkey: 0.99–0.78 Ma (Güleç et al., 1997, 2009)

Observed material: Natural History Museum, London, United Kingdom (NHMUK) 1992,42: CT scan of a skull with mandible. SNSB-ZSM 1951/0062, 0162; 1952/0248; 1953/0197; 1954/0046; 1955/0160, 0160a; 1971/0613; 1972/0331: 9 skulls with mandibles bred in Munich.

Anatomical references: (Hír, 1992b; Kryštufek & Shenbrot, 2025; Kryštufek & Vohralík, 2009; Pradel, 1989; Shehab et al., 1999; Suata-Alpaslan & Dinçarslan, 2011)

Mesocricetus brandti (Nehring, 1898c)

Geographical distribution of fossil occurrences: Armenia, Georgia, Iran, Turkey

Type locality: near Tbilisi, Georgia

Extant distribution: Turkey, Armenia, Georgia, Azerbaijan, South to North-Western Iran, and Dagestan, Russia

Oldest fossil occurrence: Kaldar Cave 5, Iran: 0.071–0.057 Ma: MIS 4 (Bazgir et al., 2022; Rey-Rodríguez et al., 2020)

Observed material: SNSB-ZSM 1970/0007: 1 skull (braincase damaged), 2 maxilla fragments, 6 mandibles from a cave South-East of Divanlar, Wilkonya, Central Anatolia, Turkey

Anatomical references: (Darvish et al., 2014; Kryštufek & Shenbrot, 2025; Kryštufek & Vohralík, 2009; Rey-Rodríguez et al., 2020; Suata-Alpaslan, 2011; Yiğit, 2003)

Mesocricetus newtoni (Nehring, 1898c)

Geographical distribution of fossil occurrences: Bulgaria, Greece, Montenegro, Romania, and Serbia

Type locality: Kolarovgrad, Bulgaria

Extant distribution: South-Eastern Romania, Northern Bulgaria

Oldest fossil occurrence: Kozarnika B1 (11a), Bulgaria: 0.75–0.62 Ma (Popov, 2018; Popov & Marinska, 2007)

Anatomical references: (Bogićević et al., 2011, 2017; Chatzopoulou, 2014; Ellerman, 1941; Kryštufek & Shenbrot, 2025; Peshev et al., 2004; Piskoulis et al., 2024; Popov, 1989; Pradel, 1989)

†*Mesocricetus primitivus* de Bruijn, Dawson & Mein, 1970

Geographical distribution of occurrences: Greece, Israel, and Turkey

Type locality: Maritsa 1, Greece

Holotype: Department Aardwetenschappen, University Utrecht, MA-113, m2 (figured in de Bruijn et al. (1970), pl. 4, fig. 10 and in Vasileiadou and Sylvestrou (2022), fig. 20e)

Age range: 5.4–1.2 Ma (4.11 Ma)

Oldest occurrences: Maritsa 1, Silata (Koufos & Vasileiadou, 2015; Vasileiadou & Sylvestrou, 2022)

Youngest occurrence: Ubeidiya (Martínez-Navarro et al., 2009; Tchernov, 1996)

Observed material: FSL: 2 M1 from Ubeidiya II-23

Anatomical references: (de Bruijn et al., 1970, 2013; Şen, 1977; Skandalos et al., 2023; van den Hoek Ostende et al., 2015; Vasileiadou et al., 2012, 2003; Vasileiadou & Sylvestrou, 2022)

Nothocricetulus Lebedev, Bannikova, Neumann, Ushakova, Ivanova & Surov, 2018

Based on molecular data, Lebedev et al. (2018) erected a new genus, *Nothocricetulus*, for the former '*Cricetulus migratorius*' (see also Neumann et al., 2006). This single species of the genus is included in the analysis:

Nothocricetulus migratorius (Pallas, 1773)

Geographical distribution of fossil occurrences: Armenia, Azerbaijan, Belgium, Bosnia and Herzegovina, Bulgaria, France, Georgia, Germany, Greece, Hungary, Iran, Iraq, Israel, Italy, Kyrgyzstan, Moldova, Montenegro, Netherlands, Poland, Romania, Russia, Serbia, Slovenia, Turkey, and Ukraine

Type locality: Ural River, South-West of Orenburg, Russia

Extant distribution: Greece, Romania, Bulgaria, Southern European Russia, Kazakhstan, Southern Mongolia, Northern China, Turkey, Israel, Jordan, Lebanon, Iraq, Iran, Afghanistan, Northern India

Oldest fossil occurrence: Tourkovounia 2, Greece: 1.9–1.5 Ma (Doukas & Papayianni, 2016; Reumer & Doukas, 1985)

Observed material: SNSB-BSPG 1977 VI: 21 mandibles, 3 maxilla fragments from the surroundings of Kabul. SNSB-ZSM 1911/2253: skull with mandible from Pakistan (originally determined as *Cricetulus phaeus*).

Anatomical references: (Hír, 1993b; Kryštufek & Shenbrot, 2025; Kryštufek & Vohralík, 2009; Lebedev et al., 2018; Nedyalkov, 2016; Peshev et al., 2004; Pradel, 1981; Topachevsky & Skorik, 1992; Xie et al., 2023)

***Phodopus* Miller, 1910**

All 3 species of this genus are included in this analysis. Recently Kryštufek and Shenbrot (2025) split the genus in two, with *Phodopus campbelli* and *P. sungorus* as *Cricetiscus* Thomas, 1917, resulting in a monospecific *Phodopus* including only *P. roborovskii* (following Neumann et al., 2006). A potential fossil species, †*P. minutus* Tjutkova, 1992, is described from the Upper Pliocene of Kazakhstan but based solely on a single m1. It is therefore neglected in this study. Other fossil specimens could not be identified at species level with certainty (see, e.g. Cai et al., 2013; Wu & Flynn, 2017).

***Phodopus campbelli* (Thomas, 1905)**

Type locality: Shaborte, Mongolia (Musser & Carleton, 2005) or Zhenglanqi, Xilin Gol, Nei Mongol, China (Kryštufek & Shenbrot, 2025)

Extant distribution: Mongolia, North-Eastern China, Transbaikalia in Russia

Observed material: SNSB-BSPG 1985 I: skull with mandible from an unknown locality. MNHN 1986-597: skull with mandible from Russia.

Anatomical reference: (Kryštufek & Shenbrot, 2025; Ross, 1995)

***Phodopus roborovskii* (Satunin, 1902)**

Type locality: upper part of Scharogol-Dzhin, China

Extant distribution: Tuva in Russia, Eastern Kazakhstan, Southern and Western Mongolia, China, Eastern India

Observed material: SNSB-BSPG 1970 I 7–8: skull with mandible from the Flaming Cliffs site ('Bain-Dzak'), Mongolia. MNHN 1960-3699, 1960-3700: 2 skulls with mandibles from Northern China; 1986-1103: skull with mandible from Touva, Tere-Khol Lake, Russia; 1997-1295, 1997-1298: 2 skulls with mandibles from Nemegt, omnogobi, Mongolia.

Anatomical reference: (Argyropulo, 1933; Kryštufek & Shenbrot, 2025; Wu & Flynn, 2017)

***Phodopus sungorus* (Pallas, 1773)**

Type locality: near Grachevskiy, Kazakhstan

Extant distribution: Northern and Eastern Kazakhstan and South-Western Siberia, Russia

Observed material: SNSB-BSPG 1985 I: skull with mandibles from an unknown locality. SNSB-ZSM 1972/0207,0477; 1974/0002: 3 skulls with mandibles bred in Munich. MNHN 1848-346: skull with one hemi-mandible from Siberia, Russia; 1958-301, 1961-883: 2 skulls with mandibles from Russia; 1986-1102: skull with mandible from Dimitrievka, Siberia, Russia.

Anatomical reference: (Horáček & Lebedová, 2022; Kryštufek & Shenbrot, 2025; Topachevsky & Skorik, 1992; Wu & Flynn, 2017)

***Tscherskia* Ognev, 1914**

The type species of the genus, *Tscherskia triton*, is included in the analysis. Recently, a second extant taxon was considered as a valid species, *T. ningshaanensis* (Song, 1985), or *T. collina* (Allen, 1925) (see Jiang et al., 2024; or Kryštufek & Shenbrot, 2025). Regarding fossil specimens, ‘†*Cricetinus varians*’ is treated as a subspecies of *T. triton* (Xie et al., 2023). Additionally, several species of the genus *Cricetulus* (or *Cricetinus* respectively) were recently tentatively transferred to *Tscherskia* (see Xie et al., 2021, 2023). They are here listed under *Cricetulus*. There is one extinct species described from the Holocene of Iran, †*T. rusa* Storch, 1974. The status of the species and its genus assignment are however questionable (Xie et al., 2023).

***Tscherskia triton* (de Winton, 1899)**

Geographical distribution of fossil occurrences: China

Type locality: Northern Shandong, China

Extant distribution: Upper Ussuri, Russia; North-Eastern China, Korean Peninsula

Oldest fossil occurrence: Zhoukoudian 9, China: 1.376–0.774 Ma: late third of the Early Pleistocene (Cheng et al., 1997; Zheng, 1984b)

Observed material: MNHN 2024-2553: skull with mandible (one m3 fell out) from Xiji, Ningxia, China; 2024-1818–1830: 13 skulls with mandibles (m3s missing in 2024-1819, all third molars missing in 2024-1830) from Ban Ban Wan, Gansu, China.

Anatomical reference: (Kryštufek & Shenbrot, 2025; Xie et al., 2023)

***Urocricetus* Satunin, 1902**

The changes in the intergeneric taxonomy over the last decades are summarized in Kryštufek and Shenbrot (2025). Currently two species are recognized: the type species *Urocricetus kamensis*, which is included here, and *U. lama* (Bonhote, 1905) (see also Pan et al., 2025). To date, there is no fossil record known.

***Urocricetus kamensis* Satunin, 1902**

Type locality: Mokchu River, Tibet, China

Extant distribution: Tibet and surrounding areas, China

Observed Material: MNHN 2024-1289–1306: 18 skulls with mandibles (2024-1305 broken) from Siqhu, Sichuan, China.

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Additional file 5 of Chapter 3

Total-evidence phylogeny reveals recent crown group radiation and biogeographical history of hamsters

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Rogue taxa identification and positions

To identify taxa that are unstable in the reconstructed phylogenies, a preliminary phylogenetic analysis was run on the morphological data (Supplemental Material S2). The analysis was done in MrBayes 3.2.7a (Ronquist et al. 2012) using the Cyber-Infrastructure for Phylogenetic Research (CIPRES) Science Gateway v.3.3 (Miller et al. 2010). We used the Mkv model (Lewis 2001) with among character rate heterogeneity under a gamma distribution (Yang 1993). Following the previous analysis of older Neogene taxa (Dirnberger et al. 2024), an uncorrelated relaxed-clock model was chosen, the IGR (independent gamma rate) model. For the prior of the base clock rate, a log-normal distribution was chosen, with the mean calculated by dividing the median tree length of a preceding non-clock analysis by the median of the root age prior ($6.02/37.1 = 0.162$, in natural log scale: -1.82) (following e.g. Simões et al. 2018). For a broad standard deviation, the exponent of the mean ($e^{0.162} = 1.176$) was chosen (following Pyron 2017). The remaining settings for the tree model and constraints are the same as described in the manuscript. The analyses were run with four independent Metropolis-Coupled Markov chain Monte Carlo runs (MCMCMC) with six chains, sampling every 1,000 of 50,000,000 generations, and a burn-in of 30%.

The resulting posterior tree sample was used to identify the rogue taxa with the R package Rogue v.2.1.6 (Smith 2022, 2023) (for the used commands, see Dirnberger et al. 2024, suppl. mat. S4). The analysis resulted in eleven identified rogue taxa, all of them extinct: *Allocricetus anterolophidens*, *Apocricetus plinii*, *Cricetulodon hartenbergeri*, *Cricetulus gritzai*, *Cricetulus*

beremendensis, *Nannocricetus mongolicus*, *Neocricetodon occidentalis*, *Neocricetodon progressus*, *Pseudocricetus polgardiensis*, *Sinocricetus progressus*, *Sinocricetus zdanskyi*. The position of these taxa in the tree is shown in Figure S5.1.

Figure S5.2 shows the phylogeny resulting from the IGR clock-model after excluding the rogue taxa.

Model comparison of different biogeography evolution models

Resulting from the BioGeoBEARS analysis (Matzke 2013, 2018) described in the main text, the different models can be compared using the corrected Akaike information criterion (see Tab. S5.1). The results of the less fitting models and the estimated ancestral states depicted as pie-charts of the DEC+J model can be seen in Figures S5.3–S5.8.

Table S5.1 Log-likelihood, corrected Akaike information criterion (best in bold) and Akaike weights of each of the six tested model for ancestral biogeographical range estimations

	LnL	AICc	AICc_wt
DEC	-173.3	350.8	0.055
DEC+J	-169.4	345.3	0.9
DIVALIKE	-186.6	377.4	9.50E-08
DIVALIKE+J	-173.8	354	0.012
BAYAREALIKE	-191.2	386.5	9.90E-10
BAYAREALIKE+J	-172.6	351.6	0.037

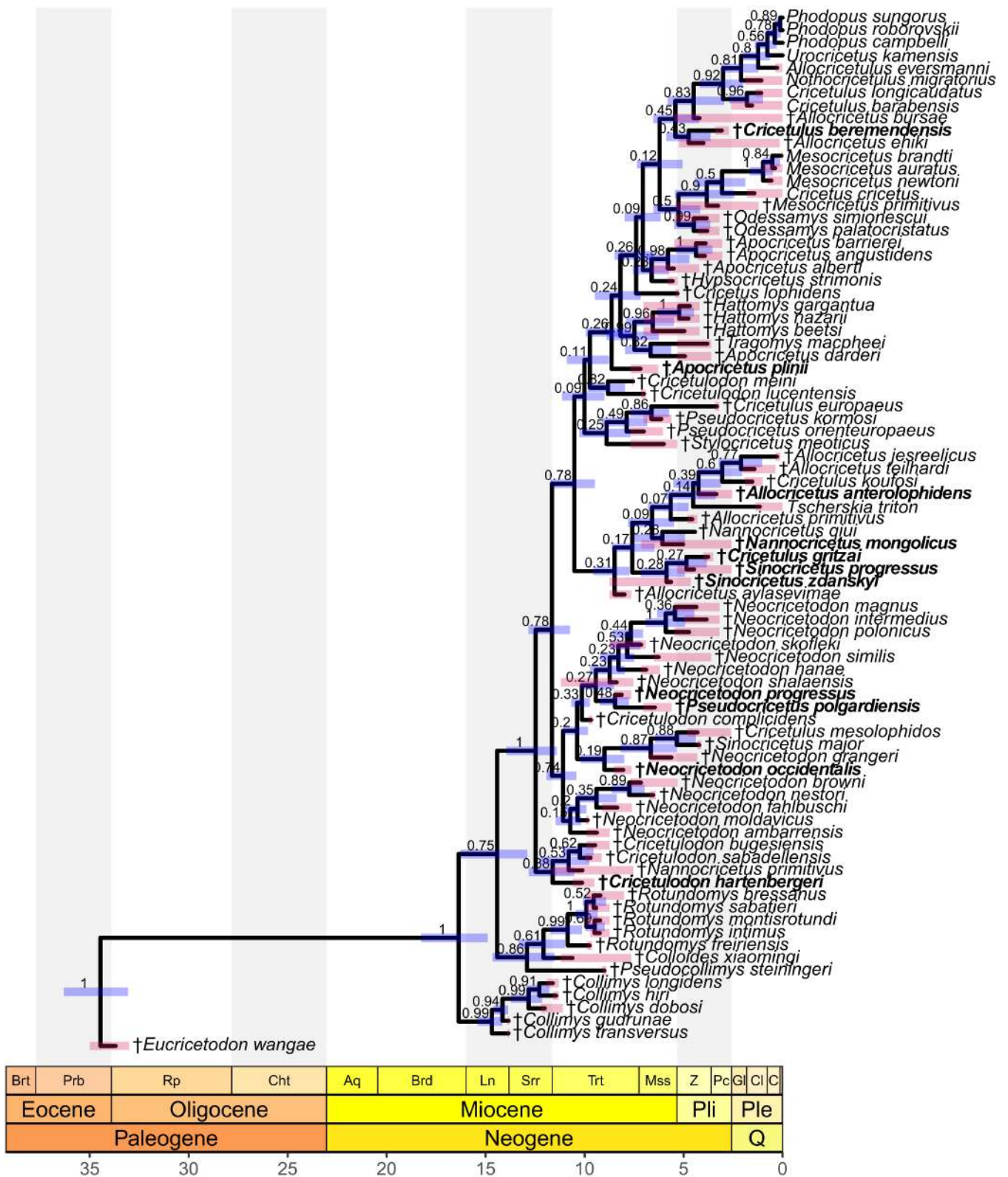


Figure S5.1. Maximum clade compatibility tree of the preliminary tip-calibrated relaxed-clock IGR Bayesian inference analysis based on the complete taxon set and only the morphological data. Identified rogue taxa shown in bold. Posterior probabilities of clades are indicated at respective nodes, blue node bars indicate the 95% highest posterior density for divergence times, red tip bars indicate the stratigraphic range of the respective taxon. The scale axis is in Ma, the chronostratigraphic chart follows Cohen et al. (2013; updated 2024/10)

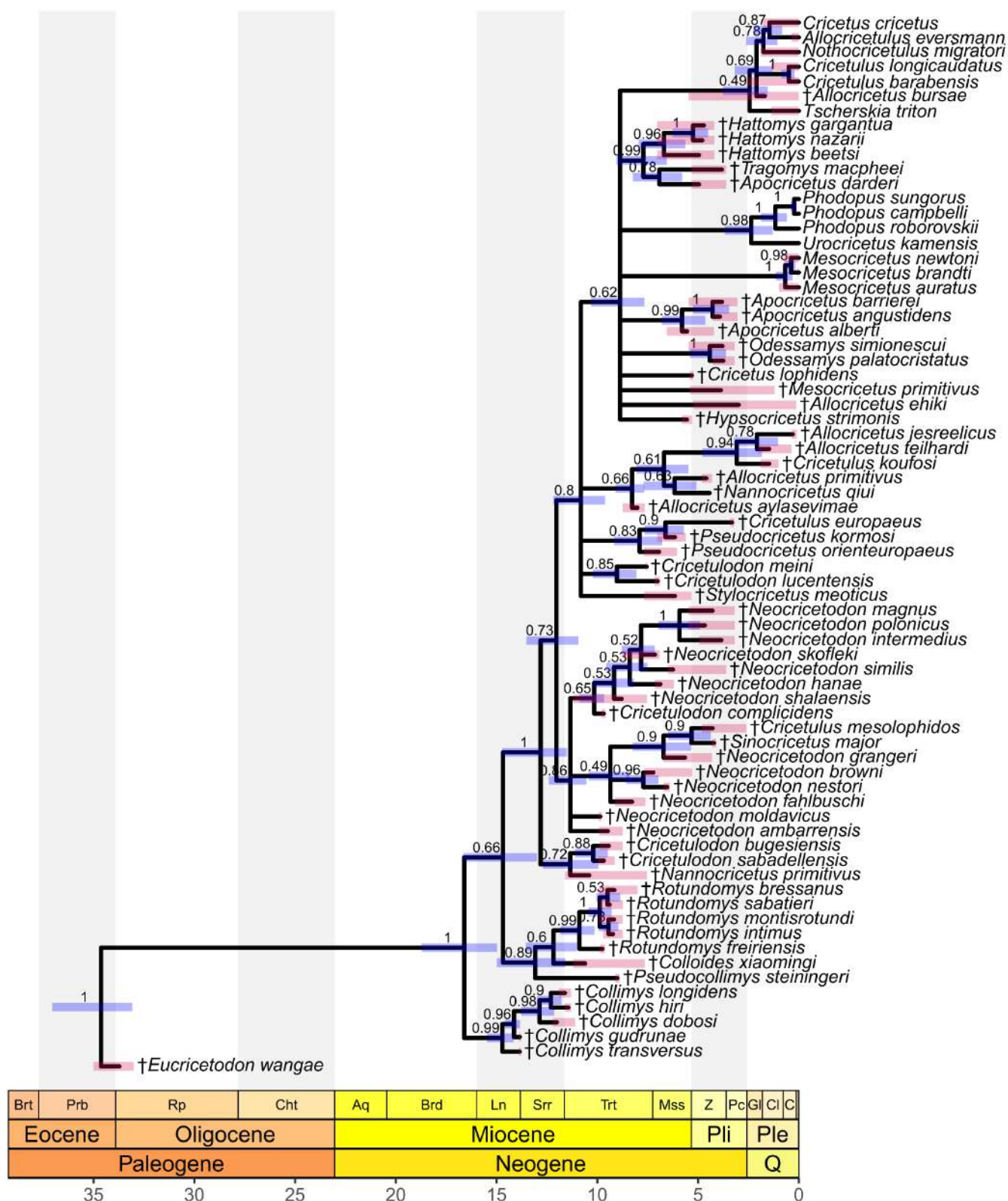


Figure S5.2 Maximum clade compatibility tree of the tip-calibrated relaxed-clock IGR Bayesian inference analysis based on the reduced taxon set and the combined data. Posterior probabilities of clades are indicated at respective nodes, blue node bars indicate the 95% highest posterior density for divergence times, red tip bars indicate the stratigraphic range of the respective taxon. The scale axis is in Ma, the chronostratigraphic chart follows Cohen et al. (2013; updated 2024/10)

ancstates: global optim, 5 areas max. d=0.1032; e=0.0616; j=0.048; LnL=-169.45

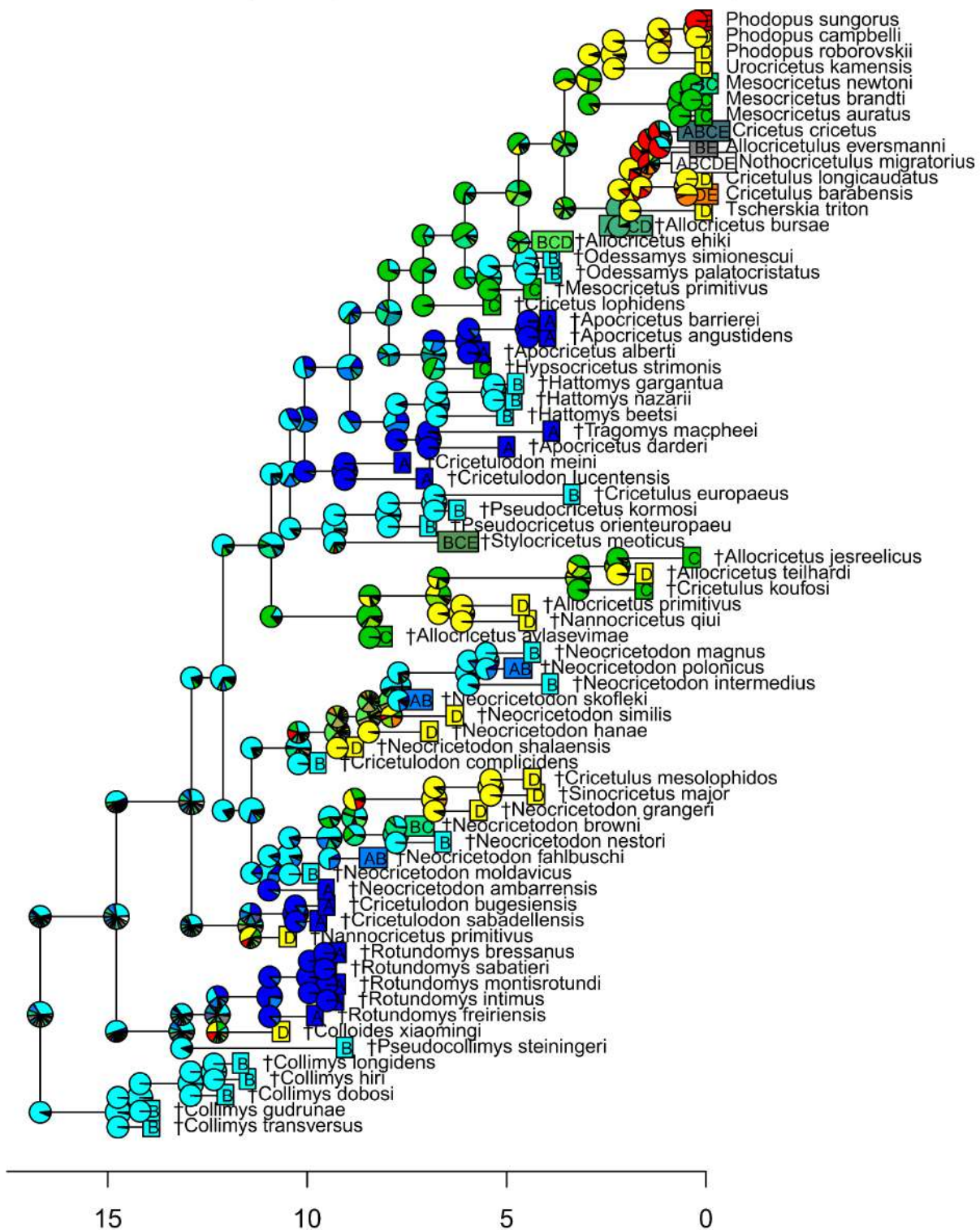


Figure S5.3 Reconstruction of the biogeographical history of Cricetinae based on the DEC+J model. Tip ranges and estimated ancestral ranges as pie-charts, are represented by colors and letters. For visual representation of the areas and most likely estimates, see Figure 2 in the main text. A: South Western Europe, B: Central and Eastern Europe, C: South Eastern Europe, South Western Asia, D: Eastern Central Asia, E: Western Central Asia. The scale axis is in Ma.

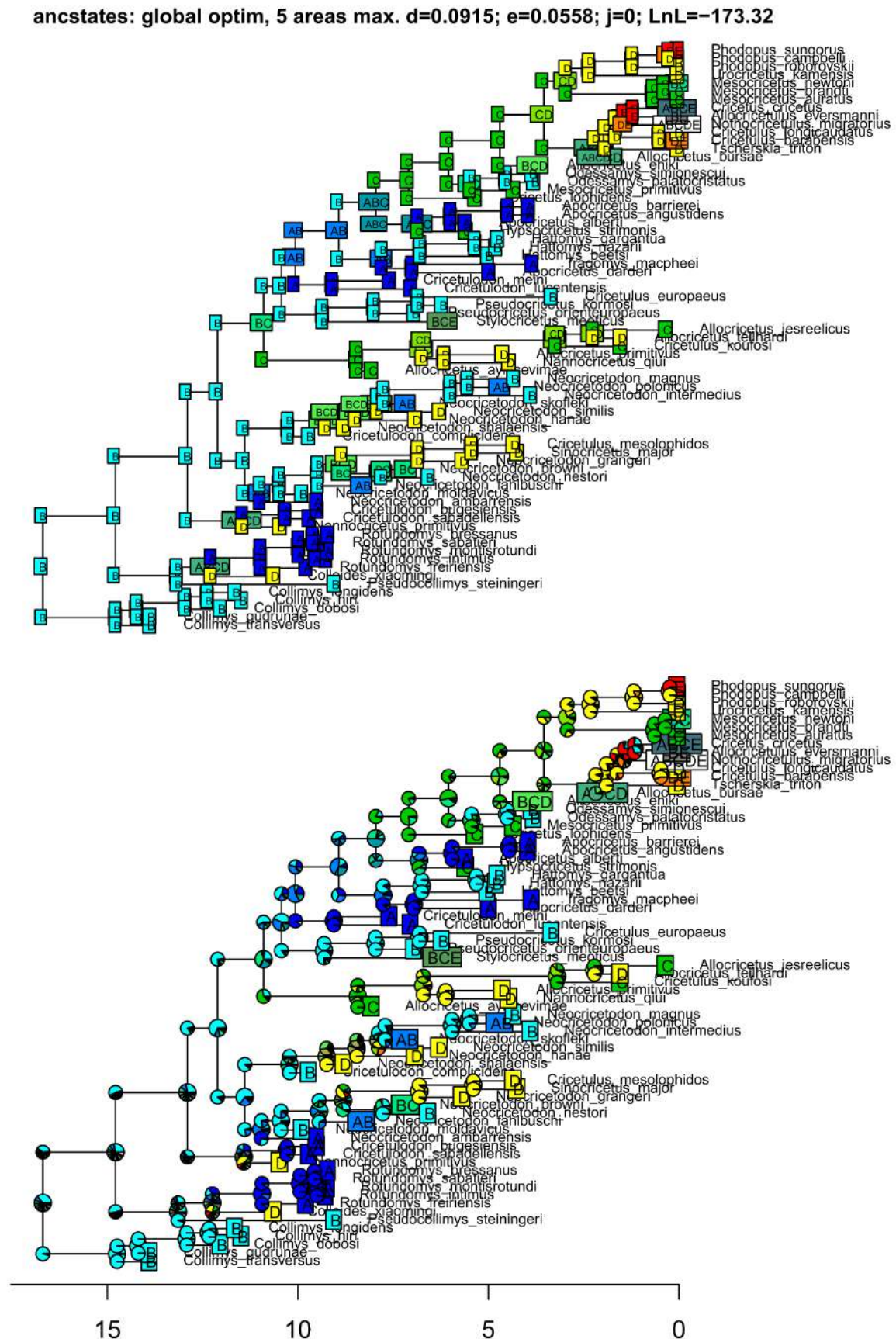


Figure S5.5 Reconstruction of the biogeographical history of Cricetinae based on the DEC model. Ancestral ranges shown as, **upper**: most likely estimate, **lower**: pie-chart. See Figure S5.3 for explanations of tip ranges and estimated ancestral ranges. Scale axis in Ma.

ancstates: global optim, 5 areas max. d=0.117; e=0.2254; j=0; LnL=-186.60

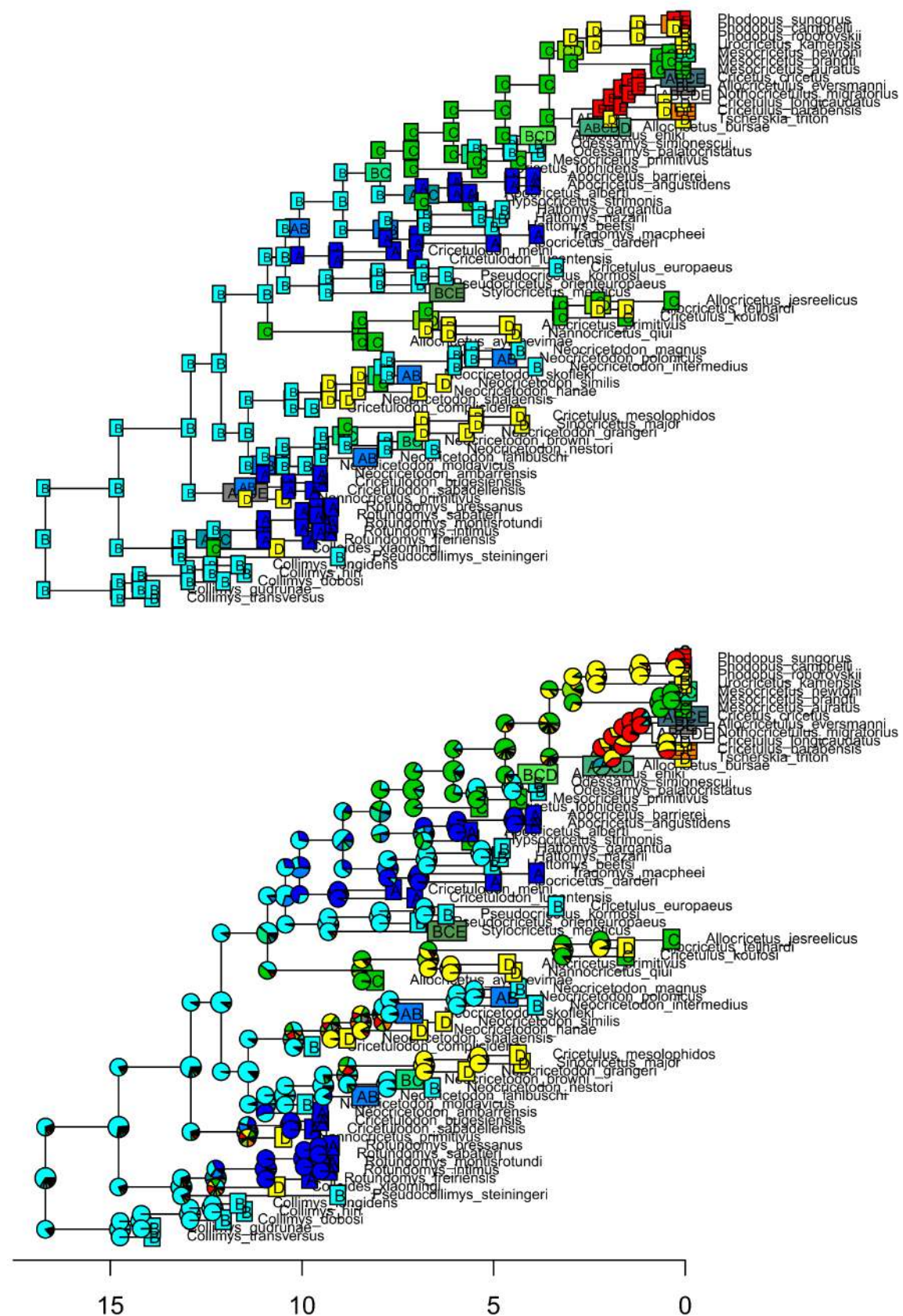


Figure S5.5 Reconstruction of the biogeographical history of Cricetinae based on the DIVALIKE model. Ancestral ranges shown as, **upper**: most likely estimate, **lower**: pie-chart. See Figure S5.3 for explanations of tip ranges and estimated ancestral ranges. Scale axis in Ma.

ancstates: global optim, 5 areas max. d=0.1347; e=0.063; j=0.0309; LnL=-173.80

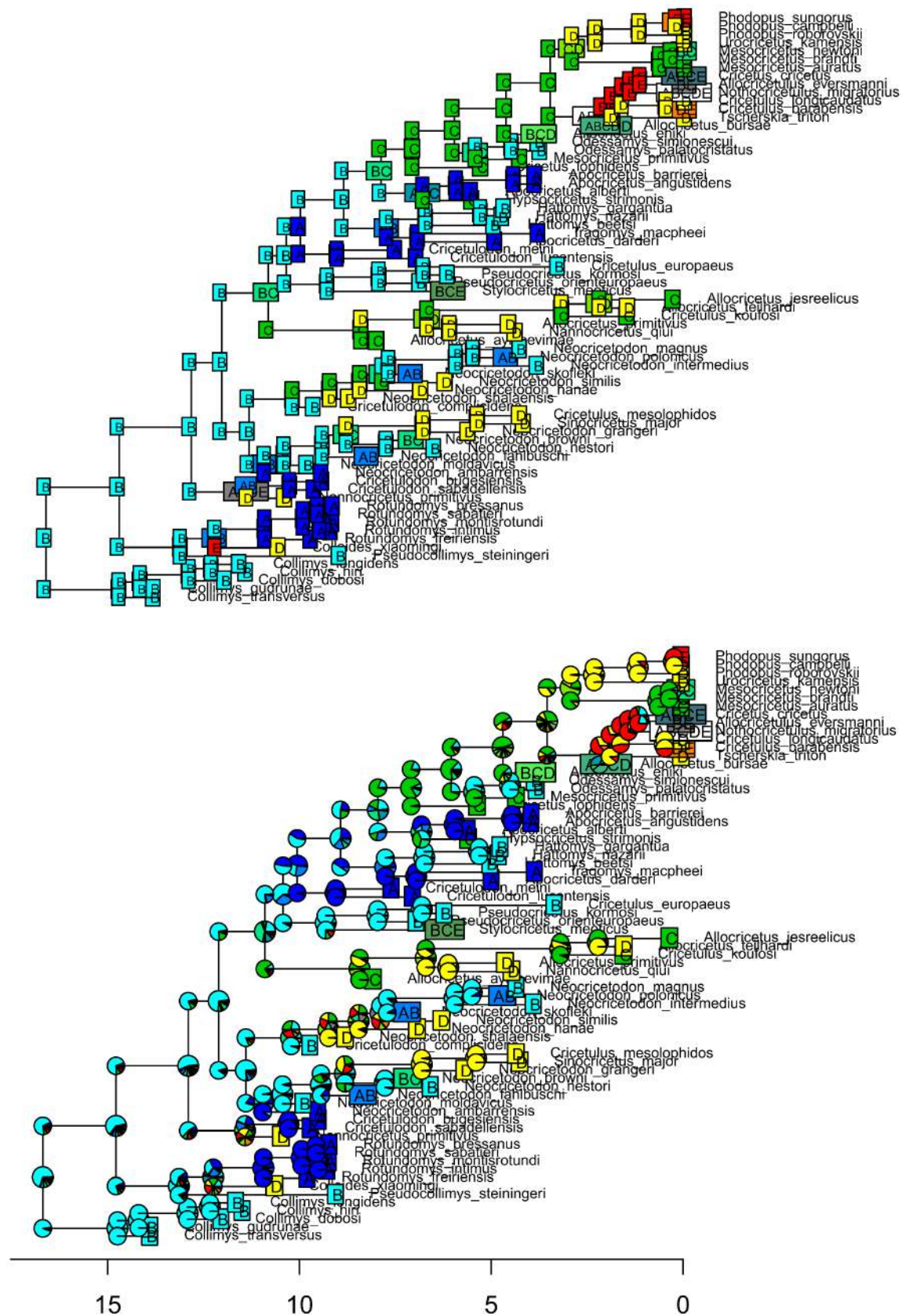


Figure S5.6 Reconstruction of the biogeographical history of Cricetinae based on the DIVALIKE+J model. Ancestral ranges shown as, **upper**: most likely estimate, **lower**: pie-chart. See Figure S5.3 for explanations of tip ranges and estimated ancestral ranges. Scale axis in Ma.

ancstates: global optim, 5 areas max. d=0.0944; e=0.3649; j=0; LnL=-191.16

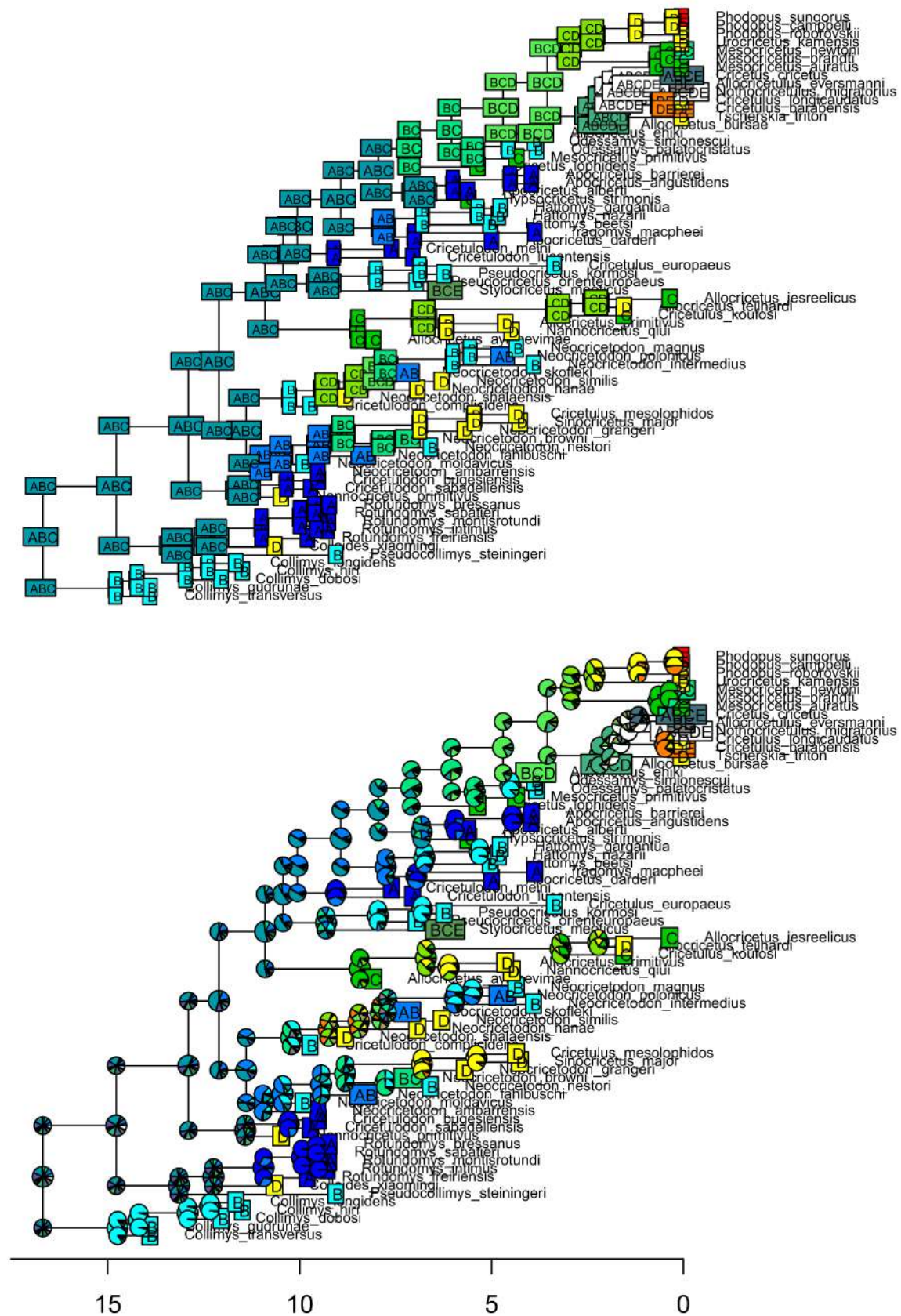
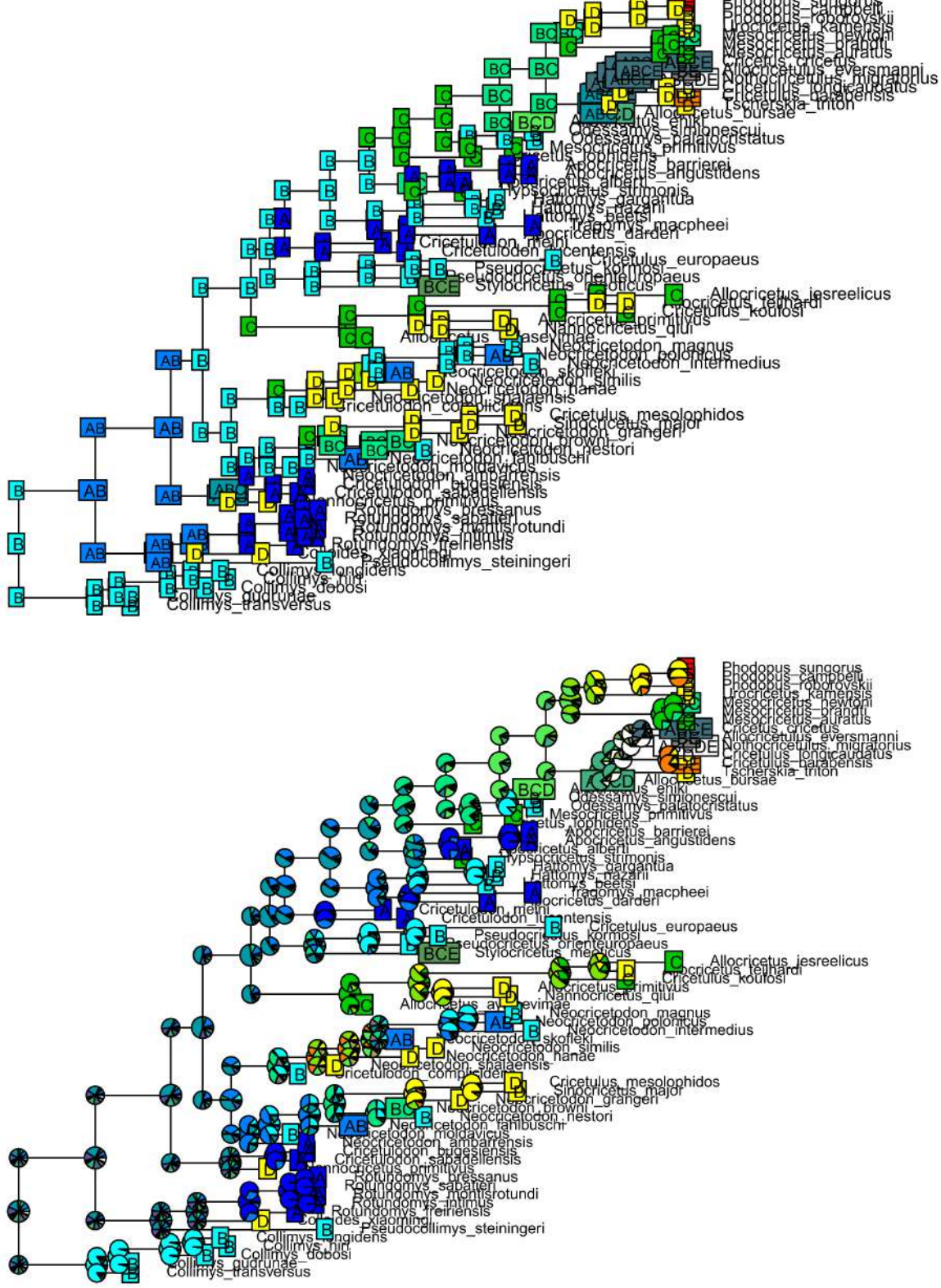


Figure S5.7 Reconstruction of the biogeographical history of Cricetinae based on the BAYAREALIKE model. Ancestral ranges shown as, **upper**: most likely estimate, **lower**: pie-chart. See Figure S5.3 for explanations of tip ranges and estimated ancestral ranges. Scale axis in Ma.

 Rhodnius sanguis



References for Additional file 5 of Chapter 3

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